

CIS CentOS Linux 7 Benchmark

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Overview

This document provides prescriptive guidance for establishing a secure configuration posture for CentOS Linux 7 systems running on x86 and x64 platforms. This document was tested against CentOS 7.9.

The guidance within broadly assumes that operations are being performed as the root user. Operations performed using sudo instead of the root user may produce unexpected results, or fail to make the intended changes to the system. Non-root users may not be able to access certain areas of the system, especially after remediation has been performed. It is advisable to verify root users path integrity and the integrity of any programs being run prior to execution of commands and scripts included in this benchmark.

To obtain the latest version of this guide, please visit <http://workbench.cisecurity.org>. If you have questions, comments, or have identified ways to improve this guide, please write us at feedback@cisecurity.org.

Intended Audience

This benchmark is intended for system and application administrators, security specialists, auditors, help desk, and platform deployment personnel who plan to develop, deploy, assess, or secure solutions that incorporate CentOS Linux 7.x on x86 or x64 platforms.

Consensus Guidance

This benchmark was created using a consensus review process comprised of subject matter experts. Consensus participants provide perspective from a diverse set of backgrounds including consulting, software development, audit and compliance, security research, operations, government, and legal.

Each CIS benchmark undergoes two phases of consensus review. The first phase occurs during initial benchmark development. During this phase, subject matter experts convene to discuss, create, and test working drafts of the benchmark. This discussion occurs until consensus has been reached on benchmark recommendations. The second phase begins after the benchmark has been published. During this phase, all feedback provided by the Internet community is reviewed by the consensus team for incorporation in the benchmark. If you are interested in participating in the consensus process, please visit <https://workbench.cisecurity.org/>.

Typographical Conventions

The following typographical conventions are used throughout this guide:

Convention	Meaning
<code>Stylized Monospace font</code>	Used for blocks of code, command, and script examples. Text should be interpreted exactly as presented.
Monospace font	Used for inline code, commands, or examples. Text should be interpreted exactly as presented.
<i><italic font in brackets></i>	Italic texts set in angle brackets denote a variable requiring substitution for a real value.
<i>Italic font</i>	Used to denote the title of a book, article, or other publication.
Note	Additional information or caveats

Assessment Status

An assessment status is included for every recommendation. The assessment status indicates whether the given recommendation can be automated or requires manual steps to implement. Both statuses are equally important and are determined and supported as defined below:

Automated

Represents recommendations for which assessment of a technical control can be fully automated and validated to a pass/fail state. Recommendations will include the necessary information to implement automation.

Manual

Represents recommendations for which assessment of a technical control cannot be fully automated and requires all or some manual steps to validate that the configured state is set as expected. The expected state can vary depending on the environment.

Profile Definitions

The following configuration profiles are defined by this Benchmark:

- **Level 1 - Server**

Items in this profile intend to:

- be practical and prudent;
- provide a clear security benefit; and
- not inhibit the utility of the technology beyond acceptable means.

This profile is intended for servers.

- **Level 2 - Server**

This profile extends the "Level 1 - Server" profile. Items in this profile exhibit one or more of the following characteristics:

- are intended for environments or use cases where security is paramount.
- acts as defense in depth measure.
- may negatively inhibit the utility or performance of the technology.

This profile is intended for servers.

- **Level 1 - Workstation**

Items in this profile intend to:

- be practical and prudent;
- provide a clear security benefit; and
- not inhibit the utility of the technology beyond acceptable means.

This profile is intended for workstations.

- **Level 2 - Workstation**

This profile extends the "Level 1 - Workstation" profile. Items in this profile exhibit one or more of the following characteristics:

- are intended for environments or use cases where security is paramount.
- acts as defense in depth measure.
- may negatively inhibit the utility or performance of the technology.

This profile is intended for workstations.

Acknowledgements

This benchmark exemplifies the great things a community of users, vendors, and subject matter experts can accomplish through consensus collaboration. The CIS community thanks the entire consensus team with special recognition to the following individuals who contributed greatly to the creation of this guide:

This benchmark is based upon previous Linux benchmarks published and would not be possible without the contributions provided over the history of all of these benchmarks. The CIS community thanks everyone who has contributed to the Linux benchmarks.

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Recommendations

1 Initial Setup

Items in this section are advised for all systems, but may be difficult or require extensive preparation after the initial setup of the system.

1.1 Filesystem Configuration

Directories that are used for system-wide functions can be further protected by placing them on separate partitions. This provides protection for resource exhaustion and enables the use of mounting options that are applicable to the directory's intended use. Users' data can be stored on separate partitions and have stricter mount options. A user partition is a filesystem that has been established for use by the users and does not contain software for system operations.

The recommendations in this section are easier to perform during initial system installation. If the system is already installed, it is recommended that a full backup be performed before repartitioning the system.

Note: If you are repartitioning a system that has already been installed, make sure the data has been copied over to the new partition, unmount it and then remove the data from the directory that was in the old partition. Otherwise it will still consume space in the old partition that will be masked when the new filesystem is mounted. For example, if a system is in single-user mode with no filesystems mounted and the administrator adds a lot of data to the `/tmp` directory, this data will still consume space in `/` once the `/tmp` filesystem is mounted unless it is removed first.

1.1.1 Disable unused filesystems

A number of uncommon filesystem types are supported under Linux. Removing support for unneeded filesystem types reduces the local attack surface of the system. If a filesystem type is not needed it should be disabled. Native Linux file systems are designed to ensure that built-in security controls function as expected. Non-native filesystems can lead to unexpected consequences to both the security and functionality of the system and should be used with caution. Many filesystems are created for niche use cases and are not maintained and supported as the operating systems are updated and patched. Users of non-native filesystems should ensure that there is attention and ongoing support for them, especially in light of frequent operating system changes.

Standard network connectivity and Internet access to cloud storage may make the use of non-standard filesystem formats to directly attach heterogeneous devices much less attractive.

Note: This should not be considered a comprehensive list of filesystems. You may wish to consider additions to those listed here for your environment.

1.1.1.1 Ensure mounting of cramfs filesystems is disabled (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `cramfs` filesystem type is a compressed read-only Linux filesystem embedded in small footprint systems. A `cramfs` image can be used without having to first decompress the image.

Rationale:

Removing support for unneeded filesystem types reduces the local attack surface of the server. If this filesystem type is not needed, disable it.

Audit:

Run the following commands and verify the output is as indicated:

```
# modprobe -n -v cramfs | grep -E '(cramfs|install)'  
install /bin/true  
  
# lsmod | grep cramfs  
<No output>
```

Remediation:

Edit or create a file in the `/etc/modprobe.d/` directory ending in `.conf`

Example: `vim /etc/modprobe.d/cramfs.conf`

and add the following line:

```
install cramfs /bin/true
```

Run the following command to unload the `cramfs` module:

```
# rmmod cramfs
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.1 Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	<u>5.1 Establish Secure Configurations</u> Maintain documented, standard security configuration standards for all authorized operating systems and software.			

1.1.1.2 Ensure mounting of squashfs filesystems is disabled (Automated)

Profile Applicability:

- Level 2 - Server
- Level 2 - Workstation

Description:

The `squashfs` filesystem type is a compressed read-only Linux filesystem embedded in small footprint systems (similar to `cramfs`). A `squashfs` image can be used without having to first decompress the image.

Rationale:

Removing support for unneeded filesystem types reduces the local attack surface of the system. If this filesystem type is not needed, disable it.

Impact:

Disabling `squashfs` will prevent the use of `snap`. `Snap` is a package manager for Linux for installing `Snap` packages.

"Snap" application packages of software are self-contained and work across a range of Linux distributions. This is unlike traditional Linux package management approaches, like `APT` or `RPM`, which require specifically adapted packages per Linux distribution on an application update and delay therefore application deployment from developers to their software's end-user. Snaps themselves have no dependency on any external store ("App store"), can be obtained from any source and can be therefore used for upstream software deployment. When snaps are deployed on versions of Linux, the Ubuntu app store is used as default back-end, but other stores can be enabled as well.

Audit:

Run the following commands and verify the output is as indicated:

```
# modprobe -n -v squashfs | grep -E '(squashfs|install)'  
  
install /bin/true  
  
# lsmod | grep squashfs  
  
<No output>
```

Remediation:

Edit or create a file in the `/etc/modprobe.d/` directory ending in `.conf`

Example: `vi /etc/modprobe.d/squashfs.conf`

and add the following line:

```
install squashfs /bin/true
```

Run the following command to unload the `squashfs` module:

```
# rmmod squashfs
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.1 Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	<u>5.1 Establish Secure Configurations</u> Maintain documented, standard security configuration standards for all authorized operating systems and software.			

1.1.1.3 Ensure mounting of udf filesystems is disabled (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `udf` filesystem type is the universal disk format used to implement ISO/IEC 13346 and ECMA-167 specifications. This is an open vendor filesystem type for data storage on a broad range of media. This filesystem type is necessary to support writing DVDs and newer optical disc formats.

Rationale:

Removing support for unneeded filesystem types reduces the local attack surface of the system. If this filesystem type is not needed, disable it.

Audit:

Run the following commands and verify the output is as indicated:

```
# modprobe -n -v udf | grep -E '(udf|install)'  
  
install /bin/true  
  
# lsmod | grep udf  
  
<No output>
```

Remediation:

Edit or create a file in the `/etc/modprobe.d/` directory ending in `.conf`

Example: `vi /etc/modprobe.d/udf.conf`

and add the following line:

```
install udf /bin/true
```

Run the following command to unload the `udf` module:

```
# rmmod udf
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.1 Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	<u>5.1 Establish Secure Configurations</u> Maintain documented, standard security configuration standards for all authorized operating systems and software.			

1.1.2 Ensure /tmp is configured (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `/tmp` directory is a world-writable directory used for temporary storage by all users and some applications.

Rationale:

Making `/tmp` its own file system allows an administrator to set the `noexec` option on the mount, making `/tmp` useless for an attacker to install executable code. It would also prevent an attacker from establishing a hardlink to a system `setuid` program and wait for it to be updated. Once the program was updated, the hardlink would be broken and the attacker would have his own copy of the program. If the program happened to have a security vulnerability, the attacker could continue to exploit the known flaw.

This can be accomplished by either mounting `tmpfs` to `/tmp`, or creating a separate partition for `/tmp`.

Impact:

Since the `/tmp` directory is intended to be world-writable, there is a risk of resource exhaustion if it is not bound to a separate partition.

Running out of `/tmp` space is a problem regardless of what kind of filesystem lies under it, but in a default installation a disk-based `/tmp` will essentially have the whole disk available, as it only creates a single `/` partition. On the other hand, a RAM-based `/tmp` as with `tmpfs` will almost certainly be much smaller, which can lead to applications filling up the filesystem much more easily.

Audit:

Run the following command and verify output shows `/tmp` to `tmpfs` or a system partition:

```
# findmnt -n /tmp

/tmp    tmpfs    tmpfs    rw,nosuid,nodev,noexec
```

If `/etc/fstab` is used: run the following command and verify that `tmpfs` has been mounted to `tmpfs`, or a system partition has been created for `/tmp`

```
# grep -E '\s/tmp\s' /etc/fstab | grep -E -v '^\s*#'

tmpfs    /tmp    tmpfs    defaults,noexec,nosuid,nodev 0    0
```

OR If `systemd tmp.mount` file is used: run the following command and verify that `tmp.mount` is enabled:

```
# systemctl show "tmp.mount" | grep -i unitfilestate

UnitFileState=enabled
```

Remediation:

Create or update an entry for `/tmp` in either `/etc/fstab` **OR** in a `systemd tmp.mount` file:

If `/etc/fstab` is used: configure `/etc/fstab` as appropriate.

Example:

```
tmpfs    /tmp    tmpfs    defaults,rw,nosuid,nodev,noexec,relatime 0 0
```

Run the following command to remount `/tmp`

```
# mount -o remount,noexec,nodev,nosuid /tmp
```

OR if `systemd tmp.mount` file is used: run the following command to create the file `/etc/systemd/system/tmp.mount` if it doesn't exist:

```
# [ ! -f /etc/systemd/system/tmp.mount ] && cp -v
/usr/lib/systemd/system/tmp.mount /etc/systemd/system/
```

Edit the file `/etc/systemd/system/tmp.mount`:

```
[Mount]
What=tmpfs
Where=/tmp
Type=tmpfs
Options=mode=1777,strictatime,noexec,nodev,nosuid
```

Run the following command to reload the systemd daemon:

```
# systemctl daemon-reload
```

Run the following command to unmask and start `tmp.mount`:

```
# systemctl --now unmask tmp.mount
```

References:

1. AJ Lewis, "LVM HOWTO", <http://tldp.org/HOWTO/LVM-HOWTO/>
2. <https://www.freedesktop.org/wiki/Software/systemd/APIFileSystems/>

Additional Information:

- If an entry for `/tmp` exists in `/etc/fstab` it will take precedence over entries in the `tmp.mount` file.
- `tmpfs` can be resized using the `size={size}` parameter in `/etc/fstab` or on the Options line in the `tmp.mount` file. If we don't specify the size, it will be half the RAM.

Resize tmpfs examples:

/etc/fstab example:

tmpfs	/tmp	tmpfs	rw,noexec,nodev,nosuid,size=2G	0	0
-------	------	-------	--------------------------------	---	---

tmp.mount example:

```
[Mount]
What=tmpfs
Where=/tmp
Type=tmpfs
Options=mode=1777,strictatime,size=2G,noexec,nodev,nosuid
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 <u>Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	9.4 <u>Apply Host-based Firewalls or Port Filtering</u> Apply host-based firewalls or port filtering tools on end systems, with a default-deny rule that drops all traffic except those services and ports that are explicitly allowed.			
v7	13 <u>Data Protection</u> Data Protection			

1.1.3 Ensure noexec option set on /tmp partition (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `noexec` mount option specifies that the filesystem cannot contain executable binaries.

Rationale:

Since the `/tmp` filesystem is only intended for temporary file storage, set this option to ensure that users cannot run executable binaries from `/tmp`.

Audit:

Run the following command to verify the `noexec` option is set:

```
# findmnt -n /tmp | grep -Ev '\bnodev\b'
```

Nothing should be returned

Remediation:

Edit the `/etc/fstab` file **OR** the `/etc/systemd/system/local-fs.target.wants/tmp.mount` file:

IF `/etc/fstab` is used to mount `/tmp`

Edit the `/etc/fstab` file and add `noexec` to the fourth field (mounting options) for the `/tmp` partition. See the `fstab(5)` manual page for more information.

Run the following command to remount `/tmp`:

```
# mount -o remount,noexec /tmp
```

OR if systemd is used to mount `/tmp`:

Edit `/etc/systemd/system/local-fs.target.wants/tmp.mount` to add `noexec` to the `/tmp` mount options:

```
[Mount]
Options=mode=1777,strictatime,noexec,nodev,nosuid
```

Run the following command to restart the systemd daemon:

```
# systemctl daemon-reload
```

Run the following command to restart `tmp.mount`

```
# systemctl restart tmp.mount
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 Establish and Maintain a Secure Configuration Process Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	2.6 Address unapproved software Ensure that unauthorized software is either removed or the inventory is updated in a timely manner			

1.1.4 Ensure nodev option set on /tmp partition (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `nodev` mount option specifies that the filesystem cannot contain special devices.

Rationale:

Since the `/tmp` filesystem is not intended to support devices, set this option to ensure that users cannot attempt to create block or character special devices in `/tmp`.

Audit:

Run the following command and verify the `nodev` option is set:

```
# findmnt -n /tmp -n | grep -Ev '\bnodev\b'
```

Nothing should be returned

Remediation:

Edit the `/etc/fstab` file **OR** the `/etc/systemd/system/local-fs.target.wants/tmp.mount` file:

IF `/etc/fstab` is used to mount `/tmp`

Edit the `/etc/fstab` file and add `nodev` to the fourth field (mounting options) for the `/tmp` partition. See the `fstab(5)` manual page for more information.

Run the following command to remount `/tmp`:

```
# mount -o remount,nodev /tmp
```

OR if `systemd` is used to mount `/tmp`:

Edit `/etc/systemd/system/local-fs.target.wants/tmp.mount` to add `nodev` to the `/tmp` mount options:

```
[Mount]
Options=mode=1777,strictatime,noexec,nodev,nosuid
```

Run the following command to restart the `systemd` daemon:

```
# systemctl daemon-reload
```

Run the following command to restart `tmp.mount`

```
# systemctl restart tmp.mount
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 Establish and Maintain a Secure Configuration Process Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	5.1 Establish Secure Configurations Maintain documented, standard security configuration standards for all authorized operating systems and software.			

1.1.5 Ensure nosuid option set on /tmp partition (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `nosuid` mount option specifies that the filesystem cannot contain `setuid` files.

Rationale:

Since the `/tmp` filesystem is only intended for temporary file storage, set this option to ensure that users cannot create `setuid` files in `/tmp`.

Audit:

Run the following command and verify the `nodev` option is set:

```
# findmnt -n /tmp -n | grep -Ev '\bnosuid\b'
```

Nothing should be returned

Remediation:

IF /etc/fstab is used to mount /tmp

Edit the `/etc/fstab` file and add `nosuid` to the fourth field (mounting options) for the `/tmp` partition. See the `fstab(5)` manual page for more information.

Run the following command to remount `/tmp` :

```
# mount -o remount,nosuid /tmp
```

OR if systemd is used to mount /tmp:

Edit `/etc/systemd/system/local-fs.target.wants/tmp.mount` to add `nosuid` to the `/tmp` mount options:

```
[Mount]
Options=mode=1777,strictatime,noexec,nodev,nosuid
```

Run the following command to restart the `systemd` daemon:

```
# systemctl daemon-reload
```

Run the following command to restart `tmp.mount`:

```
# systemctl restart tmp.mount
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 <u>Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v8	4.1 <u>Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	5.1 <u>Establish Secure Configurations</u> Maintain documented, standard security configuration standards for all authorized operating systems and software.			
v7	13 <u>Data Protection</u> Data Protection			

1.1.6 Ensure /dev/shm is configured (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

/dev/shm is a traditional shared memory concept. One program will create a memory portion, which other processes (if permitted) can access. Mounting `tmpfs` at /dev/shm is handled automatically by `systemd`.

Rationale:

Any user can upload and execute files inside the /dev/shm similar to the /tmp partition. Configuring /dev/shm allows an administrator to set the `noexec` option on the mount, making /dev/shm useless for an attacker to install executable code. It would also prevent an attacker from establishing a hardlink to a system `setuid` program and wait for it to be updated. Once the program was updated, the hardlink would be broken and the attacker would have his own copy of the program. If the program happened to have a security vulnerability, the attacker could continue to exploit the known flaw.

Audit:

Run the following command and verify output shows /dev/shm is mounted:

```
# findmnt -n /dev/shm
tmpfs on /dev/shm type tmpfs (rw,nosuid,nodev,noexec,relatime,seclabel)
```

Run the following command and verify an entry for /dev/shm exists in /etc/fstab:

```
# grep -E '\s/dev/shm\s' /etc/fstab
tmpfs      /dev/shm   tmpfs      defaults,noexec,nodev,nosuid 0 0
```

Remediation:

Edit `/etc/fstab` and add or edit the following line:

```
tmpfs /dev/shm tmpfs defaults,noexec,nodev,nosuid,seclabel 0 0
```

Run the following command to remount `/dev/shm`:

```
# mount -o remount,noexec,nodev,nosuid /dev/shm
```

Additional Information:

- An entry for `/dev/shm` in `/etc/fstab` will take precedence.
- `tmpfs` can be resized using the `size={size}` parameter in `/etc/fstab`. If we don't specify the size, it will be half the RAM.

Resize `tmpfs` example:

```
tmpfs /dev/shm tmpfs defaults,noexec,nodev,nosuid,size=2G 0 0
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 <u>Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v8	4.1 <u>Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	5.1 <u>Establish Secure Configurations</u> Maintain documented, standard security configuration standards for all authorized operating systems and software.			
v7	13 <u>Data Protection</u> Data Protection			

1.1.7 Ensure noexec option set on /dev/shm partition (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `noexec` mount option specifies that the filesystem cannot contain executable binaries.

Rationale:

Setting this option on a file system prevents users from executing programs from shared memory. This deters users from introducing potentially malicious software on the system.

Audit:

Run the following command to verify that the `noexec` option is set:

```
# findmnt -n /dev/shm | grep -Ev '\bnoexec\b'
```

Nothing should be returned

Remediation:

Edit the `/etc/fstab` file and add `noexec` to the fourth field (mounting options) for the `/dev/shm` partition. See the `fstab(5)` manual page for more information.

Run the following command to remount `/dev/shm`:

```
# mount -o remount,noexec,nodev,nosuid /dev/shm
```

Additional Information:

`/dev/shm` is mounted automatically by `systemd`. `/dev/shm` needs to be added to `/etc/fstab` to add mount options even though it is already being mounted on boot.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>3.3 Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v8	<u>4.1 Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	<u>2.6 Address unapproved software</u> Ensure that unauthorized software is either removed or the inventory is updated in a timely manner			
v7	<u>13 Data Protection</u> Data Protection			

1.1.8 Ensure nodev option set on /dev/shm partition (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `nodev` mount option specifies that the filesystem cannot contain special devices.

Rationale:

Since the `/dev/shm` filesystem is not intended to support devices, set this option to ensure that users cannot attempt to create special devices in `/dev/shm` partitions.

Audit:

Run the following command to verify that the `nodev` option is set:

```
# findmnt -n /dev/shm | grep -Ev '\bnodev\b'
```

Nothing should be returned

Remediation:

Edit the `/etc/fstab` file and add `nodev` to the fourth field (mounting options) for the `/dev/shm` partition. See the `fstab(5)` manual page for more information.

Run the following command to remount `/dev/shm`:

```
# mount -o remount,noexec,nodev,nosuid /dev/shm
```

Additional Information:

`/dev/shm` is mounted automatically by `systemd`. `/dev/shm` needs to be added to `/etc/fstab` to add mount options even though it is already being mounted on boot.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>3.3 Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v8	<u>4.1 Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	<u>5.1 Establish Secure Configurations</u> Maintain documented, standard security configuration standards for all authorized operating systems and software.			
v7	<u>13 Data Protection</u> Data Protection			

1.1.9 Ensure nosuid option set on /dev/shm partition (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `nosuid` mount option specifies that the filesystem cannot contain `setuid` files.

Rationale:

Setting this option on a file system prevents users from introducing privileged programs onto the system and allowing non-root users to execute them.

Audit:

Run the following command to verify that the `nosuid` option is set:

```
# findmnt -n /dev/shm | grep -Ev '\bnosuid\b'
```

Nothing should be returned

Remediation:

Edit the `/etc/fstab` file and add `nosuid` to the fourth field (mounting options) for the `/dev/shm` partition. See the `fstab(5)` manual page for more information.

Run the following command to remount `/dev/shm`:

```
# mount -o remount,noexec,nodev,nosuid /dev/shm
```

Additional Information:

`/dev/shm` is mounted automatically by `systemd`. `/dev/shm` needs to be added to `/etc/fstab` to add mount options even though it is already being mounted on boot.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>3.3 Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v8	<u>4.1 Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	<u>5.1 Establish Secure Configurations</u> Maintain documented, standard security configuration standards for all authorized operating systems and software.			
v7	<u>13 Data Protection</u> Data Protection			

1.1.10 Ensure separate partition exists for /var (Automated)

Profile Applicability:

- Level 2 - Server
- Level 2 - Workstation

Description:

The `/var` directory is used by daemons and other system services to temporarily store dynamic data. Some directories created by these processes may be world-writable.

Rationale:

Since the `/var` directory may contain world-writable files and directories, there is a risk of resource exhaustion if it is not bound to a separate partition.

Impact:

Resizing filesystems is a common activity in cloud-hosted servers. Separate filesystem partitions may prevent successful resizing, or may require the installation of additional tools solely for the purpose of resizing operations. The use of these additional tools may introduce their own security considerations.

Audit:

Run the following command and verify output shows `/var` is mounted:

```
# findmnt /var
TARGET SOURCE      FSTYPE OPTIONS
/var <device> <fstype> rw,relatime,attr2,inode64,noquota
```

Remediation:

For new installations, during installation create a custom partition setup and specify a separate partition for `/var`

For systems that were previously installed, create a new partition and configure `/etc/fstab` as appropriate.

References:

1. AJ Lewis, "LVM HOWTO", <http://tldp.org/HOWTO/LVM-HOWTO/>

Additional Information:

When modifying `/var` it is advisable to bring the system to emergency mode (so `auditd` is not running), rename the existing directory, mount the new file system, and migrate the data over before returning to multiuser mode.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 <u>Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	5.1 <u>Establish Secure Configurations</u> Maintain documented, standard security configuration standards for all authorized operating systems and software.			

1.1.11 Ensure separate partition exists for /var/tmp (Automated)

Profile Applicability:

- Level 2 - Server
- Level 2 - Workstation

Description:

The `/var/tmp` directory is a world-writable directory used for temporary storage by all users and some applications and is intended for temporary files that are preserved across reboots.

Rationale:

Since the `/var/tmp` directory is intended to be world-writable, there is a risk of resource exhaustion if it is not bound to a separate partition. In addition, making `/var/tmp` its own file system allows an administrator to set the `noexec` option on the mount, making `/var/tmp` useless for an attacker to install executable code. It would also prevent an attacker from establishing a hardlink to a system `setuid` program and wait for it to be updated. Once the program was updated, the hardlink would be broken and the attacker would have his own copy of the program. If the program happened to have a security vulnerability, the attacker could continue to exploit the known flaw.

Impact:

Resizing filesystems is a common activity in cloud-hosted servers. Separate filesystem partitions may prevent successful resizing, or may require the installation of additional tools solely for the purpose of resizing operations. The use of these additional tools may introduce their own security considerations.

Audit:

Run the following command and verify output shows `/var/tmp` is mounted:

```
# findmnt /var/tmp
```

TARGET	SOURCE	FSTYPE	OPTIONS
/var/tmp	<device>	<fstype>	rw,relatime,attr2,inode64,noquota

Remediation:

For new installations, during installation create a custom partition setup and specify a separate partition for `/var/tmp`

For systems that were previously installed, create a new partition and configure `/etc/fstab` as appropriate.

Additional Information:

- `tmpfs` should not be used for `/var/tmp/`
- `tmpfs` is a temporary filesystem that resides in memory and/or swap partition(s)
- Files in `tmpfs` are automatically cleared at each bootup

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 <u>Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v8	4.1 <u>Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	5.1 <u>Establish Secure Configurations</u> Maintain documented, standard security configuration standards for all authorized operating systems and software.			
v7	13 <u>Data Protection</u> Data Protection			

1.1.12 Ensure /var/tmp partition includes the noexec option (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `noexec` mount option specifies that the filesystem cannot contain executable binaries.

Rationale:

Since the `/var/tmp` filesystem is only intended for temporary file storage, set this option to ensure that users cannot run executable binaries from `/var/tmp`.

Audit:

If a `/var/tmp` partition exists, run the following command to verify that the `noexec` option is set:

```
# findmnt -n /var/tmp | grep -Ev '\bnoexec\b'
```

Nothing should be returned

Remediation:

For existing `/var/tmp` partitions, edit the `/etc/fstab` file and add `noexec` to the fourth field (mounting options) of the `/var/tmp` entry. See the `fstab(5)` manual page for more information.

Run the following command to remount `/var/tmp`:

```
# mount -o remount,noexec /var/tmp
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>3.3 Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v8	<u>4.1 Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	<u>2.6 Address unapproved software</u> Ensure that unauthorized software is either removed or the inventory is updated in a timely manner			

1.1.13 Ensure /var/tmp partition includes the nodev option (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `nodev` mount option specifies that the filesystem cannot contain special devices.

Rationale:

Since the `/var/tmp` filesystem is not intended to support devices, set this option to ensure that users cannot attempt to create block or character special devices in `/var/tmp`.

Audit:

If a `/var/tmp` partition exists, run the following command to verify that the `nodev` option is set:

```
# findmnt -n /var/tmp | grep -Ev '\bnodev\b'
```

Nothing should be returned

Remediation:

For existing `/var/tmp` partitions, edit the `/etc/fstab` file and add `nodev` to the fourth field (mounting options) of the `/var/tmp` entry. See the `fstab(5)` manual page for more information.

Run the following command to remount `/var/tmp`:

```
# mount -o remount,nodev /var/tmp
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 <u>Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v8	4.1 <u>Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	5.1 <u>Establish Secure Configurations</u> Maintain documented, standard security configuration standards for all authorized operating systems and software.			
v7	13 <u>Data Protection</u> Data Protection			

1.1.14 Ensure /var/tmp partition includes the nosuid option (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `nosuid` mount option specifies that the filesystem cannot contain `setuid` files.

Rationale:

Since the `/var/tmp` filesystem is only intended for temporary file storage, set this option to ensure that users cannot create `setuid` files in `/var/tmp`.

Audit:

If a `/var/tmp` partition exists, run the following command to verify that the `nosuid` option is set:

```
# findmnt -n /var/tmp | grep -Ev '\bnosuid\b'
```

Nothing should be returned

Remediation:

For existing `/var/tmp` partitions, edit the `/etc/fstab` file and add `nosuid` to the fourth field (mounting options) of the `/var/tmp` entry. See the `fstab(5)` manual page for more information.

Run the following command to remount `/var/tmp`:

```
# mount -o remount,nosuid /var/tmp
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>3.3 Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v8	<u>4.1 Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	<u>5.1 Establish Secure Configurations</u> Maintain documented, standard security configuration standards for all authorized operating systems and software.			
v7	<u>13 Data Protection</u> Data Protection			

1.1.15 Ensure separate partition exists for /var/log (Automated)

Profile Applicability:

- Level 2 - Server
- Level 2 - Workstation

Description:

The `/var/log` directory is used by system services to store log data.

Rationale:

There are two important reasons to ensure that system logs are stored on a separate partition: protection against resource exhaustion (since logs can grow quite large) and protection of audit data.

Impact:

Resizing filesystems is a common activity in cloud-hosted servers. Separate filesystem partitions may prevent successful resizing, or may require the installation of additional tools solely for the purpose of resizing operations. The use of these additional tools may introduce their own security considerations.

Audit:

Run the following command and verify output shows `/var/log` is mounted:

```
# findmnt /var/log

TARGET    SOURCE    FSTYPE    OPTIONS
/var/log  <device>  <fstype>  rw,relatime,attr2,inode64,noquota
```

Remediation:

For new installations, during installation create a custom partition setup and specify a separate partition for `/var/log`.

For systems that were previously installed, create a new partition and configure `/etc/fstab` as appropriate.

References:

1. AJ Lewis, "LVM HOWTO", <http://tldp.org/HOWTO/LVM-HOWTO/>

Additional Information:

When modifying `/var/log` it is advisable to bring the system to emergency mode (so auditd is not running), rename the existing directory, mount the new file system, and migrate the data over before returning to multiuser mode.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 <u>Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v8	8.3 <u>Ensure Adequate Audit Log Storage</u> Ensure that logging destinations maintain adequate storage to comply with the enterprise's audit log management process.			
v7	6.4 <u>Ensure adequate storage for logs</u> Ensure that all systems that store logs have adequate storage space for the logs generated.			

1.1.16 Ensure separate partition exists for /var/log/audit (Automated)

Profile Applicability:

- Level 2 - Server
- Level 2 - Workstation

Description:

The auditing daemon, `auditd`, stores log data in the `/var/log/audit` directory.

Rationale:

There are two important reasons to ensure that data gathered by `auditd` is stored on a separate partition: protection against resource exhaustion (since the `audit.log` file can grow quite large) and protection of audit data. The audit daemon calculates how much free space is left and performs actions based on the results. If other processes (such as `syslog`) consume space in the same partition as `auditd`, it may not perform as desired.

Impact:

Resizing filesystems is a common activity in cloud-hosted servers. Separate filesystem partitions may prevent successful resizing, or may require the installation of additional tools solely for the purpose of resizing operations. The use of these additional tools may introduce their own security considerations.

Audit:

Run the following command and verify output shows `/var/log/audit` is mounted:

```
# findmnt /var/log/audit

TARGET          SOURCE      FSTYPE     OPTIONS
/var/log/audit <device> <fstype>  rw,relatime,attr2,inode64,noquota
```

Remediation:

For new installations, during installation create a custom partition setup and specify a separate partition for `/var/log/audit`.

For systems that were previously installed, create a new partition and configure `/etc/fstab` as appropriate.

References:

1. AJ Lewis, "LVM HOWTO", <http://tldp.org/HOWTO/LVM-HOWTO/>

Additional Information:

When modifying `/var/log/audit` it is advisable to bring the system to emergency mode (so `auditd` is not running), rename the existing directory, mount the new file system, and migrate the data over before returning to multiuser mode.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.3 <u>Ensure Adequate Audit Log Storage</u> Ensure that logging destinations maintain adequate storage to comply with the enterprise's audit log management process.			
v7	6.4 <u>Ensure adequate storage for logs</u> Ensure that all systems that store logs have adequate storage space for the logs generated.			

1.1.17 Ensure separate partition exists for /home (Automated)

Profile Applicability:

- Level 2 - Server
- Level 2 - Workstation

Description:

The `/home` directory is used to support disk storage needs of local users.

Rationale:

If the system is intended to support local users, create a separate partition for the `/home` directory to protect against resource exhaustion and restrict the type of files that can be stored under `/home`.

Impact:

Resizing filesystems is a common activity in cloud-hosted servers. Separate filesystem partitions may prevent successful resizing, or may require the installation of additional tools solely for the purpose of resizing operations. The use of these additional tools may introduce their own security considerations.

Audit:

Run the following command and verify output shows `/home` is mounted:

```
# findmnt /home

TARGET SOURCE      FSTYPE  OPTIONS
/home  <device> <fstype> rw,relatime,attr2,inode64,noquota
```

Remediation:

For new installations, during installation create a custom partition setup and specify a separate partition for `/home`.

For systems that were previously installed, create a new partition and configure `/etc/fstab` as appropriate.

References:

1. AJ Lewis, "LVM HOWTO", <http://tldp.org/HOWTO/LVM-HOWTO/>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.1 Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	<u>5.1 Establish Secure Configurations</u> Maintain documented, standard security configuration standards for all authorized operating systems and software.			
v7	<u>13 Data Protection</u> Data Protection			

1.1.18 Ensure /home partition includes the nodev option (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `nodev` mount option specifies that the filesystem cannot contain special devices.

Rationale:

Since the user partitions are not intended to support devices, set this option to ensure that users cannot attempt to create block or character special devices.

Audit:

If a `/home` partition exists, run the following command to verify that the `nodev` option is set:

```
# findmnt /home | grep -Ev '\bnodev\b'
```

Nothing should be returned

Remediation:

For existing `/home` partitions, edit the `/etc/fstab` file and add `nodev` to the fourth field (mounting options) of the `/home` entry. See the `fstab(5)` manual page for more information.

Run the following command to remount `/home`:

```
# mount -o remount,nodev /home
```

Additional Information:

The actions in this recommendation refer to the `/home` partition, which is the default user partition. If you have created other user partitions, it is recommended that the Remediation and Audit steps be applied to these partitions as well.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>3.3 Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v8	<u>4.1 Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	<u>5.1 Establish Secure Configurations</u> Maintain documented, standard security configuration standards for all authorized operating systems and software.			
v7	<u>13 Data Protection</u> Data Protection			

1.1.19 Ensure removable media partitions include noexec option (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `noexec` mount option specifies that the filesystem cannot contain executable binaries.

Rationale:

Setting this option on a file system prevents users from executing programs from the removable media. This deters users from being able to introduce potentially malicious software on the system.

Audit:

Run the following script and verify that the `noexec` option is set on all removable media partitions.

```
#!/usr/bin/bash

for rmpo in $(lsblk -o RM,MOUNTPOINT | awk -F " " '{print $2}'); do
    findmnt -n "$rmpo" | grep -Ev "\bnoexec\b"
done
```

Nothing should be returned

Remediation:

Edit the `/etc/fstab` file and add `noexec` to the fourth field (mounting options) of all removable media partitions. Look for entries that have mount points that contain words such as `floppy` or `cdrom`. See the `fstab(5)` manual page for more information.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>3.3 Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v7	<u>2.6 Address unapproved software</u> Ensure that unauthorized software is either removed or the inventory is updated in a timely manner			
v7	<u>13 Data Protection</u> Data Protection			

1.1.20 Ensure nodev option set on removable media partitions (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `nodev` mount option specifies that the filesystem cannot contain special devices.

Rationale:

Removable media containing character and block special devices could be used to circumvent security controls by allowing non-root users to access sensitive device files such as `/dev/kmem` or the raw disk partitions.

Audit:

Run the following script and verify that the `nodev` option is set on all removable media partitions.

```
#!/usr/bin/bash

for rmpo in $(lsblk -o RM,MOUNTPOINT | awk -F " " '{print $2}'); do
    findmnt -n "$rmpo" | grep -Ev "\bnodev\b"
done
```

Nothing should be returned

Remediation:

Edit the `/etc/fstab` file and add `nodev` to the fourth field (mounting options) of all removable media partitions. Look for entries that have mount points that contain words such as `floppy` or `cdrom`. See the `fstab(5)` manual page for more information.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>3.3 Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v8	<u>4.1 Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	<u>5.1 Establish Secure Configurations</u> Maintain documented, standard security configuration standards for all authorized operating systems and software.			
v7	<u>13 Data Protection</u> Data Protection			

1.1.21 Ensure nosuid option set on removable media partitions (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `nosuid` mount option specifies that the filesystem cannot contain `setuid` files.

Rationale:

Setting this option on a file system prevents users from introducing privileged programs onto the system and allowing non-root users to execute them.

Audit:

Run the following command and verify that the `nosuid` option is set on all removable media partitions.

```
#!/usr/bin/bash

for rmpo in $(lsblk -o RM,MOUNTPOINT | awk -F " " '{print $2}'); do
    findmnt -n "$rmpo" | grep -Ev "\bnosuid\b"
done
```

Nothing should be returned

Remediation:

Edit the `/etc/fstab` file and add `nosuid` to the fourth field (mounting options) of all removable media partitions. Look for entries that have mount points that contain words such as `floppy` or `cdrom`. See the `fstab(5)` manual page for more information.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 <u>Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v8	4.1 <u>Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	5.1 <u>Establish Secure Configurations</u> Maintain documented, standard security configuration standards for all authorized operating systems and software.			
v7	13 <u>Data Protection</u> Data Protection			

1.1.22 Ensure sticky bit is set on all world-writable directories (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

Setting the sticky bit on world writable directories prevents users from deleting or renaming files in that directory that are not owned by them.

Rationale:

This feature prevents the ability to delete or rename files in world writable directories (such as `/tmp`) that are owned by another user.

Audit:

Run the following command to verify no world writable directories exist without the sticky bit set:

```
# df --local -P 2> /dev/null | awk '{if (NR!=1) print $6}' | xargs -I '{}' find '{}' -xdev -type d \( -perm -0002 -a ! -perm -1000 \) 2>/dev/null
```

No output should be returned.

Remediation:

Run the following command to set the sticky bit on all world writable directories:

```
# df --local -P | awk '{if (NR!=1) print $6}' | xargs -I '{}' find '{}' -xdev -type d \( -perm -0002 -a ! -perm -1000 \) 2>/dev/null | xargs -I '{}' chmod a+t '{}'
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 <u>Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v8	4.1 <u>Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	5.1 <u>Establish Secure Configurations</u> Maintain documented, standard security configuration standards for all authorized operating systems and software.			
v7	13 <u>Data Protection</u> Data Protection			

1.1.23 Disable Automounting (Automated)

Profile Applicability:

- Level 1 - Server
- Level 2 - Workstation

Description:

`autofs` allows automatic mounting of devices, typically including CD/DVDs and USB drives.

Rationale:

With automounting enabled anyone with physical access could attach a USB drive or disc and have its contents available in system even if they lacked permissions to mount it themselves.

Impact:

The use of portable hard drives is very common for workstation users. If your organization allows the use of portable storage or media on workstations and physical access controls to workstations is considered adequate there is little value add in turning off automounting.

Audit:

Run the following command to verify `autofs` is not enabled:

```
# systemctl show "autofs.service" | grep -i unitfilestate=enabled  
Nothing should be returned
```

Remediation:

Run the following command to mask `autofs`:

```
# systemctl --now mask autofs
```

OR run the following command to remove `autofs`

```
# yum remove autofs
```

Additional Information:

- Additional methods of disabling a service exist. Consult your distribution documentation for appropriate methods.
- This control should align with the tolerance of the use of portable drives and optical media in the organization.
 - On a server requiring an admin to manually mount media can be part of defense-in-depth to reduce the risk of unapproved software or information being introduced or proprietary software or information being exfiltrated.
 - If admins commonly use flash drives and Server access has sufficient physical controls, requiring manual mounting may not increase security.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	10.3 <u>Disable Autorun and Autoplay for Removable Media</u> Disable autorun and autoplay auto-execute functionality for removable media.			
v7	8.4 <u>Configure Anti-Malware Scanning of Removable Devices</u> Configure devices so that they automatically conduct an anti-malware scan of removable media when inserted or connected.			
v7	8.5 <u>Configure Devices Not To Auto-run Content</u> Configure devices to not auto-run content from removable media.			

1.1.24 Disable USB Storage (Automated)

Profile Applicability:

- Level 1 - Server
- Level 2 - Workstation

Description:

USB storage provides a means to transfer and store files insuring persistence and availability of the files independent of network connection status. Its popularity and utility has led to USB-based malware being a simple and common means for network infiltration and a first step to establishing a persistent threat within a networked environment.

Rationale:

Restricting USB access on the system will decrease the physical attack surface for a device and diminish the possible vectors to introduce malware.

Audit:

Run the following commands and verify the output is as indicated:

```
# modprobe -n -v usb-storage  
  
install /bin/true  
  
# lsmod | grep usb-storage  
  
<No output>
```

Remediation:

Edit or create a file in the `/etc/modprobe.d/` directory ending in `.conf`

Example: `vim /etc/modprobe.d/usb_storage.conf`

Add the following line:

```
install usb-storage /bin/true
```

Run the following command to unload the usb-storage module:

```
rmmod usb-storage
```

Additional Information:

- An alternative solution to disabling the usb-storage module may be found in USBGuard.
- Use of USBGuard and construction of USB device policies should be done in alignment with site policy.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.1 Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	<u>8.4 Configure Anti-Malware Scanning of Removable Devices</u> Configure devices so that they automatically conduct an anti-malware scan of removable media when inserted or connected.			
v7	<u>8.5 Configure Devices Not To Auto-run Content</u> Configure devices to not auto-run content from removable media.			

1.2 Configure Software Updates

Most distributions use a package manager such as yum, apt, or zypper to install and update software packages. Patch management procedures may vary widely between enterprises. Large enterprises may choose to install a local updates server that can be used in place of their distributions servers, whereas a single deployment of a system may prefer to get updates directly. Updates can be performed automatically or manually, depending on the site's policy for patch management. Many large enterprises prefer to test patches on a non-production system before rolling out to production.

For the purpose of this benchmark, the requirement is to ensure that a patch management system is configured and maintained. The specifics on patch update procedures are left to the organization.

1.2.1 Ensure GPG keys are configured (Manual)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

Most packages managers implement GPG key signing to verify package integrity during installation.

Rationale:

It is important to ensure that updates are obtained from a valid source to protect against spoofing that could lead to the inadvertent installation of malware on the system.

Audit:

Verify GPG keys are configured correctly for your package manager. Depending on the package management in use one of the following command groups may provide the needed information:

```
# rpm -q gpg-pubkey --qf '%{name}-%{version}-%{release} --> %{summary}\n'
```

Remediation:

Update your package manager GPG keys in accordance with site policy.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>7.3 Perform Automated Operating System Patch Management</u> Perform operating system updates on enterprise assets through automated patch management on a monthly, or more frequent, basis.			
v8	<u>7.4 Perform Automated Application Patch Management</u> Perform application updates on enterprise assets through automated patch management on a monthly, or more frequent, basis.			
v7	<u>3.4 Deploy Automated Operating System Patch Management Tools</u> Deploy automated software update tools in order to ensure that the operating systems are running the most recent security updates provided by the software vendor.			
v7	<u>3.5 Deploy Automated Software Patch Management Tools</u> Deploy automated software update tools in order to ensure that third-party software on all systems is running the most recent security updates provided by the software vendor.			

1.2.2 Ensure package manager repositories are configured (Manual)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

Systems need to have package manager repositories configured to ensure they receive the latest patches and updates.

Rationale:

If a system's package repositories are misconfigured important patches may not be identified or a rogue repository could introduce compromised software.

Audit:

Run the following command to verify repositories are configured correctly:

```
# yum repolist
```

Remediation:

Configure your package manager repositories according to site policy.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>7.3 Perform Automated Operating System Patch Management</u> Perform operating system updates on enterprise assets through automated patch management on a monthly, or more frequent, basis.			
v8	<u>7.4 Perform Automated Application Patch Management</u> Perform application updates on enterprise assets through automated patch management on a monthly, or more frequent, basis.			
v7	<u>3.4 Deploy Automated Operating System Patch Management Tools</u> Deploy automated software update tools in order to ensure that the operating systems are running the most recent security updates provided by the software vendor.			
v7	<u>3.5 Deploy Automated Software Patch Management Tools</u> Deploy automated software update tools in order to ensure that third-party software on all systems is running the most recent security updates provided by the software vendor.			

1.2.3 Ensure gpgcheck is globally activated (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `gpgcheck` option, found in the main section of the `/etc/yum.conf` and individual `/etc/yum/repos.d/*.repo` files determines if an RPM package's signature is checked prior to its installation.

Rationale:

It is important to ensure that an RPM's package signature is always checked prior to installation to ensure that the software is obtained from a trusted source.

Audit:

Run the following command and verify `gpgcheck` is set to 1:

```
# grep ^\s*gpgcheck /etc/yum.conf
gpgcheck=1
```

Run the following command and verify that all instances of `gpgcheck` are set to 1:

```
# grep -P '^h*gpgcheck=[^1\n\r]+\b(\h+.)?$', /etc/yum.conf
/etc/yum/repos.d/*.repo
Nothing should be returned
```

Remediation:

Edit `/etc/yum.conf` and set `'gpgcheck=1'` in the `[main]` section.

Edit any failing files in `/etc/yum/repos.d/*.repo` and set all instances of `gpgcheck` to 1.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>7.3 Perform Automated Operating System Patch Management</u> Perform operating system updates on enterprise assets through automated patch management on a monthly, or more frequent, basis.			
v7	<u>3.4 Deploy Automated Operating System Patch Management Tools</u> Deploy automated software update tools in order to ensure that the operating systems are running the most recent security updates provided by the software vendor.			

1.3 Filesystem Integrity Checking

AIDE is a file integrity checking tool, similar in nature to Tripwire. While it cannot prevent intrusions, it can detect unauthorized changes to configuration files by alerting when the files are changed. When setting up AIDE, decide internally what the site policy will be concerning integrity checking. Review the AIDE quick start guide and AIDE documentation before proceeding.

1.3.1 Ensure AIDE is installed (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

AIDE takes a snapshot of filesystem state including modification times, permissions, and file hashes which can then be used to compare against the current state of the filesystem to detect modifications to the system.

Note: The prelinking feature can interfere with AIDE because it alters binaries to speed up their start up times. Run `prelink -ua` to restore the binaries to their prelinked state, thus avoiding false positives from AIDE.

Rationale:

By monitoring the filesystem state compromised files can be detected to prevent or limit the exposure of accidental or malicious misconfigurations or modified binaries.

Audit:

Run the following command and verify `aide` is installed:

```
# rpm -q aide
aide-<version>
```

Remediation:

Run the following command to install AIDE:

```
# yum install aide
```

Configure AIDE as appropriate for your environment. Consult the AIDE documentation for options.

Initialize AIDE:

Run the following commands:

```
# aide --init
# mv /var/lib/aide/aide.db.new.gz /var/lib/aide/aide.db.gz
```

References:

1. AIDE stable manual: <http://aide.sourceforge.net/stable/manual.html>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.14 <u>Log Sensitive Data Access</u> Log sensitive data access, including modification and disposal.			●
v7	14.9 <u>Enforce Detail Logging for Access or Changes to Sensitive Data</u> Enforce detailed audit logging for access to sensitive data or changes to sensitive data (utilizing tools such as File Integrity Monitoring or Security Information and Event Monitoring).			●

1.3.2 Ensure filesystem integrity is regularly checked (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

Periodic checking of the filesystem integrity is needed to detect changes to the filesystem.

Rationale:

Periodic file checking allows the system administrator to determine on a regular basis if critical files have been changed in an unauthorized fashion.

Audit:

Run the following commands to verify a cron job scheduled to run the aide check.

```
# grep -Ers '^(#[^#]+\s+)?(\/usr\/s?bin\/|^\/s*)aide(\/.wrapper)?\s(--?\S+\s)*(-
-(check|update)|\$AIDEARGS)\b' /etc/cron.* /etc/crontab /var/spool/cron/
```

Ensure a cron job in compliance with site policy is returned.

OR run the following commands to verify that aidcheck.service and aidcheck.timer are enabled and aidcheck.timer is running

```
# systemctl is-enabled aidecheck.service
# systemctl is-enabled aidecheck.timer
# systemctl status aidecheck.timer
```

Remediation:

If cron will be used to schedule and run aide check

Run the following command:

```
# crontab -u root -e
```

Add the following line to the crontab:

```
0 5 * * * /usr/sbin/aide --check
```

OR if aidecheck.service and aidecheck.timer will be used to schedule and run aide check:

Create or edit the file `/etc/systemd/system/aidecheck.service` and add the following lines:

```
[Unit]
Description=Aide Check

[Service]
Type=simple
ExecStart=/usr/sbin/aide --check

[Install]
WantedBy=multi-user.target
```

Create or edit the file `/etc/systemd/system/aidecheck.timer` and add the following lines:

```
[Unit]
Description=Aide check every day at 5AM

[Timer]
OnCalendar=*-*-* 05:00:00
Unit=aidecheck.service

[Install]
WantedBy=multi-user.target
```

Run the following commands:

```
# chown root:root /etc/systemd/system/aidecheck.*
# chmod 0644 /etc/systemd/system/aidecheck.*

# systemctl daemon-reload

# systemctl enable aidecheck.service
# systemctl --now enable aidecheck.timer
```

References:

1. <https://github.com/konstruktoid/hardening/blob/master/config/aidecheck.service>
2. <https://github.com/konstruktoid/hardening/blob/master/config/aidecheck.timer>

Additional Information:

The checking in this recommendation occurs every day at 5am. Alter the frequency and time of the checks in compliance with site policy.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.14 <u>Log Sensitive Data Access</u> Log sensitive data access, including modification and disposal.			●
v7	14.9 <u>Enforce Detail Logging for Access or Changes to Sensitive Data</u> Enforce detailed audit logging for access to sensitive data or changes to sensitive data (utilizing tools such as File Integrity Monitoring or Security Information and Event Monitoring).			●

1.4 Secure Boot Settings

The recommendations in this section focus on securing the bootloader and settings involved in the boot process directly.

1.4.1 Ensure bootloader password is set (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

Setting the boot loader password will require that anyone rebooting the system must enter a password before being able to set command line boot parameters

Rationale:

Requiring a boot password upon execution of the boot loader will prevent an unauthorized user from entering boot parameters or changing the boot partition. This prevents users from weakening security (e.g. turning off SELinux at boot time).

Impact:

- If password protection is enabled, only the designated superuser can edit a Grub 2 menu item by pressing "e" or access the GRUB 2 command line by pressing "c"
- If GRUB 2 is set up to boot automatically to a password-protected menu entry the user has no option to back out of the password prompt to select another menu entry. Holding the SHIFT key will not display the menu in this case. The user must enter the correct username and password. If unable, the configuration files will have to be edited via the LiveCD or other means to fix the problem
- You can add --unrestricted to the menu entries to allow the system to boot without entering a password. Password will still be required to edit menu items.

Audit:

Run the following script to verify the bootloader password has been set:

```
#!/bin/bash

tst1="" tst2="" output=""
efidir=$(find /boot/efi/EFI/* -type d -not -name 'BOOT')
gbdir=$(find /boot -maxdepth 1 -type d -name 'grub*')
if [ -f "$efidir/grub.cfg" ]; then
    grubdir="$efidir" && grubfile="$efidir/grub.cfg"
elif [ -f "$gbdir/grub.cfg" ]; then
    grubdir="$gbdir" && grubfile="$gbdir/grub.cfg"
fi
userfile="$grubdir/user.cfg"
[ -f "$userfile" ] && grep -Pq '^h*GRUB2_PASSWORD\h*=\h*\.+$' "$userfile" &&
output="\n PASSED: bootloader password set in \"$userfile\" \n\n"
if [ -z "$output" ] && [ -f "$grubfile" ]; then
    grep -Piq '^h*set\h+superusers\h*=\h*"?"[^"\n\r]+"?(\h+\.*)?$(?)"$grubfile'
    && tst1=pass
    grep -Piq '^h*password\h+\H+\h+\.+$' "$grubfile" && tst2=pass
    [ "$tst1" = pass ] && [ "$tst2" = pass ] && output="\n\n*** PASSED:
bootloader password set in \"$grubfile\" ***\n\n"
fi
[ -n "$output" ] && echo -e "$output" || echo -e "\n\n *** FAILED: bootloader
password is not set ***\n\n"
```

Remediation:

For newer grub2 based systems (Release 7.2 and newer), create an encrypted password with `grub2-setpassword`:

```
# grub2-setpassword  
Enter password: <password>  
Confirm password: <password>
```

OR

For older grub2 based systems, create an encrypted password with `grub2-mkpasswd-pbkdf2`:

```
# grub2-mkpasswd-pbkdf2  
Enter password: <password>  
Reenter password: <password>  
  
Your PBKDF2 is <encrypted-password>
```

Add the following into `/etc/grub.d/01_users` or a custom `/etc/grub.d` configuration file:

```
cat <<EOF  
set superusers="<username>"  
password_pbkdf2 <username> <encrypted-password>  
EOF
```

Note:

- *If placing the information in a custom file, do not include the "cat << EOF" and "EOF" lines as the content is automatically added from these files*
- *The superuser/user information and password should not be contained in the `/etc/grub.d/00_header` file. The information can be placed in any `/etc/grub.d` file as long as that file is incorporated into `grub.cfg`. It is preferable to enter this data into a custom file, such as `/etc/grub.d/40_custom`, so it is not overwritten should the Grub package be updated*

Run the following command to update the grub2 configuration:

```
# grub2-mkconfig -o /boot/grub2/grub.cfg
```

Additional Information:

The `older` method will also work on Release 7.2 and newer systems

This recommendation is designed around the grub2 bootloader, if LILO or another bootloader is in use in your environment enact equivalent settings. Replace `/boot/grub2/grub.cfg` with the appropriate grub configuration file for your environment

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 <u>Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	5.1 <u>Establish Secure Configurations</u> Maintain documented, standard security configuration standards for all authorized operating systems and software.			

1.4.2 Ensure permissions on bootloader config are configured (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The grub configuration file contains information on boot settings and passwords for unlocking boot options. The grub2 configuration is usually `grub.cfg`. On newer grub2 systems the encrypted bootloader password is contained in `user.cfg`.

If the system uses UEFI, `/boot/efi` is a vfat filesystem. The vfat filesystem itself doesn't have the concept of permissions but can be mounted under Linux with whatever permissions desired.

Rationale:

Setting the permissions to read and write for root only prevents non-root users from seeing the boot parameters or changing them. Non-root users who read the boot parameters may be able to identify weaknesses in security upon boot and be able to exploit them.

Audit:

Run the following script to verify correct permissions, ownership, and group for `grub.cfg` and if it exists, `user.cfg`:

```
#!/bin/bash

tst1="" tst2="" tst3="" tst4="" test1="" test2="" efidir="" gbdir=""
grubdir="" grubfile="" userfile=""
efidir=$(find /boot/efi/EFI/* -type d -not -name 'BOOT')
gbdir=$(find /boot -maxdepth 1 -type d -name 'grub*')
for file in "$efidir"/grub.cfg "$efidir"/grub.conf; do
    [ -f "$file" ] && grubdir="$efidir" && grubfile=$file
done
if [ -z "$grubdir" ]; then
    for file in "$gbdir"/grub.cfg "$gbdir"/grub.conf; do
        [ -f "$file" ] && grubdir="$gbdir" && grubfile=$file
    done
fi
userfile="$grubdir/user.cfg"
stat -c "%a" "$grubfile" | grep -Pq '^\\h*[0-7]00$' && tst1=pass
output="Permissions on \"$grubfile\" are \"$(stat -c "%a" "$grubfile")\" \"\"
stat -c "%u:%g" "$grubfile" | grep -Pq '^\\h*0:0$' && tst2=pass
output2="\"$grubfile\" is owned by \"$(stat -c "%U" "$grubfile")\" and
belongs to group \"$(stat -c "%G" "$grubfile")\" \"\"
[ "$tst1" = pass ] && [ "$tst2" = pass ] && test1=pass
if [ -f "$userfile" ]; then
    stat -c "%a" "$userfile" | grep -Pq '^\\h*[0-7]00$' && tst3=pass
    output3="Permissions on \"$userfile\" are \"$(stat -c "%a" "$userfile")\" \"\"
    stat -c "%u:%g" "$userfile" | grep -Pq '^\\h*0:0$' && tst4=pass
    output4="\"$userfile\" is owned by \"$(stat -c "%U" "$userfile")\" and
belongs to group \"$(stat -c "%G" "$userfile")\" \"\"
    [ "$tst3" = pass ] && [ "$tst4" = pass ] && test2=pass
else
    test2=pass
fi
[ "$test1" = pass ] && [ "$test2" = pass ] && passing=true
# If passing is true we pass
if [ "$passing" = true ]; then
    echo "PASSED:"
    echo "$output"
    echo "$output2"
    [ -n "$output3" ] && echo "$output3"
    [ -n "$output4" ] && echo "$output4"
else
    # print the reason why we are failing
    echo "FAILED:"
    echo "$output"
    echo "$output2"
    [ -n "$output3" ] && echo "$output3"
    [ -n "$output4" ] && echo "$output4"
fi
```

Remediation:

Run the following commands to set ownership and permissions on your grub configuration file(s):

```
# chown root:root /boot/grub2/grub.cfg
# test -f /boot/grub2/user.cfg && chown root:root /boot/grub2/user.cfg
# chmod og-rwx /boot/grub2/grub.cfg
# test -f /boot/grub2/user.cfg && chmod og-rwx /boot/grub2/user.cfg
```

OR If the system uses UEFI, edit `/etc/fstab` and add the `fmask=0077` option:

Example:

```
<device> /boot/efi vfat defaults,umask=0027,fmask=0077,uid=0,gid=0 0 0
```

Note: This may require a re-boot to enable the change

Additional Information:

- This recommendation is designed around the grub2 bootloader.
- If LILO or another bootloader is in use in your environment:
 - Enact equivalent settings
 - Replace `/boot/grub2/grub.cfg` and `/boot/grub2/user.cfg` with the appropriate boot configuration files for your environment

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 <u>Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	5.1 <u>Establish Secure Configurations</u> Maintain documented, standard security configuration standards for all authorized operating systems and software.			

1.4.3 Ensure authentication required for single user mode (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

Single user mode (rescue mode) is used for recovery when the system detects an issue during boot or by manual selection from the bootloader.

Note: The systemctl option `--fail` is synonymous with `--job-mode=fail`. Using either is acceptable.

Rationale:

Requiring authentication in single user mode (rescue mode) prevents an unauthorized user from rebooting the system into single user to gain root privileges without credentials.

Audit:

Run the following commands and verify that `/sbin/sullogin` or `/usr/sbin/sullogin` is used as shown:

```
# grep /sbin/sullogin /usr/lib/systemd/system/rescue.service

ExecStart=-/bin/sh -c "/sbin/sullogin; /usr/bin/systemctl --fail --no-block
default"

# grep /sbin/sullogin /usr/lib/systemd/system/emergency.service

ExecStart=-/bin/sh -c "/sbin/sullogin; /usr/bin/systemctl --fail --no-block
default"
```

Remediation:

Edit `/usr/lib/systemd/system/rescue.service` and `/usr/lib/systemd/system/emergency.service` and set `ExecStart` to use `/sbin/sullogin` or `/usr/sbin/sullogin`:

```
ExecStart=-/bin/sh -c "/sbin/sullogin; /usr/bin/systemctl --fail --no-block
default"
```


CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.1 Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	<u>5.1 Establish Secure Configurations</u> Maintain documented, standard security configuration standards for all authorized operating systems and software.			

1.5 Additional Process Hardening

1.5.1 Ensure core dumps are restricted (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

A core dump is the memory of an executable program. It is generally used to determine why a program aborted. It can also be used to glean confidential information from a core file. The system provides the ability to set a soft limit for core dumps, but this can be overridden by the user.

Rationale:

Setting a hard limit on core dumps prevents users from overriding the soft variable. If core dumps are required, consider setting limits for user groups (see `limits.conf(5)`). In addition, setting the `fs.suid_dumpable` variable to 0 will prevent setuid programs from dumping core.

Audit:

Run the following commands and verify output matches:

```
# grep -E "^s*\s+hard\s+core" /etc/security/limits.conf
/etc/security/limits.d/*

* hard core 0

# sysctl fs.suid_dumpable
fs.suid_dumpable = 0

# grep "fs.suid_dumpable" /etc/sysctl.conf /etc/sysctl.d/*
fs.suid_dumpable = 0
```

Run the following command to check if systemd-coredump is installed:

```
# systemctl is-enabled coredump.service
```

If `enabled` or `disabled` is returned `systemd-coredump` is installed

Remediation:

Add the following line to `/etc/security/limits.conf` or a `/etc/security/limits.d/*` file:

```
* hard core 0
```

Set the following parameter in `/etc/sysctl.conf` or a `/etc/sysctl.d/*` file:

```
fs.suid_dumpable = 0
```

Run the following command to set the active kernel parameter:

```
# sysctl -w fs.suid_dumpable=0
```

If `systemd-coredump` is installed:

edit `/etc/systemd/coredump.conf` and add/modify the following lines:

```
Storage=none  
ProcessSizeMax=0
```

Run the command:

```
systemctl daemon-reload
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.1 Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	<u>5.1 Establish Secure Configurations</u> Maintain documented, standard security configuration standards for all authorized operating systems and software.			

1.5.2 Ensure XD/NX support is enabled (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

Recent processors in the x86 family support the ability to prevent code execution on a per memory page basis. Generically and on AMD processors, this ability is called No Execute (NX), while on Intel processors it is called Execute Disable (XD). This ability can help prevent exploitation of buffer overflow vulnerabilities and should be activated whenever possible. Extra steps must be taken to ensure that this protection is enabled, particularly on 32-bit x86 systems. Other processors, such as Itanium and POWER, have included such support since inception and the standard kernel for those platforms supports the feature.

Rationale:

Enabling any feature that can protect against buffer overflow attacks enhances the security of the system.

Note: Ensure your system supports the XD or NX bit and has PAE support before implementing this recommendation as this may prevent it from booting if these are not supported by your hardware.

Audit:

Run the following command and verify your kernel has identified and activated NX/XD protection.

```
# journalctl | grep 'protection: active'

kernel: NX (Execute Disable) protection: active
```

OR

on systems without journalctl:

```
# [[ -n $(grep noexec[0-9]*=off /proc/cmdline) || -z $(grep -E -i ' (pae|nx)' /proc/cpuinfo) || -n $(grep '\sNX\s.*\sprotection:\s' /var/log/dmesg | grep -v active) ]] && echo "NX Protection is not active"

Nothing should be returned
```

Remediation:

On 32 bit systems install a kernel with PAE support, no installation is required on 64 bit systems:

If necessary configure your bootloader to load the new kernel and reboot the system.

You may need to enable NX or XD support in your bios.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	10.5 <u>Enable Anti-Exploitation Features</u> Enable anti-exploitation features on enterprise assets and software, where possible, such as Microsoft® Data Execution Prevention (DEP), Windows® Defender Exploit Guard (WDEG), or Apple® System Integrity Protection (SIP) and Gatekeeper™.		●	●
v7	8.3 <u>Enable Operating System Anti-Exploitation Features/</u> <u>Deploy Anti-Exploit Technologies</u> Enable anti-exploitation features such as Data Execution Prevention (DEP) or Address Space Layout Randomization (ASLR) that are available in an operating system or deploy appropriate toolkits that can be configured to apply protection to a broader set of applications and executables.		●	●

1.5.3 Ensure address space layout randomization (ASLR) is enabled (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

Address space layout randomization (ASLR) is an exploit mitigation technique which randomly arranges the address space of key data areas of a process.

Rationale:

Randomly placing virtual memory regions will make it difficult to write memory page exploits as the memory placement will be consistently shifting.

Audit:

Run the following commands and verify output matches:

```
# sysctl kernel.randomize_va_space
kernel.randomize_va_space = 2
# grep "kernel\.randomize_va_space" /etc/sysctl.conf /etc/sysctl.d/*
kernel.randomize_va_space = 2
```

Remediation:

Set the following parameter in `/etc/sysctl.conf` or a `/etc/sysctl.d/*` file:

```
kernel.randomize_va_space = 2
```

Run the following command to set the active kernel parameter:

```
# sysctl -w kernel.randomize_va_space=2
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	10.5 <u>Enable Anti-Exploitation Features</u> Enable anti-exploitation features on enterprise assets and software, where possible, such as Microsoft® Data Execution Prevention (DEP), Windows® Defender Exploit Guard (WDEG), or Apple® System Integrity Protection (SIP) and Gatekeeper™.			
v7	8.3 <u>Enable Operating System Anti-Exploitation Features/Deploy Anti-Exploit Technologies</u> Enable anti-exploitation features such as Data Execution Prevention (DEP) or Address Space Layout Randomization (ASLR) that are available in an operating system or deploy appropriate toolkits that can be configured to apply protection to a broader set of applications and executables.			

1.5.4 Ensure prelink is not installed (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

`prelink` is a program that modifies ELF shared libraries and ELF dynamically linked binaries in such a way that the time needed for the dynamic linker to perform relocations at startup significantly decreases.

Rationale:

The prelinking feature can interfere with the operation of AIDE, because it changes binaries. Prelinking can also increase the vulnerability of the system if a malicious user is able to compromise a common library such as `libc`.

Audit:

Verify `prelink` is not installed.

Run the following command:

```
# rpm -q prelink
package prelink is not installed
```

Remediation:

Run the following command to restore binaries to normal:

```
# prelink -ua
```

Run the following command to uninstall `prelink`:

```
# yum remove prelink
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.1 Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	<u>14.9 Enforce Detail Logging for Access or Changes to Sensitive Data</u> Enforce detailed audit logging for access to sensitive data or changes to sensitive data (utilizing tools such as File Integrity Monitoring or Security Information and Event Monitoring).			

1.6 Mandatory Access Control

Mandatory Access Control (MAC) provides an additional layer of access restrictions to processes on top of the base Discretionary Access Controls. By restricting how processes can access files and resources on a system the potential impact from vulnerabilities in the processes can be reduced.

Impact: Mandatory Access Control limits the capabilities of applications and daemons on a system, while this can prevent unauthorized access the configuration of MAC can be complex and difficult to implement correctly preventing legitimate access from occurring.

1.6.1 Configure SELinux

SELinux provides a Mandatory Access Control (MAC) system that greatly augments the default Discretionary Access Control (DAC) model. Under SELinux, every process and every object (files, sockets, pipes) on the system is assigned a security context, a label that includes detailed type information about the object. The kernel allows processes to access objects only if that access is explicitly allowed by the policy in effect. The policy defines transitions, so that a user can be allowed to run software, but the software can run under a different context than the user's default. This automatically limits the damage that the software can do to files accessible by the calling user. The user does not need to take any action to gain this benefit. For an action to occur, both the traditional DAC permissions must be satisfied as well as the SELinux MAC rules. The action will not be allowed if either one of these models does not permit the action. In this way, SELinux rules can only make a system's permissions more restrictive and secure. SELinux requires a complex policy to allow all the actions required of a system under normal operation. Three such policies have been designed for use with RHEL7 and are included with the system: `targeted`, `strict`, and `mls`. These are described as follows:

- `targeted`: consists mostly of Type Enforcement (TE) rules, and a small number of Role-Based Access Control (RBAC) rules. Targeted restricts the actions of many types of programs, but leaves interactive users largely unaffected.
- `strict`: also uses TE and RBAC rules, but on more programs and more aggressively.
- `mls`: implements Multi-Level Security (MLS), which introduces even more kinds of labels (sensitivity and category) and rules that govern access based on these.

Notes:

- *This section provides guidance for the configuration of the `targeted` policy.*
- *This section **only** applies if SELinux is in use on the system.*
- *Additional Mandatory Access Control systems exist. If a different MAC is used, such as AppArmor, configure the MAC according to its security guidance.*

References:

- NSA SELinux resources:
 - <http://www.nsa.gov/research/selinux>
 - <http://www.nsa.gov/research/selinux/list.shtml>
- Fedora SELinux resources:
 - FAQ: <http://docs.fedoraproject.org/selinux-faq>
 - User Guide: <http://docs.fedoraproject.org/selinux-user-guide>
 - Managing Services Guide: <http://docs.fedoraproject.org/selinux-managing-confined-services-guide>
- SELinux Project web page and wiki:
 - <http://www.selinuxproject.org>
- Chapters 43-45 of Red Hat Enterprise Linux 5: Deployment Guide (Frank Mayer, Karl MacMillan and David Caplan),
- SELinux by Example: Using Security Enhanced Linux (Prentice Hall, August 6, 2006)

1.6.1.1 Ensure SELinux is installed (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

SELinux provides Mandatory Access Control.

Rationale:

Without a Mandatory Access Control system installed only the default Discretionary Access Control system will be available.

Audit:

Verify SELinux is installed.

Run the following command:

```
# rpm -q libselinux  
libselinux-<version>
```

Remediation:

Run the following command to install SELinux:

```
# yum install libselinux
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 <u>Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v7	14.6 <u>Protect Information through Access Control Lists</u> Protect all information stored on systems with file system, network share, claims, application, or database specific access control lists. These controls will enforce the principle that only authorized individuals should have access to the information based on their need to access the information as a part of their responsibilities.			

1.6.1.2 Ensure SELinux is not disabled in bootloader configuration (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

Configure SELINUX to be enabled at boot time and verify that it has not been overwritten by the grub boot parameters.

Note: This recommendation is designed around the grub 2 bootloader, if LILO or another bootloader is in use in your environment enact equivalent settings.

Rationale:

SELinux must be enabled at boot time in your grub configuration to ensure that the controls it provides are not overridden.

Audit:

Run the following script to verify that no `linux` line has the `selinux=0` or `enforcing=0` parameters set:

```
#!/bin/bash

# IF check passes return PASSED
efidir=$(find /boot/efi/EFI/* -type d -not -name 'BOOT')
gbdir=$(find /boot -maxdepth 1 -type d -name 'grub*')
if [ -f "$efidir/grub.cfg" ]; then
    grep "^s*linux" "$efidir/grub.cfg" | grep -Eq "(selinux=0|enforcing=0)"
    && echo "FAILED: \"$()\" exists" || echo "PASSED"
elif [ -f "$gbdir/grub.cfg" ]; then
    grep "^s*linux" "$gbdir/grub.cfg" | grep -Eq "(selinux=0|enforcing=0)" &&
    echo "FAILED: \"$()\" exists" || echo "PASSED"
else
    echo "FAILED"
fi
```

Remediation:

Edit `/etc/default/grub` and remove all instances of `selinux=0` and `enforcing=0` from all `CMDLINE_LINUX` parameters:

```
GRUB_CMDLINE_LINUX_DEFAULT="quiet"
GRUB_CMDLINE_LINUX=""
```

Run the following command to update the `grub2` configuration:

```
# grub2-mkconfig -o /boot/grub2/grub.cfg
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 <u>Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v7	14.6 <u>Protect Information through Access Control Lists</u> Protect all information stored on systems with file system, network share, claims, application, or database specific access control lists. These controls will enforce the principle that only authorized individuals should have access to the information based on their need to access the information as a part of their responsibilities.			

1.6.1.3 Ensure SELinux policy is configured (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

Configure SELinux to meet or exceed the default targeted policy, which constrains daemons and system software only.

Note: If your organization requires stricter policies, ensure that they are set in the `/etc/selinux/config` file.

Rationale:

Security configuration requirements vary from site to site. Some sites may mandate a policy that is stricter than the default policy, which is perfectly acceptable. This item is intended to ensure that at least the default recommendations are met.

Audit:

Run the following commands and ensure output matches either "targeted" or "mls":

```
# grep SELINUXTYPE= /etc/selinux/config
SELINUXTYPE=targeted
# sestatus | grep 'Loaded policy'
Loaded policy name:          targeted
```

Remediation:

Edit the `/etc/selinux/config` file to set the SELINUXTYPE parameter:

```
SELINUXTYPE=targeted
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>3.3 Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v7	<u>14.6 Protect Information through Access Control Lists</u> Protect all information stored on systems with file system, network share, claims, application, or database specific access control lists. These controls will enforce the principle that only authorized individuals should have access to the information based on their need to access the information as a part of their responsibilities.			

1.6.1.4 Ensure the SELinux mode is enforcing or permissive (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

SELinux can run in one of three modes: disabled, permissive, or enforcing:

- Enforcing - Is the default, and recommended, mode of operation; in enforcing mode SELinux operates normally, enforcing the loaded security policy on the entire system.
- Permissive - The system acts as if SELinux is enforcing the loaded security policy, including labeling objects and emitting access denial entries in the logs, but it does not actually deny any operations. While not recommended for production systems, permissive mode can be helpful for SELinux policy development.
- Disabled - Is strongly discouraged; not only does the system avoid enforcing the SELinux policy, it also avoids labeling any persistent objects such as files, making it difficult to enable SELinux in the future

Note: you can set individual domains to permissive mode while the system runs in enforcing mode. For example, to make the httpd_t domain permissive:

```
# semanage permissive -a httpd_t
```

Rationale:

Running SELinux in disabled mode is strongly discouraged; not only does the system avoid enforcing the SELinux policy, it also avoids labeling any persistent objects such as files, making it difficult to enable SELinux in the future.

Audit:

Run the following commands and ensure output matches:

Run the following command to verify SELinux's current mode:

```
# getenforce  
  
Enforcing  
-OR-  
Permissive
```

Run the following command to verify SELinux's configured mode:

```
# grep -Ei '^s*SELINUX=(enforcing|permissive)' /etc/selinux/config  
  
SELINUX=enforcing  
-OR-  
SELINUX=permissive
```

Remediation:

Run one of the following commands to set SELinux's running mode:

To set SELinux mode to `Enforcing`:

```
# setenforce 1
```

OR

To set SELinux mode to `Permissive`:

```
# setenforce 0
```

Edit the `/etc/selinux/config` file to set the SELINUX parameter:

For `Enforcing` mode:

```
SELINUX=enforcing
```

OR

For `Permissive` mode:

```
SELINUX=permissive
```

References:

1. https://access.redhat.com/documentation/en-us/red_hat_enterprise_linux/7/html/selinux_users_and_administrators_guide/sect-security-enhanced-linux-introduction-selinux-modes

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>3.3 Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v7	<u>14.6 Protect Information through Access Control Lists</u> Protect all information stored on systems with file system, network share, claims, application, or database specific access control lists. These controls will enforce the principle that only authorized individuals should have access to the information based on their need to access the information as a part of their responsibilities.			

1.6.1.5 Ensure the SELinux mode is enforcing (Automated)

Profile Applicability:

- Level 2 - Server
- Level 2 - Workstation

Description:

SELinux can run in one of three modes: disabled, permissive, or enforcing:

- Enforcing - Is the default, and recommended, mode of operation; in enforcing mode SELinux operates normally, enforcing the loaded security policy on the entire system.
- Permissive - The system acts as if SELinux is enforcing the loaded security policy, including labeling objects and emitting access denial entries in the logs, but it does not actually deny any operations. While not recommended for production systems, permissive mode can be helpful for SELinux policy development.
- Disabled - Is strongly discouraged; not only does the system avoid enforcing the SELinux policy, it also avoids labeling any persistent objects such as files, making it difficult to enable SELinux in the future

Note: you can set individual domains to permissive mode while the system runs in enforcing mode. For example, to make the httpd_t domain permissive:

```
# semanage permissive -a httpd_t
```

Rationale:

Running SELinux in disabled mode the system not only avoids enforcing the SELinux policy, it also avoids labeling any persistent objects such as files, making it difficult to enable SELinux in the future.

Running SELinux in Permissive mode, though helpful for developing SELinux policy, only logs access denial entries, but does not deny any operations.

Audit:

Run the following commands and ensure output matches:

Run the following command to verify SELinux's current mode:

```
# getenforce
Enforcing
```

Run the following command to verify SELinux's configured mode:

```
# grep -i SELINUX=enforcing /etc/selinux/config
SELINUX=enforcing
```

Remediation:

Run the following command to set SELinux's running mode:

```
# setenforce 1
```

Edit the `/etc/selinux/config` file to set the SELINUX parameter:

For Enforcing mode:

```
SELINUX=enforcing
```

References:

1. https://access.redhat.com/documentation/en-us/red_hat_enterprise_linux/7/html/selinux_users_and_administrators_guide/sect-security-enhanced-linux-introduction-selinux-modes

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 Configure Data Access Control Lists Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v7	14.6 Protect Information through Access Control Lists Protect all information stored on systems with file system, network share, claims, application, or database specific access control lists. These controls will enforce the principle that only authorized individuals should have access to the information based on their need to access the information as a part of their responsibilities.			

1.6.1.6 Ensure no unconfined services exist (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

Unconfined processes run in unconfined domains

Note: Occasionally certain daemons such as backup or centralized management software may require running unconfined. Any such software should be carefully analyzed and documented before such an exception is made.

Rationale:

For unconfined processes, SELinux policy rules are applied, but policy rules exist that allow processes running in unconfined domains almost all access. Processes running in unconfined domains fall back to using DAC rules exclusively. If an unconfined process is compromised, SELinux does not prevent an attacker from gaining access to system resources and data, but of course, DAC rules are still used. SELinux is a security enhancement on top of DAC rules – it does not replace them

Audit:

Run the following command and verify not output is produced:

```
# ps -eZ | grep unconfined_service_t
```

Remediation:

Investigate any unconfined processes found during the audit action. They may need to have an existing security context assigned to them or a policy built for them.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>3.3 Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v7	<u>9.2 Ensure Only Approved Ports, Protocols and Services Are Running</u> Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.			

1.6.1.7 Ensure SETroubleshoot is not installed (Automated)

Profile Applicability:

- Level 1 - Server

Description:

The SETroubleshoot service notifies desktop users of SELinux denials through a user-friendly interface. The service provides important information around configuration errors, unauthorized intrusions, and other potential errors.

Rationale:

The SETroubleshoot service is an unnecessary daemon to have running on a server, especially if X Windows is disabled.

Audit:

Verify `setroubleshoot` is not installed.

Run the following command:

```
# rpm -q setroubleshoot  
package setroubleshoot is not installed
```

Remediation:

Run the following command to Uninstall `setroubleshoot`:

```
# yum remove setroubleshoot
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software</u> Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.			
v7	<u>14.6 Protect Information through Access Control Lists</u> Protect all information stored on systems with file system, network share, claims, application, or database specific access control lists. These controls will enforce the principle that only authorized individuals should have access to the information based on their need to access the information as a part of their responsibilities.			

1.6.1.8 Ensure the MCS Translation Service (mcstrans) is not installed (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `mcstransd` daemon provides category label information to client processes requesting information. The label translations are defined in `/etc/selinux/targeted/setrans.conf`

Rationale:

Since this service is not used very often, remove it to reduce the amount of potentially vulnerable code running on the system.

Audit:

Verify `mcstrans` is not installed.

Run the following command:

```
# rpm -q mcstrans  
package mcstrans is not installed
```

Remediation:

Run the following command to uninstall `mcstrans`:

```
# yum remove mcstrans
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software</u> Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.			
v7	<u>9.2 Ensure Only Approved Ports, Protocols and Services Are Running</u> Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.			

1.7 Command Line Warning Banners

Presenting a warning message prior to the normal user login may assist in the prosecution of trespassers on the computer system. Changing some of these login banners also has the side effect of hiding OS version information and other detailed system information from attackers attempting to target specific exploits at a system.

Guidelines published by the US Department of Defense require that warning messages include at least the name of the organization that owns the system, the fact that the system is subject to monitoring and that such monitoring is in compliance with local statutes, and that use of the system implies consent to such monitoring. It is important that the organization's legal counsel review the content of all messages before any system modifications are made, as these warning messages are inherently site-specific. More information (including citations of relevant case law) can be found at <http://www.justice.gov/criminal/cybercrime/>

Note: The text provided in the remediation actions for these items is intended as an example only. Please edit to include the specific text for your organization as approved by your legal department.

1.7.1 Ensure message of the day is configured properly (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The contents of the `/etc/motd` file are displayed to users after login and function as a message of the day for authenticated users.

Unix-based systems have typically displayed information about the OS release and patch level upon logging in to the system. This information can be useful to developers who are developing software for a particular OS platform. If `mingetty(8)` supports the following options, they display operating system information: `\m` - machine architecture `\r` - operating system release `\s` - operating system name `\v` - operating system version

Rationale:

Warning messages inform users who are attempting to login to the system of their legal status regarding the system and must include the name of the organization that owns the system and any monitoring policies that are in place. Displaying OS and patch level information in login banners also has the side effect of providing detailed system information to attackers attempting to target specific exploits of a system. Authorized users can easily get this information by running the "`uname -a`" command once they have logged in.

Audit:

Run the following command and verify that the contents match site policy:

```
# cat /etc/motd
```

Run the following command and verify no results are returned:

```
# grep -E -i "(\\v|\\r|\\m|\\s|$(grep '^ID=' /etc/os-release | cut -d= -f2 | sed -e 's/"/g'))" /etc/motd
```

Remediation:

Edit the `/etc/motd` file with the appropriate contents according to your site policy, remove any instances of `\m`, `\r`, `\s`, `\v` or references to the OS platform

OR

If the motd is not used, this file can be removed.

Run the following command to remove the motd file:

```
# rm /etc/motd
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 <u>Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	5.1 <u>Establish Secure Configurations</u> Maintain documented, standard security configuration standards for all authorized operating systems and software.			

1.7.2 Ensure local login warning banner is configured properly (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The contents of the `/etc/issue` file are displayed to users prior to login for local terminals.

Unix-based systems have typically displayed information about the OS release and patch level upon logging in to the system. This information can be useful to developers who are developing software for a particular OS platform. If `mingetty(8)` supports the following options, they display operating system information: `\m` - machine architecture `\r` - operating system release `\s` - operating system name `\v` - operating system version - or the operating system's name

Rationale:

Warning messages inform users who are attempting to login to the system of their legal status regarding the system and must include the name of the organization that owns the system and any monitoring policies that are in place. Displaying OS and patch level information in login banners also has the side effect of providing detailed system information to attackers attempting to target specific exploits of a system. Authorized users can easily get this information by running the "`uname -a`" command once they have logged in.

Audit:

Run the following command and verify that the contents match site policy:

```
# cat /etc/issue
```

Run the following command and verify no results are returned:

```
# grep -E -i "(\\v|\\r|\\m|\\s|$(grep '^ID=' /etc/os-release | cut -d= -f2 | sed -e 's/"/g'))" /etc/issue
```

Remediation:

Edit the `/etc/issue` file with the appropriate contents according to your site policy, remove any instances of `\m`, `\r`, `\s`, `\v` or references to the OS platform

```
# echo "Authorized uses only. All activity may be monitored and reported." > /etc/issue
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.1 Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	<u>5.1 Establish Secure Configurations</u> Maintain documented, standard security configuration standards for all authorized operating systems and software.			

1.7.3 Ensure remote login warning banner is configured properly (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The contents of the `/etc/issue.net` file are displayed to users prior to login for remote connections from configured services.

Unix-based systems have typically displayed information about the OS release and patch level upon logging in to the system. This information can be useful to developers who are developing software for a particular OS platform. If `mingetty(8)` supports the following options, they display operating system information: `\m` - machine architecture `\r` - operating system release `\s` - operating system name `\v` - operating system version

Rationale:

Warning messages inform users who are attempting to login to the system of their legal status regarding the system and must include the name of the organization that owns the system and any monitoring policies that are in place. Displaying OS and patch level information in login banners also has the side effect of providing detailed system information to attackers attempting to target specific exploits of a system. Authorized users can easily get this information by running the "`uname -a`" command once they have logged in.

Audit:

Run the following command and verify that the contents match site policy:

```
# cat /etc/issue.net
```

Run the following command and verify no results are returned:

```
# grep -E -i "(\v|r|m|s|$(grep '^ID=' /etc/os-release | cut -d= -f2 | sed -e 's/"/g'))" /etc/issue.net
```

Remediation:

Edit the `/etc/issue.net` file with the appropriate contents according to your site policy, remove any instances of `\m`, `\r`, `\s`, `\v` or references to the OS platform

```
# echo "Authorized uses only. All activity may be monitored and reported." > /etc/issue.net
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.1 Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	<u>5.1 Establish Secure Configurations</u> Maintain documented, standard security configuration standards for all authorized operating systems and software.			

1.7.4 Ensure permissions on /etc/motd are configured (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The contents of the `/etc/motd` file are displayed to users after login and function as a message of the day for authenticated users.

Rationale:

If the `/etc/motd` file does not have the correct ownership it could be modified by unauthorized users with incorrect or misleading information.

Audit:

Run the following command and verify `Uid` and `Gid` are both `0/root` and `Access` is `644` :

```
# stat /etc/motd
Access: (0644/-rw-r--r--)  Uid: (   0/   root)   Gid: (   0/   root)
```

Remediation:

Run the following commands to set permissions on `/etc/motd` :

```
# chown root:root /etc/motd
# chmod u-x,go-wx /etc/motd
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 <u>Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v7	14.6 <u>Protect Information through Access Control Lists</u> Protect all information stored on systems with file system, network share, claims, application, or database specific access control lists. These controls will enforce the principle that only authorized individuals should have access to the information based on their need to access the information as a part of their responsibilities.			

1.7.5 Ensure permissions on /etc/issue are configured (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The contents of the `/etc/issue` file are displayed to users prior to login for local terminals.

Rationale:

If the `/etc/issue` file does not have the correct ownership it could be modified by unauthorized users with incorrect or misleading information.

Audit:

Run the following command and verify `Uid` and `Gid` are both `0/root` and `Access` is `644` :

```
# stat /etc/issue
Access: (0644/-rw-r--r--)  Uid: (   0/   root)   Gid: (   0/   root)
```

Remediation:

Run the following commands to set permissions on `/etc/issue` :

```
# chown root:root /etc/issue
# chmod u-x,go-wx /etc/issue
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 <u>Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v7	14.6 <u>Protect Information through Access Control Lists</u> Protect all information stored on systems with file system, network share, claims, application, or database specific access control lists. These controls will enforce the principle that only authorized individuals should have access to the information based on their need to access the information as a part of their responsibilities.			

1.7.6 Ensure permissions on /etc/issue.net are configured (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The contents of the `/etc/issue.net` file are displayed to users prior to login for remote connections from configured services.

Rationale:

If the `/etc/issue.net` file does not have the correct ownership it could be modified by unauthorized users with incorrect or misleading information.

Audit:

Run the following command and verify `Uid` and `Gid` are both `0/root` and Access is `644` :

```
# stat /etc/issue.net
Access: (0644/-rw-r--r--)  Uid: (   0/   root)   Gid: (   0/   root)
```

Remediation:

Run the following commands to set permissions on `/etc/issue.net` :

```
# chown root:root /etc/issue.net
# chmod u-x,go-wx /etc/issue.net
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 <u>Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v7	14.6 <u>Protect Information through Access Control Lists</u> Protect all information stored on systems with file system, network share, claims, application, or database specific access control lists. These controls will enforce the principle that only authorized individuals should have access to the information based on their need to access the information as a part of their responsibilities.			

1.8 GNOME Display Manager

The GNOME Display Manager (GDM) is a program that manages graphical display servers and handles graphical user logins.

The system will need to be re-booted, or brought down to run level 3 and back to run level 5 for changes to the GDM configuration to take effect.

Note: If GDM is not installed on the system, this section can be skipped

1.8.1 Ensure GNOME Display Manager is removed (Manual)

Profile Applicability:

- Level 2 - Server

Description:

The GNOME Display Manager (GDM) is a program that manages graphical display servers and handles graphical user logins.

Rationale:

If a Graphical User Interface (GUI) is not required, it should be removed to reduce the attack surface of the system.

Impact:

Removing the GNOME Display manager will remove the GUI from the system.

Audit:

Run the following command and verify the output:

```
# rpm -q gdm  
package gdm is not installed
```

Remediation:

Run the following command to remove the `gdm` package

```
# yum remove gdm
```

References:

1. <https://wiki.gnome.org/Projects/GDM>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software</u> Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.			
v7	<u>2.6 Address unapproved software</u> Ensure that unauthorized software is either removed or the inventory is updated in a timely manner			

1.8.2 Ensure GDM login banner is configured (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

GDM is the GNOME Display Manager which handles graphical login for GNOME based systems.

Rationale:

Warning messages inform users who are attempting to login to the system of their legal status regarding the system and must include the name of the organization that owns the system and any monitoring policies that are in place.

Note: If a graphical login is not required, it should be removed to reduce the attack surface of the system.

Audit:

Verify that `/etc/dconf/profile/gdm` exists and includes the following:

```
user-db:user
system-db:gdm
file-db:/usr/share/gdm/greeter-dconf-defaults
```

Verify that a file exists in `/etc/dconf/db/gdm.d/` and includes the following: *(This is typically `/etc/dconf/db/gdm.d/01-banner-message`)*

```
[org/gnome/login-screen]
banner-message-enable=true
banner-message-text='<banner message>'
```

Remediation:

Edit or create the file `/etc/dconf/profile/gdm` and add the following:

```
user-db:user
system-db:gdm
file-db:/usr/share/gdm/greeter-dconf-defaults
```

Edit or create the file `/etc/dconf/db/gdm.d/` and add the following: *(This is typically `/etc/dconf/db/gdm.d/01-banner-message`)*

```
[org/gnome/login-screen]
banner-message-enable=true
banner-message-text='<banner message>'
```

Example Banner Text: 'Authorized users only. All activity may be monitored and reported.'
Run the following command to update the system databases:

```
# dconf update
```

Additional Information:

Additional options and sections may appear in the `/etc/dconf/db/gdm.d/01-banner-message` file.

If a different GUI login service is in use and required on the system, consult your documentation to disable displaying the last logged on user and apply an equivalent banner.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 Establish and Maintain a Secure Configuration Process Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	5.1 Establish Secure Configurations Maintain documented, standard security configuration standards for all authorized operating systems and software.			

1.8.3 Ensure last logged in user display is disabled (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

GDM is the GNOME Display Manager which handles graphical login for GNOME based systems.

Rationale:

Displaying the last logged in user eliminates half of the Userid/Password equation that an unauthorized person would need to log on.

Warning messages inform users who are attempting to login to the system of their legal status regarding the system and must include the name of the organization that owns the system and any monitoring policies that are in place.

Notes:

- *If a graphical login is not required, it should be removed to reduce the attack surface of the system.*
- *If a different GUI login service is in use and required on the system, consult your documentation to disable displaying the last logged on user*

Audit:

Verify that `/etc/dconf/profile/gdm` exists and includes the following:

```
user-db:user
system-db:gdm
file-db:/usr/share/gdm/greeter-dconf-defaults
```

Verify that a file exists in `/etc/dconf/db/gdm.d/` and includes the following: *(This is typically `/etc/dconf/db/gdm.d/00-login-screen`)*

```
[org/gnome/login-screen]
disable-user-list=true
```


Remediation:

Edit or create the file `/etc/dconf/profile/gdm` and add the following:

```
user-db:user
system-db:gdm
file-db:/usr/share/gdm/greeter-dconf-defaults
```

Edit or create the file `/etc/dconf/db/gdm.d/` and add the following: *(This is typically `/etc/dconf/db/gdm.d/00-login-screen`)*

```
[org/gnome/login-screen]
# Do not show the user list
disable-user-list=true
```

Run the following command to update the system databases:

```
# dconf update
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.1 Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	<u>5.1 Establish Secure Configurations</u> Maintain documented, standard security configuration standards for all authorized operating systems and software.			

1.8.4 Ensure XDMCP is not enabled (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

X Display Manager Control Protocol (XDMCP) is designed to provide authenticated access to display management services for remote displays

Rationale:

XDMCP is inherently insecure.

- XDMCP is not a ciphered protocol. This may allow an attacker to capture keystrokes entered by a user
- XDMCP is vulnerable to man-in-the-middle attacks. This may allow an attacker to steal the credentials of legitimate users by impersonating the XDMCP server.

Audit:

Run the following command and verify the output:

```
# grep -Eis '^\\s*Enable\\s*=\\s*true' /etc/gdm/custom.conf  
Nothing should be returned
```

Remediation:

Edit the file `/etc/gdm/custom.conf` and remove the line

```
Enable=true
```

Default Value:

false (This is denoted by no `Enabled=` entry in the file `/etc/gdm/custom.conf` in the `[xdmcp]` section)

References:

1. <https://help.gnome.org/admin/gdm/2.32/configuration.html.en>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software</u> Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.			
v7	<u>9.2 Ensure Only Approved Ports, Protocols and Services Are Running</u> Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.			

1.9 Ensure updates, patches, and additional security software are installed (Manual)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

Periodically patches are released for included software either due to security flaws or to include additional functionality.

Note: Site policy may mandate a testing period before install onto production systems for available updates.

Rationale:

Newer patches may contain security enhancements that would not be available through the latest full update. As a result, it is recommended that the latest software patches be used to take advantage of the latest functionality. As with any software installation, organizations need to determine if a given update meets their requirements and verify the compatibility and supportability of any additional software against the update revision that is selected.

Audit:

Run the following command to verify there are no updates or patches to install.

```
# yum check-update
```

Remediation:

Use your package manager to update all packages on the system according to site policy. The following command will install all available packages

```
# yum update
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>7.3 Perform Automated Operating System Patch Management</u> Perform operating system updates on enterprise assets through automated patch management on a monthly, or more frequent, basis.			
v8	<u>7.4 Perform Automated Application Patch Management</u> Perform application updates on enterprise assets through automated patch management on a monthly, or more frequent, basis.			
v7	<u>3.4 Deploy Automated Operating System Patch Management Tools</u> Deploy automated software update tools in order to ensure that the operating systems are running the most recent security updates provided by the software vendor.			
v7	<u>3.5 Deploy Automated Software Patch Management Tools</u> Deploy automated software update tools in order to ensure that third-party software on all systems is running the most recent security updates provided by the software vendor.			

2 Services

While applying system updates and patches helps correct known vulnerabilities, one of the best ways to protect the system against as yet unreported vulnerabilities is to disable all services that are not required for normal system operation. This prevents the exploitation of vulnerabilities discovered at a later date. If a service is not enabled, it cannot be exploited. The actions in this section of the document provide guidance on some services which can be safely disabled and under which circumstances, greatly reducing the number of possible threats to the resulting system. Additionally some services which should remain enabled but with secure configuration are covered as well as insecure service clients.

2.1 inetd Services

inetd is a super-server daemon that provides internet services and passes connections to configured services. While not commonly used inetd and any unneeded inetd based services should be disabled if possible.

2.1.1 Ensure xinetd is not installed (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The eXtended InterNET Daemon (`xinetd`) is an open source super daemon that replaced the original `inetd` daemon. The `xinetd` daemon listens for well known services and dispatches the appropriate daemon to properly respond to service requests.

Rationale:

If there are no `xinetd` services required, it is recommended that the package be removed to reduce the attack surface are of the system.

Note: If an `xinetd` service or services are required, ensure that any `xinetd` service not required is stopped and disabled

Audit:

Run the following command to verify `xinetd` is not installed:

```
# rpm -q xinetd  
package xinetd is not installed
```

Remediation:

Run the following command to remove `xinetd`:

```
# yum remove xinetd
```


CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software</u> Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.			
v7	<u>2.6 Address unapproved software</u> Ensure that unauthorized software is either removed or the inventory is updated in a timely manner			
v7	<u>9.2 Ensure Only Approved Ports, Protocols and Services Are Running</u> Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.			

2.2 Special Purpose Services

This section describes services that are installed on systems that specifically need to run these services. If any of these services are not required, it is recommended that the package be removed, or the service be masked to reduce the potential attack surface.

2.2.1 Time Synchronization

It is recommended that physical systems and virtual guests lacking direct access to the physical host's clock be configured to synchronize their time using a service such as NTP or chrony.

2.2.1.1 *Ensure time synchronization is in use (Manual)*

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

System time should be synchronized between all systems in an environment. This is typically done by establishing an authoritative time server or set of servers and having all systems synchronize their clocks to them.

Note:

- If another method for time synchronization is being used, this section may be skipped.
- Only **one** time synchronization package should be installed

Rationale:

Time synchronization is important to support time sensitive security mechanisms like Kerberos and also ensures log files have consistent time records across the enterprise, which aids in forensic investigations.

Audit:

Run the following commands to verify that a time synchronization packages is installed:

```
# rpm -q chrony ntp
chrony-<version>
# rpm -q ntp
ntp-<version>
```

Remediation:

Run **One** of the following commands to install chrony **or** NTP:

To install chrony, run the following command:

```
# yum install chrony
```

OR

To install ntp, run the following command:

```
# yum install ntp
```

Note: On systems where host based time synchronization is available consult your virtualization software documentation and setup host based synchronization.

Additional Information:

- On systems where host based time synchronization is not available, verify that chrony *or* NTP is installed.
- On systems where host based time synchronization is available consult your documentation and verify that host based synchronization is in use.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.4 <u>Standardize Time Synchronization</u> Standardize time synchronization. Configure at least two synchronized time sources across enterprise assets, where supported.			
v7	6.1 <u>Utilize Three Synchronized Time Sources</u> Use at least three synchronized time sources from which all servers and network devices retrieve time information on a regular basis so that timestamps in logs are consistent.			

2.2.1.2 Ensure chrony is configured (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

`chrony` is a daemon which implements the Network Time Protocol (NTP) and is designed to synchronize system clocks across a variety of systems and use a source that is highly accurate. More information on `chrony` can be found at <http://chrony.tuxfamily.org/>. `chrony` can be configured to be a client and/or a server.

Rationale:

If `chrony` is in use on the system proper configuration is vital to ensuring time synchronization is working properly.

Note: This recommendation only applies if `chrony` is in use on the system.

Audit:

IF `chrony` is installed on the system:

Run the following command and verify remote server is configured properly:

```
# grep -E "^(server|pool)" /etc/chrony.conf  
server <remote-server>
```

Multiple servers may be configured.

Run the following command and verify `OPTIONS` includes `'-u chrony'`:

```
# grep ^OPTIONS /etc/sysconfig/chronyd  
OPTIONS="-u chrony"
```

Additional options may be present.

Remediation:

Add or edit server or pool lines to `/etc/chrony.conf` as appropriate:

```
server <remote-server>
```

Add or edit the `OPTIONS` in `/etc/sysconfig/chronyd` to include `'-u chrony'`:

```
OPTIONS="-u chrony"
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.4 <u>Standardize Time Synchronization</u> Standardize time synchronization. Configure at least two synchronized time sources across enterprise assets, where supported.			
v7	6.1 <u>Utilize Three Synchronized Time Sources</u> Use at least three synchronized time sources from which all servers and network devices retrieve time information on a regular basis so that timestamps in logs are consistent.			

2.2.1.3 Ensure ntp is configured (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

ntp is a daemon which implements the Network Time Protocol (NTP). It is designed to synchronize system clocks across a variety of systems and use a source that is highly accurate. More information on NTP can be found at <http://www.ntp.org>. ntp can be configured to be a client and/or a server.

Note: This recommendation only applies if ntp is in use on the system.

Rationale:

If ntp is in use on the system proper configuration is vital to ensuring time synchronization is working properly.

Audit:

IF NTP is installed on the system:

Run the following command and verify ntpd is enabled:

```
# systemctl is-enabled ntpd  
  
enabled
```

Run the following command and verify output matches:

```
# grep "^restrict" /etc/ntp.conf  
  
restrict -4 default kod nomodify notrap nopeer noquery  
restrict -6 default kod nomodify notrap nopeer noquery
```

The -4 in the first line is optional and options after default can appear in any order.

Additional restriction lines may exist.

Run the following command and verify remote server is configured properly:

```
# grep -E "^(server|pool)" /etc/ntp.conf  
  
server <remote-server>
```

Multiple servers may be configured

Run the following commands and verify that '-u ntp:ntp' is included in `OPTIONS` OR `ExecStart` as listed:

```
# grep "^OPTIONS" /etc/sysconfig/ntpd  
  
OPTIONS="-u ntp:ntp"
```

OR

```
# grep "^ExecStart" /usr/lib/systemd/system/ntpd.service  
  
ExecStart=/usr/sbin/ntpd -u ntp:ntp $OPTIONS
```

Additional options may be present.

Remediation:

Add or edit restrict lines in `/etc/ntp.conf` to match the following:

```
restrict -4 default kod nomodify notrap nopeer noquery
restrict -6 default kod nomodify notrap nopeer noquery
```

Add or edit server or pool lines to `/etc/ntp.conf` as appropriate:

```
server <remote-server>
```

Add or edit the `OPTIONS` in `/etc/sysconfig/ntpd` to include `'-u ntp:ntp'`:

```
OPTIONS="-u ntp:ntp"
```

Reload the systemd daemon:

```
systemctl daemon-reload
```

Enable and start the ntp service:

```
systemctl --now enable ntpd
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.4 <u>Standardize Time Synchronization</u> Standardize time synchronization. Configure at least two synchronized time sources across enterprise assets, where supported.		●	●
v7	6.1 <u>Utilize Three Synchronized Time Sources</u> Use at least three synchronized time sources from which all servers and network devices retrieve time information on a regular basis so that timestamps in logs are consistent.		●	●

2.2.2 Ensure X11 Server components are not installed (Automated)

Profile Applicability:

- Level 1 - Server

Description:

The X Window System provides a Graphical User Interface (GUI) where users can have multiple windows in which to run programs and various add on. The X Windows system is typically used on workstations where users login, but not on servers where users typically do not login.

Rationale:

Unless your organization specifically requires graphical login access via X Windows, remove it to reduce the potential attack surface.

Impact:

Many Linux systems run applications which require a Java runtime. Some Linux Java packages have a dependency on specific X Windows xorg-x11-fonts. One workaround to avoid this dependency is to use the "headless" Java packages for your specific Java runtime.

Audit:

Run the following command to Verify X Windows Server is not installed.

```
# rpm -qa xorg-x11-server*
```

Remediation:

Run the following command to remove the X Windows Server packages:

```
# yum remove xorg-x11-server*
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software</u> Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.			
v7	<u>2.6 Address unapproved software</u> Ensure that unauthorized software is either removed or the inventory is updated in a timely manner			

2.2.3 Ensure Avahi Server is not installed (Automated)

Profile Applicability:

- Level 1 - Server
- Level 2 - Workstation

Description:

Avahi is a free zeroconf implementation, including a system for multicast DNS/DNS-SD service discovery. Avahi allows programs to publish and discover services and hosts running on a local network with no specific configuration. For example, a user can plug a computer into a network and Avahi automatically finds printers to print to, files to look at and people to talk to, as well as network services running on the machine.

Rationale:

Automatic discovery of network services is not normally required for system functionality. It is recommended to remove this package to reduce the potential attack surface.

Audit:

Run one of the following command to verify `avahi-autoipd` and `avahi` are not installed:

```
# rpm -q avahi-autoipd avahi
package avahi-autoipd is not installed
package avahi is not installed
```

Remediation:

Run the following commands to stop, mask and remove `avahi-autoipd` and `avahi`:

```
# systemctl stop avahi-daemon.socket avahi-daemon.service
# yum remove avahi-autoipd avahi
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.8 <u>Uninstall or Disable Unnecessary Services on Enterprise Assets and Software</u> Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.			
v7	9.2 <u>Ensure Only Approved Ports, Protocols and Services Are Running</u> Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.			

2.2.4 Ensure CUPS is not installed (Automated)

Profile Applicability:

- Level 1 - Server

Description:

The Common Unix Print System (CUPS) provides the ability to print to both local and network printers. A system running CUPS can also accept print jobs from remote systems and print them to local printers. It also provides a web based remote administration capability.

Rationale:

If the system does not need to print jobs or accept print jobs from other systems, it is recommended that CUPS be removed to reduce the potential attack surface.

Note: Removing CUPS will prevent printing from the system

Impact:

Disabling CUPS will prevent printing from the system, a common task for workstation systems.

Audit:

Run the following command to verify `cups` is not installed:

```
# rpm -q cups  
package cups is not installed
```

Remediation:

Run the following command to remove `cups`:

```
# yum remove cups
```

References:

1. More detailed documentation on CUPS is available at the project homepage at <http://www.cups.org>.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software</u> Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.			
v7	<u>9.2 Ensure Only Approved Ports, Protocols and Services Are Running</u> Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.			

2.2.5 Ensure DHCP Server is not installed (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The Dynamic Host Configuration Protocol (DHCP) is a service that allows machines to be dynamically assigned IP addresses.

Rationale:

Unless a system is specifically set up to act as a DHCP server, it is recommended that the dhcp package be removed to reduce the potential attack surface.

Audit:

Run the following command to verify dhcp is not installed:

```
# rpm -q dhcp  
package dhcp is not installed
```

Remediation:

Run the following command to remove dhcp:

```
# yum remove dhcp
```

References:

1. dhcpd(8)

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software</u> Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.			
v7	<u>9.2 Ensure Only Approved Ports, Protocols and Services Are Running</u> Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.			

2.2.6 Ensure LDAP server is not installed (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The Lightweight Directory Access Protocol (LDAP) was introduced as a replacement for NIS/YP. It is a service that provides a method for looking up information from a central database.

Rationale:

If the system will not need to act as an LDAP server, it is recommended that the software be removed to reduce the potential attack surface.

Audit:

Run the following command to verify `openldap-servers` is not installed:

```
# rpm -q openldap-servers
package openldap-servers is not installed
```

Remediation:

Run the following command to remove `openldap-servers`:

```
# yum remove openldap-servers
```

References:

1. For more detailed documentation on OpenLDAP, go to the project homepage at <http://www.openldap.org>.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software</u> Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.			
v7	<u>9.2 Ensure Only Approved Ports, Protocols and Services Are Running</u> Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.			

2.2.7 Ensure DNS Server is not installed (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The Domain Name System (DNS) is a hierarchical naming system that maps names to IP addresses for computers, services and other resources connected to a network.

Rationale:

Unless a system is specifically designated to act as a DNS server, it is recommended that the package be removed to reduce the potential attack surface.

Audit:

Run one of the following commands to verify `bind` is not installed:

```
# rpm -q bind  
package bind is not installed
```

Remediation:

Run the following command to remove `bind`:

```
# yum remove bind
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software</u> Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.			
v7	<u>9.2 Ensure Only Approved Ports, Protocols and Services Are Running</u> Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.			

2.2.8 Ensure FTP Server is not installed (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

FTP (File Transfer Protocol) is a traditional and widely used standard tool for transferring files between a server and clients over a network, especially where no authentication is necessary (permits anonymous users to connect to a server).

Rationale:

FTP does not protect the confidentiality of data or authentication credentials. It is recommended SFTP be used if file transfer is required. Unless there is a need to run the system as a FTP server (for example, to allow anonymous downloads), it is recommended that the package be removed to reduce the potential attack surface.

Note: Additional FTP servers also exist and should be removed if not required.

Audit:

Run the following command to verify `vsftpd` is not installed:

```
# rpm -q vsftpd  
package vsftpd is not installed
```

Remediation:

Run the following command to remove `vsftpd`:

```
# yum remove vsftpd
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.8 <u>Uninstall or Disable Unnecessary Services on Enterprise Assets and Software</u> Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.			
v7	9.2 <u>Ensure Only Approved Ports, Protocols and Services Are Running</u> Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.			

2.2.9 Ensure HTTP server is not installed (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

HTTP or web servers provide the ability to host web site content.

Rationale:

Unless there is a need to run the system as a web server, it is recommended that the package be removed to reduce the potential attack surface.

Notes:

- *Several http servers exist. `apache`, `apache2`, `lighttpd`, and `nginx` are example packages that provide an HTTP server.*
- *These and other packages should also be audited, and removed if not required.*

Audit:

Run the following command to verify `httpd` is not installed:

```
# rpm -q httpd  
package httpd is not installed
```

Remediation:

Run the following command to remove `httpd`:

```
# yum remove httpd
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software</u> Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.			
v7	<u>9.2 Ensure Only Approved Ports, Protocols and Services Are Running</u> Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.			

2.2.10 Ensure IMAP and POP3 server is not installed (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

`dovecot` is an open source IMAP and POP3 server for Linux based systems.

Rationale:

Unless POP3 and/or IMAP servers are to be provided by this system, it is recommended that the package be removed to reduce the potential attack surface.

Notes:

- *Several IMAP/POP3 servers exist and can use other service names. `courier-imap` and `cyrus-imap` are example services that provide a mail server.*
- *These and other services should also be audited and the packages removed if not required.*

Audit:

Run the following command to verify `dovecot` is not installed:

```
# rpm -q dovecot  
package dovecot is not installed
```

Remediation:

Run the following command to remove `dovecot`:

```
# yum remove dovecot
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software</u> Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.			
v7	<u>9.2 Ensure Only Approved Ports, Protocols and Services Are Running</u> Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.			

2.2.11 Ensure Samba is not installed (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The Samba daemon allows system administrators to configure their Linux systems to share file systems and directories with Windows desktops. Samba will advertise the file systems and directories via the Server Message Block (SMB) protocol. Windows desktop users will be able to mount these directories and file systems as letter drives on their systems.

Rationale:

If there is no need to mount directories and file systems to Windows systems, then this package can be removed to reduce the potential attack surface.

Audit:

Run the following command to verify `samba` is not installed:

```
# rpm -q samba  
package samba is not installed
```

Remediation:

Run the following command to remove `samba`:

```
# yum remove samba
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software</u> Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.			
v7	<u>9.2 Ensure Only Approved Ports, Protocols and Services Are Running</u> Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.			

2.2.12 Ensure HTTP Proxy Server is not installed (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

Squid is a standard proxy server used in many distributions and environments.

Rationale:

Unless a system is specifically set up to act as a proxy server, it is recommended that the squid package be removed to reduce the potential attack surface.

Note: Several HTTP proxy servers exist. These should be checked and removed unless required.

Audit:

Run the following command to verify `squid` is not installed:

```
# rpm -q squid  
package squid is not installed
```

Remediation:

Run the following command to remove the `squid` package:

```
# yum remove squid
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software</u> Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.			
v7	<u>9.2 Ensure Only Approved Ports, Protocols and Services Are Running</u> Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.			

2.2.13 Ensure net-snmp is not installed (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

Simple Network Management Protocol (SNMP) is a widely used protocol for monitoring the health and welfare of network equipment, computer equipment and devices like UPSs.

Net-SNMP is a suite of applications used to implement SNMPv1 (RFC 1157), SNMPv2 (RFCs 1901-1908), and SNMPv3 (RFCs 3411-3418) using both IPv4 and IPv6.

Support for SNMPv2 classic (a.k.a. "SNMPv2 historic" - RFCs 1441-1452) was dropped with the 4.0 release of the UCD-snmp package.

The Simple Network Management Protocol (SNMP) server is used to listen for SNMP commands from an SNMP management system, execute the commands or collect the information and then send results back to the requesting system.

Rationale:

The SNMP server can communicate using `SNMPv1`, which transmits data in the clear and does not require authentication to execute commands. `SNMPv3` replaces the simple/clear text password sharing used in `SNMPv2` with more securely encoded parameters. If the the SNMP service is not required, the `net-snmp` package should be removed to reduce the attack surface of the system.

Note: If SNMP is required:

- *The server should be configured for `SNMP v3` only. User Authentication and Message Encryption should be configured.*
- *If `SNMP v2` is **absolutely** necessary, modify the community strings' values.*

Audit:

Run the following command to verify `net-snmp` is not installed:

```
# rpm -q net-snmp  
package net-snmp is not installed
```

Remediation:

Run the following command to remove `net-snmpd`:

```
# yum remove net-snmp
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.8 <u>Uninstall or Disable Unnecessary Services on Enterprise Assets and Software</u> Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.		●	●
v7	2.6 <u>Address unapproved software</u> Ensure that unauthorized software is either removed or the inventory is updated in a timely manner	●	●	●
v7	9.2 <u>Ensure Only Approved Ports, Protocols and Services Are Running</u> Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.		●	●

2.2.14 Ensure NIS server is not installed (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `ypserv` package provides the Network Information Service (NIS). This service, formally known as Yellow Pages, is a client-server directory service protocol for distributing system configuration files. The NIS server is a collection of programs that allow for the distribution of configuration files.

Rationale:

The NIS service is inherently an insecure system that has been vulnerable to DOS attacks, buffer overflows and has poor authentication for querying NIS maps. NIS generally has been replaced by such protocols as Lightweight Directory Access Protocol (LDAP). It is recommended that the `ypserv` package be removed, and if required a more secure services be used.

Audit:

Run the following command to verify `ypserv` is not installed:

```
# rpm -q ypserv  
package ypserv is not installed
```

Remediation:

Run the following command to remove `ypserv`:

```
# yum remove ypserv
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software</u> Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.		●	●
v7	<u>2.6 Address unapproved software</u> Ensure that unauthorized software is either removed or the inventory is updated in a timely manner	●	●	●
v7	<u>9.2 Ensure Only Approved Ports, Protocols and Services Are Running</u> Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.		●	●

2.2.15 Ensure telnet-server is not installed (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `telnet-server` package contains the `telnet` daemon, which accepts connections from users from other systems via the `telnet` protocol.

Rationale:

The `telnet` protocol is insecure and unencrypted. The use of an unencrypted transmission medium could allow a user with access to sniff network traffic the ability to steal credentials. The `ssh` package provides an encrypted session and stronger security.

Audit:

Run the following command to verify the `telnet-server` package is not installed:

```
rpm -q telnet-server  
package telnet-server is not installed
```

Remediation:

Run the following command to remove the `telnet-server` package:

```
# yum remove telnet-server
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.8 <u>Uninstall or Disable Unnecessary Services on Enterprise Assets and Software</u> Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.			
v7	2.6 <u>Address unapproved software</u> Ensure that unauthorized software is either removed or the inventory is updated in a timely manner			
v7	9.2 <u>Ensure Only Approved Ports, Protocols and Services Are Running</u> Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.			

2.2.16 Ensure mail transfer agent is configured for local-only mode (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

Mail Transfer Agents (MTA), such as sendmail and Postfix, are used to listen for incoming mail and transfer the messages to the appropriate user or mail server. If the system is not intended to be a mail server, it is recommended that the MTA be configured to only process local mail.

Rationale:

The software for all Mail Transfer Agents is complex and most have a long history of security issues. While it is important to ensure that the system can process local mail messages, it is not necessary to have the MTA's daemon listening on a port unless the server is intended to be a mail server that receives and processes mail from other systems.

Notes:

- *This recommendation is designed around the postfix mail server.*
- *Depending on your environment you may have an alternative MTA installed such as sendmail. If this is the case consult the documentation for your installed MTA to configure the recommended state.*

Audit:

Run the following command to verify that the MTA is not listening on any non-loopback address (127.0.0.1 or ::1)
Nothing should be returned

```
# ss -ltnu | grep -E ':25\s' | grep -E -v '\s(127.0.0.1|\[?::1\])? :25\s'
```

Remediation:

Edit `/etc/postfix/main.cf` and add the following line to the RECEIVING MAIL section. If the line already exists, change it to look like the line below:

```
inet_interfaces = loopback-only
```

Run the following command to restart postfix:

```
# systemctl restart postfix
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software</u> Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.			
v7	<u>9.2 Ensure Only Approved Ports, Protocols and Services Are Running</u> Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.			

2.2.17 Ensure *nfs-utils* is not installed or the *nfs-server* service is masked (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The Network File System (NFS) is one of the first and most widely distributed file systems in the UNIX environment. It provides the ability for systems to mount file systems of other servers through the network.

Rationale:

If the system does not require network shares, it is recommended that the *nfs-utils* package be removed to reduce the attack surface of the system.

Audit:

Run the following command to verify *nfs-utils* is not installed:

```
# rpm -q nfs-utils
package nfs-utils is not installed
```

OR

If the *nfs*-package is required as a dependency, run the following command to verify that the *nfs-server* service is masked:

```
# systemctl is-enabled nfs-server
masked
```

Remediation:

Run the following command to remove `nfs-utils`:

```
# yum remove nfs-utils
```

OR

If the `nfs`-package is required as a dependency, run the following command to stop and mask the `nfs-server` service:

```
# systemctl --now mask nfs-server
```

Additional Information:

many of the libvirt packages used by Enterprise Linux virtualization are dependent on the `nfs-utils` package. If the `nfs`-package is required as a dependency, the `nfs-server` should be disabled and masked to reduce the attack surface of the system.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.8 <u>Uninstall or Disable Unnecessary Services on Enterprise Assets and Software</u> Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.			
v7	9.2 <u>Ensure Only Approved Ports, Protocols and Services Are Running</u> Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.			

2.2.18 Ensure rpcbind is not installed or the rpcbind services are masked (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The rpcbind utility maps RPC services to the ports on which they listen. RPC processes notify rpcbind when they start, registering the ports they are listening on and the RPC program numbers they expect to serve. The client system then contacts rpcbind on the server with a particular RPC program number. The rpcbind service redirects the client to the proper port number so it can communicate with the requested service

Portmapper is an RPC service, which always listens on tcp and udp 111, and is used to map other RPC services (such as nfs, nlockmgr, quotad, mountd, etc.) to their corresponding port number on the server. When a remote host makes an RPC call to that server, it first consults with portmap to determine where the RPC server is listening.

Rationale:

A small request (~82 bytes via UDP) sent to the Portmapper generates a large response (7x to 28x amplification), which makes it a suitable tool for DDoS attacks. If rpcbind is not required, it is recommended that the rpcbind package be removed to reduce the attack surface of the system.

Audit:

Run the following command to verify `rpcbind` is not installed:

```
# rpm -q rpcbind  
package rpcbind is not installed
```

OR

If the `rpcbind` package is required as a dependency, run the following commands to verify that the `rpcbind` and `rpcbind.socket` services are masked:

```
# systemctl is-enabled rpcbind  
masked  
  
# systemctl is-enabled rpcbind.socket  
masked
```

Remediation:

Run the following command to remove `nfs-utils`:

```
# yum remove rpcbind
```

OR

If the `rpcbind` package is required as a dependency, run the following commands to stop and mask the `rpcbind` and `rpcbind.socket` services:

```
# systemctl --now mask rpcbind  
# systemctl --now mask rpcbind.socket
```

Additional Information:

Many of the `libvirt` packages used by Enterprise Linux virtualization, and the `nfs-utils` package used for The Network File System (NFS), are dependent on the `rpcbind` package. If the `rpcbind` is required as a dependency, the services `rpcbind.service` and `rpcbind.socket` should be stopped and masked to reduce the attack surface of the system.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software</u> Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.			
v7	<u>9.2 Ensure Only Approved Ports, Protocols and Services Are Running</u> Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.			

2.2.19 Ensure rsync is not installed or the rsyncd service is masked (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `rsyncd` service can be used to synchronize files between systems over network links.

Rationale:

Unless required, the `rsync` package should be removed to reduce the attack surface area of the system.

The `rsyncd` service presents a security risk as it uses unencrypted protocols for communication.

Note: If a required dependency exists for the `rsync` package, but the `rsyncd` service is not required, the service should be masked.

Impact:

There are packages that are dependent on the `rsync` package. If the `rsync` package is removed, these packages will be removed as well.

Before removing the `rsync` package, review any dependent packages to determine if they are required on the system. If a dependent package is required, mask the `rsyncd` service and leave the `rsync` package installed.

Audit:

Run the following command to verify that `rsync` is not installed:

```
# rpm -q rsync  
package rsync is not installed
```

OR

Run the following command to verify the `rsyncd` service is masked:

```
# systemctl is-enabled rsyncd  
masked
```

Remediation:

Run the following command to remove the `rsync` package:

```
# yum remove rsync
```

OR

Run the following command to mask the `rsyncd` service:

```
# systemctl --now mask rsyncd
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.8 <u>Uninstall or Disable Unnecessary Services on Enterprise Assets and Software</u> Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.			
v7	9.2 <u>Ensure Only Approved Ports, Protocols and Services Are Running</u> Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.			

2.3 Service Clients

A number of insecure services exist. While disabling the servers prevents a local attack against these services, it is advised to remove their clients unless they are required.

Note: This should not be considered a comprehensive list of insecure service clients. You may wish to consider additions to those listed here for your environment.

2.3.1 Ensure NIS Client is not installed (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The Network Information Service (NIS), formerly known as Yellow Pages, is a client-server directory service protocol used to distribute system configuration files. The NIS client (`ypbind`) was used to bind a machine to an NIS server and receive the distributed configuration files.

Rationale:

The NIS service is inherently an insecure system that has been vulnerable to DOS attacks, buffer overflows and has poor authentication for querying NIS maps. NIS generally has been replaced by such protocols as Lightweight Directory Access Protocol (LDAP). It is recommended that the service be removed.

Impact:

Many insecure service clients are used as troubleshooting tools and in testing environments. Uninstalling them can inhibit capability to test and troubleshoot. If they are required it is advisable to remove the clients after use to prevent accidental or intentional misuse.

Audit:

Run the following command to verify that the `ypbind` package is not installed:

```
# rpm -q ypbind
package ypbind is not installed
```

Remediation:

Run the following command to remove the `ypbind` package:

```
# yum remove ypbind
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software</u> Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.		●	●
v7	<u>2.6 Address unapproved software</u> Ensure that unauthorized software is either removed or the inventory is updated in a timely manner	●	●	●

2.3.2 Ensure rsh client is not installed (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `rsh` package contains the client commands for the `rsh` services.

Rationale:

These legacy clients contain numerous security exposures and have been replaced with the more secure SSH package. Even if the server is removed, it is best to ensure the clients are also removed to prevent users from inadvertently attempting to use these commands and therefore exposing their credentials. Note that removing the `rsh` package removes the clients for `rsh`, `rcp` and `rlogin`.

Impact:

Many insecure service clients are used as troubleshooting tools and in testing environments. Uninstalling them can inhibit capability to test and troubleshoot. If they are required it is advisable to remove the clients after use to prevent accidental or intentional misuse.

Audit:

Run the following command to verify that the `rsh` package is not installed:

```
# rpm -q rsh
package rsh is not installed
```

Remediation:

Run the following command to remove the `rsh` package:

```
# yum remove rsh
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software</u> Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.			
v7	<u>2.6 Address unapproved software</u> Ensure that unauthorized software is either removed or the inventory is updated in a timely manner			

2.3.3 Ensure talk client is not installed (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `talk` software makes it possible for users to send and receive messages across systems through a terminal session. The `talk` client, which allows initialization of talk sessions, is installed by default.

Rationale:

The software presents a security risk as it uses unencrypted protocols for communication.

Impact:

Many insecure service clients are used as troubleshooting tools and in testing environments. Uninstalling them can inhibit capability to test and troubleshoot. If they are required it is advisable to remove the clients after use to prevent accidental or intentional misuse.

Audit:

Run the following command to verify that the `talk` package is not installed:

```
# rpm -q talk  
package talk is not installed
```

Remediation:

Run the following command to remove the `talk` package:

```
# yum remove talk
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software</u> Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.			
v7	<u>2.6 Address unapproved software</u> Ensure that unauthorized software is either removed or the inventory is updated in a timely manner			

2.3.4 Ensure telnet client is not installed (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `telnet` package contains the `telnet` client, which allows users to start connections to other systems via the telnet protocol.

Rationale:

The `telnet` protocol is insecure and unencrypted. The use of an unencrypted transmission medium could allow an unauthorized user to steal credentials. The `ssh` package provides an encrypted session and stronger security and is included in most Linux distributions.

Impact:

Many insecure service clients are used as troubleshooting tools and in testing environments. Uninstalling them can inhibit capability to test and troubleshoot. If they are required it is advisable to remove the clients after use to prevent accidental or intentional misuse.

Audit:

Run the following command to verify that the `telnet` package is not installed:

```
# rpm -q telnet  
package telnet is not installed
```

Remediation:

Run the following command to remove the `telnet` package:

```
# yum remove telnet
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software</u> Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.			
v7	<u>2.6 Address unapproved software</u> Ensure that unauthorized software is either removed or the inventory is updated in a timely manner			

2.3.5 Ensure LDAP client is not installed (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The Lightweight Directory Access Protocol (LDAP) was introduced as a replacement for NIS/YP. It is a service that provides a method for looking up information from a central database.

Rationale:

If the system will not need to act as an LDAP client, it is recommended that the software be removed to reduce the potential attack surface.

Impact:

Removing the LDAP client will prevent or inhibit using LDAP for authentication in your environment.

Audit:

Run the following command to verify that the `openldap-clients` package is not installed:

```
# rpm -q openldap-clients
package openldap-clients is not installed
```

Remediation:

Run the following command to remove the `openldap-clients` package:

```
# yum remove openldap-clients
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software</u> Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.		●	●
v7	<u>2.6 Address unapproved software</u> Ensure that unauthorized software is either removed or the inventory is updated in a timely manner	●	●	●

2.4 Ensure nonessential services are removed or masked (Manual)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

A network port is identified by its number, the associated IP address, and the type of the communication protocol such as TCP or UDP.

A listening port is a network port on which an application or process listens on, acting as a communication endpoint.

Each listening port can be open or closed (filtered) using a firewall. In general terms, an open port is a network port that accepts incoming packets from remote locations.

Rationale:

Services listening on the system pose a potential risk as an attack vector. These services should be reviewed, and if not required, the service should be stopped, and the package containing the service should be removed. If required packages have a dependency, the service should be stopped and masked to reduce the attack surface of the system.

Audit:

Run the following command:

```
# lsof -i -P -n | grep -v "(ESTABLISHED)"
```

Review the output to ensure that all services listed are required on the system. If a listed service is not required, remove the package containing the service. If the package containing the service is required, stop and mask the service

Remediation:

Run the following command to remove the package containing the service:

```
# yum remove <package_name>
```

OR If required packages have a dependency:

Run the following command to stop and mask the service:

```
# systemctl --now mask <service_name>
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.8 <u>Uninstall or Disable Unnecessary Services on Enterprise Assets and Software</u> Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.			
v7	9.2 <u>Ensure Only Approved Ports, Protocols and Services Are Running</u> Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.			

3 Network Configuration

This section provides guidance on for securing the network configuration of the system through kernel parameters, access list control, and firewall settings.

3.1 Disable unused network protocols and devices

To reduce the attack surface of a system, unused network protocols and devices should be disabled.

3.1.1 Disable IPv6 (Manual)

Profile Applicability:

- Level 2 - Server
- Level 2 - Workstation

Description:

Although IPv6 has many advantages over IPv4, not all organizations have IPv6 or dual stack configurations implemented.

Rationale:

If IPv6 or dual stack is not to be used, it is recommended that IPv6 be disabled to reduce the attack surface of the system.

Impact:

If IPv6 is disabled through `sysctl` config, `SSH X11forwarding` may no longer function as expected. We recommend that `SSH X11forwarding` be disabled, but if required, the following will allow for `SSH X11forwarding` with IPv6 disabled through `sysctl` config:

Add the following line the `/etc/ssh/sshd_config` file:

```
AddressFamily inet
```

Run the following command to re-start the openSSH server:

```
# systemctl restart sshd
```

Audit:

Run the following commands to verify that one of the following methods has been used to disable IPv6:

IF IPv6 is disabled through the GRUB2 config:

Run the following command and verify no lines should be returned.

```
# grep "^s*linux" /boot/grub2/grub.cfg | grep -v ipv6.disable=1
```

OR

IF IPv6 is disabled through sysctl settings:

Run the following commands:

```
# sysctl net.ipv6.conf.all.disable_ipv6

net.ipv6.conf.all.disable_ipv6 = 1

# sysctl net.ipv6.conf.default.disable_ipv6

net.ipv6.conf.default.disable_ipv6 = 1

# grep -E
'^s*net\.ipv6\.conf\.(all|default)\.disable_ipv6\s*=\s*1\b(\s+#.*)?$',
/etc/sysctl.conf /etc/sysctl.d/*.conf | cut -d: -f2

net.ipv6.conf.all.disable_ipv6 = 1
net.ipv6.conf.default.disable_ipv6 = 1
```


Remediation:

Use **one** of the two following methods to disable IPv6 on the system:

To disable IPv6 through the GRUB2 config:

Edit `/etc/default/grub` and add `ipv6.disable=1` to the `GRUB_CMDLINE_LINUX` parameters:

```
GRUB_CMDLINE_LINUX="ipv6.disable=1"
```

Run the following command to update the `grub2` configuration:

```
# grub2-mkconfig -o /boot/grub2/grub.cfg
```

OR

To disable IPv6 through sysctl settings:

Set the following parameters in `/etc/sysctl.conf` or a `/etc/sysctl.d/*` file:

```
net.ipv6.conf.all.disable_ipv6 = 1
net.ipv6.conf.default.disable_ipv6 = 1
```

Run the following commands to set the active kernel parameters:

```
# sysctl -w net.ipv6.conf.all.disable_ipv6=1
# sysctl -w net.ipv6.conf.default.disable_ipv6=1
# sysctl -w net.ipv6.route.flush=1
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.		●	●
v7	9.4 Apply Host-based Firewalls or Port Filtering Apply host-based firewalls or port filtering tools on end systems, with a default-deny rule that drops all traffic except those services and ports that are explicitly allowed.	●	●	●

3.1.2 Ensure wireless interfaces are disabled (Automated)

Profile Applicability:

- Level 1 - Server
- Level 2 - Workstation

Description:

Wireless networking is used when wired networks are unavailable.

Rationale:

If wireless is not to be used, wireless devices should be disabled to reduce the potential attack surface.

Impact:

Many if not all laptop workstations and some desktop workstations will connect via wireless requiring these interfaces be enabled.

Audit:

Run the following script to verify no wireless interfaces are active on the system:

```
#!/bin/bash

if command -v nmcli >/dev/null 2>&1 ; then
    if nmcli radio all | grep -Eq '\s*\S+\s+disabled\s+\S+\s+disabled\b'; then
        echo "Wireless is not enabled"
    else
        nmcli radio all
    fi
elif [ -n "$(find /sys/class/net/*/ -type d -name wireless)" ]; then
    t=0
    mname=$(for driverdir in $(find /sys/class/net/*/ -type d -name wireless |
xargs -0 dirname); do basename "$(readlink -f
"$driverdir"/device/driver/module)";done | sort -u)
    for dm in $mname; do
        if grep -Eq "^s*install\s+$dm\s+/bin/(true|false)"
/etc/modprobe.d/*.conf; then
            /bin/true
        else
            echo "$dm is not disabled"
            t=1
        fi
    done
    [ "$t" -eq 0 ] && echo "Wireless is not enabled"
else
    echo "Wireless is not enabled"
fi
```

Output should be:

```
Wireless is not enabled
```

Remediation:

Run the following script to disable any wireless interfaces:

```
#!/bin/bash

if command -v nmcli >/dev/null 2>&1 ; then
    nmcli radio all off
else
    if [ -n "$(find /sys/class/net/*/ -type d -name wireless)" ]; then
        mname=$(for driverdir in $(find /sys/class/net/*/ -type d -name
wireless | xargs -0 dirname); do basename "$(readlink -f
"$driverdir"/device/driver/module)";done | sort -u)
        for dm in $mname; do
            echo "install $dm /bin/true" >>
/etc/modprobe.d/disable_wireless.conf
        done
    fi
fi
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.8 <u>Uninstall or Disable Unnecessary Services on Enterprise Assets and Software</u> Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.			
v7	15.4 <u>Disable Wireless Access on Devices if Not Required</u> Disable wireless access on devices that do not have a business purpose for wireless access.			
v7	15.5 <u>Limit Wireless Access on Client Devices</u> Configure wireless access on client machines that do have an essential wireless business purpose, to allow access only to authorized wireless networks and to restrict access to other wireless networks.			

3.2 Network Parameters (Host Only)

The following network parameters are intended for use if the system is to act as a host only. A system is considered host only if the system has a single interface, or has multiple interfaces but will not be configured as a router.

Note:

- sysctl settings are defined through files in `/usr/lib/sysctl.d/`, `/run/sysctl.d/`, and `/etc/sysctl.d/`.
- Files must have the the ".conf" extension.
- Vendors settings live in `/usr/lib/sysctl.d/`
- To override a whole file, create a new file with the same name in `/etc/sysctl.d/` and put new settings there.
- To override only specific settings, add a file with a lexically later name in `/etc/sysctl.d/` and put new settings there.
- The paths where sysctl preload files usually exist
 - `/run/sysctl.d/*.conf`
 - `/etc/sysctl.d/*.conf`
 - `/usr/local/lib/sysctl.d/*.conf`
 - `/usr/lib/sysctl.d/*.conf`
 - `/lib/sysctl.d/*.conf`
 - `/etc/sysctl.conf`

3.2.1 Ensure IP forwarding is disabled (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `net.ipv4.ip_forward` and `net.ipv6.conf.all.forwarding` flags are used to tell the system whether it can forward packets or not.

Rationale:

Setting the flags to 0 ensures that a system with multiple interfaces (for example, a hard proxy), will never be able to forward packets, and therefore, never serve as a router.

Audit:

Run the following commands and verify output matches:

```
# sysctl net.ipv4.ip forward

net.ipv4.ip_forward = 0

# grep -E -s "^s*net\.ipv4\.ip_forward\s*=\s*1" /etc/sysctl.conf
/etc/sysctl.d/*.conf /usr/lib/sysctl.d/*.conf /run/sysctl.d/*.conf

No value should be returned
```

IFIPv6 is enabled:

Run the following commands and verify output matches:

```
# sysctl net.ipv6.conf.all.forwarding

net.ipv6.conf.all.forwarding = 0

# grep -E -s "^s*net\.ipv6\.conf\.all\.forwarding\s*=\s*1" /etc/sysctl.conf
/etc/sysctl.d/*.conf /usr/lib/sysctl.d/*.conf /run/sysctl.d/*.conf

No value should be returned
```

OR

Verify that IPv6 is disabled:

Run the following script. Output will confirm if IPv6 is disabled on the system.

```
#!/bin/bash

[ -n "$passing" ] && passing=""
[ -z "$(grep "^s*linux" /boot/grub2/grub.cfg | grep -v ipv6.disable=1)" ] &&
passing="true"
grep -Eq "^s*net\.ipv6\.conf\.all\.disable_ipv6\s*=\s*1\b(\s+#.*)?$"
/etc/sysctl.conf \
/etc/sysctl.d/*.conf && grep -Eq
"^s*net\.ipv6\.conf\.default\.disable_ipv6\s*=\s*1\b(\s+#.*)?$" \
/etc/sysctl.conf /etc/sysctl.d/*.conf && sysctl
net.ipv6.conf.all.disable_ipv6 | \
grep -Eq "^s*net\.ipv6\.conf\.all\.disable_ipv6\s*=\s*1\b(\s+#.*)?$" && \
sysctl net.ipv6.conf.default.disable_ipv6 | \
grep -Eq "^s*net\.ipv6\.conf\.default\.disable_ipv6\s*=\s*1\b(\s+#.*)?$" &&
passing="true"
if [ "$passing" = true ] ; then
    echo "IPv6 is disabled on the system"
else
    echo "IPv6 is enabled on the system"
fi
```

Remediation:

Run the following commands to restore the default parameters and set the active kernel parameters:

```
# grep -Els "^s*net\.ipv4\.ip_forward\s*=\s*1" /etc/sysctl.conf
/etc/sysctl.d/*.conf /usr/lib/sysctl.d/*.conf /run/sysctl.d/*.conf | while
read filename; do sed -ri "s/^\s*(net\.ipv4\.ip_forward\s*) (=) (\s*\S+\b).*$/#
*REMOVED* \1/" $filename; done; sysctl -w net.ipv4.ip_forward=0; sysctl -w
net.ipv4.route.flush=1

# grep -Els "^s*net\.ipv6\.conf\.all\.forwarding\s*=\s*1" /etc/sysctl.conf
/etc/sysctl.d/*.conf /usr/lib/sysctl.d/*.conf /run/sysctl.d/*.conf | while
read filename; do sed -ri
"s/^\s*(net\.ipv6\.conf\.all\.forwarding\s*) (=) (\s*\S+\b).*$/# *REMOVED* \1/"
$filename; done; sysctl -w net.ipv6.conf.all.forwarding=0; sysctl -w
net.ipv6.route.flush=1
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 Establish and Maintain a Secure Configuration Process Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	5.1 Establish Secure Configurations Maintain documented, standard security configuration standards for all authorized operating systems and software.			

3.2.2 Ensure packet redirect sending is disabled (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

ICMP Redirects are used to send routing information to other hosts. As a host itself does not act as a router (in a host only configuration), there is no need to send redirects.

Rationale:

An attacker could use a compromised host to send invalid ICMP redirects to other router devices in an attempt to corrupt routing and have users access a system set up by the attacker as opposed to a valid system.

Audit:

Run the following commands and verify output matches:

```
# sysctl net.ipv4.conf.all.send_redirects
net.ipv4.conf.all.send_redirects = 0

# sysctl net.ipv4.conf.default.send_redirects
net.ipv4.conf.default.send_redirects = 0

# grep "net\.ipv4\.conf\.all\.send_redirects" /etc/sysctl.conf
/etc/sysctl.d/*.conf /usr/lib/sysctl.d/*.conf /run/sysctl.d/*.conf

net.ipv4.conf.all.send_redirects = 0

# grep "net\.ipv4\.conf\.default\.send_redirects" /etc/sysctl.conf
/etc/sysctl.d/*.conf /usr/lib/sysctl.d/*.conf /run/sysctl.d/*.conf

net.ipv4.conf.default.send_redirects= 0
```

Remediation:

Set the following parameters in `/etc/sysctl.conf` or a `/etc/sysctl.d/*` file:

```
net.ipv4.conf.all.send_redirects = 0
net.ipv4.conf.default.send_redirects = 0
```

Run the following commands to set the active kernel parameters:

```
# sysctl -w net.ipv4.conf.all.send_redirects=0
# sysctl -w net.ipv4.conf.default.send_redirects=0
# sysctl -w net.ipv4.route.flush=1
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 <u>Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	5.1 <u>Establish Secure Configurations</u> Maintain documented, standard security configuration standards for all authorized operating systems and software.			

3.3 Network Parameters (Host and Router)

The following network parameters are intended for use on both host only and router systems. A system acts as a router if it has at least two interfaces and is configured to perform routing functions.

Note:

- sysctl settings are defined through files in `/usr/lib/sysctl.d/`, `/run/sysctl.d/`, and `/etc/sysctl.d/`.
- Files must have the the ".conf" extension.
- Vendors settings live in `/usr/lib/sysctl.d/`
- To override a whole file, create a new file with the same name in `/etc/sysctl.d/` and put new settings there.
- To override only specific settings, add a file with a lexically later name in `/etc/sysctl.d/` and put new settings there.
- The paths where sysctl preload files usually exist
 - `/run/sysctl.d/*.conf`
 - `/etc/sysctl.d/*.conf`
 - `/usr/local/lib/sysctl.d/*.conf`
 - `/usr/lib/sysctl.d/*.conf`
 - `/lib/sysctl.d/*.conf`
 - `/etc/sysctl.conf`

3.3.1 Ensure source routed packets are not accepted (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

In networking, source routing allows a sender to partially or fully specify the route packets take through a network. In contrast, non-source routed packets travel a path determined by routers in the network. In some cases, systems may not be routable or reachable from some locations (e.g. private addresses vs. Internet routable), and so source routed packets would need to be used.

Rationale:

Setting `net.ipv4.conf.all.accept_source_route`, `net.ipv4.conf.default.accept_source_route`, `net.ipv6.conf.all.accept_source_route` and `net.ipv6.conf.default.accept_source_route` to 0 disables the system from accepting source routed packets. Assume this system was capable of routing packets to Internet routable addresses on one interface and private addresses on another interface. Assume that the private addresses were not routable to the Internet routable addresses and vice versa. Under normal routing circumstances, an attacker from the Internet routable addresses could not use the system as a way to reach the private address systems. If, however, source routed packets were allowed, they could be used to gain access to the private address systems as the route could be specified, rather than rely on routing protocols that did not allow this routing.

Audit:

Run the following commands and verify output matches:

```
# sysctl net.ipv4.conf.all.accept_source_route
net.ipv4.conf.all.accept_source_route = 0

# sysctl net.ipv4.conf.default.accept_source_route
net.ipv4.conf.default.accept_source_route = 0

# grep "net\.ipv4\.conf\.all\.accept_source_route" /etc/sysctl.conf
/etc/sysctl.d/*.conf /usr/lib/sysctl.d/*.conf /run/sysctl.d/*.conf

net.ipv4.conf.all.accept_source_route= 0

# grep "net\.ipv4\.conf\.default\.accept_source_route" /etc/sysctl.conf
/etc/sysctl.d/*.conf /usr/lib/sysctl.d/*.conf /run/sysctl.d/*.conf

net.ipv4.conf.default.accept_source_route= 0
```

IF IPv6 is enabled:

Run the following commands and verify output matches:

```
# sysctl net.ipv6.conf.all.accept_source_route
net.ipv6.conf.all.accept_source_route = 0

# sysctl net.ipv6.conf.default.accept_source_route
net.ipv6.conf.default.accept_source_route = 0

# grep "net\.ipv6\.conf\.all\.accept_source_route" /etc/sysctl.conf
/etc/sysctl.d/*.conf /usr/lib/sysctl.d/*.conf /run/sysctl.d/*.conf

net.ipv4.conf.all.accept_source_route= 0

# grep "net\.ipv6\.conf\.default\.accept_source_route" /etc/sysctl.conf
/etc/sysctl.d/*.conf /usr/lib/sysctl.d/*.conf /run/sysctl.d/*.conf

net.ipv6.conf.default.accept_source_route= 0
```

OR

Verify that IPv6 is disabled:

Run the following script. Output will confirm if IPv6 is disabled on the system.

```
#!/bin/bash

[ -n "$passing" ] && passing=""
[ -z "$(grep "^s*linux" /boot/grub2/grub.cfg | grep -v ipv6.disable=1)" ] &&
passing="true"
grep -Eq "^s*net\.ipv6\.conf\.all\.disable_ipv6\s*=\s*1\b(\s+#.*)?$"
/etc/sysctl.conf \
/etc/sysctl.d/*.conf && grep -Eq
"^s*net\.ipv6\.conf\.default\.disable_ipv6\s*=\s*1\b(\s+#.*)?$" \
/etc/sysctl.conf /etc/sysctl.d/*.conf && sysctl
net.ipv6.conf.all.disable_ipv6 | \
grep -Eq "^s*net\.ipv6\.conf\.all\.disable_ipv6\s*=\s*1\b(\s+#.*)?$" && \
sysctl net.ipv6.conf.default.disable_ipv6 | \
grep -Eq "^s*net\.ipv6\.conf\.default\.disable_ipv6\s*=\s*1\b(\s+#.*)?$" &&
passing="true"
if [ "$passing" = true ] ; then
    echo "IPv6 is disabled on the system"
else
    echo "IPv6 is enabled on the system"
fi
```

Remediation:

Set the following parameters in `/etc/sysctl.conf` or a `/etc/sysctl.d/*` file:

```
net.ipv4.conf.all.accept_source_route = 0
net.ipv4.conf.default.accept_source_route = 0
```

Run the following commands to set the active kernel parameters:

```
# sysctl -w net.ipv4.conf.all.accept_source_route=0
# sysctl -w net.ipv4.conf.default.accept_source_route=0
# sysctl -w net.ipv4.route.flush=1
```

IF IPv6 is not disabled:

Set the following parameters in `/etc/sysctl.conf` or a `/etc/sysctl.d/*` file:

```
net.ipv6.conf.all.accept_source_route = 0
net.ipv6.conf.default.accept_source_route = 0
```

Run the following commands to set the active kernel parameters:

```
# sysctl -w net.ipv6.conf.all.accept_source_route=0
# sysctl -w net.ipv6.conf.default.accept_source_route=0
# sysctl -w net.ipv6.route.flush=1
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.1 Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	<u>5.1 Establish Secure Configurations</u> Maintain documented, standard security configuration standards for all authorized operating systems and software.			

3.3.2 Ensure ICMP redirects are not accepted (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

ICMP redirect messages are packets that convey routing information and tell your host (acting as a router) to send packets via an alternate path. It is a way of allowing an outside routing device to update your system routing tables. By setting `net.ipv4.conf.all.accept_redirects` and `net.ipv6.conf.all.accept_redirects` to 0, the system will not accept any ICMP redirect messages, and therefore, won't allow outsiders to update the system's routing tables.

Rationale:

Attackers could use bogus ICMP redirect messages to maliciously alter the system routing tables and get them to send packets to incorrect networks and allow your system packets to be captured.

Audit:

Run the following commands and verify output matches:

```
# sysctl net.ipv4.conf.all.accept_redirects
net.ipv4.conf.all.accept_redirects = 0

# sysctl net.ipv4.conf.default.accept_redirects
net.ipv4.conf.default.accept_redirects = 0

# grep "net\.ipv4\.conf\.all\.accept_redirects" /etc/sysctl.conf
/etc/sysctl.d/*.conf /usr/lib/sysctl.d/*.conf /run/sysctl.d/*.conf
net.ipv4.conf.all.accept_redirects= 0

# grep "net\.ipv4\.conf\.default\.accept_redirects" /etc/sysctl.conf
/etc/sysctl.d/*.conf /usr/lib/sysctl.d/*.conf /run/sysctl.d/*.conf
net.ipv4.conf.default.accept_redirects= 0
```

IF IPv6 is not disabled:

Run the following commands and verify output matches:


```
# sysctl net.ipv6.conf.all.accept_redirects

net.ipv6.conf.all.accept_redirects = 0

# sysctl net.ipv6.conf.default.accept_redirects

net.ipv6.conf.default.accept_redirects = 0

# grep "net\.ipv6\.conf\.all\.accept_redirects" /etc/sysctl.conf
/etc/sysctl.d/*.conf /usr/lib/sysctl.d/*.conf /run/sysctl.d/*.conf

net.ipv6.conf.all.accept_redirects= 0

# grep "net\.ipv6\.conf\.default\.accept_redirects" /etc/sysctl.conf
/etc/sysctl.d/*.conf /usr/lib/sysctl.d/*.conf /run/sysctl.d/*.conf

net.ipv6.conf.default.accept_redirects= 0
```

OR verify IPv6 is disabled:

Run the following script. Output will confirm if IPv6 is disabled on the system.

```
#!/bin/bash

[ -n "$passing" ] && passing=""
[ -z "$(grep "^s*linux" /boot/grub2/grub.cfg | grep -v ipv6.disable=1)" ] &&
passing="true"
grep -Eq "^s*net\.ipv6\.conf\.all\.disable_ipv6\s*=\s*1\b(\s+#.*)?$"
/etc/sysctl.conf \
/etc/sysctl.d/*.conf && grep -Eq
"^s*net\.ipv6\.conf\.default\.disable_ipv6\s*=\s*1\b(\s+#.*)?$" \
/etc/sysctl.conf /etc/sysctl.d/*.conf && sysctl
net.ipv6.conf.all.disable_ipv6 | \
grep -Eq "^s*net\.ipv6\.conf\.all\.disable_ipv6\s*=\s*1\b(\s+#.*)?$" && \
sysctl net.ipv6.conf.default.disable_ipv6 | \
grep -Eq "^s*net\.ipv6\.conf\.default\.disable_ipv6\s*=\s*1\b(\s+#.*)?$" &&
passing="true"
if [ "$passing" = true ] ; then
    echo "IPv6 is disabled on the system"
else
    echo "IPv6 is enabled on the system"
fi
```

Remediation:

Set the following parameters in `/etc/sysctl.conf` or a `/etc/sysctl.d/*` file:

```
net.ipv4.conf.all.accept_redirects = 0
net.ipv4.conf.default.accept_redirects = 0
```

Run the following commands to set the active kernel parameters:

```
# sysctl -w net.ipv4.conf.all.accept_redirects=0
# sysctl -w net.ipv4.conf.default.accept_redirects=0
# sysctl -w net.ipv4.route.flush=1
```

IF IPv6 is not disabled

Set the following parameters in `/etc/sysctl.conf` or a `/etc/sysctl.d/*` file:

```
net.ipv6.conf.all.accept_redirects = 0
net.ipv6.conf.default.accept_redirects = 0
```

Run the following commands to set the active kernel parameters:

```
# sysctl -w net.ipv6.conf.all.accept_redirects=0
# sysctl -w net.ipv6.conf.default.accept_redirects=0
# sysctl -w net.ipv6.route.flush=1
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 <u>Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	5.1 <u>Establish Secure Configurations</u> Maintain documented, standard security configuration standards for all authorized operating systems and software.			

3.3.3 Ensure secure ICMP redirects are not accepted (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

Secure ICMP redirects are the same as ICMP redirects, except they come from gateways listed on the default gateway list. It is assumed that these gateways are known to your system, and that they are likely to be secure.

Rationale:

It is still possible for even known gateways to be compromised. Setting `net.ipv4.conf.all.secure_redirects` to 0 protects the system from routing table updates by possibly compromised known gateways.

Audit:

Run the following commands and verify output matches:

```
# sysctl net.ipv4.conf.all.secure_redirects
net.ipv4.conf.all.secure_redirects = 0

# sysctl net.ipv4.conf.default.secure_redirects
net.ipv4.conf.default.secure_redirects = 0

# grep "net\.ipv4\.conf\.all\.secure_redirects" /etc/sysctl.conf
/etc/sysctl.d/*.conf /usr/lib/sysctl.d/*.conf /run/sysctl.d/*.conf

net.ipv4.conf.all.secure_redirects= 0

# grep "net\.ipv4\.conf\.default\.secure_redirects" /etc/sysctl.conf
/etc/sysctl.d/*.conf /usr/lib/sysctl.d/*.conf /run/sysctl.d/*.conf

net.ipv4.conf.default.secure_redirects= 0
```

Remediation:

Set the following parameters in `/etc/sysctl.conf` or a `/etc/sysctl.d/*` file:

```
net.ipv4.conf.all.secure_redirects = 0
net.ipv4.conf.default.secure_redirects = 0
```

Run the following commands to set the active kernel parameters:

```
# sysctl -w net.ipv4.conf.all.secure_redirects=0
# sysctl -w net.ipv4.conf.default.secure_redirects=0
# sysctl -w net.ipv4.route.flush=1
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 <u>Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	5.1 <u>Establish Secure Configurations</u> Maintain documented, standard security configuration standards for all authorized operating systems and software.			

3.3.4 Ensure suspicious packets are logged (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

When enabled, this feature logs packets with un-routable source addresses to the kernel log.

Rationale:

Enabling this feature and logging these packets allows an administrator to investigate the possibility that an attacker is sending spoofed packets to their system.

Audit:

Run the following commands and verify output matches:

```
# sysctl net.ipv4.conf.all.log_martians
net.ipv4.conf.all.log_martians = 1

# sysctl net.ipv4.conf.default.log_martians
net.ipv4.conf.default.log_martians = 1

# grep "net\.ipv4\.conf\.all\.log_martians" /etc/sysctl.conf
/etc/sysctl.d/*.conf /usr/lib/sysctl.d/*.conf /run/sysctl.d/*.conf
net.ipv4.conf.all.log_martians = 1

# grep "net\.ipv4\.conf\.default\.log_martians" /etc/sysctl.conf
/etc/sysctl.d/*.conf /usr/lib/sysctl.d/*.conf /run/sysctl.d/*.conf
net.ipv4.conf.default.log_martians = 1
```

Remediation:

Set the following parameters in `/etc/sysctl.conf` or a `/etc/sysctl.d/*` file:

```
net.ipv4.conf.all.log_martians = 1
net.ipv4.conf.default.log_martians = 1
```

Run the following commands to set the active kernel parameters:

```
# sysctl -w net.ipv4.conf.all.log_martians=1
# sysctl -w net.ipv4.conf.default.log_martians=1
# sysctl -w net.ipv4.route.flush=1
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.5 <u>Collect Detailed Audit Logs</u> Configure detailed audit logging for enterprise assets containing sensitive data. Include event source, date, username, timestamp, source addresses, destination addresses, and other useful elements that could assist in a forensic investigation.			
v7	6.2 <u>Activate audit logging</u> Ensure that local logging has been enabled on all systems and networking devices.			
v7	6.3 <u>Enable Detailed Logging</u> Enable system logging to include detailed information such as an event source, date, user, timestamp, source addresses, destination addresses, and other useful elements.			

3.3.5 Ensure broadcast ICMP requests are ignored (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

Setting `net.ipv4.icmp_echo_ignore_broadcasts` to 1 will cause the system to ignore all ICMP echo and timestamp requests to broadcast and multicast addresses.

Rationale:

Accepting ICMP echo and timestamp requests with broadcast or multicast destinations for your network could be used to trick your host into starting (or participating) in a Smurf attack. A Smurf attack relies on an attacker sending large amounts of ICMP broadcast messages with a spoofed source address. All hosts receiving this message and responding would send echo-reply messages back to the spoofed address, which is probably not routable. If many hosts respond to the packets, the amount of traffic on the network could be significantly multiplied.

Audit:

Run the following commands and verify output matches:

```
# sysctl net.ipv4.icmp_echo_ignore_broadcasts

net.ipv4.icmp_echo_ignore_broadcasts = 1

# grep "net\.ipv4\.icmp_echo_ignore_broadcasts" /etc/sysctl.conf
/etc/sysctl.d/*.conf /usr/lib/sysctl.d/*.conf /run/sysctl.d/*.conf

net.ipv4.icmp_echo_ignore_broadcasts = 1
```

Remediation:

Set the following parameters in `/etc/sysctl.conf` or a `/etc/sysctl.d/*` file:

```
net.ipv4.icmp_echo_ignore_broadcasts = 1
```

Run the following commands to set the active kernel parameters:

```
# sysctl -w net.ipv4.icmp_echo_ignore_broadcasts=1
# sysctl -w net.ipv4.route.flush=1
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.1 Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	<u>5.1 Establish Secure Configurations</u> Maintain documented, standard security configuration standards for all authorized operating systems and software.			

3.3.6 Ensure bogus ICMP responses are ignored (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

Setting `icmp_ignore_bogus_error_responses` to 1 prevents the kernel from logging bogus responses (RFC-1122 non-compliant) from broadcast retransmits, keeping file systems from filling up with useless log messages.

Rationale:

Some routers (and some attackers) will send responses that violate RFC-1122 and attempt to fill up a log file system with many useless error messages.

Audit:

Run the following commands and verify output matches:

```
# sysctl net.ipv4.icmp_ignore_bogus_error_responses
net.ipv4.icmp_ignore_bogus_error_responses = 1

# grep "net.ipv4.icmp ignore bogus error responses" /etc/sysctl.conf
/etc/sysctl.d/*.conf /usr/lib/sysctl.d/*.conf /run/sysctl.d/*.conf
net.ipv4.icmp_ignore_bogus_error_responses = 1
```

Remediation:

Set the following parameter in `/etc/sysctl.conf` or a `/etc/sysctl.d/*` file:

```
net.ipv4.icmp_ignore_bogus_error_responses = 1
```

Run the following commands to set the active kernel parameters:

```
# sysctl -w net.ipv4.icmp_ignore_bogus_error_responses=1
# sysctl -w net.ipv4.route.flush=1
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.1 Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	<u>5.1 Establish Secure Configurations</u> Maintain documented, standard security configuration standards for all authorized operating systems and software.			

3.3.7 Ensure Reverse Path Filtering is enabled (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

Setting `net.ipv4.conf.all.rp_filter` and `net.ipv4.conf.default.rp_filter` to 1 forces the Linux kernel to utilize reverse path filtering on a received packet to determine if the packet was valid. Essentially, with reverse path filtering, if the return packet does not go out the same interface that the corresponding source packet came from, the packet is dropped (and logged if `log_martians` is set).

Rationale:

Setting these flags is a good way to deter attackers from sending your system bogus packets that cannot be responded to. One instance where this feature breaks down is if asymmetrical routing is employed. This would occur when using dynamic routing protocols (bgp, ospf, etc) on your system. If you are using asymmetrical routing on your system, you will not be able to enable this feature without breaking the routing.

Audit:

Run the following commands and verify output matches:

```
# sysctl net.ipv4.conf.all.rp_filter
net.ipv4.conf.all.rp_filter = 1

# sysctl net.ipv4.conf.default.rp_filter
net.ipv4.conf.default.rp_filter = 1

# grep "net\.ipv4\.conf\.all\.rp_filter" /etc/sysctl.conf
/etc/sysctl.d/*.conf /usr/lib/sysctl.d/*.conf /run/sysctl.d/*.conf

net.ipv4.conf.all.rp_filter = 1

# grep "net\.ipv4\.conf\.default\.rp_filter" /etc/sysctl.conf
/etc/sysctl.d/*.conf /usr/lib/sysctl.d/*.conf /run/sysctl.d/*.conf

net.ipv4.conf.default.rp_filter = 1
```

Remediation:

Set the following parameters in `/etc/sysctl.conf` or a `/etc/sysctl.d/*` file:

```
net.ipv4.conf.all.rp_filter = 1
net.ipv4.conf.default.rp_filter = 1
```

Run the following commands to set the active kernel parameters:

```
# sysctl -w net.ipv4.conf.all.rp_filter=1
# sysctl -w net.ipv4.conf.default.rp_filter=1
# sysctl -w net.ipv4.route.flush=1
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 <u>Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	5.1 <u>Establish Secure Configurations</u> Maintain documented, standard security configuration standards for all authorized operating systems and software.			

3.3.8 Ensure TCP SYN Cookies is enabled (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

When `tcp_syncookies` is set, the kernel will handle TCP SYN packets normally until the half-open connection queue is full, at which time, the SYN cookie functionality kicks in. SYN cookies work by not using the SYN queue at all. Instead, the kernel simply replies to the SYN with a SYN|ACK, but will include a specially crafted TCP sequence number that encodes the source and destination IP address and port number and the time the packet was sent. A legitimate connection would send the ACK packet of the three way handshake with the specially crafted sequence number. This allows the system to verify that it has received a valid response to a SYN cookie and allow the connection, even though there is no corresponding SYN in the queue.

Rationale:

Attackers use SYN flood attacks to perform a denial of service attack on a system by sending many SYN packets without completing the three way handshake. This will quickly use up slots in the kernel's half-open connection queue and prevent legitimate connections from succeeding. SYN cookies allow the system to keep accepting valid connections, even if under a denial of service attack.

Audit:

Run the following commands and verify output matches:

```
# sysctl net.ipv4.tcp_syncookies

net.ipv4.tcp_syncookies = 1

# grep "net\.ipv4\.tcp_syncookies" /etc/sysctl.conf /etc/sysctl.d/*.conf
/usr/lib/sysctl.d/*.conf /run/sysctl.d/*.conf

net.ipv4.tcp_syncookies = 1
```

Remediation:

Set the following parameters in `/etc/sysctl.conf` or a `/etc/sysctl.d/*` file:

```
net.ipv4.tcp_syncookies = 1
```

Run the following commands to set the active kernel parameters:

```
# sysctl -w net.ipv4.tcp_syncookies=1  
# sysctl -w net.ipv4.route.flush=1
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 <u>Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	5.1 <u>Establish Secure Configurations</u> Maintain documented, standard security configuration standards for all authorized operating systems and software.			

3.3.9 Ensure IPv6 router advertisements are not accepted (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

This setting disables the system's ability to accept IPv6 router advertisements.

Rationale:

It is recommended that systems do not accept router advertisements as they could be tricked into routing traffic to compromised machines. Setting hard routes within the system (usually a single default route to a trusted router) protects the system from bad routes.

Audit:

Run the following commands and verify output matches:

```
# sysctl net.ipv6.conf.all.accept_ra
net.ipv6.conf.all.accept_ra = 0

# sysctl net.ipv6.conf.default.accept_ra
net.ipv6.conf.default.accept_ra = 0

# grep "net\.ipv6\.conf\.all\.accept_ra" /etc/sysctl.conf
/etc/sysctl.d/*.conf /usr/lib/sysctl.d/*.conf /run/sysctl.d/*.conf
net.ipv6.conf.all.accept_ra = 0

# grep "net\.ipv6\.conf\.default\.accept_ra" /etc/sysctl.conf
/etc/sysctl.d/*.conf /usr/lib/sysctl.d/*.conf /run/sysctl.d/*.conf
net.ipv6.conf.default.accept_ra = 0
```

OR Verify IPv6 is disabled:

Run the following script. Output will confirm if IPv6 is disabled on the system.

```
#!/bin/bash

[ -n "$passing" ] && passing=""
[ -z "$(grep "^s*linux" /boot/grub2/grub.cfg | grep -v ipv6.disable=1)" ] &&
passing="true"
grep -Eq "\s*net\.ipv6\.conf\.all\.disable_ipv6\s*=\s*1\b(\s+#.*)?$"
/etc/sysctl.conf \
/etc/sysctl.d/*.conf && grep -Eq
"\s*net\.ipv6\.conf\.default\.disable_ipv6\s*=\s*1\b(\s+#.*)?$" \
/etc/sysctl.conf /etc/sysctl.d/*.conf && sysctl
net.ipv6.conf.all.disable_ipv6 | \
grep -Eq "\s*net\.ipv6\.conf\.all\.disable_ipv6\s*=\s*1\b(\s+#.*)?$" && \
sysctl net.ipv6.conf.default.disable_ipv6 | \
grep -Eq "\s*net\.ipv6\.conf\.default\.disable_ipv6\s*=\s*1\b(\s+#.*)?$" &&
passing="true"
if [ "$passing" = true ] ; then
    echo "IPv6 is disabled on the system"
else
    echo "IPv6 is enabled on the system"
fi
```


Remediation:

IF IPv6 is enabled:

Set the following parameters in `/etc/sysctl.conf` or a `/etc/sysctl.d/*` file:

```
net.ipv6.conf.all.accept_ra = 0
net.ipv6.conf.default.accept_ra = 0
```

Run the following commands to set the active kernel parameters:

```
# sysctl -w net.ipv6.conf.all.accept_ra=0
# sysctl -w net.ipv6.conf.default.accept_ra=0
# sysctl -w net.ipv6.route.flush=1
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 <u>Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	5.1 <u>Establish Secure Configurations</u> Maintain documented, standard security configuration standards for all authorized operating systems and software.			

3.4 Uncommon Network Protocols

The Linux kernel modules support several network protocols that are not commonly used. If these protocols are not needed, it is recommended that they be disabled in the kernel.

Note: This should not be considered a comprehensive list of uncommon network protocols, you may wish to consider additions to those listed here for your environment.

3.4.1 Ensure DCCP is disabled (Automated)

Profile Applicability:

- Level 2 - Server
- Level 2 - Workstation

Description:

The Datagram Congestion Control Protocol (DCCP) is a transport layer protocol that supports streaming media and telephony. DCCP provides a way to gain access to congestion control, without having to do it at the application layer, but does not provide in-sequence delivery.

Rationale:

If the protocol is not required, it is recommended that the drivers not be installed to reduce the potential attack surface.

Audit:

Run the following commands and verify the output is as indicated:

```
# modprobe -n -v dccp
install /bin/true

# lsmod | grep dccp
<No output>
```

Remediation:

Edit or create a file in the `/etc/modprobe.d/` directory ending in `.conf`

Example: `vim /etc/modprobe.d/dccp.conf`

Add the following line:

```
install dccp /bin/true
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software</u> Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.			
v7	<u>9.2 Ensure Only Approved Ports, Protocols and Services Are Running</u> Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.			

3.4.2 Ensure SCTP is disabled (Automated)

Profile Applicability:

- Level 2 - Server
- Level 2 - Workstation

Description:

The Stream Control Transmission Protocol (SCTP) is a transport layer protocol used to support message oriented communication, with several streams of messages in one connection. It serves a similar function as TCP and UDP, incorporating features of both. It is message-oriented like UDP, and ensures reliable in-sequence transport of messages with congestion control like TCP.

Rationale:

If the protocol is not being used, it is recommended that kernel module not be loaded, disabling the service to reduce the potential attack surface.

Audit:

Run the following commands and verify the output is as indicated:

```
# modprobe -n -v sctp
install /bin/true

# lsmod | grep sctp
<No output>
```

Remediation:

Edit or create a file in the `/etc/modprobe.d/` directory ending in `.conf`

Example: `vim /etc/modprobe.d/sctp.conf`

Add the following line:

```
install sctp /bin/true
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software</u> Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.			
v7	<u>9.2 Ensure Only Approved Ports, Protocols and Services Are Running</u> Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.			

3.5 Firewall Configuration

A firewall is a set of rules. When a data packet moves into or out of a protected network space, its contents (in particular, information about its origin, target, and the protocol it plans to use) are tested against the firewall rules to see if it should be allowed through

To provide a Host Based Firewall, the Linux kernel includes support for:

- Netfilter - A set of hooks inside the Linux kernel that allows kernel modules to register callback functions with the network stack. A registered callback function is then called back for every packet that traverses the respective hook within the network stack. Includes the `ip_tables`, `ip6_tables`, `arp_tables`, and `ebtables` kernel modules. These modules are some of the significant parts of the Netfilter hook system.
- nftables - A subsystem of the Linux kernel providing filtering and classification of network packets/datagrams/frames. nftables is supposed to replace certain parts of Netfilter, while keeping and reusing most of it. nftables utilizes the building blocks of the Netfilter infrastructure, such as the existing hooks into the networking stack, connection tracking system, userspace queueing component, and logging subsystem. **Is available in Linux kernels 3.13 and newer.**

In order to configure firewall rules for Netfilter or nftables, a firewall utility needs to be installed. Guidance has been included for the following firewall utilities:

- FirewallD - Provides firewall features by acting as a front-end for the Linux kernel's netfilter framework via the iptables backend. Starting in v0.6.0, FirewallD added support for acting as a front-end for the Linux kernel's netfilter framework via the nftables userspace utility, acting as an alternative to the nft command line program. firewalld supports both IPv4 and IPv6 networks and can administer separate firewall zones with varying degrees of trust as defined in zone profiles.
- nftables - Includes the nft utility for configuration of the nftables subsystem of the Linux kernel
- iptables - Includes the iptables, ip6tables, arptables and ebtables utilities for configuration Netfilter and the `ip_tables`, `ip6_tables`, `arp_tables`, and `ebtables` kernel modules.

Note:

- *Only **one** method should be used to configure a firewall on the system. Use of more than one method could produce unexpected results.*
- *This section is intended only to ensure the resulting firewall rules are in place, not how they are configured.*

3.5.1 Configure firewalld

If nftables or iptables are being used in your environment, please follow the guidance in their respective section and pass-over the guidance in this section.

firewalld (Dynamic Firewall Manager) provides a dynamically managed firewall with support for network/firewall “zones” to assign a level of trust to a network and its associated connections, interfaces or sources. It has support for IPv4, IPv6, Ethernet bridges and also for IPSet firewall settings. There is a separation of the runtime and permanent configuration options. It also provides an interface for services or applications to add iptables, ip6tables and ebtables rules directly. This interface can also be used by advanced users.

In the v0.6.0 release, firewalld gained support for using nftables as a firewall back-end.

Note: Configuration of a live system's firewall directly over a remote connection will often result in being locked out.

3.5.1.1 Ensure firewalld is installed (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

firewalld is a firewall management tool for Linux operating systems. It provides firewall features by acting as a front-end for the Linux kernel's netfilter framework via the iptables backend or provides firewall features by acting as a front-end for the Linux kernel's netfilter framework via the nftables utility.

firewalld replaces iptables as the default firewall management tool. Use the firewalld utility to configure a firewall for less complex firewalls. The utility is easy to use and covers the typical use cases scenario. FirewallD supports both IPv4 and IPv6 networks and can administer separate firewall zones with varying degrees of trust as defined in zone profiles.

Note: Starting in v0.6.0, FirewallD added support for acting as a front-end for the Linux kernel's netfilter framework via the nftables userspace utility, acting as an alternative to the nft command line program.

Rationale:

A firewall utility is required to configure the Linux kernel's netfilter framework via the iptables or nftables back-end.

The Linux kernel's netfilter framework host-based firewall can protect against threats originating from within a corporate network to include malicious mobile code and poorly configured software on a host.

*Note: Only **one** firewall utility should be installed and configured. FirewallD is dependent on the iptables package.*

Impact:

Changing firewall settings while connected over the network can result in being locked out of the system.

Audit:

Run the following command to verify that FirewallD and iptables are installed:

```
# rpm -q firewalld iptables  
  
firewalld-<version>  
iptables-<version>
```

Remediation:

Run the following command to install FirewallD and iptables:

```
# yum install firewalld iptables
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.4 <u>Implement and Manage a Firewall on Servers</u> Implement and manage a firewall on servers, where supported. Example implementations include a virtual firewall, operating system firewall, or a third-party firewall agent.			
v7	9.4 <u>Apply Host-based Firewalls or Port Filtering</u> Apply host-based firewalls or port filtering tools on end systems, with a default-deny rule that drops all traffic except those services and ports that are explicitly allowed.			

3.5.1.2 Ensure iptables-services not installed with firewalld (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `iptables-services` package contains the `iptables.service` and `ip6tables.service`. These services allow for management of the Host Based Firewall provided by the `iptables` package.

Rationale:

`iptables.service` and `ip6tables.service` are still supported and can be installed with the `iptables-services` package. Running both `firewalld` and the services included in the `iptables-services` package may lead to conflict.

Impact:

Running both `firewalld` and `iptables/ip6tables` service may lead to conflict.

Audit:

Run the following commands to verify that the `iptables-services` package is not installed

```
# rpm -q iptables-services  
package iptables-services is not installed
```

Remediation:

Run the following commands to stop the services included in the `iptables-services` package and remove the `iptables-services` package

```
# systemctl stop iptables  
# systemctl stop ip6tables  
# yum remove iptables-services
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.4 Implement and Manage a Firewall on Servers</u> Implement and manage a firewall on servers, where supported. Example implementations include a virtual firewall, operating system firewall, or a third-party firewall agent.			
v8	<u>4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software</u> Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.			
v7	<u>9.4 Apply Host-based Firewalls or Port Filtering</u> Apply host-based firewalls or port filtering tools on end systems, with a default-deny rule that drops all traffic except those services and ports that are explicitly allowed.			

3.5.1.3 Ensure nftables either not installed or masked with firewalld (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

nftables is a subsystem of the Linux kernel providing filtering and classification of network packets/datagrams/frames and is the successor to iptables.

_Note: Support for using nftables as the back-end for firewalld was added in release v0.6.0. In Fedora 19 Linux derivatives, firewalld utilizes iptables as its back-end by default.

Rationale:

Running both firewalld and nftables may lead to conflict.

Note: firewalld may configured as the front-end to nftables. If this case, nftables should be stopped and masked instead of removed.

Audit:

Run the following command to verify that nftables is not installed:

```
# rpm -q nftables
package nftables is not installed
```

OR

Run the following commands to verify that nftables is stopped:

```
# systemctl status nftables | grep "Active: " | grep -E " active
\((running|exited)\) "
```

No output should be returned

Run the following command to verify nftables.service is masked:

```
# systemctl is-enabled nftables
masked
```

Remediation:

Run the following command to remove `nftables`:

```
# yum remove nftables
```

OR

Run the following command to stop and mask `nftables`:

```
systemctl --now mask nftables
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.4 <u>Implement and Manage a Firewall on Servers</u> Implement and manage a firewall on servers, where supported. Example implementations include a virtual firewall, operating system firewall, or a third-party firewall agent.			
v8	4.8 <u>Uninstall or Disable Unnecessary Services on Enterprise Assets and Software</u> Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.			
v7	9.4 <u>Apply Host-based Firewalls or Port Filtering</u> Apply host-based firewalls or port filtering tools on end systems, with a default-deny rule that drops all traffic except those services and ports that are explicitly allowed.			

3.5.1.4 Ensure firewalld service enabled and running (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

`firewalld.service` enables the enforcement of firewall rules configured through `firewalld`

Rationale:

Ensure that the `firewalld.service` is enabled and running to enforce firewall rules configured through `firewalld`

Impact:

Changing firewall settings while connected over network can result in being locked out of the system.

Audit:

Run the following command to verify that `firewalld` is enabled:

```
# systemctl is-enabled firewalld  
enabled
```

Run the following command to verify that `firewalld` is running

```
# firewall-cmd --state  
running
```

Remediation:

Run the following command to unmask `firewalld`

```
# systemctl unmask firewalld
```

Run the following command to enable and start `firewalld`

```
# systemctl --now enable firewalld
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.4 <u>Implement and Manage a Firewall on Servers</u> Implement and manage a firewall on servers, where supported. Example implementations include a virtual firewall, operating system firewall, or a third-party firewall agent.			
v7	9.4 <u>Apply Host-based Firewalls or Port Filtering</u> Apply host-based firewalls or port filtering tools on end systems, with a default-deny rule that drops all traffic except those services and ports that are explicitly allowed.			

3.5.1.5 Ensure firewalld default zone is set (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

A firewall zone defines the trust level for a connection, interface or source address binding. This is a one to many relation, which means that a connection, interface or source can only be part of one zone, but a zone can be used for many network connections, interfaces and sources.

- The default zone is the zone that is used for everything that is not explicitly bound/assigned to another zone.
- If no zone assigned to a connection, interface or source, only the default zone is used.
- The default zone is not always listed as being used for an interface or source as it will be used for it either way. This depends on the manager of the interfaces.

Connections handled by NetworkManager are listed as NetworkManager requests to add the zone binding for the interface used by the connection. Also interfaces under control of the network service are listed also because the service requests it.

Note:

- *A firewalld zone configuration file contains the information for a zone.*
 - *These are the zone description, services, ports, protocols, icmp-blocks, masquerade, forward-ports and rich language rules in an XML file format.*
 - *The file name has to be `zone_name.xml` where length of `zone_name` is currently limited to 17 chars.*
- *NetworkManager binds interfaces to zones automatically*

Rationale:

Because the default zone is the zone that is used for everything that is not explicitly bound/assigned to another zone, it is important for the default zone to set

Audit:

Run the following command and verify that the default zone adheres to company policy:

```
# firewall-cmd --get-default-zone
```

Remediation:

Run the following command to set the default zone:

```
# firewall-cmd --set-default-zone=<NAME_OF_ZONE>
```

Example:

```
# firewall-cmd --set-default-zone=public
```

References:

1. <https://firewalld.org/documentation>
2. <https://firewalld.org/documentation/man-pages/firewalld.zone>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.4 Implement and Manage a Firewall on Servers Implement and manage a firewall on servers, where supported. Example implementations include a virtual firewall, operating system firewall, or a third-party firewall agent.			
v7	9.4 Apply Host-based Firewalls or Port Filtering Apply host-based firewalls or port filtering tools on end systems, with a default-deny rule that drops all traffic except those services and ports that are explicitly allowed.			

3.5.1.6 Ensure network interfaces are assigned to appropriate zone (Manual)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

firewall zones define the trust level of network connections or interfaces.

Rationale:

A network interface not assigned to the appropriate zone can allow unexpected or undesired network traffic to be accepted on the interface.

Impact:

Changing firewall settings while connected over network can result in being locked out of the system.

Audit:

Run the following and verify that the interface(s) follow site policy for zone assignment

```
# find /sys/class/net/* -maxdepth 1 | awk -F"/" '{print $NF}' | while read -r netint; do [ "$netint" != "lo" ] && firewall-cmd --get-active-zones | grep -B1 $netint; done
```

Example output:

```
<custom zone>  
eth0
```

Remediation:

Run the following command to assign an interface to the appropriate zone.

```
# firewall-cmd --zone=<Zone NAME> --change-interface=<INTERFACE NAME>
```

Example:

```
# firewall-cmd --zone=customzone --change-interface=eth0
```

Default Value:

default zone defined in the firewalld configuration

References:

1. <https://firewalld.org/documentation/zone/connections-interfaces-and-sources.html>

Additional Information:

The firewall in the Linux kernel is not able to handle network connections with the name shown by NetworkManager, it can only handle the network interfaces used by the connection. Because of this NetworkManager tells firewalld to assign the network interface that is used for this connection to the zone defined in the configuration of that connection. This assignment happens before the interface is used. The configuration of the connection can either be the NetworkManager configuration or also an `ifcfg`.

Example: If the zone is not set in the configuration file, the interfaces will be assigned to the default zone defined in the firewalld configuration. If a connection has more than one interface, all of them will be supplied to firewalld. Also changes in the names of interfaces will be handled by NetworkManager and supplied to firewalld.

If the zone is not set in the configuration file, the interfaces will be assigned to the default zone defined in the firewalld configuration

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.4 Implement and Manage a Firewall on Servers Implement and manage a firewall on servers, where supported. Example implementations include a virtual firewall, operating system firewall, or a third-party firewall agent.			
v7	9.4 Apply Host-based Firewalls or Port Filtering Apply host-based firewalls or port filtering tools on end systems, with a default-deny rule that drops all traffic except those services and ports that are explicitly allowed.			

3.5.1.7 Ensure firewalld drops unnecessary services and ports (Manual)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

Services and ports can be accepted or explicitly rejected or dropped by a zone.

For every zone, you can set a default behavior that handles incoming traffic that is not further specified. Such behavior is defined by setting the target of the zone. There are three options - default, ACCEPT, REJECT, and DROP.

- ACCEPT - you accept all incoming packets except those disabled by a specific rule.
- REJECT - you disable all incoming packets except those that you have allowed in specific rules and the source machine is informed about the rejection.
- DROP - you disable all incoming packets except those that you have allowed in specific rules and no information sent to the source machine.

Rationale:

To reduce the attack surface of a system, all services and ports should be blocked unless required

Audit:

Run the following command and review output to ensure that listed services and ports follow site policy.

```
# firewall-cmd --get-active-zones | awk '!/:/ {print $1}' | while read ZN; do  
firewall-cmd --list-all --zone=$ZN; done
```

Remediation:

Run the following command to remove an unnecessary service:

```
# firewall-cmd --remove-service=<service>
```

Example:

```
# firewall-cmd --remove-service=cockpit
```

Run the following command to remove an unnecessary port:

```
# firewall-cmd --remove-port=<port-number>/<port-type>
```

Example:

```
# firewall-cmd --remove-port=25/tcp
```

Run the following command to make new settings persistent:

```
# firewall-cmd --runtime-to-permanent
```

References:

1. firewall.service(5)
2. https://access.redhat.com/documentation/en-us/red_hat_enterprise_linux/8/html/securing_networks/using-and-configuring-firewalls_securing-networks

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.4 Implement and Manage a Firewall on Servers Implement and manage a firewall on servers, where supported. Example implementations include a virtual firewall, operating system firewall, or a third-party firewall agent.			
v7	9.4 Apply Host-based Firewalls or Port Filtering Apply host-based firewalls or port filtering tools on end systems, with a default-deny rule that drops all traffic except those services and ports that are explicitly allowed.			

3.5.2 Configure nftables

If firewalld or iptables are being used in your environment, please follow the guidance in their respective section and pass-over the guidance in this section.

nftables is a subsystem of the Linux kernel providing filtering and classification of network packets/datagrams/frames and is the successor to iptables. The biggest change with the successor nftables is its simplicity. With iptables, we have to configure every single rule and use the syntax which can be compared with normal commands. With nftables, the simpler syntax, much like BPF (Berkely Packet Filter) means shorter lines and less repetition. Support for nftables should also be compiled into the kernel, together with the related nftables modules. It is available in Linux kernels ≥ 3.13 . **Please ensure that your kernel supports nftables before choosing this option.**

This section broadly assumes starting with an empty nftables firewall ruleset (established by flushing the rules with `nft flush ruleset`). Remediation steps included only affect the live system, you will also need to configure your default firewall configuration to apply on boot. **Configuration of a live systems firewall directly over a remote connection will often result in being locked out.** It is advised to have a known good firewall configuration set to run on boot and to configure an entire firewall structure in a script that is then run and tested before saving to boot.

Note: *Configuration of a live systems firewall directly over a remote connection will often result in being locked out. It is advised to have a known good firewall configuration set to run on boot and to configure an entire firewall structure in a script that is then run and tested before saving to boot.*

The following will implement the firewall rules of this section and open ICMP, IGMP, and port 22(ssh) from anywhere. Opening the ports for ICMP, IGMP, and port 22(ssh) needs to be updated in accordance with local site policy. Allow port 22(ssh) needs to be updated to only allow systems requiring ssh connectivity to connect, as per site policy.

Save the script bellow as `/etc/nftables/nftables.rules`

```
#!/sbin/nft -f

# This nftables.rules config should be saved as /etc/nftables/nftables.rules

# flush nftables ruleset
flush ruleset

# Load nftables ruleset

# nftables config with inet table named filter
table inet filter {
    # Base chain for input hook named input (Filters inbound network packets)
    chain input {
        type filter hook input priority 0; policy drop;

        # Ensure loopback traffic is configured
        iif "lo" accept
        ip saddr 127.0.0.0/8 counter packets 0 bytes 0 drop
        ip6 saddr ::1 counter packets 0 bytes 0 drop

        # Ensure established connections are configured
        ip protocol tcp ct state established accept
        ip protocol udp ct state established accept
        ip protocol icmp ct state established accept

        # Accept port 22(SSH) traffic from anywhere
        tcp dport ssh accept

        # Accept ICMP and IGMP from anywhere
        icmpv6 type { destination-unreachable, packet-too-big, time-exceeded,
parameter-problem, mld-listener-query, mld-listener-report, mld-listener-done, nd-
router-solicit, nd-router-advert, nd-neighbor-solicit, nd-neighbor-advert, ind-
neighbor-solicit, ind-neighbor-advert, mld2-listener-report } accept
        icmp type { destination-unreachable, router-advertisement, router-
solicitation, time-exceeded, parameter-problem } accept
        ip protocol igmp accept
    }

    # Base chain for hook forward named forward (Filters forwarded network
packets)
    chain forward {
        type filter hook forward priority 0; policy drop;
    }

    # Base chain for hook output named output (Filters outbound network packets)
    chain output {
        type filter hook output priority 0; policy drop;
        # Ensure outbound and established connections are configured
        ip protocol tcp ct state established,related,new accept
        ip protocol udp ct state established,related,new accept
        ip protocol icmp ct state established,related,new accept
    }
}
```

Run the following command to load the file into nftables

```
# nft -f /etc/nftables/nftables.rules
```

All changes in the nftables subsections are temporary

To make these changes permanent:

Run the following command to create the nftables.rules file

```
nft list ruleset > /etc/nftables/nftables.rules
```

Add the following line to /etc/sysconfig/nftables.conf

```
include "/etc/nftables/nftables.rules"
```

3.5.2.1 Ensure nftables is installed (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

nftables provides a new in-kernel packet classification framework that is based on a network-specific Virtual Machine (VM) and a new nft userspace command line tool. nftables reuses the existing Netfilter subsystems such as the existing hook infrastructure, the connection tracking system, NAT, userspace queuing and logging subsystem.

Note:

- *nftables is available in Linux kernel 3.13 and newer.*
- *Only **one** firewall utility should be installed and configured.*

Rationale:

nftables is a subsystem of the Linux kernel that can protect against threats originating from within a corporate network to include malicious mobile code and poorly configured software on a host.

Impact:

Changing firewall settings while connected over the network can result in being locked out of the system.

Audit:

Run the following command to verify that `nftables` is installed:

```
# rpm -q nftables  
nftables-<version>
```

Remediation:

Run the following command to install `nftables`

```
# yum install nftables
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.4 <u>Implement and Manage a Firewall on Servers</u> Implement and manage a firewall on servers, where supported. Example implementations include a virtual firewall, operating system firewall, or a third-party firewall agent.			
v7	9.4 <u>Apply Host-based Firewalls or Port Filtering</u> Apply host-based firewalls or port filtering tools on end systems, with a default-deny rule that drops all traffic except those services and ports that are explicitly allowed.			

3.5.2.2 Ensure firewalld is either not installed or masked with nftables (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

firewalld (Dynamic Firewall Manager) provides a dynamically managed firewall with support for network/firewall “zones” to assign a level of trust to a network and its associated connections, interfaces or sources. It has support for IPv4, IPv6, Ethernet bridges and also for IPSet firewall settings. There is a separation of the runtime and permanent configuration options.

Rationale:

Running both `nftables.service` and `firewalld.service` may lead to conflict and unexpected results.

Audit:

Run the following command to verify that `firewalld` is not installed:

```
# rpm -q firewalld  
package firewalld is not installed
```

OR

Run the following command to verify that FirewallD is not running

```
command -v firewall-cmd >/dev/null && firewall-cmd --state | grep 'running'  
not running
```

Run the following command to verify that FirewallD is masked

```
# systemctl is-enabled firewalld  
masked
```

Remediation:

Run the following command to remove `firewalld`

```
# yum remove firewalld
```

OR

Run the following command to stop and mask `firewalld`

```
# systemctl --now mask firewalld
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.4 <u>Implement and Manage a Firewall on Servers</u> Implement and manage a firewall on servers, where supported. Example implementations include a virtual firewall, operating system firewall, or a third-party firewall agent.			
v8	4.8 <u>Uninstall or Disable Unnecessary Services on Enterprise Assets and Software</u> Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.			
v7	9.4 <u>Apply Host-based Firewalls or Port Filtering</u> Apply host-based firewalls or port filtering tools on end systems, with a default-deny rule that drops all traffic except those services and ports that are explicitly allowed.			

3.5.2.3 Ensure iptables-services not installed with nftables (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `iptables-services` package contains the `iptables.service` and `ip6tables.service`. These services allow for management of the Host Based Firewall provided by the `iptables` package.

Rationale:

`iptables.service` and `ip6tables.service` are still supported and can be installed with the `iptables-services` package. Running both `nftables` and the services included in the `iptables-services` package may lead to conflict.

Audit:

Run the following commands to verify that the `iptables-services` package is not installed

```
# rpm -q iptables-services  
package iptables-services is not installed
```

Remediation:

Run the following commands to stop the services included in the `iptables-services` package and remove the `iptables-services` package

```
# systemctl stop iptables  
# systemctl stop ip6tables  
# yum remove iptables-services
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.4 Implement and Manage a Firewall on Servers</u> Implement and manage a firewall on servers, where supported. Example implementations include a virtual firewall, operating system firewall, or a third-party firewall agent.			
v8	<u>4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software</u> Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.			
v7	<u>9.4 Apply Host-based Firewalls or Port Filtering</u> Apply host-based firewalls or port filtering tools on end systems, with a default-deny rule that drops all traffic except those services and ports that are explicitly allowed.			

3.5.2.4 Ensure iptables are flushed with nftables (Manual)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

nftables is a replacement for iptables, ip6tables, ebtables and arptables

Rationale:

It is possible to mix iptables and nftables. However, this increases complexity and also the chance to introduce errors. For simplicity flush out all iptables rules, and ensure it is not loaded

Audit:

Run the following commands to ensure not iptables rules exist

For iptables:

```
# iptables -L  
No rules should be returned
```

For ip6tables:

```
# ip6tables -L  
No rules should be returned
```

Remediation:

Run the following commands to flush iptables:

For iptables:

```
# iptables -F
```

For ip6tables:

```
# ip6tables -F
```


CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.4 <u>Implement and Manage a Firewall on Servers</u> Implement and manage a firewall on servers, where supported. Example implementations include a virtual firewall, operating system firewall, or a third-party firewall agent.			
v7	9.4 <u>Apply Host-based Firewalls or Port Filtering</u> Apply host-based firewalls or port filtering tools on end systems, with a default-deny rule that drops all traffic except those services and ports that are explicitly allowed.			

3.5.2.5 Ensure an nftables table exists (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

Tables hold chains. Each table only has one address family and only applies to packets of this family. Tables can have one of five families.

Rationale:

nftables doesn't have any default tables. Without a table being build, nftables will not filter network traffic.

Impact:

Adding rules to a running nftables can cause loss of connectivity to the system

Audit:

Run the following command to verify that a nftables table exists:

```
# nft list tables
```

Return should include a list of nftables:

Example:

```
table inet filter
```

Remediation:

Run the following command to create a table in nftables

```
# nft create table inet <table name>
```

Example:

```
# nft create table inet filter
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.4 <u>Implement and Manage a Firewall on Servers</u> Implement and manage a firewall on servers, where supported. Example implementations include a virtual firewall, operating system firewall, or a third-party firewall agent.			
v7	9.4 <u>Apply Host-based Firewalls or Port Filtering</u> Apply host-based firewalls or port filtering tools on end systems, with a default-deny rule that drops all traffic except those services and ports that are explicitly allowed.			

3.5.2.6 Ensure nftables base chains exist (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

Chains are containers for rules. They exist in two kinds, base chains and regular chains. A base chain is an entry point for packets from the networking stack, a regular chain may be used as jump target and is used for better rule organization.

Rationale:

If a base chain doesn't exist with a hook for input, forward, and delete, packets that would flow through those chains will not be touched by nftables.

Impact:

If configuring nftables over ssh, creating a base chain with a policy of drop will cause loss of connectivity.

Ensure that a rule allowing ssh has been added to the base chain prior to setting the base chain's policy to drop

Audit:

Run the following commands and verify that base chains exist for INPUT, FORWARD, and OUTPUT.

```
# nft list ruleset | grep 'hook input'
type filter hook input priority 0;
# nft list ruleset | grep 'hook forward'
type filter hook forward priority 0;
# nft list ruleset | grep 'hook output'
type filter hook output priority 0;
```

Remediation:

Run the following command to create the base chains:

```
# nft create chain inet <table name> <base chain name> { type filter hook  
<(input|forward|output)> priority 0 \; }
```

Example:

```
# nft create chain inet filter input { type filter hook input priority 0 \; }  
# nft create chain inet filter forward { type filter hook forward priority 0  
\; }  
# nft create chain inet filter output { type filter hook output priority 0 \;  
}
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.4 <u>Implement and Manage a Firewall on Servers</u> Implement and manage a firewall on servers, where supported. Example implementations include a virtual firewall, operating system firewall, or a third-party firewall agent.			
v7	9.4 <u>Apply Host-based Firewalls or Port Filtering</u> Apply host-based firewalls or port filtering tools on end systems, with a default-deny rule that drops all traffic except those services and ports that are explicitly allowed.			

3.5.2.7 Ensure nftables loopback traffic is configured (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

Configure the loopback interface to accept traffic. Configure all other interfaces to deny traffic to the loopback network

Rationale:

Loopback traffic is generated between processes on machine and is typically critical to operation of the system. The loopback interface is the only place that loopback network traffic should be seen, all other interfaces should ignore traffic on this network as an anti-spoofing measure.

Audit:

Run the following commands to verify that the loopback interface is configured:

```
# nft list ruleset | awk '/hook input/,/}/' | grep 'iif "lo" accept'
iif "lo" accept
# nft list ruleset | awk '/hook input/,/}/' | grep 'ip saddr'
ip saddr 127.0.0.0/8 counter packets 0 bytes 0 drop
```

IF IPv6 is enabled, run the following command to verify that the IPv6 loopback interface is configured:

```
# nft list ruleset | awk '/hook input/,/}/' | grep 'ip6 saddr'
ip6 saddr ::1 counter packets 0 bytes 0 drop
```

OR

Verify that IPv6 is disabled:

Run the following script. Output will confirm if IPv6 is disabled on the system.

```
#!/bin/bash

[ -n "$passing" ] && passing=""
[ -z "$(grep "^s*linux" /boot/grub2/grub.cfg | grep -v ipv6.disabled=1)" ]
&& passing="true"
grep -Eq "^s*net\.ipv6\.conf\.all\.disable_ipv6\s*=\s*1\b(\s+#.*)?$"
/etc/sysctl.conf \
/etc/sysctl.d/*.conf && grep -Eq
"^s*net\.ipv6\.conf\.default\.disable_ipv6\s*=\s*1\b(\s+#.*)?$" \
/etc/sysctl.conf /etc/sysctl.d/*.conf && sysctl
net.ipv6.conf.all.disable_ipv6 | \
grep -Eq "^s*net\.ipv6\.conf\.all\.disable_ipv6\s*=\s*1\b(\s+#.*)?$" && \
sysctl net.ipv6.conf.default.disable_ipv6 | \
grep -Eq "^s*net\.ipv6\.conf\.default\.disable_ipv6\s*=\s*1\b(\s+#.*)?$" &&
passing="true"
if [ "$passing" = true ] ; then
    echo "IPv6 is disabled on the system"
else
    echo "IPv6 is enabled on the system"
fi
```

Remediation:

Run the following commands to implement the loopback rules:

```
# nft add rule inet filter input iif lo accept
# nft create rule inet filter input ip saddr 127.0.0.0/8 counter drop
```

IF IPv6 is enabled:

Run the following command to implement the IPv6 loopback rules:

```
# nft add rule inet filter input ip6 saddr ::1 counter drop
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.4 <u>Implement and Manage a Firewall on Servers</u> Implement and manage a firewall on servers, where supported. Example implementations include a virtual firewall, operating system firewall, or a third-party firewall agent.			
v7	9.4 <u>Apply Host-based Firewalls or Port Filtering</u> Apply host-based firewalls or port filtering tools on end systems, with a default-deny rule that drops all traffic except those services and ports that are explicitly allowed.			

3.5.2.8 Ensure nftables outbound and established connections are configured (Manual)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

Configure the firewall rules for new outbound and established connections

Rationale:

If rules are not in place for new outbound and established connections, all packets will be dropped by the default policy preventing network usage.

Audit:

Run the following commands and verify all rules for established incoming connections match site policy: site policy:

```
# nft list ruleset | awk '/hook input/,/}/' | grep -E 'ip protocol (tcp|udp|icmp) ct state'
```

Output should be similar to:

```
ip protocol tcp ct state established accept
ip protocol udp ct state established accept
ip protocol icmp ct state established accept
```

Run the following command and verify all rules for new and established outbound connections match site policy

```
# nft list ruleset | awk '/hook output/,/}/' | grep -E 'ip protocol (tcp|udp|icmp) ct state'
```

Output should be similar to:

```
ip protocol tcp ct state established,related,new accept
ip protocol udp ct state established,related,new accept
ip protocol icmp ct state established,related,new accept
```


Remediation:

Configure nftables in accordance with site policy. The following commands will implement a policy to allow all outbound connections and all established connections:

```
# nft add rule inet filter input ip protocol tcp ct state established accept
# nft add rule inet filter input ip protocol udp ct state established accept
# nft add rule inet filter input ip protocol icmp ct state established accept
# nft add rule inet filter output ip protocol tcp ct state
new,related,established accept
# nft add rule inet filter output ip protocol udp ct state
new,related,established accept
# nft add rule inet filter output ip protocol icmp ct state
new,related,established accept
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.4 <u>Implement and Manage a Firewall on Servers</u> Implement and manage a firewall on servers, where supported. Example implementations include a virtual firewall, operating system firewall, or a third-party firewall agent.			
v7	9.4 <u>Apply Host-based Firewalls or Port Filtering</u> Apply host-based firewalls or port filtering tools on end systems, with a default-deny rule that drops all traffic except those services and ports that are explicitly allowed.			

3.5.2.9 Ensure nftables default deny firewall policy (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

Base chain policy is the default verdict that will be applied to packets reaching the end of the chain.

Rationale:

There are two policies: accept (Default) and drop. If the policy is set to `accept`, the firewall will accept any packet that is not configured to be denied and the packet will continue traversing the network stack.

It is easier to white list acceptable usage than to black list unacceptable usage.

Note: Changing firewall settings while connected over the network can result in being locked out of the system.

Impact:

If configuring nftables over ssh, creating a base chain with a policy of drop will cause loss of connectivity.

Ensure that a rule allowing `ssh` has been added to the base chain prior to setting the base chain's policy to drop

Audit:

Run the following commands and verify that base chains contain a policy of `DROP`.

```
# nft list ruleset | grep 'hook input'
type filter hook input priority 0; policy drop;
# nft list ruleset | grep 'hook forward'
type filter hook forward priority 0; policy drop;
# nft list ruleset | grep 'hook output'
type filter hook output priority 0; policy drop;
```

Remediation:

Run the following command for the base chains with the input, forward, and output hooks to implement a default DROP policy:

```
# nft chain <table family> <table name> <chain name> { policy drop \; }
```

Example:

```
# nft chain inet filter input { policy drop \; }  
# nft chain inet filter forward { policy drop \; }  
# nft chain inet filter output { policy drop \; }
```

Default Value:

accept

References:

1. Manual Page nft

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.4 <u>Implement and Manage a Firewall on Servers</u> Implement and manage a firewall on servers, where supported. Example implementations include a virtual firewall, operating system firewall, or a third-party firewall agent.			
v7	9.4 <u>Apply Host-based Firewalls or Port Filtering</u> Apply host-based firewalls or port filtering tools on end systems, with a default-deny rule that drops all traffic except those services and ports that are explicitly allowed.			

3.5.2.10 Ensure nftables service is enabled (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The nftables service allows for the loading of nftables rulesets during boot, or starting on the nftables service

Rationale:

The nftables service restores the nftables rules from the rules files referenced in the `/etc/sysconfig/nftables.conf` file during boot or the starting of the nftables service

Audit:

Run the following command and verify that the nftables service is enabled:

```
# systemctl is-enabled nftables
enabled
```

Remediation:

Run the following command to enable the nftables service:

```
# systemctl enable nftables
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.4 <u>Implement and Manage a Firewall on Servers</u> Implement and manage a firewall on servers, where supported. Example implementations include a virtual firewall, operating system firewall, or a third-party firewall agent.			
v7	9.4 <u>Apply Host-based Firewalls or Port Filtering</u> Apply host-based firewalls or port filtering tools on end systems, with a default-deny rule that drops all traffic except those services and ports that are explicitly allowed.			

3.5.2.11 Ensure nftables rules are permanent (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

nftables is a subsystem of the Linux kernel providing filtering and classification of network packets/datagrams/frames.

The nftables service reads the `/etc/sysconfig/nftables.conf` file for a nftables file or files to include in the nftables ruleset.

A nftables ruleset containing the input, forward, and output base chains allow network traffic to be filtered.

Rationale:

Changes made to nftables ruleset only affect the live system, you will also need to configure the nftables ruleset to apply on boot

Audit:

Run the following commands to verify that input, forward, and output base chains are configured to be applied to a nftables ruleset on boot:

Run the following command to verify the input base chain:

```
# awk '/hook input/,/}/' $(awk '$1 ~ /^s*include/ { gsub("\"","",$2);print $2 }' /etc/sysconfig/nftables.conf)
```

Output should be similar to:

```
type filter hook input priority 0; policy drop;

# Ensure loopback traffic is configured
iif "lo" accept
ip saddr 127.0.0.0/8 counter packets 0 bytes 0 drop
ip6 saddr ::1 counter packets 0 bytes 0 drop

# Ensure established connections are configured
ip protocol tcp ct state established accept
ip protocol udp ct state established accept
ip protocol icmp ct state established accept

# Accept port 22 (SSH) traffic from anywhere
tcp dport ssh accept

# Accept ICMP and IGMP from anywhere
icmpv6 type { destination-unreachable, packet-too-big, time-
exceeded, parameter-problem, mld-listener-query, mld-listener-report, mld-
listener-done, nd-router-solicit, nd-router-advert, nd-neighbor-solicit, nd-
neighbor-advert, ind-neighbor-solicit, ind-neighbor-advert, mld2-listener-
report } accept
```

Note: Review the input base chain to ensure that it follows local site policy

Run the following command to verify the forward base chain:

```
# awk '/hook forward/,/}/' $(awk '$1 ~ /^s*include/ { gsub("\"","",$2);print $2 }' /etc/sysconfig/nftables.conf)
```

Output should be similar to:

```
# Base chain for hook forward named forward (Filters forwarded
network packets)
chain forward {
    type filter hook forward priority 0; policy drop;
}
```

Note: Review the forward base chain to ensure that it follows local site policy.

Run the following command to verify the forward base chain:

```
# awk '/hook output/,/}/' $(awk '$1 ~ /^s*include/ { gsub("\\"", "\"", $2); print $2 }' /etc/sysconfig/nftables.conf)
```

Output should be similar to:

```
# Base chain for hook output named output (Filters outbound network packets)
chain output {
    type filter hook output priority 0; policy drop;
    # Ensure outbound and established connections are configured
    ip protocol tcp ct state established,related,new accept
    ip protocol tcp ct state established,related,new accept
    ip protocol udp ct state established,related,new accept
    ip protocol icmp ct state established,related,new accept
}
```

Note: Review the output base chain to ensure that it follows local site policy.

Remediation:

Edit the `/etc/sysconfig/nftables.conf` file and un-comment or add a line with `include <Absolute path to nftables rules file>` for each nftables file you want included in the nftables ruleset on boot:

Example:

```
include "/etc/nftables/nftables.rules"
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.4 <u>Implement and Manage a Firewall on Servers</u> Implement and manage a firewall on servers, where supported. Example implementations include a virtual firewall, operating system firewall, or a third-party firewall agent.			
v7	9.4 <u>Apply Host-based Firewalls or Port Filtering</u> Apply host-based firewalls or port filtering tools on end systems, with a default-deny rule that drops all traffic except those services and ports that are explicitly allowed.			

3.5.3 Configure iptables

If firewalld or nftables are being used in your environment, please follow the guidance in their respective section and pass-over the guidance in this section.

Iptables is an application that allows a system administrator to configure the IPv4 and IPv6 tables, chains and rules provided by the Linux kernel firewall. While several methods of configuration exist this section is intended only to ensure the resulting IPtables rules are in place, not how they are configured. If IPv6 is in use in your environment, similar settings should be applied to the IP6tables as well.

Note: Configuration of a live systems firewall directly over a remote connection will often result in being locked out. It is advised to have a known good firewall configuration set to run on boot and to configure an entire firewall structure in a script that is then run and tested before saving to boot.

3.5.3.1 Configure iptables software

This section provides guidance for installing, enabling, removing, and disabling software packages necessary for using IPTables as the method for configuring and maintaining a Host Based Firewall on the system.

Note: *Using more than one method to configure and maintain a Host Based Firewall can cause unexpected results. If FirewallD or NFTables are being used for configuration and maintenance, this section should be skipped and the guidance in their respective section followed.*

3.5.3.1.1 Ensure iptables packages are installed (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

iptables is a utility program that allows a system administrator to configure the tables provided by the Linux kernel firewall, implemented as different Netfilter modules, and the chains and rules it stores. Different kernel modules and programs are used for different protocols; iptables applies to IPv4, ip6tables to IPv6, arptables to ARP, and ebtables to Ethernet frames.

Rationale:

A method of configuring and maintaining firewall rules is necessary to configure a Host Based Firewall.

Audit:

Run the following command to verify that iptables and iptables-services are installed:

```
rpm -q iptables iptables-services  
  
iptables-<version>  
iptables-services-<version>
```

Remediation:

Run the following command to install iptables and iptables-services

```
# yum install iptables iptables-services
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.4 <u>Implement and Manage a Firewall on Servers</u> Implement and manage a firewall on servers, where supported. Example implementations include a virtual firewall, operating system firewall, or a third-party firewall agent.			
v7	9.4 <u>Apply Host-based Firewalls or Port Filtering</u> Apply host-based firewalls or port filtering tools on end systems, with a default-deny rule that drops all traffic except those services and ports that are explicitly allowed.			

3.5.3.1.2 Ensure *nftables* is not installed with *iptables* (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

nftables is a subsystem of the Linux kernel providing filtering and classification of network packets/datagrams/frames and is the successor to *iptables*.

Rationale:

Running both *iptables* and *nftables* may lead to conflict.

Audit:

Run the following command to verify that *nftables* is not installed:

```
# rpm -q nftables  
package nftables is not installed
```

Remediation:

Run the following command to remove *nftables*:

```
# yum remove nftables
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.4 Implement and Manage a Firewall on Servers</u> Implement and manage a firewall on servers, where supported. Example implementations include a virtual firewall, operating system firewall, or a third-party firewall agent.			
v8	<u>4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software</u> Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.			
v7	<u>9.4 Apply Host-based Firewalls or Port Filtering</u> Apply host-based firewalls or port filtering tools on end systems, with a default-deny rule that drops all traffic except those services and ports that are explicitly allowed.			

3.5.3.1.3 Ensure firewalld is either not installed or masked with iptables (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

firewalld (Dynamic Firewall Manager) provides a dynamically managed firewall with support for network/firewall “zones” to assign a level of trust to a network and its associated connections, interfaces or sources. It has support for IPv4, IPv6, Ethernet bridges and also for IPSet firewall settings. There is a separation of the runtime and permanent configuration options.

Rationale:

Running `iptables.service` and/or `ip6tables.service` with `firewalld.service` may lead to conflict and unexpected results.

Audit:

Run the following command to verify that `firewalld` is not installed:

```
# rpm -q firewalld  
package firewalld is not installed
```

OR

Run the following commands to verify that `firewalld` is stopped and masked

```
# systemctl status firewalld | grep "Active: " | grep -v "active (running) "  
No output should be returned  
  
# systemctl is-enabled firewalld  
masked
```

Remediation:

Run the following command to remove `firewalld`

```
# yum remove firewalld
```

OR

Run the following command to stop and mask `firewalld`

```
# systemctl --now mask firewalld
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.4 <u>Implement and Manage a Firewall on Servers</u> Implement and manage a firewall on servers, where supported. Example implementations include a virtual firewall, operating system firewall, or a third-party firewall agent.			
v8	4.8 <u>Uninstall or Disable Unnecessary Services on Enterprise Assets and Software</u> Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.			
v7	9.4 <u>Apply Host-based Firewalls or Port Filtering</u> Apply host-based firewalls or port filtering tools on end systems, with a default-deny rule that drops all traffic except those services and ports that are explicitly allowed.			

3.5.3.2 Configure IPv4 iptables

IPTables is used to set up, maintain, and inspect the tables of IP packet filter rules in the Linux kernel. Several different tables may be defined. Each table contains a number of built-in chains and may also contain user-defined chains.

Each chain is a list of rules which can match a set of packets. Each rule specifies what to do with a packet that matches. This is called a 'target', which may be a jump to a user-defined chain in the same table.

Note:

- *This section broadly assumes starting with an empty IPTables firewall ruleset (established by flushing the rules with iptables -F).*
- *Configuration of a live systems firewall directly over a remote connection will often result in being locked out.*
- *It is advised to have a known good firewall configuration set to run on boot and to configure an entire firewall structure in a script that is then run and tested before saving to boot.*

The following script will implement the firewall rules of this section and open port 22(ssh) from anywhere. *This needs to be updated to only allow systems requiring ssh connectivity to connect as per site policy.*

```
#!/bin/bash

# Flush IPtables rules
iptables -F

# Ensure default deny firewall policy
iptables -P INPUT DROP
iptables -P OUTPUT DROP
iptables -P FORWARD DROP

# Ensure loopback traffic is configured
iptables -A INPUT -i lo -j ACCEPT

iptables -A OUTPUT -o lo -j ACCEPT
iptables -A INPUT -s 127.0.0.0/8 -j DROP

# Ensure outbound and established connections are configured
iptables -A OUTPUT -p tcp -m state --state NEW,ESTABLISHED -j ACCEPT
iptables -A OUTPUT -p udp -m state --state NEW,ESTABLISHED -j ACCEPT
iptables -A OUTPUT -p icmp -m state --state NEW,ESTABLISHED -j ACCEPT
iptables -A INPUT -p tcp -m state --state ESTABLISHED -j ACCEPT
iptables -A INPUT -p udp -m state --state ESTABLISHED -j ACCEPT
iptables -A INPUT -p icmp -m state --state ESTABLISHED -j ACCEPT

# Open inbound ssh(tcp port 22) connections
iptables -A INPUT -p tcp --dport 22 -m state --state NEW -j ACCEPT
```


3.5.3.2.1 Ensure iptables loopback traffic is configured (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

Configure the loopback interface to accept traffic. Configure all other interfaces to deny traffic to the loopback network (127.0.0.0/8).

Rationale:

Loopback traffic is generated between processes on machine and is typically critical to operation of the system. The loopback interface is the only place that loopback network (127.0.0.0/8) traffic should be seen, all other interfaces should ignore traffic on this network as an anti-spoofing measure.

Note: Changing firewall settings while connected over network can result in being locked out of the system.

Audit:

Run the following commands and verify output includes the listed rules in order (packet and byte counts may differ):

```
# iptables -L INPUT -v -n
Chain INPUT (policy DROP 0 packets, 0 bytes)
  pkts bytes target     prot opt in     out     source
destination
    0    0 ACCEPT     all  --  lo      *       0.0.0.0/0
    0    0 DROP       all  --  *       *       127.0.0.0/8
                                     0.0.0.0/0

# iptables -L OUTPUT -v -n
Chain OUTPUT (policy DROP 0 packets, 0 bytes)
  pkts bytes target     prot opt in     out     source
destination
    0    0 ACCEPT     all  --  *       lo      0.0.0.0/0
                                     0.0.0.0/0
```

Remediation:

Run the following commands to implement the loopback rules:

```
# iptables -A INPUT -i lo -j ACCEPT
# iptables -A OUTPUT -o lo -j ACCEPT
# iptables -A INPUT -s 127.0.0.0/8 -j DROP
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.4 <u>Implement and Manage a Firewall on Servers</u> Implement and manage a firewall on servers, where supported. Example implementations include a virtual firewall, operating system firewall, or a third-party firewall agent.			
v7	9.4 <u>Apply Host-based Firewalls or Port Filtering</u> Apply host-based firewalls or port filtering tools on end systems, with a default-deny rule that drops all traffic except those services and ports that are explicitly allowed.			

3.5.3.2.2 Ensure iptables outbound and established connections are configured (Manual)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

Configure the firewall rules for new outbound, and established connections.

Rationale:

If rules are not in place for new outbound, and established connections all packets will be dropped by the default policy preventing network usage.

Note: Changing firewall settings while connected over network can result in being locked out of the system.

Audit:

Run the following command and verify all rules for new outbound, and established connections match site policy:

```
# iptables -L -v -n
```

Remediation:

Configure iptables in accordance with site policy. The following commands will implement a policy to allow all outbound connections and all established connections:

```
# iptables -A OUTPUT -p tcp -m state --state NEW,ESTABLISHED -j ACCEPT
# iptables -A OUTPUT -p udp -m state --state NEW,ESTABLISHED -j ACCEPT
# iptables -A OUTPUT -p icmp -m state --state NEW,ESTABLISHED -j ACCEPT
# iptables -A INPUT -p tcp -m state --state ESTABLISHED -j ACCEPT
# iptables -A INPUT -p udp -m state --state ESTABLISHED -j ACCEPT
# iptables -A INPUT -p icmp -m state --state ESTABLISHED -j ACCEPT
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.4 <u>Implement and Manage a Firewall on Servers</u> Implement and manage a firewall on servers, where supported. Example implementations include a virtual firewall, operating system firewall, or a third-party firewall agent.			
v7	9.4 <u>Apply Host-based Firewalls or Port Filtering</u> Apply host-based firewalls or port filtering tools on end systems, with a default-deny rule that drops all traffic except those services and ports that are explicitly allowed.			

3.5.3.2.3 Ensure iptables rules exist for all open ports (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

Any ports that have been opened on non-loopback addresses need firewall rules to govern traffic.

Rationale:

Without a firewall rule configured for open ports default firewall policy will drop all packets to these ports.

Note:

- *Changing firewall settings while connected over network can result in being locked out of the system.*
- *The remediation command opens up the port to traffic from all sources. Consult iptables documentation and set any restrictions in compliance with site policy.*

Audit:

Run the following command to determine open ports:

```
# ss -4tuln
```

Netid	State	Recv-Q	Send-Q	Local Address:Port	Peer
Address:Port					
udp	UNCONN	0	0	*:68	
:					
udp	UNCONN	0	0	*:123	
:					
tcp	LISTEN	0	128	*:22	
:					

Run the following command to determine firewall rules:

```
# iptables -L INPUT -v -n
```

Chain INPUT (policy DROP 0 packets, 0 bytes)	pkts	bytes	target	prot	opt	in	out	source	destination
	0	0	ACCEPT	all	--	lo	*	0.0.0.0/0	0.0.0.0/0
	0	0	DROP	all	--	*	*	127.0.0.0/8	0.0.0.0/0
	0	0	ACCEPT	tcp	--	*	*	0.0.0.0/0	0.0.0.0/0

```
tcp dpt:22 state NEW
```

Verify all open ports listening on non-localhost addresses have at least one firewall rule.

Note: The last line identified by the "tcp dpt:22 state NEW" identifies it as a firewall rule for new connections on tcp port 22.

Remediation:

For each port identified in the audit which does not have a firewall rule establish a proper rule for accepting inbound connections:

```
# iptables -A INPUT -p <protocol> --dport <port> -m state --state NEW -j ACCEPT
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.4 <u>Implement and Manage a Firewall on Servers</u> Implement and manage a firewall on servers, where supported. Example implementations include a virtual firewall, operating system firewall, or a third-party firewall agent.			
v7	9.2 <u>Ensure Only Approved Ports, Protocols and Services Are Running</u> Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.			
v7	9.4 <u>Apply Host-based Firewalls or Port Filtering</u> Apply host-based firewalls or port filtering tools on end systems, with a default-deny rule that drops all traffic except those services and ports that are explicitly allowed.			

3.5.3.2.4 Ensure iptables default deny firewall policy (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

A default deny all policy on connections ensures that any unconfigured network usage will be rejected.

Rationale:

With a default accept policy the firewall will accept any packet that is not configured to be denied. It is easier to white list acceptable usage than to black list unacceptable usage.

Note: Changing firewall settings while connected over network can result in being locked out of the system.

Audit:

Run the following command and verify that the policy for the `INPUT` , `OUTPUT` , and `FORWARD` chains is `DROP` or `REJECT` :

```
# iptables -L  
  
Chain INPUT (policy DROP)  
Chain FORWARD (policy DROP)  
Chain OUTPUT (policy DROP)
```

Remediation:

Run the following commands to implement a default DROP policy:

```
# iptables -P INPUT DROP  
# iptables -P OUTPUT DROP  
# iptables -P FORWARD DROP
```


CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.4 <u>Implement and Manage a Firewall on Servers</u> Implement and manage a firewall on servers, where supported. Example implementations include a virtual firewall, operating system firewall, or a third-party firewall agent.			
v7	9.4 <u>Apply Host-based Firewalls or Port Filtering</u> Apply host-based firewalls or port filtering tools on end systems, with a default-deny rule that drops all traffic except those services and ports that are explicitly allowed.			

3.5.3.2.5 Ensure iptables rules are saved (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `iptables-services` package includes the `/etc/sysconfig/iptables` file. The `iptables` rules in this file will be loaded by the `iptables.service` during boot, or when it is started or re-loaded.

Rationale:

If the `iptables` rules are not saved and a system re-boot occurs, the `iptables` rules will be lost.

Audit:

Review the file `/etc/sysconfig/iptables` and ensure it contains the complete correct rule-set.

Example: `/etc/sysconfig/iptables`

```
# sample configuration for iptables service
# you can edit this manually or use system-config-firewall
# Generated by iptables-save v1.4.21 on Wed Mar 25 14:23:37 2020
*filter
:INPUT DROP [4:463]
:FORWARD DROP [0:0]
:OUTPUT DROP [0:0]
-A INPUT -i lo -j ACCEPT
-A INPUT -s 127.0.0.0/8 -j DROP
-A INPUT -p tcp -m state --state ESTABLISHED -j ACCEPT
-A INPUT -p udp -m state --state ESTABLISHED -j ACCEPT
-A INPUT -p icmp -m state --state ESTABLISHED -j ACCEPT
-A INPUT -p tcp -m tcp --dport 22 -m state --state NEW -j ACCEPT
-A OUTPUT -o lo -j ACCEPT
-A OUTPUT -p tcp -m state --state NEW,ESTABLISHED -j ACCEPT
-A OUTPUT -p udp -m state --state NEW,ESTABLISHED -j ACCEPT
-A OUTPUT -p icmp -m state --state NEW,ESTABLISHED -j ACCEPT
COMMIT
# Completed on Wed Mar 25 14:23:37 2020
```

Remediation:

Run the following commands to create or update the `/etc/sysconfig/iptables` file:

Run the following command to review the current running `iptables` configuration:

```
# iptables -L
```

Output should include:

```
Chain INPUT (policy DROP)
target     prot opt source                destination
ACCEPT     all  --  anywhere              anywhere
DROP       all  --  loopback/8           anywhere
ACCEPT     tcp  --  anywhere              anywhere    state
ESTABLISHED
ACCEPT     udp  --  anywhere              anywhere    state
ESTABLISHED
ACCEPT     icmp --  anywhere              anywhere    state
ESTABLISHED
ACCEPT     tcp  --  anywhere              anywhere    tcp dpt:ssh
state NEW

Chain FORWARD (policy DROP)
target     prot opt source                destination

Chain OUTPUT (policy DROP)
target     prot opt source                destination
ACCEPT     all  --  anywhere              anywhere
ACCEPT     tcp  --  anywhere              anywhere    state
NEW,ESTABLISHED
ACCEPT     udp  --  anywhere              anywhere    state
NEW,ESTABLISHED
ACCEPT     icmp --  anywhere              anywhere    state
NEW,ESTABLISHED
```

Run the following command to save the verified running configuration to the file

`/etc/sysconfig/iptables`:

```
# service iptables save
iptables: Saving firewall rules to /etc/sysconfig/iptables:[ OK ]
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.4 <u>Implement and Manage a Firewall on Servers</u> Implement and manage a firewall on servers, where supported. Example implementations include a virtual firewall, operating system firewall, or a third-party firewall agent.			
v7	9.4 <u>Apply Host-based Firewalls or Port Filtering</u> Apply host-based firewalls or port filtering tools on end systems, with a default-deny rule that drops all traffic except those services and ports that are explicitly allowed.			

3.5.3.2.6 Ensure iptables is enabled and running (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

`iptables.service` is a utility for configuring and maintaining `iptables`.

Rationale:

`iptables.service` will load the `iptables` rules saved in the file `/etc/sysconfig/iptables` at boot, otherwise the `iptables` rules will be cleared during a re-boot of the system.

Audit:

Run the following commands to verify `iptables` is enabled:

```
# systemctl is-enabled iptables
enabled
```

Run the following command to verify `iptables.service` is active and running or exited

```
# systemctl status iptables | grep -E " Active: active \((running|exited)\) "
Active: active (exited) since <day date and time>
```

Remediation:

Run the following command to enable and start `iptables`:

```
# systemctl --now enable iptables
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.4 <u>Implement and Manage a Firewall on Servers</u> Implement and manage a firewall on servers, where supported. Example implementations include a virtual firewall, operating system firewall, or a third-party firewall agent.			
v7	9.4 <u>Apply Host-based Firewalls or Port Filtering</u> Apply host-based firewalls or port filtering tools on end systems, with a default-deny rule that drops all traffic except those services and ports that are explicitly allowed.			

3.5.3.3 Configure IPv6 iptables

If IPv6 is not enabled on the system, this section can be skipped.

Iptables is used to set up, maintain, and inspect the tables of IPv6 packet filter rules in the Linux kernel. Several different tables may be defined. Each table contains a number of built-in chains and may also contain user-defined chains. Each chain is a list of rules which can match a set of packets. Each rule specifies what to do with a packet that matches. This is called a 'target', which may be a jump to a user-defined chain in the same table.

Note:

- This section broadly assumes starting with an empty iptables firewall ruleset (established by flushing the rules with iptables -F).
- Configuration of a live systems firewall directly over a remote connection will often result in being locked out. It is advised to have a known good firewall configuration set to run on boot and to configure an entire firewall structure in a script that is then run and tested before saving to boot.

The following script will implement the firewall rules of this section and open port 22(ssh) from anywhere. This needs to be updated to only allow systems requiring ssh connectivity to connect as per site policy.

```
#!/bin/bash

# Flush iptables rules
iptables -F

# Ensure default deny firewall policy
iptables -P INPUT DROP
iptables -P OUTPUT DROP
iptables -P FORWARD DROP

# Ensure loopback traffic is configured
iptables -A INPUT -i lo -j ACCEPT
iptables -A OUTPUT -o lo -j ACCEPT
iptables -A INPUT -s ::1 -j DROP

# Ensure outbound and established connections are configured
iptables -A OUTPUT -p tcp -m state --state NEW,ESTABLISHED -j ACCEPT
iptables -A OUTPUT -p udp -m state --state NEW,ESTABLISHED -j ACCEPT
iptables -A OUTPUT -p icmp -m state --state NEW,ESTABLISHED -j ACCEPT
iptables -A INPUT -p tcp -m state --state ESTABLISHED -j ACCEPT
iptables -A INPUT -p udp -m state --state ESTABLISHED -j ACCEPT
iptables -A INPUT -p icmp -m state --state ESTABLISHED -j ACCEPT

# Open inbound ssh(tcp port 22) connections
iptables -A INPUT -p tcp --dport 22 -m state --state NEW -j ACCEPT
```

3.5.3.3.1 Ensure iptables loopback traffic is configured (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

Configure the loopback interface to accept traffic. Configure all other interfaces to deny traffic to the loopback network (::1).

Rationale:

Loopback traffic is generated between processes on machine and is typically critical to operation of the system. The loopback interface is the only place that loopback network (::1) traffic should be seen, all other interfaces should ignore traffic on this network as an anti-spoofing measure.

Note: Changing firewall settings while connected over network can result in being locked out of the system.

Audit:

Run the following commands and verify output includes the listed rules in order (packet and byte counts may differ):

```
# iptables -L INPUT -v -n
Chain INPUT (policy DROP 0 packets, 0 bytes)
pkts bytes target      prot opt in      out     source
destination
    0     0 ACCEPT      all  --  lo      *      ::/0
    0     0 DROP        all  --  *       *      :::1

# iptables -L OUTPUT -v -n
Chain OUTPUT (policy DROP 0 packets, 0 bytes)
pkts bytes target      prot opt in      out     source
destination
    0     0 ACCEPT      all  --  *       lo      ::/0
```

OR verify IPv6 is disabled:

Run the following script. Output will confirm if IPv6 is disabled on the system.


```
#!/bin/bash

[ -n "$passing" ] && passing=""
[ -z "$(grep "^s*linux" /boot/grub2/grub.cfg | grep -v ipv6.disable=1)" ] &&
passing="true"
grep -Eq "\s*net\.ipv6\.conf\.all\.disable_ipv6\s*=\s*1\b(\s+#.*)?$"
/etc/sysctl.conf \
/etc/sysctl.d/*.conf && grep -Eq
"\s*net\.ipv6\.conf\.default\.disable_ipv6\s*=\s*1\b(\s+#.*)?$" \
/etc/sysctl.conf /etc/sysctl.d/*.conf && sysctl
net.ipv6.conf.all.disable_ipv6 | \
grep -Eq "\s*net\.ipv6\.conf\.all\.disable_ipv6\s*=\s*1\b(\s+#.*)?$" && \
sysctl net.ipv6.conf.default.disable_ipv6 | \
grep -Eq "\s*net\.ipv6\.conf\.default\.disable_ipv6\s*=\s*1\b(\s+#.*)?$" &&
passing="true"
if [ "$passing" = true ] ; then
    echo "IPv6 is disabled on the system"
else
    echo "IPv6 is enabled on the system"
fi
```

Remediation:

Run the following commands to implement the loopback rules:

```
# ip6tables -A INPUT -i lo -j ACCEPT
# ip6tables -A OUTPUT -o lo -j ACCEPT
# ip6tables -A INPUT -s ::1 -j DROP
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.4 <u>Implement and Manage a Firewall on Servers</u> Implement and manage a firewall on servers, where supported. Example implementations include a virtual firewall, operating system firewall, or a third-party firewall agent.			
v7	9.4 <u>Apply Host-based Firewalls or Port Filtering</u> Apply host-based firewalls or port filtering tools on end systems, with a default-deny rule that drops all traffic except those services and ports that are explicitly allowed.			

3.5.3.3.2 Ensure iptables outbound and established connections are configured (Manual)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

Configure the firewall rules for new outbound, and established IPv6 connections.

Rationale:

If rules are not in place for new outbound, and established connections all packets will be dropped by the default policy preventing network usage.

Note: Changing firewall settings while connected over network can result in being locked out of the system.

Audit:

Run the following command and verify all rules for new outbound, and established connections match site policy:

```
# iptables -L -v -n
```

OR verify IPv6 is disabled:

Run the following script. Output will confirm if IPv6 is disabled on the system.

```
#!/bin/bash

[ -n "$passing" ] && passing=""
[ -z "$(grep "^s*linux" /boot/grub2/grub.cfg | grep -v ipv6.disable=1)" ] &&
passing="true"
grep -Eq "\s*net\.ipv6\.conf\.all\.disable_ipv6\s*=\s*1\b(\s+#.*)?$"
/etc/sysctl.conf \
/etc/sysctl.d/*.conf && grep -Eq
"\s*net\.ipv6\.conf\.default\.disable_ipv6\s*=\s*1\b(\s+#.*)?$" \
/etc/sysctl.conf /etc/sysctl.d/*.conf && sysctl
net.ipv6.conf.all.disable_ipv6 | \
grep -Eq "\s*net\.ipv6\.conf\.all\.disable_ipv6\s*=\s*1\b(\s+#.*)?$" && \
sysctl net.ipv6.conf.default.disable_ipv6 | \
grep -Eq "\s*net\.ipv6\.conf\.default\.disable_ipv6\s*=\s*1\b(\s+#.*)?$" &&
passing="true"
if [ "$passing" = true ] ; then
    echo "IPv6 is disabled on the system"
else
    echo "IPv6 is enabled on the system"
fi
```

Remediation:

Configure iptables in accordance with site policy. The following commands will implement a policy to allow all outbound connections and all established connections:

```
# iptables -A OUTPUT -p tcp -m state --state NEW,ESTABLISHED -j ACCEPT
# iptables -A OUTPUT -p udp -m state --state NEW,ESTABLISHED -j ACCEPT
# iptables -A OUTPUT -p icmp -m state --state NEW,ESTABLISHED -j ACCEPT
# iptables -A INPUT -p tcp -m state --state ESTABLISHED -j ACCEPT
# iptables -A INPUT -p udp -m state --state ESTABLISHED -j ACCEPT
# iptables -A INPUT -p icmp -m state --state ESTABLISHED -j ACCEPT
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.4 Implement and Manage a Firewall on Servers Implement and manage a firewall on servers, where supported. Example implementations include a virtual firewall, operating system firewall, or a third-party firewall agent.			
v7	9.4 Apply Host-based Firewalls or Port Filtering Apply host-based firewalls or port filtering tools on end systems, with a default-deny rule that drops all traffic except those services and ports that are explicitly allowed.			

3.5.3.3.3 Ensure iptables firewall rules exist for all open ports (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

Any ports that have been opened on non-loopback addresses need firewall rules to govern traffic.

Rationale:

Without a firewall rule configured for open ports default firewall policy will drop all packets to these ports.

Note:

- Changing firewall settings while connected over network can result in being locked out of the system.
- The remediation command opens up the port to traffic from all sources. Consult iptables documentation and set any restrictions in compliance with site policy.

Audit:

Run the following command to determine open ports:

# ss -6tuln						
Netid	State	Recv-Q	Send-Q	Local Address:Port	Peer	
udp	UNCONN	0	0	:::1:123		
:::*						
udp	UNCONN	0	0	:::123		
:::*						
tcp	LISTEN	0	128	:::22		
:::*						
tcp	LISTEN	0	20	:::1:25		
:::*						

Run the following command to determine firewall rules:

```
# ip6tables -L INPUT -v -n

Chain INPUT (policy DROP 0 packets, 0 bytes)
  pkts bytes target     prot opt in     out     source
destination
    0      0 ACCEPT     all  --  lo      *       ::/0      ::/0
    0      0 DROP       all  --  *       *       :::1      ::/0
    0      0 ACCEPT     tcp  --  *       *       ::/0      ::/0
tcp dpt:22 state NEW
```

Verify all open ports listening on non-localhost addresses have at least one firewall rule. The last line identified by the "tcp dpt:22 state NEW" identifies it as a firewall rule for new connections on tcp port 22.

OR verify IPv6 is disabled:

Run the following script. Output will confirm if IPv6 is disabled on the system.

```
#!/bin/bash

[ -n "$passing" ] && passing=""
[ -z "$(grep "^s*linux" /boot/grub2/grub.cfg | grep -v ipv6.disable=1)" ] &&
passing="true"
grep -Eq "^s*net\.ipv6\.conf\.all\.disable_ipv6\s*=\s*1\b(\s+#.*)?$"
/etc/sysctl.conf \
/etc/sysctl.d/*.conf && grep -Eq
"^s*net\.ipv6\.conf\.default\.disable_ipv6\s*=\s*1\b(\s+#.*)?$" \
/etc/sysctl.conf /etc/sysctl.d/*.conf && sysctl
net.ipv6.conf.all.disable_ipv6 | \
grep -Eq "^s*net\.ipv6\.conf\.all\.disable_ipv6\s*=\s*1\b(\s+#.*)?$" && \
sysctl net.ipv6.conf.default.disable_ipv6 | \
grep -Eq "^s*net\.ipv6\.conf\.default\.disable_ipv6\s*=\s*1\b(\s+#.*)?$" &&
passing="true"
if [ "$passing" = true ] ; then
    echo "IPv6 is disabled on the system"
else
    echo "IPv6 is enabled on the system"
fi
```

Remediation:

For each port identified in the audit which does not have a firewall rule establish a proper rule for accepting inbound connections:

```
# ip6tables -A INPUT -p <protocol> --dport <port> -m state --state NEW -j
ACCEPT
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.4 <u>Implement and Manage a Firewall on Servers</u> Implement and manage a firewall on servers, where supported. Example implementations include a virtual firewall, operating system firewall, or a third-party firewall agent.			
v7	9.4 <u>Apply Host-based Firewalls or Port Filtering</u> Apply host-based firewalls or port filtering tools on end systems, with a default-deny rule that drops all traffic except those services and ports that are explicitly allowed.			

3.5.3.3.4 Ensure iptables default deny firewall policy (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

A default deny all policy on connections ensures that any unconfigured network usage will be rejected.

Rationale:

With a default accept policy the firewall will accept any packet that is not configured to be denied. It is easier to white list acceptable usage than to black list unacceptable usage.

Note: Changing firewall settings while connected over network can result in being locked out of the system.

Audit:

Run the following command and verify that the policy for the INPUT, OUTPUT, and FORWARD chains is DROP or REJECT:

```
# iptables -L
Chain INPUT (policy DROP)
Chain FORWARD (policy DROP)
Chain OUTPUT (policy DROP)
```

OR

Verify IPv6 is disabled:

Run the following script. Output will confirm if IPv6 is disabled on the system.

```
#!/bin/bash

[ -n "$passing" ] && passing=""
[ -z "$(grep "^s*linux" /boot/grub2/grub.cfg | grep -v ipv6.disable=1)" ] &&
passing="true"
grep -Eq "\s*net\.ipv6\.conf\.all\.disable_ipv6\s*=\s*1\b(\s+#.*)?$"
/etc/sysctl.conf \
/etc/sysctl.d/*.conf && grep -Eq
"\s*net\.ipv6\.conf\.default\.disable_ipv6\s*=\s*1\b(\s+#.*)?$" \
/etc/sysctl.conf /etc/sysctl.d/*.conf && sysctl
net.ipv6.conf.all.disable_ipv6 | \
grep -Eq "\s*net\.ipv6\.conf\.all\.disable_ipv6\s*=\s*1\b(\s+#.*)?$" && \
sysctl net.ipv6.conf.default.disable_ipv6 | \
grep -Eq "\s*net\.ipv6\.conf\.default\.disable_ipv6\s*=\s*1\b(\s+#.*)?$" &&
passing="true"
if [ "$passing" = true ] ; then
    echo "IPv6 is disabled on the system"
else
    echo "IPv6 is enabled on the system"
fi
```

Remediation:

Run the following commands to implement a default DROP policy:

```
# ip6tables -P INPUT DROP
# ip6tables -P OUTPUT DROP
# ip6tables -P FORWARD DROP
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.4 <u>Implement and Manage a Firewall on Servers</u> Implement and manage a firewall on servers, where supported. Example implementations include a virtual firewall, operating system firewall, or a third-party firewall agent.			
v7	9.4 <u>Apply Host-based Firewalls or Port Filtering</u> Apply host-based firewalls or port filtering tools on end systems, with a default-deny rule that drops all traffic except those services and ports that are explicitly allowed.			

3.5.3.3.5 Ensure iptables rules are saved (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `iptables-services` package includes the `/etc/sysconfig/iptables` file. The `iptables` rules in this file will be loaded by the `iptables.service` during boot, or when it is started or re-loaded.

Rationale:

If the `iptables` rules are not saved and a system re-boot occurs, the `iptables` rules will be lost.

Audit:

Review the file `/etc/sysconfig/iptables` and ensure it contains the complete correct rule-set.

Example: `/etc/sysconfig/iptables`

```
# sample configuration for iptables service
# you can edit this manually or use system-config-firewall
# Generated by iptables-save v1.4.21 on Wed Mar 25 14:23:37 2020
*filter
:INPUT DROP [0:0]
:FORWARD DROP [0:0]
:OUTPUT DROP [0:0]
-A INPUT -i lo -j ACCEPT
-A INPUT -s ::1/128 -j DROP
-A INPUT -p tcp -m state --state ESTABLISHED -j ACCEPT
-A INPUT -p udp -m state --state ESTABLISHED -j ACCEPT
-A INPUT -p icmp -m state --state ESTABLISHED -j ACCEPT
-A INPUT -p tcp -m tcp --dport 22 -m state --state NEW -j ACCEPT
-A OUTPUT -o lo -j ACCEPT
-A OUTPUT -p tcp -m state --state NEW,ESTABLISHED -j ACCEPT
-A OUTPUT -p udp -m state --state NEW,ESTABLISHED -j ACCEPT
-A OUTPUT -p icmp -m state --state NEW,ESTABLISHED -j ACCEPT
COMMIT
# Completed on Wed Mar 25 14:58:32 2020
```

OR

Verify IPv6 is disabled:

Run the following script. Output will confirm if IPv6 is disabled on the system.

```
#!/bin/bash

[ -n "$passing" ] && passing=""
[ -z "$(grep "^s*linux" /boot/grub2/grub.cfg | grep -v ipv6.disable=1)" ] &&
passing="true"
grep -Eq "^s*net\.ipv6\.conf\.all\.disable_ipv6\s*=\s*1\b(\s+#.*)?$"
/etc/sysctl.conf \
/etc/sysctl.d/*.conf && grep -Eq
"^s*net\.ipv6\.conf\.default\.disable_ipv6\s*=\s*1\b(\s+#.*)?$" \
/etc/sysctl.conf /etc/sysctl.d/*.conf && sysctl
net.ipv6.conf.all.disable_ipv6 | \
grep -Eq "^s*net\.ipv6\.conf\.all\.disable_ipv6\s*=\s*1\b(\s+#.*)?$" && \
sysctl net.ipv6.conf.default.disable_ipv6 | \
grep -Eq "^s*net\.ipv6\.conf\.default\.disable_ipv6\s*=\s*1\b(\s+#.*)?$" &&
passing="true"
if [ "$passing" = true ] ; then
    echo "IPv6 is disabled on the system"
else
    echo "IPv6 is enabled on the system"
fi
```

Remediation:

Run the following commands to create or update the `/etc/sysconfig/ip6tables` file:

Run the following command to review the current running `iptables` configuration:

```
# iptables -L
```

Output should include:

```
Chain INPUT (policy DROP)
target     prot opt source                destination
ACCEPT     all  --  anywhere              anywhere
DROP       all  --  localhost             anywhere
ACCEPT     tcp  --  anywhere              anywhere    state
ESTABLISHED
ACCEPT     udp  --  anywhere              anywhere    state
ESTABLISHED
ACCEPT     icmp --  anywhere              anywhere    state
ESTABLISHED
ACCEPT     tcp  --  anywhere              anywhere    tcp dpt:ssh
state NEW

Chain FORWARD (policy DROP)
target     prot opt source                destination

Chain OUTPUT (policy DROP)
target     prot opt source                destination
ACCEPT     all  --  anywhere              anywhere
ACCEPT     tcp  --  anywhere              anywhere    state
NEW,ESTABLISHED
ACCEPT     udp  --  anywhere              anywhere    state
NEW,ESTABLISHED
ACCEPT     icmp --  anywhere              anywhere    state
NEW,ESTABLISHED
```

Run the following command to save the verified running configuration to the file

`/etc/sysconfig/ip6tables`:

```
# service iptables save
iptables: Saving firewall rules to /etc/sysconfig/ip6table[ OK ]
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.4 <u>Implement and Manage a Firewall on Servers</u> Implement and manage a firewall on servers, where supported. Example implementations include a virtual firewall, operating system firewall, or a third-party firewall agent.			
v7	9.4 <u>Apply Host-based Firewalls or Port Filtering</u> Apply host-based firewalls or port filtering tools on end systems, with a default-deny rule that drops all traffic except those services and ports that are explicitly allowed.			

3.5.3.3.6 Ensure ip6tables is enabled and running (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

`ip6tables.service` is a utility for configuring and maintaining `ip6tables`.

Rationale:

`ip6tables.service` will load the `iptables` rules saved in the file `/etc/sysconfig/ip6tables` at boot, otherwise the `ip6tables` rules will be cleared during a re-boot of the system.

Audit:

Run the following commands to verify `ip6tables` is enabled:

```
# systemctl is-enabled ip6tables
enabled
```

Run the following command to verify `ip6tables.service` is active and running or exited

```
# systemctl status ip6tables | grep -E " Active: active \((running|exited)\)"
Active: active (exited) since <day date and time>
```

OR verify IPv6 is disabled:

Run the following script. Output will confirm if IPv6 is disabled on the system.

```
#!/bin/bash

[ -n "$passing" ] && passing=""
[ -z "$(grep "^s*linux" /boot/grub2/grub.cfg | grep -v ipv6.disable=1)" ] &&
passing="true"
grep -Eq "^s*net\.ipv6\.conf\.all\.disable_ipv6\s*=\s*1\b(\s+#.*)?$"
/etc/sysctl.conf \
/etc/sysctl.d/*.conf && grep -Eq
"^s*net\.ipv6\.conf\.default\.disable_ipv6\s*=\s*1\b(\s+#.*)?$" \
/etc/sysctl.conf /etc/sysctl.d/*.conf && sysctl
net.ipv6.conf.all.disable_ipv6 | \
grep -Eq "^s*net\.ipv6\.conf\.all\.disable_ipv6\s*=\s*1\b(\s+#.*)?$" && \
sysctl net.ipv6.conf.default.disable_ipv6 | \
grep -Eq "^s*net\.ipv6\.conf\.default\.disable_ipv6\s*=\s*1\b(\s+#.*)?$" &&
passing="true"
if [ "$passing" = true ] ; then
    echo "IPv6 is disabled on the system"
else
    echo "IPv6 is enabled on the system"
fi
```

Remediation:

Run the following command to enable and start ip6tables:

```
# systemctl --now start ip6tables
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.4 <u>Implement and Manage a Firewall on Servers</u> Implement and manage a firewall on servers, where supported. Example implementations include a virtual firewall, operating system firewall, or a third-party firewall agent.			
v7	9.4 <u>Apply Host-based Firewalls or Port Filtering</u> Apply host-based firewalls or port filtering tools on end systems, with a default-deny rule that drops all traffic except those services and ports that are explicitly allowed.			

4 Logging and Auditing

The items in this section describe how to configure logging, log monitoring, and auditing, using tools included in most distributions.

It is recommended that `rsyslog` be used for logging (with `logwatch` providing summarization) and `auditd` be used for auditing (with `aureport` providing summarization) to automatically monitor logs for intrusion attempts and other suspicious system behavior.

In addition to the local log files created by the steps in this section, it is also recommended that sites collect copies of their system logs on a secure, centralized log server via an encrypted connection. Not only does centralized logging help sites correlate events that may be occurring on multiple systems, but having a second copy of the system log information may be critical after a system compromise where the attacker has modified the local log files on the affected system(s). If a log correlation system is deployed, configure it to process the logs described in this section.

Because it is often necessary to correlate log information from many different systems (particularly after a security incident) it is recommended that the time be synchronized among systems and devices connected to the local network. The standard Internet protocol for time synchronization is the Network Time Protocol (NTP), which is supported by most network-ready devices. See the `ntpd(8)` manual page for more information on configuring NTP.

It is important that all logs described in this section be monitored on a regular basis and correlated to determine trends. A seemingly innocuous entry in one log could be more significant when compared to an entry in another log.

Note on log file permissions: There really isn't a "one size fits all" solution to the permissions on log files. Many sites utilize group permissions so that administrators who are in a defined security group, such as "wheel" do not have to elevate privileges to root in order to read log files. Also, if a third party log aggregation tool is used, it may need to have group permissions to read the log files, which is preferable to having it run setuid to root. Therefore, there are two remediation and audit steps for log file permissions. One is for systems that do not have a secured group method implemented that only permits root to read the log files (`root:root 600`). The other is for sites that do have such a setup and are designated as `root:securegrp 640` where `securegrp` is the defined security group (in some cases `wheel`).

4.1 Configure System Accounting (auditd)

System auditing, through `auditd`, allows system administrators to monitor their systems such that they can detect unauthorized access or modification of data. By default, `auditd` will audit system logins, account modifications, and authentication events. Events will be logged to `/var/log/audit/audit.log`. The recording of these events will use a modest amount of disk space on a system. If significantly more events are captured, additional on system or off system storage may need to be allocated.

Note:

- *The recommendations in this section implement an audit policy that produces large quantities of logged data. In some environments it can be challenging to store or process these logs and as such they are marked as Level 2 for both Servers and Workstations.*
- *For 64 bit systems that have `arch` as a rule parameter, you will need two rules: one for 64 bit and one for 32 bit systems. For 32 bit systems, only one rule is needed.*
- *Several recommendations in this section filter based off of `auditd>=500` for unprivileged non-system users. Some systems may have a non-default `UID_MIN` setting, consult the `UID_MIN` setting in `/etc/login.defs` to determine the `UID_MIN` setting for your system.*
- *Once all audit rules have been added to a file or files in the `/etc/audit/rules.d/` directory, **the auditd service must be re-started, or the system rebooted, for the new rules to be included.***
- *The audit and remediation in this section look for a 'key' value. **The 'key' value may be different for the audit settings on your system.***

4.1.1 Ensure auditing is enabled

The capturing of system events provides system administrators with information to allow them to determine if unauthorized access to their system is occurring.

4.1.1.1 Ensure auditd is installed (Automated)

Profile Applicability:

- Level 2 - Server
- Level 2 - Workstation

Description:

auditd is the userspace component to the Linux Auditing System. It's responsible for writing audit records to the disk

Rationale:

The capturing of system events provides system administrators with information to allow them to determine if unauthorized access to their system is occurring.

Audit:

Run the following command and verify auditd is installed:

```
# rpm -q audit audit-libs  
audit-<version>  
audit-libs-<version>
```

Remediation:

Run the following command to Install auditd

```
# yum install audit audit-libs
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.2 <u>Collect Audit Logs</u> Collect audit logs. Ensure that logging, per the enterprise's audit log management process, has been enabled across enterprise assets.			
v7	6.2 <u>Activate audit logging</u> Ensure that local logging has been enabled on all systems and networking devices.			
v7	6.3 <u>Enable Detailed Logging</u> Enable system logging to include detailed information such as an event source, date, user, timestamp, source addresses, destination addresses, and other useful elements.			

4.1.1.2 Ensure auditd service is enabled and running (Automated)

Profile Applicability:

- Level 2 - Server
- Level 2 - Workstation

Description:

Turn on the `auditd` daemon to record system events.

Rationale:

The capturing of system events provides system administrators with information to allow them to determine if unauthorized access to their system is occurring.

Audit:

Run the following command to verify `auditd` is enabled:

```
# systemctl is-enabled auditd  
enabled
```

Run the following command to verify that `auditd` is running:

```
# systemctl status auditd | grep 'Active: active (running) '  
Active: active (running) since <time and date>
```

Remediation:

Run the following command to enable and start `auditd`:

```
# systemctl --now enable auditd
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.2 <u>Collect Audit Logs</u> Collect audit logs. Ensure that logging, per the enterprise's audit log management process, has been enabled across enterprise assets.			
v7	6.2 <u>Activate audit logging</u> Ensure that local logging has been enabled on all systems and networking devices.			
v7	6.3 <u>Enable Detailed Logging</u> Enable system logging to include detailed information such as an event source, date, user, timestamp, source addresses, destination addresses, and other useful elements.			

4.1.1.3 Ensure auditing for processes that start prior to auditd is enabled (Automated)

Profile Applicability:

- Level 2 - Server
- Level 2 - Workstation

Description:

Configure `grub` so that processes that are capable of being audited can be audited even if they start up prior to `auditd` startup.

Rationale:

Audit events need to be captured on processes that start up prior to `auditd`, so that potential malicious activity cannot go undetected.

Note: This recommendation is designed around the `grub2` bootloader, if `LILO` or another bootloader is in use in your environment enact equivalent settings.

Audit:

Run the following script to verify that each linux line has the `audit=1` parameter set:

```
#!/bin/bash

# IF check passes return PASSED
efidir=$(find /boot/efi/EFI/* -type d -not -name 'BOOT')
gbdir=$(find /boot -maxdepth 1 -type d -name 'grub*')
if [ -f "$efidir/grub.cfg" ]; then
    grep "^s*linux" "$efidir/grub.cfg" | grep -Evq "audit=1\b" && echo
    "FAILED" || echo "PASSED"
elif [ -f "$gbdir/grub.cfg" ]; then
    grep "^s*linux" "$gbdir/grub.cfg" | grep -Evq "audit=1\b" && echo
    "FAILED" || echo "PASSED"
else
    echo "FAILED"
fi
```

Remediation:

Edit `/etc/default/grub` and add `audit=1` to `GRUB_CMDLINE_LINUX`:

```
GRUB_CMDLINE_LINUX="audit=1"
```

Run the following command to update the `grub2` configuration:

```
# grub2-mkconfig -o /boot/grub2/grub.cfg
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.2 <u>Collect Audit Logs</u> Collect audit logs. Ensure that logging, per the enterprise's audit log management process, has been enabled across enterprise assets.			
v7	6.2 <u>Activate audit logging</u> Ensure that local logging has been enabled on all systems and networking devices.			
v7	6.3 <u>Enable Detailed Logging</u> Enable system logging to include detailed information such as an event source, date, user, timestamp, source addresses, destination addresses, and other useful elements.			

4.1.2 Configure Data Retention

When auditing, it is important to carefully configure the storage requirements for audit logs. By default, auditd will max out the log files at 5MB and retain only 4 copies of them. Older versions will be deleted. It is possible on a system that the 20 MBs of audit logs may fill up the system causing loss of audit data. While the recommendations here provide guidance, check your site policy for audit storage requirements.

4.1.2.1 Ensure audit log storage size is configured (Automated)

Profile Applicability:

- Level 2 - Server
- Level 2 - Workstation

Description:

Configure the maximum size of the audit log file. Once the log reaches the maximum size, it will be rotated and a new log file will be started.

Notes:

- *The `max_log_file` parameter is measured in megabytes.*
- *Other methods of log rotation may be appropriate based on site policy. One example is time-based rotation strategies which don't have native support in `auditd` configurations. Manual audit of custom configurations should be evaluated for effectiveness and completeness.*

Rationale:

It is important that an appropriate size is determined for log files so that they do not impact the system and audit data is not lost.

Audit:

Run the following command and ensure output is in compliance with site policy:

```
# grep max_log_file /etc/audit/auditd.conf  
max_log_file = <MB>
```

Remediation:

Set the following parameter in `/etc/audit/auditd.conf` in accordance with site policy:

```
max_log_file = <MB>
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.3 <u>Ensure Adequate Audit Log Storage</u> Ensure that logging destinations maintain adequate storage to comply with the enterprise's audit log management process.			
v7	6.4 <u>Ensure adequate storage for logs</u> Ensure that all systems that store logs have adequate storage space for the logs generated.			

4.1.2.2 Ensure audit logs are not automatically deleted (Automated)

Profile Applicability:

- Level 2 - Server
- Level 2 - Workstation

Description:

The `max_log_file_action` setting determines how to handle the audit log file reaching the max file size. A value of `keep_logs` will rotate the logs but never delete old logs.

Rationale:

In high security contexts, the benefits of maintaining a long audit history exceed the cost of storing the audit history.

Audit:

Run the following command and verify output matches:

```
# grep max_log_file_action /etc/audit/auditd.conf  
max_log_file_action = keep_logs
```

Remediation:

Set the following parameter in `/etc/audit/auditd.conf`:

```
max_log_file_action = keep_logs
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.3 <u>Ensure Adequate Audit Log Storage</u> Ensure that logging destinations maintain adequate storage to comply with the enterprise's audit log management process.			
v7	6.2 <u>Activate audit logging</u> Ensure that local logging has been enabled on all systems and networking devices.			
v7	6.4 <u>Ensure adequate storage for logs</u> Ensure that all systems that store logs have adequate storage space for the logs generated.			

4.1.2.3 Ensure system is disabled when audit logs are full (Automated)

Profile Applicability:

- Level 2 - Server
- Level 2 - Workstation

Description:

The `auditd` daemon can be configured to halt the system when the audit logs are full.

Rationale:

In high security contexts, the risk of detecting unauthorized access or nonrepudiation exceeds the benefit of the system's availability.

Audit:

Run the following commands and verify output matches:

```
# grep space_left_action /etc/audit/auditd.conf
space_left_action = email
# grep action_mail_acct /etc/audit/auditd.conf
action_mail_acct = root
# grep admin_space_left_action /etc/audit/auditd.conf
admin_space_left_action = halt
```

Remediation:

Set the following parameters in `/etc/audit/auditd.conf`:

```
space_left_action = email
action_mail_acct = root
admin_space_left_action = halt
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.2 <u>Collect Audit Logs</u> Collect audit logs. Ensure that logging, per the enterprise's audit log management process, has been enabled across enterprise assets.			
v8	8.3 <u>Ensure Adequate Audit Log Storage</u> Ensure that logging destinations maintain adequate storage to comply with the enterprise's audit log management process.			
v7	6.2 <u>Activate audit logging</u> Ensure that local logging has been enabled on all systems and networking devices.			
v7	6.4 <u>Ensure adequate storage for logs</u> Ensure that all systems that store logs have adequate storage space for the logs generated.			

4.1.2.4 Ensure audit_backlog_limit is sufficient (Automated)

Profile Applicability:

- Level 2 - Server
- Level 2 - Workstation

Description:

The backlog limit has a default setting of 64

Rationale:

During boot if audit=1, then the backlog will hold 64 records. If more than 64 records are created during boot, auditd records will be lost and potential malicious activity could go undetected.

Audit:

Run the following script to verify the audit_backlog_limit= parameter is set to an appropriate size for your organization

```
#!/bin/bash

# IF check passes return PASSED
efidir=$(find /boot/efi/EFI/* -type d -not -name 'BOOT')
gbdir=$(find /boot -maxdepth 1 -type d -name 'grub*')
if [ -f "$efidir"/grub.cfg ]; then
    grep "^s*linux" "$efidir"/grub.cfg | grep -Evq
    "audit_backlog_limit=\S+\b" && echo -e "\n\nFAILED" || echo -e "\n\nPASSED:\n
    \"$(grep "audit_backlog_limit=" "$gbdir"/grub.cfg)\"
elif [ -f "$gbdir"/grub.cfg ]; then
    grep "^s*linux" "$gbdir"/grub.cfg | grep -Evq "audit_backlog_limit=\S+\b"
    && echo -e "\n\nFAILED" || echo -e "\n\nPASSED:\n \"$(grep
    "audit_backlog_limit=" "$gbdir"/grub.cfg)\"
else
    echo "FAILED"
fi
```

Ensure the the returned value complies with local site policy. It's recommended that this value be 8192 or larger.

Remediation:

Edit /etc/default/grub and add `audit_backlog_limit=<BACKLOG SIZE>` to
GRUB_CMDLINE_LINUX:

Example:

```
GRUB_CMDLINE_LINUX="audit_backlog_limit=8192"
```

Run the following command to update the grub2 configuration:

```
# grub2-mkconfig -o /boot/grub2/grub.cfg
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.2 Collect Audit Logs Collect audit logs. Ensure that logging, per the enterprise's audit log management process, has been enabled across enterprise assets.			
v8	8.3 Ensure Adequate Audit Log Storage Ensure that logging destinations maintain adequate storage to comply with the enterprise's audit log management process.			
v7	6.2 Activate audit logging Ensure that local logging has been enabled on all systems and networking devices.			
v7	6.3 Enable Detailed Logging Enable system logging to include detailed information such as an event source, date, user, timestamp, source addresses, destination addresses, and other useful elements.			

4.1.3 Ensure events that modify date and time information are collected (Automated)

Profile Applicability:

- Level 2 - Server
- Level 2 - Workstation

Description:

Capture events where the system date and/or time has been modified. The parameters in this section are set to determine if the `adjtimex` (tune kernel clock), `settimeofday` (Set time, using `timeval` and `timezone` structures) `stime` (using seconds since 1/1/1970) or `clock_settime` (allows for the setting of several internal clocks and timers) system calls have been executed and always write an audit record to the `/var/log/audit.log` file upon exit, tagging the records with the identifier "time-change"

Note: Reloading the auditd config to set active settings requires the `auditd` service to be restarted, and may require a system reboot.

Rationale:

Unexpected changes in system date and/or time could be a sign of malicious activity on the system.

Audit:

On a 32 bit system run the following commands:

```
# grep time-change /etc/audit/rules.d/*.rules
# auditctl -l | grep time-change
```

Verify output of both matches:

```
-a always,exit -F arch=b32 -S adjtimex -S settimeofday -S stime -k time-
change
-a always,exit -F arch=b32 -S clock_settime -k time-change
-w /etc/localtime -p wa -k time-change
```

On a 64 bit system run the following commands:

```
# grep time-change /etc/audit/rules.d/*.rules
# auditctl -l | grep time-change
```

Verify output of both matches:

```
-a always,exit -F arch=b64 -S adjtimex -S settimeofday -k time-change
-a always,exit -F arch=b32 -S adjtimex -S settimeofday -S stime -k time-
change
-a always,exit -F arch=b64 -S clock_settime -k time-change
-a always,exit -F arch=b32 -S clock_settime -k time-change
-w /etc/localtime -p wa -k time-change
```

Remediation:

For 32 bit systems Edit or create a file in the /etc/audit/rules.d/ directory ending in .rules

Example: vi /etc/audit/rules.d/50-time_change.rules

Add the following lines:

```
-a always,exit -F arch=b32 -S adjtimex -S settimeofday -S stime -k time-change
-a always,exit -F arch=b32 -S clock_settime -k time-change
-w /etc/localtime -p wa -k time-change
```

For 64 bit systems Edit or create a file in the /etc/audit/rules.d/ directory ending in .rules

Example: vi /etc/audit/rules.d/50-time_change.rules

Add the following lines:

```
-a always,exit -F arch=b64 -S adjtimex -S settimeofday -k time-change
-a always,exit -F arch=b32 -S adjtimex -S settimeofday -S stime -k time-change
-a always,exit -F arch=b64 -S clock_settime -k time-change
-a always,exit -F arch=b32 -S clock_settime -k time-change
-w /etc/localtime -p wa -k time-change
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.5 Collect Detailed Audit Logs Configure detailed audit logging for enterprise assets containing sensitive data. Include event source, date, username, timestamp, source addresses, destination addresses, and other useful elements that could assist in a forensic investigation.			
v7	5.5 Implement Automated Configuration Monitoring Systems Utilize a Security Content Automation Protocol (SCAP) compliant configuration monitoring system to verify all security configuration elements, catalog approved exceptions, and alert when unauthorized changes occur.			

4.1.4 Ensure events that modify user/group information are collected (Automated)

Profile Applicability:

- Level 2 - Server
- Level 2 - Workstation

Description:

Record events affecting the `group`, `passwd` (user IDs), `shadow` and `gshadow` (passwords) or `/etc/security/opasswd` (old passwords, based on remember parameter in the PAM configuration) files. The parameters in this section will watch the files to see if they have been opened for write or have had attribute changes (e.g. permissions) and tag them with the identifier "identity" in the audit log file.

Note: Reloading the `auditd` config to set active settings may require a system reboot.

Rationale:

Unexpected changes to these files could be an indication that the system has been compromised and that an unauthorized user is attempting to hide their activities or compromise additional accounts.

Audit:

Run the following command to check the `auditd .rules` files:

```
# grep identity /etc/audit/rules.d/*.rules
```

Verify the output matches:

```
-w /etc/group -p wa -k identity
-w /etc/passwd -p wa -k identity
-w /etc/gshadow -p wa -k identity
-w /etc/shadow -p wa -k identity
-w /etc/security/opasswd -p wa -k identity
```

Run the following command to check loaded `auditd` rules:

```
# auditctl -l | grep identity
```

Verify the output matches:

```
-w /etc/group -p wa -k identity
-w /etc/passwd -p wa -k identity
-w /etc/gshadow -p wa -k identity
-w /etc/shadow -p wa -k identity
-w /etc/security/opasswd -p wa -k identity
```

Remediation:

Edit or create a file in the `/etc/audit/rules.d/` directory ending in `.rules`

Example: `vi /etc/audit/rules.d/50-identity.rules`

Add the following lines:

```
-w /etc/group -p wa -k identity
-w /etc/passwd -p wa -k identity
-w /etc/gshadow -p wa -k identity
-w /etc/shadow -p wa -k identity
-w /etc/security/opasswd -p wa -k identity
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.5 <u>Collect Detailed Audit Logs</u> Configure detailed audit logging for enterprise assets containing sensitive data. Include event source, date, username, timestamp, source addresses, destination addresses, and other useful elements that could assist in a forensic investigation.			
v7	4.8 <u>Log and Alert on Changes to Administrative Group Membership</u> Configure systems to issue a log entry and alert when an account is added to or removed from any group assigned administrative privileges.			

4.1.5 Ensure events that modify the system's network environment are collected (Automated)

Profile Applicability:

- Level 2 - Server
- Level 2 - Workstation

Description:

Record changes to network environment files or system calls. The below parameters monitor the `sethostname` (set the systems host name) or `setdomainname` (set the systems domainname) system calls, and write an audit event on system call exit. The other parameters monitor the `/etc/issue` and `/etc/issue.net` files (messages displayed pre-login), `/etc/hosts` (file containing host names and associated IP addresses) and `/etc/sysconfig/network` (directory containing network interface scripts and configurations) files.

Note: Reloading the auditd config to set active settings requires the `auditd` service to be restarted, and may require a system reboot.

Rationale:

Monitoring `sethostname` and `setdomainname` will identify potential unauthorized changes to host and domainname of a system. The changing of these names could potentially break security parameters that are set based on those names. The `/etc/hosts` file is monitored for changes in the file that can indicate an unauthorized intruder is trying to change machine associations with IP addresses and trick users and processes into connecting to unintended machines. Monitoring `/etc/issue` and `/etc/issue.net` is important, as intruders could put disinformation into those files and trick users into providing information to the intruder. Monitoring `/etc/sysconfig/network` is important as it can show if network interfaces or scripts are being modified in a way that can lead to the machine becoming unavailable or compromised. All audit records will be tagged with the identifier "system-locale."

Audit:

On a 32 bit system run the following commands:

```
# grep system-locale /etc/audit/rules.d/*.rules
# auditctl -l | grep system-locale
```

Verify output of both matches:

```
-a always,exit -F arch=b32 -S sethostname -S setdomainname -k system-locale
-w /etc/issue -p wa -k system-locale
-w /etc/issue.net -p wa -k system-locale
-w /etc/hosts -p wa -k system-locale
-w /etc/sysconfig/network -p wa -k system-locale
```

On a 64 bit system run the following commands:

```
# grep system-locale /etc/audit/rules.d/*.rules
# auditctl -l | grep system-locale
```

Verify output of both matches:

```
-a always,exit -F arch=b64 -S sethostname -S setdomainname -k system-locale
-a always,exit -F arch=b32 -S sethostname -S setdomainname -k system-locale
-w /etc/issue -p wa -k system-locale
-w /etc/issue.net -p wa -k system-locale
-w /etc/hosts -p wa -k system-locale
-w /etc/sysconfig/network -p wa -k system-locale
```

Remediation:

For 32 bit systems edit or create a file in the `/etc/audit/rules.d/` directory ending in `.rules`

Example: `vi /etc/audit/rules.d/50-system_local.rules`

Add the following lines:

```
-a always,exit -F arch=b32 -S sethostname -S setdomainname -k system-locale
-w /etc/issue -p wa -k system-locale
-w /etc/issue.net -p wa -k system-locale
-w /etc/hosts -p wa -k system-locale
-w /etc/sysconfig/network -p wa -k system-locale
```

For 64 bit systems Edit or create a file in the `/etc/audit/rules.d/` directory ending in `.rules`

Example: `vi /etc/audit/rules.d/50-system_local.rules`

Add the following lines:

```
-a always,exit -F arch=b64 -S sethostname -S setdomainname -k system-locale
-a always,exit -F arch=b32 -S sethostname -S setdomainname -k system-locale
-w /etc/issue -p wa -k system-locale
-w /etc/issue.net -p wa -k system-locale
-w /etc/hosts -p wa -k system-locale
-w /etc/sysconfig/network -p wa -k system-locale
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.5 <u>Collect Detailed Audit Logs</u> Configure detailed audit logging for enterprise assets containing sensitive data. Include event source, date, username, timestamp, source addresses, destination addresses, and other useful elements that could assist in a forensic investigation.			
v7	5.5 <u>Implement Automated Configuration Monitoring Systems</u> Utilize a Security Content Automation Protocol (SCAP) compliant configuration monitoring system to verify all security configuration elements, catalog approved exceptions, and alert when unauthorized changes occur.			
v7	6.2 <u>Activate audit logging</u> Ensure that local logging has been enabled on all systems and networking devices.			

4.1.6 Ensure events that modify the system's Mandatory Access Controls are collected (Automated)

Profile Applicability:

- Level 2 - Server
- Level 2 - Workstation

Description:

Monitor SELinux mandatory access controls. The parameters below monitor any write access (potential additional, deletion or modification of files in the directory) or attribute changes to the `/etc/selinux/` and `/usr/share/selinux/` directories.

Note:

- *If a different Mandatory Access Control method is used, changes to the corresponding directories should be audited.*
- *Reloading the auditd config to set active settings requires the `auditd` service to be restarted, and may require a system reboot.*

Rationale:

Changes to files in the `/etc/selinux/` and `/usr/share/selinux/` directories could indicate that an unauthorized user is attempting to modify access controls and change security contexts, leading to a compromise of the system.

Audit:

Run the following commands:

```
# grep MAC-policy /etc/audit/rules.d/*.rules
# auditctl -l | grep MAC-policy
```

Verify output of both matches:

```
-w /etc/selinux/ -p wa -k MAC-policy
-w /usr/share/selinux/ -p wa -k MAC-policy
```

Remediation:

Edit or create a file in the `/etc/audit/rules.d/` directory ending in `.rules`

Example: `vi /etc/audit/rules.d/50-MAC_policy.rules`

Add the following lines:

```
-w /etc/selinux/ -p wa -k MAC-policy  
-w /usr/share/selinux/ -p wa -k MAC-policy
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.5 <u>Collect Detailed Audit Logs</u> Configure detailed audit logging for enterprise assets containing sensitive data. Include event source, date, username, timestamp, source addresses, destination addresses, and other useful elements that could assist in a forensic investigation.			
v7	5.5 <u>Implement Automated Configuration Monitoring Systems</u> Utilize a Security Content Automation Protocol (SCAP) compliant configuration monitoring system to verify all security configuration elements, catalog approved exceptions, and alert when unauthorized changes occur.			

4.1.7 Ensure login and logout events are collected (Automated)

Profile Applicability:

- Level 2 - Server
- Level 2 - Workstation

Description:

Monitor login and logout events. The parameters below track changes to files associated with login/logout events.

- The file `/var/log/lastlog` maintain records of the last time a user successfully logged in.
- The `/var/run/faillock/` directory maintains records of login failures via the `pam_faillock` module.

Note: Reloading the `auditd` config to set active settings requires the `auditd` service to be restarted, and may require a system reboot.

Rationale:

Monitoring login/logout events could provide a system administrator with information associated with brute force attacks against user logins.

Audit:

Run the following commands:

```
# grep logins /etc/audit/rules.d/*.rules
# auditctl -l | grep logins
```

Verify output of both includes:

```
-w /var/log/lastlog -p wa -k logins
-w /var/run/faillock/ -p wa -k logins
```

Remediation:

Edit or create a file in the `/etc/audit/rules.d/` directory ending in `.rules`

Example: `vi /etc/audit/rules.d/50-logins.rules`

Add the following lines:

```
-w /var/log/lastlog -p wa -k logins
-w /var/run/faillock/ -p wa -k logins
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.5 <u>Collect Detailed Audit Logs</u> Configure detailed audit logging for enterprise assets containing sensitive data. Include event source, date, username, timestamp, source addresses, destination addresses, and other useful elements that could assist in a forensic investigation.		●	●
v7	4.9 <u>Log and Alert on Unsuccessful Administrative Account Login</u> Configure systems to issue a log entry and alert on unsuccessful logins to an administrative account.		●	●
v7	16.11 <u>Lock Workstation Sessions After Inactivity</u> Automatically lock workstation sessions after a standard period of inactivity.	●	●	●
v7	16.13 <u>Alert on Account Login Behavior Deviation</u> Alert when users deviate from normal login behavior, such as time-of-day, workstation location and duration.			●

4.1.8 Ensure session initiation information is collected (Automated)

Profile Applicability:

- Level 2 - Server
- Level 2 - Workstation

Description:

Monitor session initiation events. The parameters in this section track changes to the files associated with session events. The file `/var/run/utmp` tracks all currently logged in users. All audit records will be tagged with the identifier "session." The `/var/log/wtmp` file tracks logins, logouts, shutdown, and reboot events. The file `/var/log/btmp` keeps track of failed login attempts and can be read by entering the command `/usr/bin/last -f /var/log/btmp`. All audit records will be tagged with the identifier "logins."

Notes:

- *The `last` command can be used to read `/var/log/wtmp` (`last` with no parameters) and `/var/run/utmp` (`last -f /var/run/utmp`)*
- *Reloading the `auditd` config to set active settings requires the `auditd` service to be restarted, and may require a system reboot.*

Rationale:

Monitoring these files for changes could alert a system administrator to logins occurring at unusual hours, which could indicate intruder activity (i.e. a user logging in at a time when they do not normally log in).

Audit:

Run the following command to check the auditd `.rules` files:

```
# grep -E '(session|logins)' /etc/audit/rules.d/*.rules
```

Verify output includes:

```
-w /var/run/utmp -p wa -k session  
-w /var/log/wtmp -p wa -k logins  
-w /var/log/btmp -p wa -k logins
```

Run the following command to check loaded auditd rules:

```
# auditctl -l | grep -E '(session|logins)'
```

Verify output includes:

```
-w /var/run/utmp -p wa -k session  
-w /var/log/wtmp -p wa -k logins  
-w /var/log/btmp -p wa -k logins
```

Remediation:

Edit or create a file in the `/etc/audit/rules.d/` directory ending in `.rules`

Example: `vi /etc/audit/rules.d/50-session.rules`

Add the following lines:

```
-w /var/run/utmp -p wa -k session  
-w /var/log/wtmp -p wa -k logins  
-w /var/log/btmp -p wa -k logins
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.5 <u>Collect Detailed Audit Logs</u> Configure detailed audit logging for enterprise assets containing sensitive data. Include event source, date, username, timestamp, source addresses, destination addresses, and other useful elements that could assist in a forensic investigation.		●	●
v7	4.9 <u>Log and Alert on Unsuccessful Administrative Account Login</u> Configure systems to issue a log entry and alert on unsuccessful logins to an administrative account.		●	●
v7	16.11 <u>Lock Workstation Sessions After Inactivity</u> Automatically lock workstation sessions after a standard period of inactivity.	●	●	●
v7	16.13 <u>Alert on Account Login Behavior Deviation</u> Alert when users deviate from normal login behavior, such as time-of-day, workstation location and duration.			●

4.1.9 Ensure discretionary access control permission modification events are collected (Automated)

Profile Applicability:

- Level 2 - Server
- Level 2 - Workstation

Description:

Monitor changes to file permissions, attributes, ownership and group. The parameters in this section track changes for system calls that affect file permissions and attributes. The `chmod`, `fchmod` and `fchmodat` system calls affect the permissions associated with a file. The `chown`, `fchown`, `fchownat` and `lchown` system calls affect owner and group attributes on a file. The `setxattr`, `lsetxattr`, `fsetxattr` (set extended file attributes) and `removexattr`, `lremovexattr`, `fremovexattr` (remove extended file attributes) control extended file attributes. In all cases, an audit record will only be written for non-system user ids (audit ≥ 1000) and will ignore Daemon events (audit = 4294967295). All audit records will be tagged with the identifier "perm_mod."

Note: Systems may have been customized to change the default UID_MIN. To confirm the UID_MIN for your system, run the following command:

```
# awk '/^\s*UID_MIN/{print $2}' /etc/login.defs
```

If your systems' UID_MIN is not 1000, replace `audit $\geq$ 1000` with `audit $\geq$ <UID_MIN for your system>` in the Audit and Remediation procedures.

Reloading the auditd config to set active settings may require a system reboot.

Rationale:

Monitoring for changes in file attributes could alert a system administrator to activity that could indicate intruder activity or policy violation.

Audit:

On a 32 bit system run the following commands:

Run the following command to check the auditd .rules files:

```
# grep perm_mod /etc/audit/rules.d/*.rules
```

Verify output matches:


```
-a always,exit -F arch=b32 -S chmod -S fchmod -S fchmodat -F auid>=1000 -F auid!=4294967295 -k perm_mod
-a always,exit -F arch=b32 -S chown -S fchown -S fchownat -S lchown -F auid>=1000 -F auid!=4294967295 -k perm_mod
-a always,exit -F arch=b32 -S setxattr -S lsetxattr -S fsetxattr -S removexattr -S lremovexattr -S fremovexattr -F auid>=1000 -F auid!=4294967295 -k perm_mod
```

Run the following command to check loaded auditd rules:

```
# auditctl -l | grep perm_mod
```

Verify output matches:

```
-a always,exit -F arch=b32 -S chmod,fchmod,fchmodat -F auid>=1000 -F auid!=-1 -F key=perm_mod
-a always,exit -F arch=b32 -S lchown,fchown,chown,fchownat -F auid>=1000 -F auid!=-1 -F key=perm_mod
-a always,exit -F arch=b32 -S setxattr,lsetxattr,fsetxattr,removexattr,lremovexattr,fremovexattr -F auid>=1000 -F auid!=-1 -F key=perm_mod
```

On a 64 bit system run the following commands:

Run the following command to check the auditd .rules files:

```
# grep perm_mod /etc/audit/rules.d/*.rules
```

Verify output matches:

```
-a always,exit -F arch=b64 -S chmod -S fchmod -S fchmodat -F auid>=1000 -F auid!=4294967295 -k perm_mod
-a always,exit -F arch=b32 -S chmod -S fchmod -S fchmodat -F auid>=1000 -F auid!=4294967295 -k perm_mod
-a always,exit -F arch=b64 -S chown -S fchown -S fchownat -S lchown -F auid>=1000 -F auid!=4294967295 -k perm_mod
-a always,exit -F arch=b32 -S chown -S fchown -S fchownat -S lchown -F auid>=1000 -F auid!=4294967295 -k perm_mod
-a always,exit -F arch=b64 -S setxattr -S lsetxattr -S fsetxattr -S removexattr -S lremovexattr -S fremovexattr -F auid>=1000 -F auid!=4294967295 -k perm_mod
-a always,exit -F arch=b32 -S setxattr -S lsetxattr -S fsetxattr -S removexattr -S lremovexattr -S fremovexattr -F auid>=1000 -F auid!=4294967295 -k perm_mod
```

Run the following command to check loaded auditd rules:

```
# auditctl -l | grep auditctl -l | grep perm_mod
```

Verify output matches:

```
-a always,exit -F arch=b64 -S chmod,fchmod,fchmodat -F auid>=1000 -F auid!=-1
-F key=perm_mod
-a always,exit -F arch=b32 -S chmod,fchmod,fchmodat -F auid>=1000 -F auid!=-1
-F key=perm_mod
-a always,exit -F arch=b64 -S chown,fchown,lchown,fchownat -F auid>=1000 -F
auid!=-1 -F key=perm_mod
-a always,exit -F arch=b32 -S lchown,fchown,chown,fchownat -F auid>=1000 -F
auid!=-1 -F key=perm_mod
-a always,exit -F arch=b64 -S
setxattr,lsetxattr,fsetxattr,removexattr,lremovexattr,fremovexattr -F
auid>=1000 -F auid!=-1 -F key=perm_mod
-a always,exit -F arch=b32 -S
setxattr,lsetxattr,fsetxattr,removexattr,lremovexattr,fremovexattr -F
auid>=1000 -F auid!=-1 -F key=perm_mod
```

Remediation:

For 32 bit systems edit or create a file in the `/etc/audit/rules.d/` directory ending in `.rules`

Example: `vi /etc/audit/rules.d/50-perm_mod.rules`

Add the following lines:

```
-a always,exit -F arch=b32 -S chmod -S fchmod -S fchmodat -F auid>=1000 -F
auid!=4294967295 -k perm_mod
-a always,exit -F arch=b32 -S chown -S fchown -S fchownat -S lchown -F
auid>=1000 -F auid!=4294967295 -k perm_mod
-a always,exit -F arch=b32 -S setxattr -S lsetxattr -S fsetxattr -S
removexattr -S lremovexattr -S fremovexattr -F auid>=1000 -F auid!=4294967295
-k perm_mod
```

For 64 bit systems Edit or create a file in the `/etc/audit/rules.d/` directory ending in `.rules`

Example: `vi /etc/audit/rules.d/50-perm_mod.rules`

Add the following lines:

```
-a always,exit -F arch=b64 -S chmod -S fchmod -S fchmodat -F auid>=1000 -F
auid!=4294967295 -k perm_mod
-a always,exit -F arch=b32 -S chmod -S fchmod -S fchmodat -F auid>=1000 -F
auid!=4294967295 -k perm_mod
-a always,exit -F arch=b64 -S chown -S fchown -S fchownat -S lchown -F
auid>=1000 -F auid!=4294967295 -k perm_mod
-a always,exit -F arch=b32 -S chown -S fchown -S fchownat -S lchown -F
auid>=1000 -F auid!=4294967295 -k perm_mod
-a always,exit -F arch=b64 -S setxattr -S lsetxattr -S fsetxattr -S
removexattr -S lremovexattr -S fremovexattr -F auid>=1000 -F auid!=4294967295
-k perm_mod
-a always,exit -F arch=b32 -S setxattr -S lsetxattr -S fsetxattr -S
removexattr -S lremovexattr -S fremovexattr -F auid>=1000 -F auid!=4294967295
-k perm_mod
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.5 <u>Collect Detailed Audit Logs</u> Configure detailed audit logging for enterprise assets containing sensitive data. Include event source, date, username, timestamp, source addresses, destination addresses, and other useful elements that could assist in a forensic investigation.			
v7	5.5 <u>Implement Automated Configuration Monitoring Systems</u> Utilize a Security Content Automation Protocol (SCAP) compliant configuration monitoring system to verify all security configuration elements, catalog approved exceptions, and alert when unauthorized changes occur.			

4.1.10 Ensure unsuccessful unauthorized file access attempts are collected (Automated)

Profile Applicability:

- Level 2 - Server
- Level 2 - Workstation

Description:

Monitor for unsuccessful attempts to access files. The parameters below are associated with system calls that control creation (`creat`), opening (`open` , `openat`) and truncation (`truncate` , `ftruncate`) of files. An audit log record will only be written if the user is a non-privileged user (`auid>=1000`), is not a Daemon event (`auid=4294967295`) and if the system call returned `EACCES` (permission denied to the file) or `EPERM` (some other permanent error associated with the specific system call). All audit records will be tagged with the identifier "access."

Note: Systems may have been customized to change the default `UID_MIN`. To confirm the `UID_MIN` for your system, run the following command:

```
# awk '/^\s*UID_MIN/{print $2}' /etc/login.defs
```

If your systems' `UID_MIN` is not `1000`, replace `audit>=1000` with `audit>=<UID_MIN for your system>` in the Audit and Remediation procedures.

Reloading the `auditd` config to set active settings may require a system reboot.

Rationale:

Failed attempts to open, create or truncate files could be an indication that an individual or process is trying to gain unauthorized access to the system.

Audit:

On a 32 bit system run the following commands:

Run the following command to check the `auditd .rules` files:

```
# grep access /etc/audit/rules.d/*.rules
```

Verify output matches:

```
-a always,exit -F arch=b32 -S creat -S open -S openat -S truncate -S
ftruncate -F exit=-EACCES -F auid>=1000 -F auid!=4294967295 -k access
-a always,exit -F arch=b32 -S creat -S open -S openat -S truncate -S
ftruncate -F exit=-EPERM -F auid>=1000 -F auid!=4294967295 -k access
```

Run the following command to check loaded auditd rules:

```
# auditctl -l | grep access
```

Verify output matches:

```
-a always,exit -F arch=b32 -S creat -S open -S openat -S truncate -S
ftruncate -F exit=-EACCES -F auid>=1000 -F auid!=-1 -k access
-a always,exit -F arch=b32 -S creat -S open -S openat -S truncate -S
ftruncate -F exit=-EPERM -F auid>=1000 -F auid!=-1 -k access
```

On a 64 bit system run the following commands:

Run the following command to check the auditd .rules files:

```
# grep access /etc/audit/rules.d/*.rules
```

Verify output matches:

```
-a always,exit -F arch=b64 -S creat -S open -S openat -S truncate -S
ftruncate -F exit=-EACCES -F auid>=1000 -F auid!=4294967295 -k access
-a always,exit -F arch=b32 -S creat -S open -S openat -S truncate -S
ftruncate -F exit=-EACCES -F auid>=1000 -F auid!=4294967295 -k access
-a always,exit -F arch=b64 -S creat -S open -S openat -S truncate -S
ftruncate -F exit=-EPERM -F auid>=1000 -F auid!=4294967295 -k access
-a always,exit -F arch=b32 -S creat -S open -S openat -S truncate -S
ftruncate -F exit=-EPERM -F auid>=1000 -F auid!=4294967295 -k access
```

Run the following command to check loaded auditd rules:

```
# auditctl -l | grep access
```

Verify output matches:

```
-a always,exit -F arch=b64 -S creat -S open -S openat -S truncate -S
ftruncate -F exit=-EACCES -F auid>=1000 -F auid!=-1 -k access
-a always,exit -F arch=b32 -S creat -S open -S openat -S truncate -S
ftruncate -F exit=-EACCES -F auid>=1000 -F auid!=-1 -k access
-a always,exit -F arch=b64 -S creat -S open -S openat -S truncate -S
ftruncate -F exit=-EPERM -F auid>=1000 -F auid!=-1 -k access
-a always,exit -F arch=b32 -S creat -S open -S openat -S truncate -S
ftruncate -F exit=-EPERM -F auid>=1000 -F auid!=-1 -k access
```

Remediation:

For 32 bit systems Edit or create a file in the `/etc/audit/rules.d/` directory ending in `.rules`

Example: `vi /etc/audit/rules.d/50-access.rules`

Add the following lines:

```
-a always,exit -F arch=b32 -S creat -S open -S openat -S truncate -S
ftruncate -F exit=-EACCES -F auid>=1000 -F auid!=4294967295 -k access
-a always,exit -F arch=b32 -S creat -S open -S openat -S truncate -S
ftruncate -F exit=-EPERM -F auid>=1000 -F auid!=4294967295 -k access
```

For 64 bit systems Edit or create a file in the `/etc/audit/rules.d/` directory ending in `.rules`

Example: `vi /etc/audit/rules.d/50-access.rules`

Add the following lines:

```
-a always,exit -F arch=b64 -S creat -S open -S openat -S truncate -S
ftruncate -F exit=-EACCES -F auid>=1000 -F auid!=4294967295 -k access
-a always,exit -F arch=b32 -S creat -S open -S openat -S truncate -S
ftruncate -F exit=-EACCES -F auid>=1000 -F auid!=4294967295 -k access
-a always,exit -F arch=b64 -S creat -S open -S openat -S truncate -S
ftruncate -F exit=-EPERM -F auid>=1000 -F auid!=4294967295 -k access
-a always,exit -F arch=b32 -S creat -S open -S openat -S truncate -S
ftruncate -F exit=-EPERM -F auid>=1000 -F auid!=4294967295 -k access
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.5 <u>Collect Detailed Audit Logs</u> Configure detailed audit logging for enterprise assets containing sensitive data. Include event source, date, username, timestamp, source addresses, destination addresses, and other useful elements that could assist in a forensic investigation.			
v7	14.9 <u>Enforce Detail Logging for Access or Changes to Sensitive Data</u> Enforce detailed audit logging for access to sensitive data or changes to sensitive data (utilizing tools such as File Integrity Monitoring or Security Information and Event Monitoring).			

4.1.11 Ensure use of privileged commands is collected (Automated)

Profile Applicability:

- Level 2 - Server
- Level 2 - Workstation

Description:

Monitor privileged programs (those that have the setuid and/or setgid bit set on execution) to determine if unprivileged users are running these commands.

Note: Systems may have been customized to change the default UID_MIN. To confirm the UID_MIN for your system, run the following command:

```
# awk '/^\s*UID_MIN/{print $2}' /etc/login.defs
```

If your systems' UID_MIN is not 1000, replace `audit>=1000` with `audit>=<UID_MIN for your system>` in the Audit and Remediation procedures.

Reloading the auditd config to set active settings may require a system reboot.

Rationale:

Execution of privileged commands by non-privileged users could be an indication of someone trying to gain unauthorized access to the system.

Audit:

Run the following command replacing `<partition>` with a list of partitions where programs can be executed from on your system:

```
# find <partition> -xdev \( -perm -4000 -o -perm -2000 \) -type f | awk  
'{print "-a always,exit -F path=" $1 " -F perm=x -F auid>=" $(awk  
'/\s*UID_MIN/{print $2}' /etc/login.defs) "' -F auid!=4294967295 -k  
privileged" }'
```

Verify all resulting lines are a `.rules` file in `/etc/audit/rules.d/` and the output of `auditctl -l`.

Note: The `.rules` file output will be `auid!=-1` not `auid!=4294967295`

Remediation:

To remediate this issue, the system administrator will have to execute a find command to locate all the privileged programs and then add an audit line for each one of them.

The audit parameters associated with this are as follows:

- `-F path=" $1 "` - will populate each file name found through the find command and processed by awk.
- `-F perm=x` - will write an audit record if the file is executed.
- `-F audit>=1000` - will write a record if the user executing the command is not a privileged user.
- `-F auid!= 4294967295` - will ignore Daemon events

All audit records should be tagged with the identifier "privileged".

Run the following command replacing with a list of partitions where programs can be executed from on your system:

```
# find <partition> -xdev \( -perm -4000 -o -perm -2000 \) -type f | awk '{print "-a always,exit -F path=" $1 " -F perm=x -F auid>=" $(awk '/^\s*UID_MIN/{print $2}' /etc/login.defs) "' -F auid!=4294967295 -k privileged" }'
```

Edit or create a file in the `/etc/audit/rules.d/` directory ending in `.rules` and add all resulting lines to the file.

Example:

```
# find / -xdev \( -perm -4000 -o -perm -2000 \) -type f | awk '{print "-a always,exit -F path=" $1 " -F perm=x -F auid>=" $(awk '/^\s*UID_MIN/{print $2}' /etc/login.defs) "' -F auid!=4294967295 -k privileged" }' >> /etc/audit/rules.d/50-privileged.rules
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.5 <u>Collect Detailed Audit Logs</u> Configure detailed audit logging for enterprise assets containing sensitive data. Include event source, date, username, timestamp, source addresses, destination addresses, and other useful elements that could assist in a forensic investigation.		●	●
v7	6.2 <u>Activate audit logging</u> Ensure that local logging has been enabled on all systems and networking devices.	●	●	●

4.1.12 Ensure successful file system mounts are collected (Automated)

Profile Applicability:

- Level 2 - Server
- Level 2 - Workstation

Description:

Monitor the use of the `mount` system call. The `mount` (and `umount`) system call controls the mounting and unmounting of file systems. The parameters below configure the system to create an audit record when the mount system call is used by a non-privileged user

Note: Systems may have been customized to change the default `UID_MIN`. To confirm the `UID_MIN` for your system, run the following command:

```
# awk '/^s*UID_MIN/{print $2}' /etc/login.defs
```

If your systems' `UID_MIN` is not `1000`, replace `audit>=1000` with `audit>=<UID_MIN for your system>` in the Audit and Remediation procedures.

Reloading the `auditd` config to set active settings may require a system reboot.

Rationale:

It is highly unusual for a non privileged user to `mount` file systems to the system. While tracking `mount` commands gives the system administrator evidence that external media may have been mounted (based on a review of the source of the mount and confirming it's an external media type), it does not conclusively indicate that data was exported to the media. System administrators who wish to determine if data were exported, would also have to track successful `open`, `creat` and `truncate` system calls requiring write access to a file under the mount point of the external media file system. This could give a fair indication that a write occurred. The only way to truly prove it, would be to track successful writes to the external media. Tracking write system calls could quickly fill up the audit log and is not recommended. Recommendations on configuration options to track data export to media is beyond the scope of this document.

Audit:

On a 32 bit system run the following commands:

Run the following command to check the auditd .rules files:

```
# grep mounts /etc/audit/rules.d/*.rules
```

Verify output matches:

```
-a always,exit -F arch=b32 -S mount -F auid>=1000 -F auid!=4294967295 -k  
mounts
```

Run the following command to check loaded auditd rules:

```
# auditctl -l | grep mounts
```

Verify output matches:

```
-a always,exit -F arch=b32 -S mount -F auid>=1000 -F auid!=-1 -k mounts
```

On a 64 bit system run the following commands:

Run the following command to check the auditd .rules files:

```
# grep mounts /etc/audit/rules.d/*.rules
```

Verify output matches:

```
-a always,exit -F arch=b64 -S mount -F auid>=1000 -F auid!=4294967295 -k  
mounts  
-a always,exit -F arch=b32 -S mount -F auid>=1000 -F auid!=4294967295 -k  
mounts
```

Run the following command to check loaded auditd rules:

```
# auditctl -l | grep mounts
```

Verify output matches:

```
-a always,exit -F arch=b64 -S mount -F auid>=1000 -F auid!=-1 -k mounts  
-a always,exit -F arch=b32 -S mount -F auid>=1000 -F auid!=-1 -k mounts
```

Remediation:

For 32 bit systems edit or create a file in the `/etc/audit/rules.d/` directory ending in `.rules`

Example: `vi /etc/audit/rules.d/50-mounts.rules`

Add the following lines:

```
-a always,exit -F arch=b32 -S mount -F auid>=1000 -F auid!=4294967295 -k mounts
```

For 64 bit systems Edit or create a file in the `/etc/audit/rules.d/` directory ending in `.rules`

Example: `vi /etc/audit/rules.d/50-mounts.rules`

Add the following lines:

```
-a always,exit -F arch=b64 -S mount -F auid>=1000 -F auid!=4294967295 -k mounts
-a always,exit -F arch=b32 -S mount -F auid>=1000 -F auid!=4294967295 -k mounts
```

Additional Information:

This tracks successful and unsuccessful mount commands.

File system mounts do not have to come from external media and this action still does not verify write (e.g. CD ROMS).

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.5 <u>Collect Detailed Audit Logs</u> Configure detailed audit logging for enterprise assets containing sensitive data. Include event source, date, username, timestamp, source addresses, destination addresses, and other useful elements that could assist in a forensic investigation.			
v7	6.2 <u>Activate audit logging</u> Ensure that local logging has been enabled on all systems and networking devices.			

4.1.13 Ensure file deletion events by users are collected (Automated)

Profile Applicability:

- Level 2 - Server
- Level 2 - Workstation

Description:

Monitor the use of system calls associated with the deletion or renaming of files and file attributes. This configuration statement sets up monitoring for following system calls and tags them with the identifier "delete":

- `unlink` - remove a file
- `unlinkat` - remove a file attribute
- `rename` - rename a file
- `renameat` - rename a file attribute

Note: Systems may have been customized to change the default `UID_MIN`. To confirm the `UID_MIN` for your system, run the following command:

```
# awk '/^\s*UID_MIN/{print $2}' /etc/login.defs
```

If your systems' `UID_MIN` is not `1000`, replace `audit>=1000` with `audit>=<UID_MIN for your system>` in the Audit and Remediation procedures.

Reloading the `auditd` config to set active settings may require a system reboot.

Rationale:

Monitoring these calls from non-privileged users could provide a system administrator with evidence that inappropriate removal of files and file attributes associated with protected files is occurring. While this audit option will look at all events, system administrators will want to look for specific privileged files that are being deleted or altered.

Audit:

On a 32 bit system run the following commands:

Run the following command to check the auditd .rules files:

```
# grep delete /etc/audit/rules.d/*.rules
```

Verify output matches:

```
-a always,exit -F arch=b32 -S unlink -S unlinkat -S rename -S renameat -F  
auid>=1000 -F auid!=4294967295 -k delete
```

Run the following command to check loaded auditd rules:

```
# auditctl -l | grep delete
```

Verify output matches:

```
-a always,exit -F arch=b32 -S unlink -S unlinkat -S rename -S renameat -F  
auid>=1000 -F auid!=4294967295 -k delete
```

On a 64 bit system run the following commands:

Run the following command to check the auditd .rules files:

```
# grep delete /etc/audit/rules.d/*.rules
```

Verify output matches:

```
-a always,exit -F arch=b64 -S unlink -S unlinkat -S rename -S renameat -F  
auid>=1000 -F auid!=4294967295 -k delete  
-a always,exit -F arch=b32 -S unlink -S unlinkat -S rename -S renameat -F  
auid>=1000 -F auid!=4294967295 -k delete
```

Run the following command to check loaded auditd rules:

```
# auditctl -l | grep delete
```

Verify output matches:

```
-a always,exit -F arch=b64 -S unlink -S unlinkat -S rename -S renameat -F  
auid>=1000 -F auid!=-1 -k delete  
-a always,exit -F arch=b32 -S unlink -S unlinkat -S rename -S renameat -F  
auid>=1000 -F auid!=-1 -k delete
```

Remediation:

For 32 bit systems edit or create a file in the `/etc/audit/rules.d/` directory ending in `.rules`

Example: `vi /etc/audit/rules.d/50-deletion.rules`

Add the following lines:

```
-a always,exit -F arch=b32 -S unlink -S unlinkat -S rename -S renameat -F
auid>=1000 -F auid!=4294967295 -k delete
```

For 64 bit systems edit or create a file in the `/etc/audit/rules.d/` directory ending in `.rules`

Example: `vi /etc/audit/rules.d/50-deletion.rules`

Add the following lines:

```
-a always,exit -F arch=b64 -S unlink -S unlinkat -S rename -S renameat -F
auid>=1000 -F auid!=4294967295 -k delete
-a always,exit -F arch=b32 -S unlink -S unlinkat -S rename -S renameat -F
auid>=1000 -F auid!=4294967295 -k delete
```

Additional Information:

At a minimum, configure the audit system to collect file deletion events for all users and root.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.5 <u>Collect Detailed Audit Logs</u> Configure detailed audit logging for enterprise assets containing sensitive data. Include event source, date, username, timestamp, source addresses, destination addresses, and other useful elements that could assist in a forensic investigation.			
v7	6.2 <u>Activate audit logging</u> Ensure that local logging has been enabled on all systems and networking devices.			
v7	13 <u>Data Protection</u> Data Protection			

4.1.14 Ensure changes to system administration scope (sudoers) is collected (Automated)

Profile Applicability:

- Level 2 - Server
- Level 2 - Workstation

Description:

Monitor scope changes for system administrators. If the system has been properly configured to force system administrators to log in as themselves first and then use the `sudo` command to execute privileged commands, it is possible to monitor changes in scope. The file `/etc/sudoers` or a file in the `/etc/sudoers.d` directory will be written to when the file or its attributes have changed.

Note: Reloading the auditd config to set active settings may require a system reboot.

Rationale:

Changes in the `/etc/sudoers` file, or a file in the `/etc/sudoers.d/` directory can indicate that an unauthorized change has been made to scope of system administrator activity.

Audit:

Run the following command to check the auditd `.rules` files:

```
# grep scope /etc/audit/rules.d/*.rules
```

Verify output of matches:

```
-w /etc/sudoers -p wa -k scope
-w /etc/sudoers.d/ -p wa -k scope
```

Run the following command to check loaded auditd rules:

```
# auditctl -l | grep scope
```

Verify output matches:

```
-w /etc/sudoers -p wa -k scope
-w /etc/sudoers.d -p wa -k scope
```

Remediation:

Edit or create a file in the `/etc/audit/rules.d/` directory ending in `.rules`

_Example: `vi /etc/audit/rules.d/50-scope.rules`

Add the following lines:

```
-w /etc/sudoers -p wa -k scope  
-w /etc/sudoers.d/ -p wa -k scope
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.5 <u>Collect Detailed Audit Logs</u> Configure detailed audit logging for enterprise assets containing sensitive data. Include event source, date, username, timestamp, source addresses, destination addresses, and other useful elements that could assist in a forensic investigation.			
v7	4.8 <u>Log and Alert on Changes to Administrative Group Membership</u> Configure systems to issue a log entry and alert when an account is added to or removed from any group assigned administrative privileges.			

4.1.15 Ensure system administrator command executions (sudo) are collected (Automated)

Profile Applicability:

- Level 2 - Server
- Level 2 - Workstation

Description:

`sudo` provides users with temporary elevated privileges to perform operations. Monitor the administrator with temporary elevated privileges and the operation(s) they performed.

Rationale:

creating an audit log of administrators with temporary elevated privileges and the operation(s) they performed is essential to reporting. Administrators will want to correlate the events written to the audit trail with the records written to `sudo logfile` to verify if unauthorized commands have been executed.

Note: Systems may have been customized to change the default `UID_MIN`. To confirm the `UID_MIN` for your system, run the following command:

```
# awk '/^\s*UID_MIN/{print $2}' /etc/login.defs
```

If your systems' `UID_MIN` is not `1000`, replace `audit>=1000` with `audit>=<UID_MIN for your system>` in the Audit and Remediation procedures.

Reloading the `auditd` config to set active settings may require a system reboot.

Audit:

On a 32 bit system run the following commands:

Run the following command to verify the rules are contained in a `.rules` file in the `/etc/audit/rules.d/` directory:

```
# grep actions /etc/audit/rules.d/*.rules
```

Verify the output includes:

```
/etc/audit/rules.d/cis.rules:-a exit,always -F arch=b32 -C euid!=uid -F euid=0 -F auid>=1000 -F auid!=4294967295 -S execve -k actions
```

Run the following command to verify that rules are in the running `auditd` config:

```
# auditctl -l | grep actions
```

Verify the output includes:

```
-a always,exit -F arch=b32 -S execve -C uid!=euid -F euid=0 -F auid>=1000 -F auid!=-1 -F key=actions
```

On a 64 bit system run the following commands:

Run the following command to verify the rules are contained in a `.rules` file in the `/etc/audit/rules.d/` directory:

```
# grep actions /etc/audit/rules.d/*.rules
```

Verify the output includes:

```
-a exit,always -F arch=b64 -C euid!=uid -F euid=0 -F auid>=1000 -F auid!=4294967295 -S execve -k actions  
-a exit,always -F arch=b32 -C euid!=uid -F euid=0 -F auid>=1000 -F auid!=4294967295 -S execve -k actions
```

Run the following command to verify that rules are in the running `auditd` config:

```
# auditctl -l | grep actions
```

Verify the output includes:

```
-a always,exit -F arch=b64 -S execve -C uid!=euid -F euid=0 -F auid>=1000 -F auid!=-1 -F key=actions  
-a always,exit -F arch=b32 -S execve -C uid!=euid -F euid=0 -F auid>=1000 -F auid!=-1 -F key=actions
```

Remediation:

For 32 bit systems Edit or create a file in the `/etc/audit/rules.d/` directory ending in `.rules`:

Example: `vi /etc/audit/rules.d/50-actions.rules`

Add the following line:

```
-a exit,always -F arch=b32 -C euid!=uid -F euid=0 -F auid>=1000 -F
auid!=4294967295 -S execve -k actions
```

For 64 bit systems Edit or create a file in the `/etc/audit/rules.d/` directory ending in `.rules`:

Example: `vi /etc/audit/rules.d/50-actions.rules`

Add the following lines:

```
-a always,exit -F arch=b64 -C euid!=uid -F euid=0 -F auid>=1000 -F
auid!=4294967295 -S execve -k actions
-a always,exit -F arch=b32 -C euid!=uid -F euid=0 -F auid>=1000 -F
auid!=4294967295 -S execve -k actions
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.5 <u>Collect Detailed Audit Logs</u> Configure detailed audit logging for enterprise assets containing sensitive data. Include event source, date, username, timestamp, source addresses, destination addresses, and other useful elements that could assist in a forensic investigation.			
v7	4.9 <u>Log and Alert on Unsuccessful Administrative Account Login</u> Configure systems to issue a log entry and alert on unsuccessful logins to an administrative account.			

4.1.16 Ensure kernel module loading and unloading is collected (Automated)

Profile Applicability:

- Level 2 - Server
- Level 2 - Workstation

Description:

Monitor the loading and unloading of kernel modules. The programs `insmod` (install a kernel module), `rmmod` (remove a kernel module), and `modprobe` (a more sophisticated program to load and unload modules, as well as some other features) control loading and unloading of modules. The `init_module` (load a module) and `delete_module` (delete a module) system calls control loading and unloading of modules. Any execution of the loading and unloading module programs and system calls will trigger an audit record with an identifier of "modules".

Note: Reloading the auditd config to set active settings requires the `auditd` service to be restarted, and may require a system reboot.

Rationale:

Monitoring the use of `insmod`, `rmmod` and `modprobe` could provide system administrators with evidence that an unauthorized user loaded or unloaded a kernel module, possibly compromising the security of the system. Monitoring of the `init_module` and `delete_module` system calls would reflect an unauthorized user attempting to use a different program to load and unload modules.

Audit:

On a 32 bit system run the following commands:

Run the following command to check the auditd .rules files:

```
# grep modules /etc/audit/rules.d/*.rules
```

Verify output matches:

```
-w /sbin/insmod -p x -k modules
-w /sbin/rmmod -p x -k modules
-w /sbin/modprobe -p x -k modules
-a always,exit -F arch=b32 -S init_module -S delete_module -k modules
```

Run the following command to check loaded auditd rules:

```
# auditctl -l | grep modules
```

Verify output matches:

```
-w /sbin/insmod -p x -k modules
-w /sbin/rmmod -p x -k modules
-w /sbin/modprobe -p x -k modules
-a always,exit -F arch=b32 -S init_module,delete_module -F key=modules
```

On a 64 bit system run the following commands:

Run the following command to check the auditd .rules files:

```
# grep modules /etc/audit/rules.d/*.rules
```

Verify output matches:

```
-w /sbin/insmod -p x -k modules
-w /sbin/rmmod -p x -k modules
-w /sbin/modprobe -p x -k modules
-a always,exit -F arch=b64 -S init_module -S delete_module -k modules
```

Run the following command to check loaded auditd rules:

```
# auditctl -l | grep modules
```

Verify output matches:

```
-w /sbin/insmod -p x -k modules
-w /sbin/rmmod -p x -k modules
-w /sbin/modprobe -p x -k modules
-a always,exit -F arch=b64 -S init_module,delete_module -F key=modules
```

Remediation:

For 32 bit systems Edit or create a file in the `/etc/audit/rules.d/` directory ending in `.rules`

Example: `vi /etc/audit/rules.d/50-modules.rules`

Add the following lines:

```
-w /sbin/insmod -p x -k modules
-w /sbin/rmmod -p x -k modules
-w /sbin/modprobe -p x -k modules
-a always,exit -F arch=b32 -S init_module -S delete_module -k modules
```

For 64 bit systems Edit or create a file in the `/etc/audit/rules.d/` directory ending in `.rules`

Example: `vi /etc/audit/rules.d/50-modules.rules`

Add the following lines:

```
-w /sbin/insmod -p x -k modules
-w /sbin/rmmod -p x -k modules
-w /sbin/modprobe -p x -k modules
-a always,exit -F arch=b64 -S init_module -S delete_module -k modules
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.5 <u>Collect Detailed Audit Logs</u> Configure detailed audit logging for enterprise assets containing sensitive data. Include event source, date, username, timestamp, source addresses, destination addresses, and other useful elements that could assist in a forensic investigation.			
v7	5.1 <u>Establish Secure Configurations</u> Maintain documented, standard security configuration standards for all authorized operating systems and software.			

4.1.17 Ensure the audit configuration is immutable (Automated)

Profile Applicability:

- Level 2 - Server
- Level 2 - Workstation

Description:

Set system audit so that audit rules cannot be modified with `auditctl`. Setting the flag "-e 2" forces audit to be put in immutable mode. Audit changes can only be made on system reboot.

Note: This setting will require the system to be rebooted to update the active `auditd` configuration settings.

Rationale:

In immutable mode, unauthorized users cannot execute changes to the audit system to potentially hide malicious activity and then put the audit rules back. Users would most likely notice a system reboot and that could alert administrators of an attempt to make unauthorized audit changes.

Audit:

Run the following command and verify output matches:

```
# grep "^s*[^#]" /etc/audit/rules.d/*.rules | tail -1  
-e 2
```

Remediation:

Edit or create the file `/etc/audit/rules.d/99-finalize.rules` and add the following line at the end of the file:

```
-e 2
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 <u>Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v7	6.2 <u>Activate audit logging</u> Ensure that local logging has been enabled on all systems and networking devices.			
v7	6.3 <u>Enable Detailed Logging</u> Enable system logging to include detailed information such as an event source, date, user, timestamp, source addresses, destination addresses, and other useful elements.			

4.2 Configure Logging

Logging services should be configured to prevent information leaks and to aggregate logs on a remote server so that they can be reviewed in the event of a system compromise and ease log analysis.

4.2.1 Configure rsyslog

The `rsyslog` software is recommended as a replacement for the `syslogd` daemon and provides improvements over `syslogd`, such as connection-oriented (i.e. TCP) transmission of logs, the option to log to database formats, and the encryption of log data en route to a central logging server.

Notes:

- *This section only applies if `rsyslog` is installed on the system*
- *If another logging software like `syslog-ng` is in use on the system:*
 - *This section may be skipped*
 - *Ensure that logging software is configured in accordance with local site policy*

4.2.1.1 Ensure rsyslog is installed (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `rsyslog` software is a recommended replacement to the original `syslogd` daemon.

`rsyslog` provides improvements over `syslogd`, including:

- connection-oriented (i.e. TCP) transmission of logs
- The option to log to database formats
- Encryption of log data en route to a central logging server

Rationale:

The security enhancements of `rsyslog` such as connection-oriented (i.e. TCP) transmission of logs, the option to log to database formats, and the encryption of log data en route to a central logging server) justify installing and configuring the package.

Audit:

Run the following command to Verify `rsyslog` is installed:

```
# rpm -q rsyslog
rsyslog-<version>
```

Remediation:

Run the following command to install `rsyslog`:

```
# yum install rsyslog
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.2 <u>Collect Audit Logs</u> Collect audit logs. Ensure that logging, per the enterprise's audit log management process, has been enabled across enterprise assets.			
v7	6.2 <u>Activate audit logging</u> Ensure that local logging has been enabled on all systems and networking devices.			
v7	6.3 <u>Enable Detailed Logging</u> Enable system logging to include detailed information such as an event source, date, user, timestamp, source addresses, destination addresses, and other useful elements.			

4.2.1.2 Ensure rsyslog Service is enabled and running (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

`rsyslog` needs to be enabled and running to perform logging

Rationale:

If the `rsyslog` service is not activated the system may default to the `syslogd` service or lack logging instead.

Audit:

Run one of the following commands to verify `rsyslog` is enabled:

```
# systemctl is-enabled rsyslog
enabled
```

Run the following command to verify that `rsyslog` is running:

```
# systemctl status rsyslog | grep 'active (running) '
Active: active (running) since <Day date time>
```

Remediation:

Run the following command to enable and start `rsyslog`:

```
# systemctl --now enable rsyslog
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.2 <u>Collect Audit Logs</u> Collect audit logs. Ensure that logging, per the enterprise's audit log management process, has been enabled across enterprise assets.			
v7	6.2 <u>Activate audit logging</u> Ensure that local logging has been enabled on all systems and networking devices.			
v7	6.3 <u>Enable Detailed Logging</u> Enable system logging to include detailed information such as an event source, date, user, timestamp, source addresses, destination addresses, and other useful elements.			

4.2.1.3 Ensure rsyslog default file permissions configured (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

`rsyslog` will create logfiles that do not already exist on the system. This setting controls what permissions will be applied to these newly created files.

The `$FileCreateMode` parameter specifies the file creation mode with which `rsyslogd` creates new files. If not specified, the value `0644` is used.

Notes:

- *The value given must always be a 4-digit octal number, with the initial digit being zero.*
- *This setting can be overridden by a less restrictive setting in any file ending in `.conf` in the `/etc/rsyslog.d/` directory*

Rationale:

It is important to ensure that log files have the correct permissions to ensure that sensitive data is archived and protected.

Audit:

Run the following command and verify that `$FileCreateMode` is `0640` or more restrictive:

```
# grep ^\$FileCreateMode /etc/rsyslog.conf /etc/rsyslog.d/*.conf
$FileCreateMode 0640
```

Verify that no results return with a less restrictive file mode

Remediation:

Edit the `/etc/rsyslog.conf` and `/etc/rsyslog.d/*.conf` files and set `$FileCreateMode` to `0640` or more restrictive:

```
$FileCreateMode 0640
```

References:

1. See the rsyslog.conf(5) man page for more information.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 <u>Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v7	5.1 <u>Establish Secure Configurations</u> Maintain documented, standard security configuration standards for all authorized operating systems and software.			

4.2.1.4 Ensure logging is configured (Manual)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `/etc/rsyslog.conf` and `/etc/rsyslog.d/*.conf` files specifies rules for logging and which files are to be used to log certain classes of messages.

Rationale:

A great deal of important security-related information is sent via `rsyslog` (e.g., successful and failed su attempts, failed login attempts, root login attempts, etc.).

Audit:

Review the contents of the `/etc/rsyslog.conf` and `/etc/rsyslog.d/*.conf` files to ensure appropriate logging is set. In addition, run the following command and verify that the log files are logging information:

```
# ls -l /var/log/
```

Remediation:

Edit the following lines in the `/etc/rsyslog.conf` and `/etc/rsyslog.d/*.conf` files as appropriate for your environment:

```
*.emerg                                :omusrmsg:*
auth,authpriv.*                       /var/log/secure
mail.*                                -/var/log/mail
mail.info                              -/var/log/mail.info
mail.warning                           -/var/log/mail.warn
mail.err                               /var/log/mail.err
news.crit                              -/var/log/news/news.crit
news.err                              -/var/log/news/news.err
news.notice                           -/var/log/news/news.notice
*.=warning;*.=err                     -/var/log/warn
*.crit                                /var/log/warn
*.*;mail.none;news.none               -/var/log/messages
local0,local1.*                       -/var/log/localmessages
local2,local3.*                       -/var/log/localmessages
local4,local5.*                       -/var/log/localmessages
local6,local7.*                       -/var/log/localmessages
```

Run the following command to reload the `rsyslogd` configuration:

```
# systemctl restart rsyslog
```

References:

1. See the `rsyslog.conf(5)` man page for more information.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.2 Collect Audit Logs Collect audit logs. Ensure that logging, per the enterprise's audit log management process, has been enabled across enterprise assets.			
v7	6.2 Activate audit logging Ensure that local logging has been enabled on all systems and networking devices.			
v7	6.3 Enable Detailed Logging Enable system logging to include detailed information such as an event source, date, user, timestamp, source addresses, destination addresses, and other useful elements.			

4.2.1.5 Ensure rsyslog is configured to send logs to a remote log host (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `rsyslog` utility supports the ability to send logs it gathers to a remote log host running `syslogd(8)` or to receive messages from remote hosts, reducing administrative overhead.

Note: Ensure that the selection of logfiles being sent follows local site policy

Rationale:

Storing log data on a remote host protects log integrity from local attacks. If an attacker gains root access on the local system, they could tamper with or remove log data that is stored on the local system

Audit:

Review the `/etc/rsyslog.conf` and `/etc/rsyslog.d/*.conf` files and verify that logs are sent to a central host.

```
# grep -E '^\\s*([\\^#]+\\s+)?action\\((([\\^#]+\\s+)?\\btarg\\btarget=\\\"?[\\^#\"'+\\\"'?\\b' /etc/rsyslog.conf /etc/rsyslog.d/*.conf
```

Output should include `target=<FQDN or IP of remote loghost>`

OR

```
# grep -E '^\\[\\^#]\\s*\\S+\\.\\.\\s+@' /etc/rsyslog.conf /etc/rsyslog.d/*.conf
```

Output should include either the FQDN or the IP of the remote loghost

Remediation:

Edit the `/etc/rsyslog.conf` and `/etc/rsyslog.d/*.conf` files and add one of the following lines:

Newer syntax:

```
<files to sent to the remote log server> action(type="omfwd" target="<FQDN or  
ip of loghost>" port="<port number>" protocol="tcp"  
  
action.resumeRetryCount="<number of re-tries>"  
queue.type="LinkedList"  
queue.size=<number of messages to queue>")
```

Example:

```
*.* action(type="omfwd" target="192.168.2.100" port="514" protocol="tcp"  
        action.resumeRetryCount="100"  
        queue.type="LinkedList" queue.size="1000")
```

Older syntax:

```
*.* @@<FQDN or ip of loghost>
```

Example:

```
*.* @@192.168.2.100
```

Run the following command to reload the `rsyslog` configuration:

```
# systemctl restart rsyslog
```

References:

1. See the `rsyslog.conf(5)` man page for more information.

Additional Information:

The double "at" sign (`@@`) directs `rsyslog` to use TCP to send log messages to the server, which is a more reliable transport mechanism than the default UDP protocol

The `*.*` is a "wildcard" to send all logs to the remote loghost

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.9 Centralize Audit Logs Centralize, to the extent possible, audit log collection and retention across enterprise assets.			
v7	6.6 Deploy SIEM or Log Analytic tool Deploy Security Information and Event Management (SIEM) or log analytic tool for log correlation and analysis.			
v7	6.8 Regularly Tune SIEM On a regular basis, tune your SIEM system to better identify actionable events and decrease event noise.			

4.2.1.6 Ensure remote rsyslog messages are only accepted on designated log hosts. (Manual)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

By default, `rsyslog` does not listen for log messages coming in from remote systems. The `ModLoad` tells `rsyslog` to load the `imtcp.so` module so it can listen over a network via TCP. The `InputTCPServerRun` option instructs `rsyslogd` to listen on the specified TCP port.

Note: The `$ModLoad imtcp` line can have the `.so` extension added to the end of the module, or use the full path to the module.

Rationale:

The guidance in the section ensures that remote log hosts are configured to only accept `rsyslog` data from hosts within the specified domain and that those systems that are not designed to be log hosts do not accept any remote `rsyslog` messages. This provides protection from spoofed log data and ensures that system administrators are reviewing reasonably complete syslog data in a central location.

Audit:

Run the following commands and verify the resulting lines are uncommented on designated log hosts and commented or removed on all others:

```
# grep '$ModLoad imtcp' /etc/rsyslog.conf /etc/rsyslog.d/*.conf
$ModLoad imtcp
# grep '$InputTCPServerRun' /etc/rsyslog.conf /etc/rsyslog.d/*.conf
$InputTCPServerRun 514
```

Remediation:

For hosts that are designated as log hosts, edit the `/etc/rsyslog.conf` file and uncomment or add the following lines:

```
$ModLoad imtcp
$InputTCPServerRun 514
```

For hosts that are not designated as log hosts, edit the `/etc/rsyslog.conf` file and comment or remove the following lines:

```
# $ModLoad imtcp
# $InputTCPServerRun 514
```

Run the following command to reload the `rsyslogd` configuration:

```
# systemctl restart rsyslog
```

References:

1. See the `rsyslog(8)` man page for more information.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 <u>Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v7	9.2 <u>Ensure Only Approved Ports, Protocols and Services Are Running</u> Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.			

4.2.2 Configure journald

systemd-journald is a system service that collects and stores logging data. It creates and maintains structured, indexed journals based on logging information that is received from a variety of sources: Kernel log messages, via kmsg

Note: Any changes made to the systemd-journald configuration will require a re-start of systemd-journald

4.2.2.1 Ensure journald is configured to send logs to rsyslog (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

Data from journald may be stored in volatile memory or persisted locally on the server. Utilities exist to accept remote export of journald logs, however, use of the rsyslog service provides a consistent means of log collection and export.

Notes:

- *This recommendation assumes that recommendation 4.2.1.5, "Ensure rsyslog is configured to send logs to a remote log host" has been implemented.*
- *The main configuration file `/etc/systemd/journald.conf` is read before any of the custom `*.conf` files. If there are custom configs present, they override the main configuration parameters*
- *As noted in the journald man pages: journald logs may be exported to rsyslog either through the process mentioned here, or through a facility like `systemd-journald.service`. There are trade-offs involved in each implementation, where `ForwardToSyslog` will immediately capture all events (and forward to an external log server, if properly configured), but may not capture all boot-up activities. Mechanisms such as `systemd-journald.service`, on the other hand, will record bootup events, but may delay sending the information to rsyslog, leading to the potential for log manipulation prior to export. Be aware of the limitations of all tools employed to secure a system.*

Rationale:

Storing log data on a remote host protects log integrity from local attacks. If an attacker gains root access on the local system, they could tamper with or remove log data that is stored on the local system.

Audit:

Review `/etc/systemd/journald.conf` and verify that logs are forwarded to syslog

```
# grep -E ^\s*ForwardToSyslog /etc/systemd/journald.conf
ForwardToSyslog=yes
```

Remediation:

Edit the `/etc/systemd/journald.conf` file and add the following line:

```
ForwardToSyslog=yes
```

References:

1. <https://github.com/konstruktoid/hardening/blob/master/systemd.adoc#etcssystemdjournaldconf>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.9 Centralize Audit Logs Centralize, to the extent possible, audit log collection and retention across enterprise assets.			
v7	6.5 Central Log Management Ensure that appropriate logs are being aggregated to a central log management system for analysis and review.			

4.2.2.2 Ensure journald is configured to compress large log files (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The journald system includes the capability of compressing overly large files to avoid filling up the system with logs or making the logs unmanageably large.

Note: The main configuration file `/etc/systemd/journald.conf` is read before any of the custom `*.conf` files. If there are custom configs present, they override the main configuration parameters

Rationale:

Uncompressed large files may unexpectedly fill a filesystem leading to resource unavailability. Compressing logs prior to write can prevent sudden, unexpected filesystem impacts.

Audit:

Review `/etc/systemd/journald.conf` and verify that large files will be compressed:

```
# grep -E ^\s*Compress /etc/systemd/journald.conf
Compress=yes
```

Remediation:

Edit the `/etc/systemd/journald.conf` file and add the following line:

```
Compress=yes
```

References:

1. <https://github.com/konstruktoid/hardening/blob/master/systemd.adoc#etcsystemdjournaldconf>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.3 <u>Ensure Adequate Audit Log Storage</u> Ensure that logging destinations maintain adequate storage to comply with the enterprise's audit log management process.		●	●
v7	6.4 <u>Ensure adequate storage for logs</u> Ensure that all systems that store logs have adequate storage space for the logs generated.		●	●

4.2.2.3 Ensure journald is configured to write logfiles to persistent disk (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

Data from journald may be stored in volatile memory or persisted locally on the server. Logs in memory will be lost upon a system reboot. By persisting logs to local disk on the server they are protected from loss.

Note: The main configuration file `/etc/systemd/journald.conf` is read before any of the custom `*.conf` files. If there are custom configs present, they override the main configuration parameters

Rationale:

Writing log data to disk will provide the ability to forensically reconstruct events which may have impacted the operations or security of a system even after a system crash or reboot.

Audit:

Review `/etc/systemd/journald.conf` and verify that logs are persisted to disk:

```
# grep -E ^\s*Storage /etc/systemd/journald.conf
Storage=persistent
```

Remediation:

Edit the `/etc/systemd/journald.conf` file and add the following line:

```
Storage=persistent
```

References:

1. <https://github.com/konstruktoid/hardening/blob/master/systemd.adoc#etcsystemdjournaldconf>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.2 <u>Collect Audit Logs</u> Collect audit logs. Ensure that logging, per the enterprise's audit log management process, has been enabled across enterprise assets.			
v7	6.2 <u>Activate audit logging</u> Ensure that local logging has been enabled on all systems and networking devices.			
v7	6.3 <u>Enable Detailed Logging</u> Enable system logging to include detailed information such as an event source, date, user, timestamp, source addresses, destination addresses, and other useful elements.			

4.2.3 Ensure permissions on all logfiles are configured (Manual)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

Log files stored in /var/log/ contain logged information from many services on the system, or on log hosts others as well.

Rationale:

It is important to ensure that log files have the correct permissions to ensure that sensitive data is archived and protected. Other/world should not have the ability to view this information. Group should not have the ability to modify this information.

Audit:

Run the following command and verify that other has no permissions on any files and group does not have write or execute permissions on any files:

```
# find /var/log -type f -perm /g+wx,o+rw -exec ls -l {} \;
```

Nothing should be returned

Remediation:

Run the following commands to set permissions on all existing log files:

```
# find /var/log -type f -exec chmod g-wx,o-rwx "{}" +
```

Note: The configuration for your logging software or services may need to also be modified for any logs that had incorrect permissions, otherwise, the permissions may be reverted to the incorrect permissions

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 <u>Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v7	5.1 <u>Establish Secure Configurations</u> Maintain documented, standard security configuration standards for all authorized operating systems and software.			

4.2.4 Ensure logrotate is configured (Manual)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The system includes the capability of rotating log files regularly to avoid filling up the system with logs or making the logs unmanageably large. The file `/etc/logrotate.d/syslog` is the configuration file used to rotate log files created by `syslog` or `rsyslog`.

Note: If no `maxage` setting is set for logrotate a situation can occur where logrotate is interrupted and fails to delete rotated logfiles. It is recommended to set this to a value greater than the longest any log file should exist on your system to ensure that any such logfile is removed but standard rotation settings are not overridden.

Rationale:

By keeping the log files smaller and more manageable, a system administrator can easily archive these files to another system and spend less time looking through inordinately large log files.

Audit:

Review `/etc/logrotate.conf` and `/etc/logrotate.d/*` and verify logs are rotated according to site policy.

Remediation:

Edit `/etc/logrotate.conf` and `/etc/logrotate.d/*` to ensure logs are rotated according to site policy.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.3 <u>Ensure Adequate Audit Log Storage</u> Ensure that logging destinations maintain adequate storage to comply with the enterprise's audit log management process.			
v7	6.4 <u>Ensure adequate storage for logs</u> Ensure that all systems that store logs have adequate storage space for the logs generated.			

5 Access, Authentication and Authorization

- Access - Access control for an operating system determines how the operating system implements accesses to system resources by satisfying the security objectives of integrity, availability, and secrecy
- Authentication - Authentication is the process of recognizing a user's identity
- Authorization - Authorization is a security mechanism to determine access levels or user/client privileges related to system resources including files, services, computer programs, data and application features

5.1 Configure time-based job schedulers

`cron` is a time-based job scheduler used to schedule jobs, commands or shell scripts, to run periodically at fixed times, dates, or intervals.

`at` provides the ability to execute a command or shell script at a specified date and hour, or after a given interval of time.

Other methods exist for scheduling jobs, such as `systemd` timers. If another method is used, it should be secured in accordance with local site policy

Note: `systemd` timers are `systemd` unit files whose name ends in `.timer` that control `.service` files or events. Timers can be used as an alternative to `cron` and `at`. Timers have built-in support for calendar time events, monotonic time events, and can be run asynchronously

If `cron` and `at` are not installed, this section can be skipped.

5.1.1 Ensure cron daemon is enabled and running (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `cron` daemon is used to execute batch jobs on the system.

Rationale:

While there may not be user jobs that need to be run on the system, the system does have maintenance jobs that may include security monitoring that have to run. If another method for scheduling tasks is not being used, `cron` is used to execute them, and needs to be enabled and running.

Audit:

If `cron` is installed:

Run the following commands to verify `cron` is enabled and running:

```
# systemctl is-enabled crond
enabled

# systemctl status crond | grep 'Active: active (running) '
Active: active (running) since <Day Date Time>
```

Remediation:

Run the following command to enable and start `cron`:

```
# systemctl --now enable crond
```

OR

Run the following command to remove `cron`:

```
# yum remove cronie
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.1 Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	<u>5.1 Establish Secure Configurations</u> Maintain documented, standard security configuration standards for all authorized operating systems and software.			

5.1.2 Ensure permissions on /etc/crontab are configured (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `/etc/crontab` file is used by `cron` to control its own jobs. The commands in this item make sure that `root` is the user and group owner of the file and that only the owner can access the file.

Rationale:

This file contains information on what system jobs are run by `cron`. Write access to these files could provide unprivileged users with the ability to elevate their privileges. Read access to these files could provide users with the ability to gain insight on system jobs that run on the system and could provide them a way to gain unauthorized privileged access.

Audit:

If `cron` is installed:

Run the following command and verify `Uid` and `Gid` are both `0/root` and `Access` does not grant permissions to `group` or `other`:

```
# stat /etc/crontab
Access: (0600/-rw-----)  Uid: (    0/    root)   Gid: (    0/    root)
```

Remediation:

Run the following commands to set ownership and permissions on `/etc/crontab`:

```
# chown root:root /etc/crontab
# chmod u-x,og-rwx /etc/crontab
```

OR

Run the following command to remove `cron`:

```
# yum remove cronie
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 <u>Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v7	14.6 <u>Protect Information through Access Control Lists</u> Protect all information stored on systems with file system, network share, claims, application, or database specific access control lists. These controls will enforce the principle that only authorized individuals should have access to the information based on their need to access the information as a part of their responsibilities.			

5.1.3 Ensure permissions on /etc/cron.hourly are configured (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

This directory contains system `cron` jobs that need to run on an hourly basis. The files in this directory cannot be manipulated by the `crontab` command, but are instead edited by system administrators using a text editor. The commands below restrict read/write and search access to user and group root, preventing regular users from accessing this directory.

Rationale:

Granting write access to this directory for non-privileged users could provide them the means for gaining unauthorized elevated privileges. Granting read access to this directory could give an unprivileged user insight in how to gain elevated privileges or circumvent auditing controls.

Audit:

If `cron` is installed:

Run the following command and verify `Uid` and `Gid` are both `0/root` and `Access` does not grant permissions to `group` or `other` :

```
# stat /etc/cron.hourly/
Access: (0700/drwx-----)  Uid: (    0/    root)  Gid: (    0/    root)
```

Remediation:

Run the following commands to set ownership and permissions on the `/etc/cron.hourly/` directory:

```
# chown root:root /etc/cron.hourly/  
# chmod og-rwx /etc/cron.hourly/
```

OR

Run the following command to remove `cron`

```
# yum remove cronie
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 <u>Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v7	5.1 <u>Establish Secure Configurations</u> Maintain documented, standard security configuration standards for all authorized operating systems and software.			

5.1.4 Ensure permissions on /etc/cron.daily are configured (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `/etc/cron.daily` directory contains system cron jobs that need to run on a daily basis. The files in this directory cannot be manipulated by the `crontab` command, but are instead edited by system administrators using a text editor. The commands below restrict read/write and search access to user and group root, preventing regular users from accessing this directory.

Rationale:

Granting write access to this directory for non-privileged users could provide them the means for gaining unauthorized elevated privileges. Granting read access to this directory could give an unprivileged user insight in how to gain elevated privileges or circumvent auditing controls.

Audit:

If `cron` is installed:

Run the following command and verify `Uid` and `Gid` are both `0/root` and `Access` does not grant permissions to `group` or `other` :

```
# stat /etc/cron.daily/  
Access: (0700/drwx-----)  Uid: (    0/    root)  Gid: (    0/    root)
```

Remediation:

Run the following commands to set ownership and permissions on `/etc/cron.daily` directory:

```
# chown root:root /etc/cron.daily
# chmod og-rwx /etc/cron.daily
```

OR

Run the following command to remove `cron`:

```
# yum remove cronie
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 <u>Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v7	14.6 <u>Protect Information through Access Control Lists</u> Protect all information stored on systems with file system, network share, claims, application, or database specific access control lists. These controls will enforce the principle that only authorized individuals should have access to the information based on their need to access the information as a part of their responsibilities.			

5.1.5 Ensure permissions on /etc/cron.weekly are configured (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `/etc/cron.weekly` directory contains system cron jobs that need to run on a weekly basis. The files in this directory cannot be manipulated by the `crontab` command, but are instead edited by system administrators using a text editor. The commands below restrict read/write and search access to user and group root, preventing regular users from accessing this directory.

Rationale:

Granting write access to this directory for non-privileged users could provide them the means for gaining unauthorized elevated privileges. Granting read access to this directory could give an unprivileged user insight in how to gain elevated privileges or circumvent auditing controls.

Audit:

If `cron` is installed

Run the following command and verify `Uid` and `Gid` are both `0/root` and `Access` does not grant permissions to `group` or `other` :

```
# stat /etc/cron.weekly
Access: (0700/drwx-----)  Uid: (    0/    root)  Gid: (    0/    root)
```

Remediation:

Run the following commands to set ownership and permissions on `/etc/cron.weekly/` directory:

```
# chown root:root /etc/cron.weekly/  
# chmod og-rwx /etc/cron.weekly/
```

OR

Run the following command to remove `cron`:

```
# yum remove cronie
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 <u>Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v7	14.6 <u>Protect Information through Access Control Lists</u> Protect all information stored on systems with file system, network share, claims, application, or database specific access control lists. These controls will enforce the principle that only authorized individuals should have access to the information based on their need to access the information as a part of their responsibilities.			

5.1.6 Ensure permissions on /etc/cron.monthly are configured (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `/etc/cron.monthly` directory contains system cron jobs that need to run on a monthly basis. The files in this directory cannot be manipulated by the `crontab` command, but are instead edited by system administrators using a text editor. The commands below restrict read/write and search access to user and group root, preventing regular users from accessing this directory.

Rationale:

Granting write access to this directory for non-privileged users could provide them the means for gaining unauthorized elevated privileges. Granting read access to this directory could give an unprivileged user insight in how to gain elevated privileges or circumvent auditing controls.

Audit:

If `cron` is installed:

Run the following command and verify `Uid` and `Gid` are both `0/root` and `Access` does not grant permissions to `group` or `other`:

```
# stat /etc/cron.monthly/
Access: (0700/drwx-----)  Uid: (    0/    root)   Gid: (    0/    root)
```

Remediation:

Run the following commands to set ownership and permissions on `/etc/cron.monthly` directory:

```
# chown root:root /etc/cron.monthly
# chmod og-rwx /etc/cron.monthly
```

OR

Run the following command to remove `cron`:

```
# yum remove cronie
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 <u>Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v7	14.6 <u>Protect Information through Access Control Lists</u> Protect all information stored on systems with file system, network share, claims, application, or database specific access control lists. These controls will enforce the principle that only authorized individuals should have access to the information based on their need to access the information as a part of their responsibilities.			

5.1.7 Ensure permissions on /etc/cron.d are configured (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `/etc/cron.d/` directory contains system `cron` jobs that need to run in a similar manner to the hourly, daily weekly and monthly jobs from `/etc/crontab`, but require more granular control as to when they run. The files in this directory cannot be manipulated by the `crontab` command, but are instead edited by system administrators using a text editor. The commands below restrict read/write and search access to user and group root, preventing regular users from accessing this directory.

Rationale:

Granting write access to this directory for non-privileged users could provide them the means for gaining unauthorized elevated privileges. Granting read access to this directory could give an unprivileged user insight in how to gain elevated privileges or circumvent auditing controls.

Audit:

If `cron` is installed:

Run the following command and verify `Uid` and `Gid` are both `0/root` and `Access` does not grant permissions to `group` or `other` :

```
# stat /etc/cron.d
Access: (0700/drwx-----)  Uid: (    0/    root)  Gid: (    0/    root)
```

Remediation:

Run the following commands to set ownership and permissions on `/etc/cron.d` directory:

```
# chown root:root /etc/cron.d
# chmod og-rwx /etc/cron.d
```

OR

Run the following command to remove `cron`:

```
# yum remove cronie
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 <u>Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v7	14.6 <u>Protect Information through Access Control Lists</u> Protect all information stored on systems with file system, network share, claims, application, or database specific access control lists. These controls will enforce the principle that only authorized individuals should have access to the information based on their need to access the information as a part of their responsibilities.			

5.1.8 Ensure cron is restricted to authorized users (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

If `cron` is installed in the system, configure `/etc/cron.allow` to allow specific users to use these services. If `/etc/cron.allow` does not exist, then `/etc/cron.deny` is checked. Any user not specifically defined in those files is allowed to use cron. By removing the file, only users in `/etc/cron.allow` are allowed to use cron.

Note: Even though a given user is not listed in `cron.allow`, cron jobs can still be run as that user. The `cron.allow` file only controls administrative access to the `crontab` command for scheduling and modifying cron jobs.

Rationale:

On many systems, only the system administrator is authorized to schedule `cron` jobs. Using the `cron.allow` file to control who can run `cron` jobs enforces this policy. It is easier to manage an allow list than a deny list. In a deny list, you could potentially add a user ID to the system and forget to add it to the deny files.

Audit:

If `cron` is installed:

Run the following command and verify `/etc/cron.deny` does not exist:

```
# stat /etc/cron.deny
stat: cannot stat `/etc/cron.deny': No such file or directory
```

Run the following command and verify `Uid` and `Gid` are both `0/root` and `Access` does not grant permissions to `group` or `other` for `/etc/cron.allow`:

```
# stat /etc/cron.allow
Access: (0600/-rw-----)  Uid: (    0/    root)  Gid: (    0/    root)
```

Remediation:

Run the following command to remove `/etc/cron.deny`:

```
# rm /etc/cron.deny
```

Run the following command to create `/etc/cron.allow`

```
# touch /etc/cron.allow
```

Run the following commands to set the owner and permissions on `/etc/cron.allow`:

```
# chown root:root /etc/cron.allow
# chmod u-x,og-rwx /etc/cron.allow
```

OR

Run the following command to remove `cron`

```
# yum remove cronie
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 <u>Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v7	16 <u>Account Monitoring and Control</u> Account Monitoring and Control			

5.1.9 Ensure at is restricted to authorized users (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

If `at` is installed in the system, configure `/etc/at.allow` to allow specific users to use these services. If `/etc/at.allow` does not exist, then `/etc/at.deny` is checked. Any user not specifically defined in those files is allowed to use `at`. By removing the file, only users in `/etc/at.allow` are allowed to use `at`.

Note: Even though a given user is not listed in `at.allow`, `at` jobs can still be run as that user. The `at.allow` file only controls administrative access to the `at` command for scheduling and modifying `at` jobs.

Rationale:

On many systems, only the system administrator is authorized to schedule `at` jobs. Using the `at.allow` file to control who can run `at` jobs enforces this policy. It is easier to manage an allow list than a deny list. In a deny list, you could potentially add a user ID to the system and forget to add it to the deny files.

Audit:

If `at` is installed:

Run the following command and verify `/etc/at.deny` does not exist:

```
# stat /etc/at.deny
stat: cannot stat `/etc/at.deny': No such file or directory
```

Run the following command and verify `Uid` and `Gid` are both `0/root` and `Access` does not grant permissions to `group` or `other` for `/etc/at.allow`:

```
# stat /etc/at.allow
Access: (0600/-rw-----)  Uid: (    0/    root)  Gid: (    0/    root)
```

Remediation:

Run the following command to remove `/etc/at.deny`:

```
# rm /etc/at.deny
```

Run the following command to create `/etc/at.allow`

```
# touch /etc/at.allow
```

Run the following commands to set the owner and permissions on `/etc/at.allow`:

```
# chown root:root /etc/at.allow  
# chmod u-x,og-rwx /etc/at.allow
```

OR

Run the following command to remove `at`:

```
# yum remove at
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 <u>Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v7	16 <u>Account Monitoring and Control</u> Account Monitoring and Control			

5.2 Configure sudo

sudo allows a permitted user to execute a command as the superuser or another user, as specified by the security policy. The invoking user's real (not effective) user ID is used to determine the user name with which to query the security policy.

sudo supports a plugin architecture for security policies and input/output logging. Third parties can develop and distribute their own policy and I/O logging plugins to work seamlessly with the sudo front end. The default security policy is sudoers, which is configured via the file `/etc/sudoers`.

5.2.1 Ensure sudo is installed (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

sudo allows a permitted user to execute a command as the superuser or another user, as specified by the security policy. The invoking user's real (not effective) user ID is used to determine the user name with which to query the security policy.

Rationale:

sudo supports a plugin architecture for security policies and input/output logging. Third parties can develop and distribute their own policy and I/O logging plugins to work seamlessly with the sudo front end. The default security policy is sudoers, which is configured via the file /etc/sudoers.

The security policy determines what privileges, if any, a user has to run sudo. The policy may require that users authenticate themselves with a password or another authentication mechanism. If authentication is required, sudo will exit if the user's password is not entered within a configurable time limit. This limit is policy-specific.

Audit:

Verify that sudo is installed.

Run the following command:

```
# rpm -q sudo  
sudo-<VERSION>
```

Remediation:

Run the following command to install sudo.

```
# yum install sudo
```

References:

1. SUDO(8)

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 <u>Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	4 <u>Controlled Use of Administrative Privileges</u> Controlled Use of Administrative Privileges			

5.2.2 Ensure sudo commands use pty (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

sudo can be configured to run only from a pseudo-pty

Note: visudo edits the sudoers file in a safe fashion, analogous to vipw(8). visudo locks the sudoers file against multiple simultaneous edits, provides basic sanity checks, and checks for parse errors. If the sudoers file is currently being edited you will receive a message to try again later. The -f option allows you to tell visudo which file to edit.

Rationale:

Attackers can run a malicious program using sudo, which would again fork a background process that remains even when the main program has finished executing.

This can be mitigated by configuring sudo to run other commands only from a pseudo-pty, whether I/O logging is turned on or not.

Audit:

Verify that sudo can only run other commands from a pseudo-pty

Run the following command:

```
# grep -Ei '^s*Defaults\s+([\^#]\S+,\s*)?use_pty\b' /etc/sudoers
/etc/sudoers.d/*

Defaults use_pty
```

Remediation:

Edit the file /etc/sudoers or a file in /etc/sudoers.d/ with visudo or visudo -f <PATH TO FILE> and add the following line:

```
Defaults use_pty
```

References:

1. SUDO(8)

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 <u>Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	4 <u>Controlled Use of Administrative Privileges</u> Controlled Use of Administrative Privileges			

5.2.3 Ensure sudo log file exists (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

sudo can use a custom log file

Note: visudo edits the sudoers file in a safe fashion, analogous to vipw(8). visudo locks the sudoers file against multiple simultaneous edits, provides basic sanity checks, and checks for parse errors. If the sudoers file is currently being edited you will receive a message to try again later. The -f option allows you to tell visudo which file to edit.

Rationale:

A sudo log file simplifies auditing of sudo commands

Impact:

Editing the sudo configuration incorrectly can cause sudo to stop functioning

Audit:

Verify that sudo has a custom log file configured

Run the following command:

```
# grep -Ei '\s*Defaults\s+([\^#;]+,\s*)?logfile\s*=\s*([\^#;]+)?'\n/etc/sudoers /etc/sudoers.d/*\n\nDefaults logfile="/var/log/sudo.log"
```

Remediation:

edit the file `/etc/sudoers` or a file in `/etc/sudoers.d/` with `visudo` or `visudo -f <PATH TO FILE>` and add the following line:

```
Defaults logfile="<PATH TO CUSTOM LOG FILE>"
```

Example:

```
Defaults logfile="/var/log/sudo.log"
```

References:

1. SUDO(8)
2. VISUDO(8)

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.5 <u>Collect Detailed Audit Logs</u> Configure detailed audit logging for enterprise assets containing sensitive data. Include event source, date, username, timestamp, source addresses, destination addresses, and other useful elements that could assist in a forensic investigation.			
v7	6.3 <u>Enable Detailed Logging</u> Enable system logging to include detailed information such as an event source, date, user, timestamp, source addresses, destination addresses, and other useful elements.			

5.3 Configure SSH Server

SSH is a secure, encrypted replacement for common login services such as `telnet`, `ftp`, `rlogin`, `rsh`, and `rcp`. It is strongly recommended that sites abandon older clear-text login protocols and use SSH to prevent session hijacking and sniffing of sensitive data off the network.

Note:

- The recommendations in this section only apply if the SSH daemon is installed on the system, if remote access is **not** required the SSH daemon can be removed and this section skipped.
 - Command to remove the SSH daemon:

```
# yum remove openssh-server
```

- Once all configuration changes have been made to `/etc/ssh/sshd_config`, the `sshd` configuration must be reloaded:
 - Command to re-load the SSH daemon configuration:

```
# systemctl reload sshd
```

5.3.1 Ensure permissions on /etc/ssh/sshd_config are configured (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `/etc/ssh/sshd_config` file contains configuration specifications for `sshd`. The command below sets the owner and group of the file to `root`.

Rationale:

The `/etc/ssh/sshd_config` file needs to be protected from unauthorized changes by non-privileged users.

Audit:

Run the following command and verify `Uid` and `Gid` are both `0/root` and `Access` does not grant permissions to `group` or `other`:

```
# stat /etc/ssh/sshd_config
Access: (0600/-rw-----)  Uid: (  0/   root)  Gid: (  0/   root)
```

Remediation:

Run the following commands to set ownership and permissions on `/etc/ssh/sshd_config`:

```
# chown root:root /etc/ssh/sshd config
# chmod og-rwx /etc/ssh/sshd_config
```

Default Value:

Access: (0600/-rw-----) Uid: (0/ root) Gid: (0/ root)

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 <u>Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v7	14.6 <u>Protect Information through Access Control Lists</u> Protect all information stored on systems with file system, network share, claims, application, or database specific access control lists. These controls will enforce the principle that only authorized individuals should have access to the information based on their need to access the information as a part of their responsibilities.			

5.3.2 Ensure permissions on SSH private host key files are configured (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

An SSH private key is one of two files used in SSH public key authentication. In this authentication method, The possession of the private key is proof of identity. Only a private key that corresponds to a public key will be able to authenticate successfully. The private keys need to be stored and handled carefully, and no copies of the private key should be distributed.

Rationale:

If an unauthorized user obtains the private SSH host key file, the host could be impersonated

Audit:

Run the following command and verify Uid is 0/root and Gid is 0/root and permissions are 0600 or more restrictive:

```
# find /etc/ssh -xdev -type f -name 'ssh_host_*_key' -exec stat {} \;
```

Example Output:

```
File: '/etc/ssh/ssh_host_rsa_key'
Size: 1675          Blocks: 8          IO Block: 4096   regular file
Device: 801h/2049d Inode: 794321       Links: 1
Access: (0600/-rw-----)  Uid: (    0/   root)   Gid: (    0/   root)
Access: 2021-03-01 06:25:08.633246149 -0800
Modify: 2021-01-29 06:42:16.001324236 -0800
Change: 2021-01-29 06:42:16.001324236 -0800
Birth: -
File: '/etc/ssh/ssh_host_ecdsa_key'
Size: 227          Blocks: 8          IO Block: 4096   regular file
Device: 801h/2049d Inode: 794325       Links: 1
Access: (0600/-rw-----)  Uid: (    0/   root)   Gid: (    0/   root)
Access: 2021-03-01 06:25:08.633246149 -0800
Modify: 2021-01-29 06:42:16.173327263 -0800
Change: 2021-01-29 06:42:16.173327263 -0800
Birth: -
File: '/etc/ssh/ssh_host_ed25519_key'
Size: 399          Blocks: 8          IO Block: 4096   regular file
Device: 801h/2049d Inode: 794327       Links: 1
Access: (0600/-rw-----)  Uid: (    0/   root)   Gid: (    0/   root)
Access: 2021-03-01 06:25:08.633246149 -0800
Modify: 2021-01-29 06:42:16.185327474 -0800
Change: 2021-01-29 06:42:16.185327474 -0800
Birth: -
File: '/etc/ssh/ssh_host_dsa_key'
Size: 672          Blocks: 8          IO Block: 4096   regular file
Device: 801h/2049d Inode: 794323       Links: 1
Access: (0600/-rw-----)  Uid: (    0/   root)   Gid: (    0/   root)
Access: 2021-03-01 06:25:08.645246255 -0800
Modify: 2021-01-29 06:42:16.161327052 -0800
Change: 2021-01-29 06:42:16.161327052 -0800
Birth: -
```

Remediation:

Run the following commands to set permissions, ownership, and group on the private SSH host key files:

```
# find /etc/ssh -xdev -type f -name 'ssh_host_*_key' -exec chown root:root {} \;
# find /etc/ssh -xdev -type f -name 'ssh_host_*_key' -exec chmod u-x,go-rwx {} \;
```

Default Value:

Access: (0600/-rw-----) Uid: (0/ root) Gid: (0/ root)

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 <u>Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v7	14.6 <u>Protect Information through Access Control Lists</u> Protect all information stored on systems with file system, network share, claims, application, or database specific access control lists. These controls will enforce the principle that only authorized individuals should have access to the information based on their need to access the information as a part of their responsibilities.			

5.3.3 Ensure permissions on SSH public host key files are configured (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

An SSH public key is one of two files used in SSH public key authentication. In this authentication method, a public key is a key that can be used for verifying digital signatures generated using a corresponding private key. Only a public key that corresponds to a private key will be able to authenticate successfully.

Rationale:

If a public host key file is modified by an unauthorized user, the SSH service may be compromised.

Audit:

Run the following command and verify Access does not grant write or execute permissions to group or other for all returned files:

```
# find /etc/ssh -xdev -type f -name 'ssh_host_*_key.pub' -exec stat {} \;
```

Example Output:

```
File: '/etc/ssh/ssh_host_rsa_key.pub'
Size: 382          Blocks: 8          IO Block: 4096   regular file
Device: ca01h/51713d Inode: 8631758      Links: 1
Access: (0644/-rw-r--r--)  Uid: (   0/   root)   Gid: (   0/   root)
Access: 2018-10-22 18:24:56.861750616 +0000
Modify: 2018-10-22 18:24:56.861750616 +0000
Change: 2018-10-22 18:24:56.881750616 +0000
Birth: -
File: '/etc/ssh/ssh_host_ecdsa_key.pub'
Size: 162          Blocks: 8          IO Block: 4096   regular file
Device: ca01h/51713d Inode: 8631761      Links: 1
Access: (0644/-rw-r--r--)  Uid: (   0/   root)   Gid: (   0/   root)
Access: 2018-10-22 18:24:56.897750616 +0000
Modify: 2018-10-22 18:24:56.897750616 +0000
Change: 2018-10-22 18:24:56.917750616 +0000
Birth: -
File: '/etc/ssh/ssh_host_ed25519_key.pub'
Size: 82           Blocks: 8          IO Block: 4096   regular file
Device: ca01h/51713d Inode: 8631763      Links: 1
Access: (0644/-rw-r--r--)  Uid: (   0/   root)   Gid: (   0/   root)
Access: 2018-10-22 18:24:56.945750616 +0000
Modify: 2018-10-22 18:24:56.945750616 +0000
Change: 2018-10-22 18:24:56.961750616 +0000
Birth: -
```

Remediation:

Run the following commands to set permissions and ownership on the SSH host public key files

```
# find /etc/ssh -xdev -type f -name 'ssh_host_*_key.pub' -exec chmod u-x,go-  
wx {} \;  
# find /etc/ssh -xdev -type f -name 'ssh_host_*_key.pub' -exec chown  
root:root {} \;
```

Default Value:

Access: (0644/-rw-r--r--) Uid: (0/ root) Gid: (0/ root)

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 <u>Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v7	14.6 <u>Protect Information through Access Control Lists</u> Protect all information stored on systems with file system, network share, claims, application, or database specific access control lists. These controls will enforce the principle that only authorized individuals should have access to the information based on their need to access the information as a part of their responsibilities.			

5.3.4 Ensure SSH access is limited (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

There are several options available to limit which users and group can access the system via SSH. It is recommended that at least one of the following options be leveraged:

- `AllowUsers`:
 - The `AllowUsers` variable gives the system administrator the option of allowing specific users to `ssh` into the system. The list consists of space separated user names. Numeric user IDs are not recognized with this variable. If a system administrator wants to restrict user access further by only allowing the allowed users to log in from a particular host, the entry can be specified in the form of `user@host`.
- `AllowGroups`:
 - The `AllowGroups` variable gives the system administrator the option of allowing specific groups of users to `ssh` into the system. The list consists of space separated group names. Numeric group IDs are not recognized with this variable.
- `DenyUsers`:
 - The `DenyUsers` variable gives the system administrator the option of denying specific users to `ssh` into the system. The list consists of space separated user names. Numeric user IDs are not recognized with this variable. If a system administrator wants to restrict user access further by specifically denying a user's access from a particular host, the entry can be specified in the form of `user@host`.
- `DenyGroups`:
 - The `DenyGroups` variable gives the system administrator the option of denying specific groups of users to `ssh` into the system. The list consists of space separated group names. Numeric group IDs are not recognized with this variable.

Rationale:

Restricting which users can remotely access the system via SSH will help ensure that only authorized users access the system.

Audit:

Run the following commands and verify the output:

```
# sshd -T -C user=root -C host="$(hostname)" -C addr="$(grep $(hostname)
/etc/hosts | awk '{print $1}')" | grep -Pi
'^\h*(allow|deny) (users|groups) \h+\H+(\h+.*?)?$'

# grep -Pi '^ \h*(allow|deny) (users|groups) \h+\H+(\h+.*?)?$'
/etc/ssh/sshd_config
```

Verify that the output of both commands matches at least one of the following lines:

```
allowusers <userlist>
allowgroups <grouplist>
denyusers <userlist>
denygroups <grouplist>
```

Remediation:

Edit the `/etc/ssh/sshd_config` file to set one or more of the parameter as follows:

```
AllowUsers <userlist>
```

OR

```
AllowGroups <grouplist>
```

OR

```
DenyUsers <userlist>
```

OR

```
DenyGroups <grouplist>
```

Default Value:

None

References:

1. SSHD_CONFIG(5)

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 <u>Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v8	5.4 <u>Restrict Administrator Privileges to Dedicated Administrator Accounts</u> Restrict administrator privileges to dedicated administrator accounts on enterprise assets. Conduct general computing activities, such as internet browsing, email, and productivity suite use, from the user's primary, non-privileged account.			
v7	4.3 <u>Ensure the Use of Dedicated Administrative Accounts</u> Ensure that all users with administrative account access use a dedicated or secondary account for elevated activities. This account should only be used for administrative activities and not internet browsing, email, or similar activities.			

5.3.5 Ensure SSH LogLevel is appropriate (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

`INFO` level is the basic level that only records login activity of SSH users. In many situations, such as Incident Response, it is important to determine when a particular user was active on a system. The logout record can eliminate those users who disconnected, which helps narrow the field.

`VERBOSE` level specifies that login and logout activity as well as the key fingerprint for any SSH key used for login will be logged. This information is important for SSH key management, especially in legacy environments.

Rationale:

SSH provides several logging levels with varying amounts of verbosity. `DEBUG` is specifically **not** recommended other than strictly for debugging SSH communications since it provides so much data that it is difficult to identify important security information.

Audit:

Run the following command and verify that output matches `loglevel VERBOSE` or `loglevel INFO`:

```
# sshd -T -C user=root -C host="$(hostname)" -C addr="$(grep $(hostname) /etc/hosts | awk '{print $1}')" | grep loglevel  
  
loglevel VERBOSE or loglevel INFO
```

Run the following command and verify the output matches:

```
# grep -i 'loglevel' /etc/ssh/sshd_config | grep -Evi '(VERBOSE|INFO)'  
  
Nothing should be returned
```

Remediation:

Edit the `/etc/ssh/sshd_config` file to set the parameter as follows:

```
LogLevel VERBOSE
```

OR

```
LogLevel INFO
```

Default Value:

LogLevel INFO

References:

1. https://www.ssh.com/ssh/sshd_config/

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.2 <u>Collect Audit Logs</u> Collect audit logs. Ensure that logging, per the enterprise's audit log management process, has been enabled across enterprise assets.			
v8	8.5 <u>Collect Detailed Audit Logs</u> Configure detailed audit logging for enterprise assets containing sensitive data. Include event source, date, username, timestamp, source addresses, destination addresses, and other useful elements that could assist in a forensic investigation.			
v7	6.2 <u>Activate audit logging</u> Ensure that local logging has been enabled on all systems and networking devices.			
v7	6.3 <u>Enable Detailed Logging</u> Enable system logging to include detailed information such as an event source, date, user, timestamp, source addresses, destination addresses, and other useful elements.			

5.3.6 Ensure SSH X11 forwarding is disabled (Automated)

Profile Applicability:

- Level 1 - Workstation
- Level 2 - Server

Description:

The X11Forwarding parameter provides the ability to tunnel X11 traffic through an existing SSH shell session to enable remote graphic connections.

Rationale:

Disable X11 forwarding unless there is an operational requirement to use X11 applications directly. There is a small risk that the remote X11 servers of users who are logged in via SSH with X11 forwarding could be compromised by other users on the X11 server. Note that even if X11 forwarding is disabled, users can always install their own forwarders.

Impact:

X11 programs on the server will not be able to be forwarded to a ssh-client display.

Audit:

Run the following command and verify that output matches:

```
# sshd -T -C user=root -C host="$(hostname)" -C addr="$(grep $(hostname) /etc/hosts | awk '{print $1}')" | grep -i x11forwarding
x11forwarding no
```

Run the following command and verify that the output matches:

```
# grep -Ei '^\s*x11forwarding\s+yes' /etc/ssh/sshd_config
Nothing is returned
```

Remediation:

Edit the /etc/ssh/sshd_config file to set the parameter as follows:

```
X11Forwarding no
```

Default Value:

X11Forwarding yes

References:

1. SSHD_CONFIG(5)

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software</u> Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.			
v7	<u>9.2 Ensure Only Approved Ports, Protocols and Services Are Running</u> Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.			

5.3.7 Ensure SSH MaxAuthTries is set to 4 or less (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `MaxAuthTries` parameter specifies the maximum number of authentication attempts permitted per connection. When the login failure count reaches half the number, error messages will be written to the `syslog` file detailing the login failure.

Rationale:

Setting the `MaxAuthTries` parameter to a low number will minimize the risk of successful brute force attacks to the SSH server. While the recommended setting is 4, set the number based on site policy.

Audit:

Run the following command and verify that output `MaxAuthTries` is 4 or less:

```
# sshd -T -C user=root -C host="$(hostname)" -C addr="$(grep $(hostname) /etc/hosts | awk '{print $1}')" | grep maxauthtries
maxauthtries 4
```

Run the following command and verify that the output:

```
# grep -Ei '^s*maxauthtries\s+([5-9]|[1-9][0-9]+)' /etc/ssh/sshd_config
Nothing is returned
```

Remediation:

Edit the `/etc/ssh/sshd_config` file to set the parameter as follows:

```
MaxAuthTries 4
```

Default Value:

`MaxAuthTries 6`

References:

1. SSHD_CONFIG(5)

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.5 <u>Collect Detailed Audit Logs</u> Configure detailed audit logging for enterprise assets containing sensitive data. Include event source, date, username, timestamp, source addresses, destination addresses, and other useful elements that could assist in a forensic investigation.			
v7	16.13 <u>Alert on Account Login Behavior Deviation</u> Alert when users deviate from normal login behavior, such as time-of-day, workstation location and duration.			

5.3.8 Ensure SSH IgnoreRhosts is enabled (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `IgnoreRhosts` parameter specifies that `.rhosts` and `.shosts` files will not be used in `RhostsRSAAuthentication` OR `HostbasedAuthentication`.

Rationale:

Setting this parameter forces users to enter a password when authenticating with ssh.

Audit:

Run the following command and verify that output matches:

```
# sshd -T -C user=root -C host="$(hostname)" -C addr="$(grep $(hostname) /etc/hosts | awk '{print $1}')" | grep ignorerhosts
ignorerhosts yes
```

Run the following command and verify the output:

```
# grep -Ei '^s*ignorerhosts\s+no\b' /etc/ssh/sshd_config
Nothing should be returned
```

Remediation:

Edit the `/etc/ssh/sshd_config` file to set the parameter as follows:

```
IgnoreRhosts yes
```

Default Value:

`IgnoreRhosts yes`

References:

1. `SSHD_CONFIG(5)`

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.1 Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	<u>9.2 Ensure Only Approved Ports, Protocols and Services Are Running</u> Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.			

5.3.9 Ensure SSH HostbasedAuthentication is disabled (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `HostbasedAuthentication` parameter specifies if authentication is allowed through trusted hosts via the user of `.rhosts`, or `/etc/hosts.equiv`, along with successful public key client host authentication. This option only applies to SSH Protocol Version 2.

Rationale:

Even though the `.rhosts` files are ineffective if support is disabled in `/etc/pam.conf`, disabling the ability to use `.rhosts` files in SSH provides an additional layer of protection.

Audit:

Run the following command and verify that output matches:

```
# sshd -T -C user=root -C host="$(hostname)" -C addr="$(grep $(hostname) /etc/hosts | awk '{print $1}')" | grep hostbasedauthentication
hostbasedauthentication no
```

Run the following command and verify the output matches:

```
# grep -Ei '^s*HostbasedAuthentication\s+yes' /etc/ssh/sshd_config
Nothing should be returned
```

Remediation:

Edit the `/etc/ssh/sshd_config` file to set the parameter as follows:

```
HostbasedAuthentication no
```

Default Value:

`HostbasedAuthentication no`

References:

1. SSHD_CONFIG(5)

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.1 Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	<u>16.3 Require Multi-factor Authentication</u> Require multi-factor authentication for all user accounts, on all systems, whether managed onsite or by a third-party provider.			

5.3.10 Ensure SSH root login is disabled (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `PermitRootLogin` parameter specifies if the root user can log in using ssh. The default is no.

Rationale:

Disallowing root logins over SSH requires system admins to authenticate using their own individual account, then escalating to root via `sudo`. This in turn limits opportunity for non-repudiation and provides a clear audit trail in the event of a security incident

Audit:

Run the following command and verify that output matches:

```
# sshd -T -C user=root -C host="$(hostname)" -C addr="$(grep $(hostname) /etc/hosts | awk '{print $1}')" | grep permitrootlogin
permitrootlogin no
```

Run the following command and verify the output:

```
# grep -Ei '^s*PermitRootLogin\s+yes' /etc/ssh/sshd_config
Nothing should be returned
```

Remediation:

Edit the `/etc/ssh/sshd_config` file to set the parameter as follows:

```
PermitRootLogin no
```

Default Value:

`PermitRootLogin without-password`

References:

1. SSHD_CONFIG(5)

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>5.4 Restrict Administrator Privileges to Dedicated Administrator Accounts</u> Restrict administrator privileges to dedicated administrator accounts on enterprise assets. Conduct general computing activities, such as internet browsing, email, and productivity suite use, from the user's primary, non-privileged account.			
v7	<u>4.3 Ensure the Use of Dedicated Administrative Accounts</u> Ensure that all users with administrative account access use a dedicated or secondary account for elevated activities. This account should only be used for administrative activities and not internet browsing, email, or similar activities.			

5.3.11 Ensure SSH PermitEmptyPasswords is disabled (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `PermitEmptyPasswords` parameter specifies if the SSH server allows login to accounts with empty password strings.

Rationale:

Disallowing remote shell access to accounts that have an empty password reduces the probability of unauthorized access to the system

Audit:

Run the following command and verify that output matches:

```
# sshd -T -C user=root -C host="$(hostname)" -C addr="$(grep $(hostname) /etc/hosts | awk '{print $1}')" | grep permitemptypasswords
permitemptypasswords no
```

Run the following command and verify the output:

```
# grep -Ei '^\s*PermitEmptyPasswords\s+yes' /etc/ssh/sshd_config
Nothing should be returned
```

Remediation:

Edit the `/etc/ssh/sshd_config` file to set the parameter as follows:

```
PermitEmptyPasswords no
```

Default Value:

`PermitEmptyPasswords no`

References:

1. SSHD_CONFIG(5)

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.1 Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	<u>16.3 Require Multi-factor Authentication</u> Require multi-factor authentication for all user accounts, on all systems, whether managed onsite or by a third-party provider.			

5.3.12 Ensure SSH PermitUserEnvironment is disabled (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `PermitUserEnvironment` option allows users to present environment options to the `ssh` daemon.

Rationale:

Permitting users the ability to set environment variables through the SSH daemon could potentially allow users to bypass security controls (e.g. setting an execution path that has `ssh` executing a Trojan's programs)

Audit:

Run the following command and verify that output matches:

```
# sshd -T -C user=root -C host="$(hostname)" -C addr="$(grep $(hostname) /etc/hosts | awk '{print $1}')" | grep permituserenvironment
permituserenvironment no
```

Run the following command and verify the output:

```
# grep -Ei '^s*PermitUserEnvironment\s+yes' /etc/ssh/sshd_config
Nothing should be returned
```

Remediation:

Edit the `/etc/ssh/sshd_config` file to set the parameter as follows:

```
PermitUserEnvironment no
```

Default Value:

`PermitUserEnvironment no`

References:

1. SSHD_CONFIG(5)

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.1 Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	<u>5.1 Establish Secure Configurations</u> Maintain documented, standard security configuration standards for all authorized operating systems and software.			

5.3.13 Ensure only strong Ciphers are used (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

This variable limits the ciphers that SSH can use during communication.

Note: Some organizations may have stricter requirements for approved ciphers. Ensure that ciphers used are in compliance with site policy.

Rationale:

Weak ciphers that are used for authentication to the cryptographic module cannot be relied upon to provide confidentiality or integrity, and system data may be compromised.

- The DES, Triple DES, and Blowfish ciphers, as used in SSH, have a birthday bound of approximately four billion blocks, which makes it easier for remote attackers to obtain cleartext data via a birthday attack against a long-duration encrypted session, aka a "Sweet32" attack
- The RC4 algorithm, as used in the TLS protocol and SSL protocol, does not properly combine state data with key data during the initialization phase, which makes it easier for remote attackers to conduct plaintext-recovery attacks against the initial bytes of a stream by sniffing network traffic that occasionally relies on keys affected by the Invariance Weakness, and then using a brute-force approach involving LSB values, aka the "Bar Mitzvah" issue
- The passwords used during an SSH session encrypted with RC4 can be recovered by an attacker who is able to capture and replay the session
- Error handling in the SSH protocol; Client and Server, when using a block cipher algorithm in Cipher Block Chaining (CBC) mode, makes it easier for remote attackers to recover certain plaintext data from an arbitrary block of ciphertext in an SSH session via unknown vectors

Audit:

Run the following command and verify the output:

```
# sshd -T -C user=root -C host="$(hostname)" -C addr="$(grep $(hostname) /etc/hosts | awk '{print $1}')" | grep -Ei '^\\s*ciphers\\s+([\\^#]+,)?(3des-cbc|aes128-cbc|aes192-cbc|aes256-cbc|arcfour|arcfour128|arcfour256|blowfish-cbc|cast128-cbc|rijndael-cbc@lysator.liu.se)\\b'
```

Nothing should be returned

Run the following command and verify the output:

```
grep -Ei '^\\s*ciphers\\s+([\\^#]+,)?(3des-cbc|aes128-cbc|aes192-cbc|aes256-cbc|arcfour|arcfour128|arcfour256|blowfish-cbc|cast128-cbc|rijndael-cbc@lysator.liu.se)\\b' /etc/ssh/sshd_config
```

Nothing should be returned

Remediation:

Edit the `/etc/ssh/sshd_config` file add/modify the `Ciphers` line to contain a comma separated list of the site approved ciphers

Example:

```
Ciphers chacha20-poly1305@openssh.com,aes256-gcm@openssh.com,aes128-gcm@openssh.com,aes256-ctr,aes192-ctr,aes128-ctr
```

Default Value:

Ciphers chacha20-poly1305@openssh.com,aes128-ctr,aes192-ctr,aes256-ctr,aes128-gcm@openssh.com,aes256-gcm@openssh.com,aes128-cbc,aes192-cbc,aes256-cbc,blowfish-cbc,cast128-cbc,3des-cbc

References:

1. <https://nvd.nist.gov/vuln/detail/CVE-2016-2183>
2. <https://nvd.nist.gov/vuln/detail/CVE-2015-2808>
3. <https://www.kb.cert.org/vuls/id/565052>
4. <https://www.openssh.com/txt/cbc.adv>
5. <https://nvd.nist.gov/vuln/detail/CVE-2008-5161>
6. <https://nvd.nist.gov/vuln/detail/CVE-2013-4548>
7. <https://www.kb.cert.org/vuls/id/565052>
8. <https://www.openssh.com/txt/cbc.adv>
9. SSHD_CONFIG(5)

Additional Information:

Weak Ciphers:

```
3des-cbc
aes128-cbc
aes192-cbc
aes256-cbc
arcfour
arcfour128
arcfour256
blowfish-cbc
cast128-cbc
rijndael-cbc@lysator.liu.se
```

Ciphers supported by openSSH v7.4p1:

```
3des-cbc
aes128-cbc
aes192-cbc
aes256-cbc
aes128-ctr
aes192-ctr
aes256-ctr
aes128-gcm@openssh.com
aes256-gcm@openssh.com
arcfour
arcfour128
arcfour256
blowfish-cbc
cast128-cbc
chacha20-poly1305@openssh.com
```

Ciphers currently FIPS 140-2 approved:

```
aes256-gcm@openssh.com
aes128-gcm@openssh.com
aes256-ctr
aes192-ctr
aes128-ctr
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.10 <u>Encrypt Sensitive Data in Transit</u> Encrypt sensitive data in transit. Example implementations can include: Transport Layer Security (TLS) and Open Secure Shell (OpenSSH).			
v7	14.4 <u>Encrypt All Sensitive Information in Transit</u> Encrypt all sensitive information in transit.			

5.3.14 Ensure only strong MAC algorithms are used (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

This variable Specifies the available MAC (message authentication code) algorithms. The MAC algorithm is used in protocol version 2 for data integrity protection. Multiple algorithms must be comma-separated.

Note: Some organizations may have stricter requirements for approved MACs. Ensure that MACs used are in compliance with site policy.

Rationale:

MD5 and 96-bit MAC algorithms are considered weak and have been shown to increase exploitability in SSH downgrade attacks. Weak algorithms continue to have a great deal of attention as a weak spot that can be exploited with expanded computing power. An attacker that breaks the algorithm could take advantage of a MiTM position to decrypt the SSH tunnel and capture credentials and information

Audit:

Run the following command and verify the output:

```
# sshd -T -C user=root -C host="$(hostname)" -C addr="$(grep $(hostname) /etc/hosts | awk '{print $1}')" | grep -Ei '^\s*macs\s+([\^#]+,)?(hmac-md5|hmac-md5-96|hmac-ripemd160|hmac-sha1|hmac-sha1-96|umac-64@openssh\.com|hmac-md5-etm@openssh\.com|hmac-md5-96-etm@openssh\.com|hmac-ripemd160-etm@openssh\.com|hmac-sha1-etm@openssh\.com|hmac-sha1-96-etm@openssh\.com|umac-64-etm@openssh\.com|umac-128-etm@openssh\.com)\b'
```

Nothing should be returned

Run the following command and verify the output:

```
# grep -Ei '^\s*macs\s+([\^#]+,)?(hmac-md5|hmac-md5-96|hmac-ripemd160|hmac-sha1|hmac-sha1-96|umac-64@openssh\.com|hmac-md5-etm@openssh\.com|hmac-md5-96-etm@openssh\.com|hmac-ripemd160-etm@openssh\.com|hmac-sha1-etm@openssh\.com|hmac-sha1-96-etm@openssh\.com|umac-64-etm@openssh\.com|umac-128-etm@openssh\.com)\b' /etc/ssh/sshd_config
```

Nothing should be returned

Remediation:

Edit the `/etc/ssh/sshd_config` file and add/modify the MACs line to contain a comma separated list of the site approved MACs

Example:

```
MACs hmac-sha2-512-etm@openssh.com,hmac-sha2-256-etm@openssh.com,hmac-sha2-512,hmac-sha2-256
```

Default Value:

MACs umac-64-etm@openssh.com,umac-128-etm@openssh.com,hmac-sha2-256-etm@openssh.com,hmac-sha2-512-etm@openssh.com,hmac-sha1-etm@openssh.com,umac-64@openssh.com,umac-128@openssh.com,hmac-sha2-256,hmac-sha2-512,hmac-sha1,hmac-sha1-etm@openssh.com

References:

1. More information on SSH downgrade attacks can be found here:
<http://www.mitls.org/pages/attacks/SLOTH>
2. SSHD_CONFIG(5)

Additional Information:

Weak MAC algorithms:

```
hmac-md5
hmac-md5-96
hmac-ripemd160
hmac-sha1
hmac-sha1-96
umac-64@openssh.com
umac-128@openssh.com
hmac-md5-etm@openssh.com
hmac-md5-96-etm@openssh.com
hmac-ripemd160-etm@openssh.com
hmac-sha1-etm@openssh.com
hmac-sha1-96-etm@openssh.com
umac-64-etm@openssh.com
umac-128-etm@openssh.com
```

MAC algorithms supported by openSSH v7.4p1:

```
hmac-md5
hmac-md5-96
hmac-ripemd160
hmac-sha1
hmac-sha1-96
hmac-sha2-256
hmac-sha2-512
umac-64@openssh.com
umac-128@openssh.com
hmac-md5-etm@openssh.com
hmac-md5-96-etm@openssh.com
hmac-ripemd160-etm@openssh.com
hmac-sha1-etm@openssh.com
hmac-sha1-96-etm@openssh.com
hmac-sha2-256-etm@openssh.com
hmac-sha2-512-etm@openssh.com
umac-64-etm@openssh.com
umac-128-etm@openssh.com
```

MACs algorithms currently FIPS 140-2 approved:

```
hmac-sha2-512-etm@openssh.com
hmac-sha2-256-etm@openssh.com
hmac-sha2-256
hmac-sha2-512
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.1 Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	<u>14.4 Encrypt All Sensitive Information in Transit</u> Encrypt all sensitive information in transit.			
v7	<u>16.5 Encrypt Transmittal of Username and Authentication Credentials</u> Ensure that all account usernames and authentication credentials are transmitted across networks using encrypted channels.			

5.3.15 Ensure only strong Key Exchange algorithms are used (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

Key exchange is any method in cryptography by which cryptographic keys are exchanged between two parties, allowing use of a cryptographic algorithm. If the sender and receiver wish to exchange encrypted messages, each must be equipped to encrypt messages to be sent and decrypt messages received

Note: Some organizations may have stricter requirements for approved Key Exchange algorithms. Ensure that Key Exchange algorithms used are in compliance with site policy.

Rationale:

Key exchange methods that are considered weak should be removed. A key exchange method may be weak because too few bits are used or the hashing algorithm is considered too weak. Using weak algorithms could expose connections to man-in-the-middle attacks

Audit:

Run the following command and verify the output:

```
# sshd -T -C user=root -C host="$(hostname)" -C addr="$(grep $(hostname) /etc/hosts | awk '{print $1}')" | grep -Ei '^\\s*kexalgorithms\\s+([\\^#]+,)?(diffie-hellman-group1-sha1|diffie-hellman-group14-sha1|diffie-hellman-group-exchange-sha1)\\b'
```

Nothing should be returned

Run the following command and verify the output:

```
# grep -Ei '^\\s*kexalgorithms\\s+([\\^#]+,)?(diffie-hellman-group1-sha1|diffie-hellman-group14-sha1|diffie-hellman-group-exchange-sha1)\\b' /etc/ssh/sshd_config
```

Nothing should be returned

Remediation:

Edit the `/etc/ssh/sshd_config` file add/modify the `KexAlgorithms` line to contain a comma separated list of the site approved key exchange algorithms

Example:

```
KexAlgorithms curve25519-sha256,curve25519-sha256@libssh.org,ecdh-sha2-nistp521,ecdh-sha2-nistp384,ecdh-sha2-nistp256,diffie-hellman-group-exchange-sha256
```

Default Value:

```
kexalgorithms curve25519-sha256,curve25519-sha256@libssh.org,ecdh-sha2-nistp256,ecdh-sha2-nistp384,ecdh-sha2-nistp521,diffie-hellman-group-exchange-sha256,diffie-hellman-group14-sha1,diffie-hellman-group1-sha1
```

References:

1. SSHD_CONFIG(5)

Additional Information:

Weak Key Exchange Algorithms:

```
diffie-hellman-group1-sha1
diffie-hellman-group14-sha1
diffie-hellman-group-exchange-sha1
```

Key Exchange algorithms supported by OpenSSH 7.4p1:

```
curve25519-sha256
curve25519-sha256@libssh.org
diffie-hellman-group1-sha1
diffie-hellman-group14-sha1
diffie-hellman-group-exchange-sha1
diffie-hellman-group-exchange-sha256
ecdh-sha2-nistp256
ecdh-sha2-nistp384
ecdh-sha2-nistp521
```

Key Exchange algorithms currently FIPS 140-2 approved:

```
ecdh-sha2-nistp256,ecdh-sha2-nistp384
ecdh-sha2-nistp521
diffie-hellman-group-exchange-sha256
diffie-hellman-group16-sha512
diffie-hellman-group18-sha512
diffie-hellman-group14-sha256
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 <u>Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	14.4 <u>Encrypt All Sensitive Information in Transit</u> Encrypt all sensitive information in transit.			

5.3.16 Ensure SSH Idle Timeout Interval is configured (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The two options `ClientAliveInterval` and `ClientAliveCountMax` control the timeout of ssh sessions.

- `ClientAliveInterval` sets a timeout interval in seconds after which if no data has been received from the client, sshd will send a message through the encrypted channel to request a response from the client. The default is 0, indicating that these messages will not be sent to the client.
- `ClientAliveCountMax` sets the number of client alive messages which may be sent without sshd receiving any messages back from the client. If this threshold is reached while client alive messages are being sent, sshd will disconnect the client, terminating the session. The default value is 3.
 - The client alive messages are sent through the encrypted channel
 - Setting `ClientAliveCountMax` to 0 disables connection termination

Example: The default value is 3. If `ClientAliveInterval` is set to 15, and `ClientAliveCountMax` is left at the default, unresponsive SSH clients will be disconnected after approximately 45 seconds

Rationale:

Having no timeout value associated with a connection could allow an unauthorized user access to another user's ssh session (e.g. user walks away from their computer and doesn't lock the screen). Setting a timeout value reduces this risk.

- The recommended `ClientAliveInterval` setting is no greater than 900 seconds (15 minutes)
- The recommended `ClientAliveCountMax` setting is 0
- At the 15 minute interval, if the ssh session is inactive, the session will be terminated.

Impact:

In some cases this setting may cause termination of long-running scripts over SSH or remote automation tools which rely on SSH. In developing the local site policy, the requirements of such scripts should be considered and appropriate `ServerAliveInterval` and `ClientAliveInterval` settings should be calculated to insure operational continuity.

Audit:

Run the following commands and verify `ClientAliveInterval` is between 1 and 900:

```
# sshd -T -C user=root -C host="$(hostname)" -C addr="$(grep $(hostname) /etc/hosts | awk '{print $1}')" | grep clientaliveinterval  
clientaliveinterval 900
```

Run the following command and verify `ClientAliveCountMax` is 0:

```
# sshd -T -C user=root -C host="$(hostname)" -C addr="$(grep $(hostname) /etc/hosts | awk '{print $1}')" | grep clientalivecountmax  
clientalivecountmax 3
```

Run the following commands and verify the output:

```
# grep -Ei '^\s*ClientAliveInterval\s+(0|9[0-9][1-9]|[1-9][0-9][0-9][0-9]+|1[6-9]m|[2-9][0-9]m|[1-9][0-9][0-9]+m)\b' /etc/ssh/sshd_config  
Nothing should be returned  
  
# grep -Ei '^\s*ClientAliveCountMax\s+([1-9]|[1-9][0-9]+)\b' /etc/ssh/sshd_config  
Nothing should be returned
```

Remediation:

Edit the `/etc/ssh/sshd_config` file to set the parameters according to site policy. This should include `ClientAliveInterval` between 1 and 900 and `ClientAliveCountMax` of 0:

```
ClientAliveInterval 900  
ClientAliveCountMax 0
```

Default Value:

`ClientAliveInterval` 0

`ClientAliveCountMax` 3

References:

1. https://man.openbsd.org/sshd_config

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 <u>Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	16.11 <u>Lock Workstation Sessions After Inactivity</u> Automatically lock workstation sessions after a standard period of inactivity.			

5.3.17 Ensure SSH LoginGraceTime is set to one minute or less (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `LoginGraceTime` parameter specifies the time allowed for successful authentication to the SSH server. The longer the Grace period is the more open unauthenticated connections can exist. Like other session controls in this session the Grace Period should be limited to appropriate organizational limits to ensure the service is available for needed access.

Rationale:

Setting the `LoginGraceTime` parameter to a low number will minimize the risk of successful brute force attacks to the SSH server. It will also limit the number of concurrent unauthenticated connections. While the recommended setting is 60 seconds (1 Minute), set the number based on site policy.

Audit:

Run the following command and verify that output `LoginGraceTime` is between 1 and 60 seconds or 1m:

```
# sshd -T -C user=root -C host="$(hostname)" -C addr="$(grep $(hostname) /etc/hosts | awk '{print $1}')" | grep logingracetime  
logingracetime 60
```

Run the following command and verify the output:

```
# grep -Ei '^s*LoginGraceTime\s+(0|6[1-9]|[7-9][0-9]|[1-9][0-9][0-9]+|^[^1]m)' /etc/ssh/sshd_config  
Nothing should be returned
```

Remediation:

Edit the `/etc/ssh/sshd_config` file to set the parameter as follows:

```
LoginGraceTime 60
```

Default Value:

LoginGraceTime 2m

References:

1. SSHD_CONFIG(5)

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 <u>Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	5.1 <u>Establish Secure Configurations</u> Maintain documented, standard security configuration standards for all authorized operating systems and software.			

5.3.18 Ensure SSH warning banner is configured (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `Banner` parameter specifies a file whose contents must be sent to the remote user before authentication is permitted. By default, no banner is displayed.

Rationale:

Banners are used to warn connecting users of the particular site's policy regarding connection. Presenting a warning message prior to the normal user login may assist the prosecution of trespassers on the computer system.

Audit:

Run the following command and verify that output matches:

```
# sshd -T -C user=root -C host="$(hostname)" -C addr="$(grep $(hostname) /etc/hosts | awk '{print $1}')" | grep banner  
banner /etc/issue.net
```

Remediation:

Edit the `/etc/ssh/sshd_config` file to set the parameter as follows:

```
Banner /etc/issue.net
```

References:

1. SSHD_CONFIG(5)

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.1 Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	<u>5.1 Establish Secure Configurations</u> Maintain documented, standard security configuration standards for all authorized operating systems and software.			

5.3.19 Ensure SSH PAM is enabled (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

UsePAM Enables the Pluggable Authentication Module interface. If set to “yes” this will enable PAM authentication using ChallengeResponseAuthentication and PasswordAuthentication in addition to PAM account and session module processing for all authentication types

Rationale:

When usePAM is set to yes, PAM runs through account and session types properly. This is important if you want to restrict access to services based off of IP, time or other factors of the account. Additionally, you can make sure users inherit certain environment variables on login or disallow access to the server

Impact:

If UsePAM is enabled, you will not be able to run sshd(5) as a non-root user.

Audit:

Run the following command and verify that output matches:

```
# sshd -T -C user=root -C host="$(hostname)" -C addr="$(grep $(hostname) /etc/hosts | awk '{print $1}')" | grep -i usepam  
usepam yes
```

Run the following command and verify the output:

```
# grep -Ei '^s*UsePAM\s+no' /etc/ssh/sshd_config  
Nothing should be returned
```

Remediation:

Edit the /etc/ssh/sshd_config file to set the parameter as follows:

```
UsePAM yes
```

Default Value:

usePAM yes

References:

1. SSHD_CONFIG(5)

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 <u>Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	5.1 <u>Establish Secure Configurations</u> Maintain documented, standard security configuration standards for all authorized operating systems and software.			

5.3.20 Ensure SSH AllowTcpForwarding is disabled (Automated)

Profile Applicability:

- Level 2 - Server
- Level 2 - Workstation

Description:

SSH port forwarding is a mechanism in SSH for tunneling application ports from the client to the server, or servers to clients. It can be used for adding encryption to legacy applications, going through firewalls, and some system administrators and IT professionals use it for opening backdoors into the internal network from their home machines

Rationale:

Leaving port forwarding enabled can expose the organization to security risks and backdoors.

SSH connections are protected with strong encryption. This makes their contents invisible to most deployed network monitoring and traffic filtering solutions. This invisibility carries considerable risk potential if it is used for malicious purposes such as data exfiltration. Cybercriminals or malware could exploit SSH to hide their unauthorized communications, or to exfiltrate stolen data from the target network

Impact:

SSH tunnels are widely used in many corporate environments that employ mainframe systems as their application backends. In those environments the applications themselves may have very limited native support for security. By utilizing tunneling, compliance with SOX, HIPAA, PCI-DSS, and other standards can be achieved without having to modify the applications.

Audit:

Run the following command and verify that output matches:

```
# sshd -T -C user=root -C host="$(hostname)" -C addr="$(grep $(hostname) /etc/hosts | awk '{print $1}')" | grep -i allowtcpforwarding  
allowtcpforwarding no
```

Run the following command and verify the output:

```
# grep -Ei '^s*AllowTcpForwarding\s+yes' /etc/ssh/sshd_config  
Nothing should be returned
```

Remediation:

Edit the /etc/ssh/sshd_config file to set the parameter as follows:

```
AllowTcpForwarding no
```

Default Value:

AllowTcpForwarding yes

References:

1. <https://www.ssh.com/ssh/tunneling/example>
2. SSHD_CONFIG(5)

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.1 Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	<u>9.2 Ensure Only Approved Ports, Protocols and Services Are Running</u> Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.			
v7	<u>13.5 Monitor and Detect Any Unauthorized Use of Encryption</u> Monitor all traffic leaving the organization and detect any unauthorized use of encryption.			

5.3.21 Ensure SSH MaxStartups is configured (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `MaxStartups` parameter specifies the maximum number of concurrent unauthenticated connections to the SSH daemon.

Rationale:

To protect a system from denial of service due to a large number of pending authentication connection attempts, use the rate limiting function of `MaxStartups` to protect availability of `sshd` logins and prevent overwhelming the daemon.

Audit:

Run the following command and verify that output `MaxStartups` is `10:30:60` or more restrictive:

```
# sshd -T -C user=root -C host="$(hostname)" -C addr="$(grep $(hostname) /etc/hosts | awk '{print $1}')" | grep -i maxstartups  
  
maxstartups 10:30:60
```

Run the following command and verify the output:

```
# grep -Ei '^\s*maxstartups\s+(((1[1-9]|[1-9][0-9][0-9]+):([0-9]+):([0-9]+))|((([0-9]+):(3[1-9]|[4-9][0-9]|[1-9][0-9][0-9]+):([0-9]+))|((([0-9]+):([0-9]+):(6[1-9]|[7-9][0-9]|[1-9][0-9][0-9]+)))' /etc/ssh/sshd_config  
  
Nothing should be returned
```

Remediation:

Edit the `/etc/ssh/sshd_config` file to set the parameter as follows:

```
maxstartups 10:30:60
```

Default Value:

`MaxStartups 10:30:100`

References:

1. SSHD_CONFIG(5)

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 <u>Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	5.1 <u>Establish Secure Configurations</u> Maintain documented, standard security configuration standards for all authorized operating systems and software.			

5.3.22 Ensure SSH MaxSessions is limited (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `MaxSessions` parameter Specifies the maximum number of open sessions permitted per network connection.

Rationale:

To protect a system from denial of service due to a large number of concurrent sessions, use the rate limiting function of `MaxSessions` to protect availability of `sshd` logins and prevent overwhelming the daemon.

Audit:

Run the following command and verify that output `MaxSessions` is 10 or less:

```
# sshd -T -C user=root -C host="$(hostname)" -C addr="$(grep $(hostname) /etc/hosts | awk '{print $1}')" | grep -i maxsessions  
maxsessions 10
```

Run the following command and verify the output:

```
grep -Ei '^s*MaxSessions\s+(1[1-9]|[2-9][0-9]|[1-9][0-9][0-9]+) '  
/etc/ssh/sshd_config  
Nothing should be returned
```

Remediation:

Edit the `/etc/ssh/sshd_config` file to set the parameter as follows:

```
MaxSessions 10
```

Default Value:

`MaxSessions 10`

References:

1. SSHD_CONFIG(5)

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 <u>Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	5.1 <u>Establish Secure Configurations</u> Maintain documented, standard security configuration standards for all authorized operating systems and software.			

5.4 Configure PAM

PAM (Pluggable Authentication Modules) is a service that implements modular authentication modules on UNIX systems. PAM is implemented as a set of shared objects that are loaded and executed when a program needs to authenticate a user. Files for PAM are typically located in the `/etc/pam.d` directory. PAM must be carefully configured to secure system authentication. While this section covers some of PAM, please consult other PAM resources to fully understand the configuration capabilities.

5.4.1 Ensure password creation requirements are configured (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `pam_pwquality.so` module checks the strength of passwords. It performs checks such as making sure a password is not a dictionary word, it is a certain length, contains a mix of characters (e.g. alphabet, numeric, other) and more. The following are definitions of the `pam_pwquality.so` options.

The following options are set in the `/etc/security/pwquality.conf` file:

Password Length:

- `minlen = 14` - password must be 14 characters or more

Password complexity:

- `minclass = 4` - The minimum number of required classes of characters for the new password (digits, uppercase, lowercase, others)
OR
- `dcredit = -1` - provide at least one digit
- `ucredit = -1` - provide at least one uppercase character
- `ocredit = -1` - provide at least one special character
- `lcredit = -1` - provide at least one lowercase character

The following is set in the `/etc/pam.d/password-auth` and `/etc/pam.d/system-auth` files

- `try_first_pass` - retrieve the password from a previous stacked PAM module. If not available, then prompt the user for a password.
- `retry=3` - Allow 3 tries before sending back a failure.

The settings shown above are one possible policy. Alter these values to conform to your own organization's password policies.

Notes:

- *Settings in `/etc/security/pwquality.conf` must use spaces around the `=` symbol.*
- *Additional modules options may be set in the `/etc/pam.d/password-auth` and `/etc/pam.d/system-auth` files*

Rationale:

Strong passwords and limited attempts before locking an account protect systems from being hacked through brute force methods.

Audit:

Verify password creation requirements conform to organization policy.

Run the following command to verify the minimum password length is 14 or more characters.

```
# grep '^s*minlen\s*' /etc/security/pwquality.conf  
minlen = 14
```

Run one of the following commands to verify the required password complexity:

```
# grep '^s*minclass\s*' /etc/security/pwquality.conf  
minclass = 4
```

OR

```
# grep -E '^s*[duol]credit\s*' /etc/security/pwquality.conf  
dcredit = -1  
ucredit = -1  
lcredit = -1  
ocredit = -1
```

Run the following commands to verify the files: /etc/pam.d/password-auth and /etc/pam.d/system-auth include try_first_pass and retry=3 on the password requisite pam_pwquality.so line.

```
# grep -P  
'^s*password\s+(?:requisite|required)\s+pam_pwquality\.so\s+(?:\S+\s+)*(?!2  
) (retry=[1-3]|try_first_pass)\s+(?:\S+\s+)*(?!1) (retry=[1-  
3]|try_first_pass)\s*(?:\s+\S+\s+)*(?:\s+#.*)?$' /etc/pam.d/password-auth  
  
password      requisite      pam_pwquality.so try_first_pass retry=3  
  
# grep -P  
'^s*password\s+(?:requisite|required)\s+pam_pwquality\.so\s+(?:\S+\s+)*(?!2  
) (retry=[1-3]|try_first_pass)\s+(?:\S+\s+)*(?!1) (retry=[1-  
3]|try_first_pass)\s*(?:\s+\S+\s+)*(?:\s+#.*)?$' /etc/pam.d/system-auth  
  
password      requisite      pam_pwquality.so try_first_pass retry=3
```

Remediation:

Edit the file `/etc/security/pwquality.conf` and add or modify the following line for password length to conform to site policy

```
minlen = 14
```

Edit the file `/etc/security/pwquality.conf` and add or modify the following line for password complexity to conform to site policy

```
minclass = 4
```

OR

```
dcredit = -1  
ucredit = -1  
ocredit = -1  
lcredit = -1
```

Edit the `/etc/pam.d/password-auth` and `/etc/pam.d/system-auth` files to include the appropriate options for `pam_pwquality.so` and to conform to site policy:

```
password requisite pam_pwquality.so try_first_pass retry=3
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	5.2 <u>Use Unique Passwords</u> Use unique passwords for all enterprise assets. Best practice implementation includes, at a minimum, an 8-character password for accounts using MFA and a 14-character password for accounts not using MFA.			
v7	4.4 <u>Use Unique Passwords</u> Where multi-factor authentication is not supported (such as local administrator, root, or service accounts), accounts will use passwords that are unique to that system.			

5.4.2 Ensure logout for failed password attempts is configured (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

Lock out users after *n* unsuccessful consecutive login attempts.

These settings are commonly configured with the `pam_faillock.so` module. Some environments may continue using the `pam_tally2.so` module, where this older method may simplify automation in mixed environments.

Set the lockout number in `deny=` to the policy in effect at your site.

`unlock_time=_n_` is the number of seconds the account remains locked after the number of attempts configured in `deny=_n_` has been met.

Notes:

- *Additional module options may be set, recommendation only covers those listed here.*
- *When modifying authentication configuration using the `authconfig` utility, the `system-auth` and `password-auth` files are overwritten with the settings from the `authconfig` utility. This can be avoided by creating symbolic links in place of the configuration files, which `authconfig` recognizes and does not overwrite. These symbolic links are the default for Fedora 19 derived distributions.*
- *Use of the "audit" keyword may log credentials in the case of user error during authentication. This risk should be evaluated in the context of the site policies of your organization.*
- *If a user has been locked out because they have reached the maximum consecutive failure count defined by `deny=` in the `pam_faillock.so` or the `pam_tally2.so` module, the user can be unlocked by issuing following commands. This command sets the failed count to 0, effectively unlocking the user.*
 - *If `pam_faillock.so` is used:*
`# faillock --user <username> --reset`
 - *If `pam_tally2.so` is used:*
`# pam_tally2 -u <username> --reset`

Rationale:

Locking out user IDs after n unsuccessful consecutive login attempts mitigates brute force password attacks against your systems.

Audit:

Verify password lockouts are configured. Ensure that the `deny=_n_` follows local site policy. This should not exceed `deny=5`.

If `pam_faillock.so` is used:

Run the following commands:

```
# grep -E '^\\s*auth\\s+\\S+\\s+pam_(faillock|unix)\\.so' /etc/pam.d/system-auth
/etc/pam.d/password-auth
```

Verify the output includes the following lines:

```
/etc/pam.d/system-auth:auth          required          pam_faillock.so preauth
silent audit deny=5 unlock_time=900
/etc/pam.d/system-auth:auth          sufficient       pam_unix.so nullok
try_first_pass
/etc/pam.d/system-auth:auth          [default=die]    pam_faillock.so authfail
audit deny=5 unlock_time=900
/etc/pam.d/password-auth:auth        required          pam_faillock.so preauth
silent audit deny=5 unlock_time=900
/etc/pam.d/password-auth:auth        sufficient       pam_unix.so nullok
try_first_pass
/etc/pam.d/password-auth:auth        [default=die]    pam_faillock.so authfail
audit deny=5 unlock_time=900

# grep -E '^\\s*account\\s+required\\s+pam_faillock.so\\s*' /etc/pam.d/system-
auth /etc/pam.d/password-auth
```

Verify the output includes the following lines:

```
/etc/pam.d/system-auth:account        required          pam_faillock.so
/etc/pam.d/password-auth:account      required          pam_faillock.so
```

OR

If `pam_tally2.so` is used:

Run the following commands:

```
# grep -E '^\\s*auth\\s+\\S+\\s+pam_(tally2|unix)\\.so' /etc/pam.d/system-auth
/etc/pam.d/password-auth
```

Verify the output includes the following lines:

```

/etc/pam.d/system-auth:auth          required      pam_tally2.so deny=5
onerr=fail unlock_time=900
/etc/pam.d/system-auth:auth          sufficient    pam_unix.so nullok
try_first_pass
/etc/pam.d/password-auth:auth        required      pam_tally2.so deny=5
onerr=fail unlock_time=900
/etc/pam.d/password-auth:auth        sufficient    pam_unix.so nullok
try_first_pass

# grep -E '^\\s*account\\s+required\\s+pam_tally2.so\\s*' /etc/pam.d/system-auth
/etc/pam.d/password-auth

```

Verify the output includes the following lines:

```

/etc/pam.d/system-auth:account        required      pam_tally2.so
/etc/pam.d/password-auth:account      required      pam_tally2.so

```

Remediation:

Edit the files `/etc/pam.d/system-auth` and `/etc/pam.d/password-auth` and add the following lines:

Modify the `deny=` and `unlock_time=` parameters to conform to local site policy, Not to be greater than `deny=5`

To use `pam_faillock.so` module, add the following lines to the `auth` section:

```

auth          required      pam_faillock.so preauth silent audit deny=5
unlock_time=900
auth          [default=die] pam_faillock.so authfail audit deny=5
unlock_time=900

```

The `auth` sections should look similar to the following example:

Note: The ordering on the lines in the `auth` section is important. The `preauth` line needs to below the line `auth required pam_env.so` and above all password validation lines. The `authfail` line needs to be after all password validation lines such as `pam_sss.so`. Incorrect order can cause you to be locked out of the system

Example:

```

auth          required      pam_env.so
auth          required      pam_faillock.so preauth silent audit deny=5
unlock_time=900 # <- Under "auth required pam_env.so"
auth          sufficient    pam_unix.so nullok try_first_pass
auth          [default=die] pam_faillock.so authfail audit deny=5
unlock_time=900 # <- Last auth line before "auth requisite
pam_succeed_if.so"
auth          requisite      pam_succeed_if.so uid >= 1000 quiet_success
auth          required      pam_deny.so

```

Add the following line to the `account` section:

account	required	pam_faillock.so
---------	----------	-----------------

Example:

account	required	pam_faillock.so
account	required	pam_unix.so
account	sufficient	pam_localuser.so
account	sufficient	pam_pam_succeed_if.so uid < 1000 quiet
account	required	pam_permit.so

OR

To use the `pam_tally2.so` module, add the following line to the `auth` section:

auth	required	pam_tally2.so deny=5 onerr=fail unlock_time=900
------	----------	---

The `auth` sections should look similar to the following example:

Note: The ordering on the lines in the `auth` section is important. the additional line needs to below the line `auth required pam_env.so` and above all password validation lines.

Example:

auth	required	pam_env.so
auth	required	pam_tally2.so deny=5 onerr=fail unlock_time=900 #
<- Under "auth required pam_env.so"		
auth	sufficient	pam_unix.so nullok try_first_pass
auth	requisite	pam_succeed_if.so uid >= 1000 quiet_success
auth	required	pam_deny.so

Add the following line to the `account` section:

account	required	pam_tally2.so
---------	----------	---------------

Example:

account	required	pam_tally2.so
account	required	pam_unix.so
account	sufficient	pam_localuser.so
account	sufficient	pam_pam_succeed_if.so uid < 1000 quiet
account	required	pam_permit.so

References:

1. https://access.redhat.com/documentation/en-us/red_hat_enterprise_linux/7/html/security_guide

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.1 Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	<u>16.7 Establish Process for Revoking Access</u> Establish and follow an automated process for revoking system access by disabling accounts immediately upon termination or change of responsibilities of an employee or contractor . Disabling these accounts, instead of deleting accounts, allows preservation of audit trails.			

5.4.3 Ensure password hashing algorithm is SHA-512 (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The commands below change password encryption from `md5` to `sha512` (a much stronger hashing algorithm). All existing accounts will need to perform a password change to upgrade the stored hashes to the new algorithm.

Note:

- *These changes only apply to accounts configured on the local system.*
- *Additional module options may be set, recommendation only covers those listed here.*

Rationale:

The SHA-512 algorithm provides much stronger hashing than MD5, thus providing additional protection to the system by increasing the level of effort for an attacker to successfully determine passwords.

Audit:

Run the following command to verify the `sha512` option is included:

```
# grep -P
'^\h*password\h+(sufficient|requisite|required)\h+pam_unix\.so\h+([\#\n\r]+)?
sha512(\h+.)?$', /etc/pam.d/system-auth /etc/pam.d/password-auth

/etc/pam.d/system-auth:password      sufficient      pam_unix.so sha512 shadow
nullok try_first_pass use_authok
/etc/pam.d/password-auth:password    sufficient      pam_unix.so sha512 shadow
nullok try_first_pass use_authok
```

Remediation:

Edit the `/etc/pam.d/password-auth` and `/etc/pam.d/system-auth` files to include `sha512` option and remove the `md5` option for `pam_unix.so`:

```
password sufficient pam_unix.so sha512
```

Note:

- Any system accounts that need to be expired should be carefully done separately by the system administrator to prevent any potential problems.
- If it is determined that the password algorithm being used is not SHA-512, once it is changed, it is recommended that all user ID's be immediately expired and forced to change their passwords on next login, In accordance with local site policies.
- To accomplish this, the following command can be used.
 - This command intentionally does not affect the root account. **The root account's password will also need to be changed.**

```
# awk -F: ' ( $3<"$(awk '/^\s*UID_MIN/{print $2}' /etc/login.defs)"' && $1 !~  
/^(nfs)?nobody$/ && $1 != "root" ) { print $1 }' /etc/passwd | xargs -n 1  
chage -d 0
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.11 <u>Encrypt Sensitive Data at Rest</u> Encrypt sensitive data at rest on servers, applications, and databases containing sensitive data. Storage-layer encryption, also known as server-side encryption, meets the minimum requirement of this Safeguard. Additional encryption methods may include application-layer encryption, also known as client-side encryption, where access to the data storage device(s) does not permit access to the plain-text data.			
v7	16.4 <u>Encrypt or Hash all Authentication Credentials</u> Encrypt or hash with a salt all authentication credentials when stored.			

5.4.4 Ensure password reuse is limited (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `/etc/security/opasswd` file stores the users' old passwords and can be checked to ensure that users are not recycling recent passwords.

Note: Additional module options may be set, recommendation only covers those listed here.

Rationale:

Forcing users not to reuse their past 5 passwords make it less likely that an attacker will be able to guess the password.

Audit:

Verify remembered password history follows local site policy, not to be less than 5.

If `pam_pwhistory.so` is used:

Run the following command:

```
# grep -P
'^\s*password\s+(requisite|required)\s+pam_pwhistory\.so\s+([\^#]+\s+)*remeber=([5-9]|[1-9][0-9]+)\b' /etc/pam.d/system-auth /etc/pam.d/password-auth

/etc/pam.d/system-auth:password      required      pam_pwhistory.so remember=5
/etc/pam.d/password-auth:password    required      pam_pwhistory.so
remember=5
```

OR *If `pam_unix.so` is used:*

Run the following command:

```
# grep -P
'^\s*password\s+(sufficient|required)\s+pam_unix\.so\s+([\^#]+\s+)*remember=([5-9]|[1-9][0-9]+)\b' /etc/pam.d/system-auth /etc/pam.d/password-auth

/etc/pam.d/system-auth:password      sufficient    pam_unix.so remember=5
/etc/pam.d/password-auth:password    sufficient    pam_unix.so remember=5
```

Remediation:

Edit **both** the `/etc/pam.d/password-auth` and `/etc/pam.d/system-auth` files to include the `remember` option and conform to site policy as shown:

Note: Add or modify the line containing the `pam_pwhistory.so` **after the first occurrence of `password requisite`**:

```
password    required    pam_pwhistory.so remember=5
```

Example: (Second line is modified)

```
password    requisite    pam_pwquality.so try_first_pass local_users_only
authtok_type=
password    required    pam_pwhistory.so use_authtok remember=5 retry=3
password    sufficient    pam_unix.so sha512 shadow nullok try_first_pass
use_authtok
password    required    pam_deny.so
```

Additional Information:

- This setting only applies to local accounts.
- This option is configured with the `remember=n` module option in `/etc/pam.d/system-auth` and `/etc/pam.d/password-auth`
- This option can be set with either **one** of the two following modules:
 - `pam_pwhistory.so` - This is the newer recommended method included in the remediation section.
 - `pam_unix.so` - This is the *older* method, and is included in the audit to account for legacy configurations.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	5.2 <u>Use Unique Passwords</u> Use unique passwords for all enterprise assets. Best practice implementation includes, at a minimum, an 8-character password for accounts using MFA and a 14-character password for accounts not using MFA.			
v7	16 <u>Account Monitoring and Control</u> Account Monitoring and Control			

5.5 User Accounts and Environment

This section provides guidance on setting up secure defaults for system and user accounts and their environment.

5.5.1 Set Shadow Password Suite Parameters

While a majority of the password control parameters have been moved to PAM, some parameters are still available through the shadow password suite. Any changes made to `/etc/login.defs` will only be applied if the `usermod` command is used. If user IDs are added a different way, use the `chage` command to effect changes to individual user IDs.

5.5.1.1 Ensure password expiration is 365 days or less (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `PASS_MAX_DAYS` parameter in `/etc/login.defs` allows an administrator to force passwords to expire once they reach a defined age. It is recommended that the `PASS_MAX_DAYS` parameter be set to less than or equal to 365 days.

Notes:

- *A value of -1 will disable password expiration.*
- *The password expiration must be greater than the minimum days between password changes or users will be unable to change their password.*

Rationale:

The window of opportunity for an attacker to leverage compromised credentials via a brute force attack, using already compromised credentials, or gaining the credentials by other means, can be limited by the age of the password. Therefore, reducing the maximum age of a password can also reduce an attacker's window of opportunity.

Requiring passwords to be changed helps to mitigate the risk posed by the poor security practice of passwords being used for multiple accounts, and poorly implemented offboarding and change of responsibility policies. This should **not** be considered a replacement for proper implementation of these policies and practices.

Note: If it is believed that a user's password may have been compromised, the user's account should be locked immediately. Local policy should be followed to ensure the secure update of their password.

Audit:

Run the following command and verify `PASS_MAX_DAYS` conforms to site policy (no more than 365 days):

```
# grep ^\s*PASS_MAX_DAYS /etc/login.defs
PASS_MAX_DAYS 365
```

Run the following command and Review list of users and `PASS_MAX_DAYS` to verify that all users' `PASS_MAX_DAYS` conforms to site policy (no more than 365 days):

```
# grep -E '^[^:]+:[^!]*' /etc/shadow | cut -d: -f1,5
<user>:<PASS_MAX_DAYS>
```

Remediation:

Set the `PASS_MAX_DAYS` parameter to conform to site policy in `/etc/login.defs` :

```
PASS_MAX_DAYS 365
```

Modify user parameters for all users with a password set to match:

```
# chage --maxdays 365 <user>
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 Establish and Maintain a Secure Configuration Process Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	4.4 Use Unique Passwords Where multi-factor authentication is not supported (such as local administrator, root, or service accounts), accounts will use passwords that are unique to that system.			

5.5.1.2 Ensure minimum days between password changes is configured (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `PASS_MIN_DAYS` parameter in `/etc/login.defs` allows an administrator to prevent users from changing their password until a minimum number of days have passed since the last time the user changed their password. It is recommended that `PASS_MIN_DAYS` parameter be set to 1 or more days.

Rationale:

By restricting the frequency of password changes, an administrator can prevent users from repeatedly changing their password in an attempt to circumvent password reuse controls.

Audit:

Run the following command and verify `PASS_MIN_DAYS` conforms to site policy (no less than 1 day):

```
# grep ^\s*PASS_MIN_DAYS /etc/login.defs
PASS_MIN_DAYS 1
```

Run the following command and Review list of users and `PAS_MIN_DAYS` to Verify that all users' `PAS_MIN_DAYS` conforms to site policy (no less than 1 day):

```
# grep -E ^\[^\]:\]+:[^\!*\] /etc/shadow | cut -d: -f1,4
<user>:<PASS_MIN_DAYS>
```

Remediation:

Set the `PASS_MIN_DAYS` parameter to 1 in `/etc/login.defs` :

```
PASS_MIN_DAYS 1
```

Modify user parameters for all users with a password set to match:

```
# chage --mindays 1 <user>
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 <u>Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	4.4 <u>Use Unique Passwords</u> Where multi-factor authentication is not supported (such as local administrator, root, or service accounts), accounts will use passwords that are unique to that system.			

5.5.1.3 Ensure password expiration warning days is 7 or more (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `PASS_WARN_AGE` parameter in `/etc/login.defs` allows an administrator to notify users that their password will expire in a defined number of days. It is recommended that the `PASS_WARN_AGE` parameter be set to 7 or more days.

Rationale:

Providing an advance warning that a password will be expiring gives users time to think of a secure password. Users caught unaware may choose a simple password or write it down where it may be discovered.

Audit:

Run the following command and verify `PASS_WARN_AGE` conforms to site policy (No less than 7 days):

```
# grep ^\s*PASS_WARN_AGE /etc/login.defs
PASS_WARN_AGE 7
```

Verify all users with a password have their number of days of warning before password expires set to 7 or more:

Run the following command and Review list of users and `PASS_WARN_AGE` to verify that all users' `PASS_WARN_AGE` conforms to site policy (No less than 7 days):

```
# grep -E ^[^:]+:[^!*] /etc/shadow | cut -d: -f1,6
<user>:<PASS_WARN_AGE>
```

Remediation:

Set the `PASS_WARN_AGE` parameter to 7 in `/etc/login.defs` :

```
PASS_WARN_AGE 7
```

Modify user parameters for all users with a password set to match:

```
# chage --warndays 7 <user>
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 <u>Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	4.4 <u>Use Unique Passwords</u> Where multi-factor authentication is not supported (such as local administrator, root, or service accounts), accounts will use passwords that are unique to that system.			

5.5.1.4 Ensure inactive password lock is 30 days or less (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

User accounts that have been inactive for over a given period of time can be automatically disabled. It is recommended that accounts that are inactive for 30 days after password expiration be disabled.

Note: A value of -1 would disable this setting.

Rationale:

Inactive accounts pose a threat to system security since the users are not logging in to notice failed login attempts or other anomalies.

Audit:

Run the following command and verify `INACTIVE` conforms to site policy (no more than 30 days):

```
# useradd -D | grep INACTIVE  
  
INACTIVE=30
```

Verify all users with a password have Password inactive no more than 30 days after password expires:

Run the following command and Review list of users and `INACTIVE` to verify that all users' `INACTIVE` conforms to site policy (no more than 30 days):

```
# grep -E ^[^:]+:[^!*] /etc/shadow | cut -d: -f1,7  
  
<user>:<INACTIVE>
```

Remediation:

Run the following command to set the default password inactivity period to 30 days:

```
# useradd -D -f 30
```

Modify user parameters for all users with a password set to match:

```
# chage --inactive 30 <user>
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 <u>Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	16.9 <u>Disable Dormant Accounts</u> Automatically disable dormant accounts after a set period of inactivity.			

5.5.1.5 Ensure all users last password change date is in the past (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

All users should have a password change date in the past.

Rationale:

If a users recorded password change date is in the future then they could bypass any set password expiration.

Audit:

Run the following command and verify nothing is returned

```
# for usr in $(cut -d: -f1 /etc/shadow); do [[ $(chage --list $usr | grep '^Last password change' | cut -d: -f2) > $(date) ]] && echo "$usr :$(chage --list $usr | grep '^Last password change' | cut -d: -f2)"; done
```

Remediation:

Investigate any users with a password change date in the future and correct them. Locking the account, expiring the password, or resetting the password manually may be appropriate.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	5.2 Use Unique Passwords Use unique passwords for all enterprise assets. Best practice implementation includes, at a minimum, an 8-character password for accounts using MFA and a 14-character password for accounts not using MFA.			
v7	4.4 Use Unique Passwords Where multi-factor authentication is not supported (such as local administrator, root, or service accounts), accounts will use passwords that are unique to that system.			

5.5.2 Ensure system accounts are secured (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

There are a number of accounts provided with most distributions that are used to manage applications and are not intended to provide an interactive shell.

Rationale:

It is important to make sure that accounts that are not being used by regular users are prevented from being used to provide an interactive shell. By default, most distributions set the password field for these accounts to an invalid string, but it is also recommended that the shell field in the password file be set to the `nologin` shell. This prevents the account from potentially being used to run any commands.

Note: The `root`, `sync`, `shutdown`, and `halt` users are exempted from requiring a non-login shell.

Audit:

Run the following commands and verify no results are returned:

```
awk -F: '($1!="root" && $1!="sync" && $1!="shutdown" && $1!="halt" && $1!~/^\/ && $3<"$(awk '/^\s*UID_MIN/{print $2}' /etc/login.defs)" && $7!="$(which nologin)" && $7!="bin/false") {print}' /etc/passwd

awk -F: '($1!="root" && $1!~/^\/ && $3<"$(awk '/^\s*UID_MIN/{print $2}' /etc/login.defs)"') {print $1}' /etc/passwd | xargs -I '{}' passwd -S '{}' |
awk '($2!="L" && $2!="LK") {print $1}'
```

Remediation:

Run the commands appropriate for your distribution:

Set the shell for any accounts returned by the audit to nologin:

```
# usermod -s $(which nologin) <user>
```

Lock any non root accounts returned by the audit:

```
# usermod -L <user>
```

The following command will set all system accounts to a non login shell:

```
awk -F: '($1!="root" && $1!="sync" && $1!="shutdown" && $1!="halt" && $1!~/^\/+ && $3<"$(awk '/^\s*UID_MIN/{print $2}' /etc/login.defs)"' && $7!="$(which nologin)"' && $7!="bin/false" && $7!="usr/bin/false") {print $1}' /etc/passwd | while read -r user; do usermod -s "$(which nologin)" "$user"; done
```

The following command will automatically lock not root system accounts:

```
awk -F: '($1!="root" && $1!~/^\/+ && $3<"$(awk '/^\s*UID_MIN/{print $2}' /etc/login.defs)"') {print $1}' /etc/passwd | xargs -I '{}' passwd -S '{}' | awk '($2!="L" && $2!="LK") {print $1}' | while read -r user; do usermod -L "$user"; done
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 <u>Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v7	16 <u>Account Monitoring and Control</u> Account Monitoring and Control			

5.5.3 Ensure default group for the root account is GID 0 (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `usermod` command can be used to specify which group the root user belongs to. This affects permissions of files that are created by the root user.

Rationale:

Using GID 0 for the `root` account helps prevent `root` -owned files from accidentally becoming accessible to non-privileged users.

Audit:

Run the following command and verify the result is 0 :

```
# grep "^root:" /etc/passwd | cut -f4 -d:
0
```

Remediation:

Run the following command to set the `root` user default group to GID 0 :

```
# usermod -g 0 root
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.1 Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	<u>5.1 Establish Secure Configurations</u> Maintain documented, standard security configuration standards for all authorized operating systems and software.			

5.5.4 Ensure default user shell timeout is configured (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

`TMOUT` is an environmental setting that determines the timeout of a shell in seconds.

- `TMOUT=n` - Sets the shell timeout to *n* seconds. A setting of `TMOUT=0` disables timeout.
- `readonly TMOUT` - Sets the `TMOUT` environmental variable as readonly, preventing unwanted modification during run-time.
- `export TMOUT` - exports the `TMOUT` variable

System Wide Shell Configuration Files:

- `/etc/profile` - used to set system wide environmental variables on users shells. The variables are sometimes the same ones that are in the `.bash_profile`, however this file is used to set an initial `PATH` or `PS1` for all shell users of the system. **is only executed for interactive *login* shells, or shells executed with the `--login` parameter.**
- `/etc/profile.d` - `/etc/profile` will execute the scripts within `/etc/profile.d/*.sh`. It is recommended to place your configuration in a shell script within `/etc/profile.d` to set your own system wide environmental variables.
- `/etc/bashrc` - System wide version of `.bashrc`. In Fedora derived distributions, `etc/bashrc` also invokes `/etc/profile.d/*.sh` if *non-login* shell, but redirects output to `/dev/null` if *non-interactive*. **Is only executed for *interactive* shells or if `BASH_ENV` is set to `/etc/bashrc`.**

Rationale:

Setting a timeout value reduces the window of opportunity for unauthorized user access to another user's shell session that has been left unattended. It also ends the inactive session and releases the resources associated with that session.

Audit:

Run the following script to verify that `TMOUT` is configured to: include a timeout of no more than 900 seconds, to be `readonly`, to be `exported`, and is not being changed to a longer timeout.

```
#!/bin/bash

output1="" output2=""
[ -f /etc/bashrc ] && BRC="/etc/bashrc"
for f in "$BRC" /etc/profile /etc/profile.d/*.sh ; do
    grep -Pq '^s*([^#]+\s+)?TMOUT=(900|[1-8][0-9][0-9]|[1-9][0-9]|[1-9])\b'
"$f" && grep -Pq '^s*([^#]+\s+)?readonly\s+TMOUT(\s+|\s*;\s*$|=(900|[1-
8][0-9][0-9]|[1-9][0-9]|[1-9]))\b' "$f" && grep -Pq
'^s*([^#]+\s+)?export\s+TMOUT(\s+|\s*;\s*$|=(900|[1-8][0-9][0-9]|[1-9][0-
9]|[1-9]))\b' "$f" && output1="$f"
done
grep -Pq '^s*([^#]+\s+)?TMOUT=(9[0-9][1-9]|9[1-9][0-9]|0+|[1-9]\d{3,})\b'
/etc/profile /etc/profile.d/*.sh "$BRC" && output2=$(grep -Ps
'^s*([^#]+\s+)?TMOUT=(9[0-9][1-9]|9[1-9][0-9]|0+|[1-9]\d{3,})\b'
/etc/profile /etc/profile.d/*.sh $BRC)
if [ -n "$output1" ] && [ -z "$output2" ]; then
    echo -e "\nPASSED\n\nTMOUT is configured in: \"$output1\"\n"
else
    [ -z "$output1" ] && echo -e "\nFAILED\n\nTMOUT is not configured\n"
    [ -n "$output2" ] && echo -e "\nFAILED\n\nTMOUT is incorrectly configured
in: \"$output2\"\n"
fi
```

Remediation:

Review `/etc/bashrc`, `/etc/profile`, and all files ending in `*.sh` in the `/etc/profile.d/` directory and remove or edit all `TMOUT=_n_` entries to follow local site policy. `TMOUT` should not exceed 900 or be equal to 0.

Configure `TMOUT` in **one** of the following files:

- A file in the `/etc/profile.d/` directory ending in `.sh`
- `/etc/profile`
- `/etc/bashrc`

TMOUT configuration examples:

- As multiple lines:

```
TMOUT=900
readonly TMOUT
export TMOUT
```

- As a single line:

```
readonly TMOUT=900 ; export TMOUT
```

Additional Information:

- The audit and remediation in this recommendation apply to bash and shell. If other shells are supported on the system, it is recommended that their configuration files are also checked. Other methods of setting a timeout exist for other shells not covered here.
- Ensure that the timeout conforms to your local policy.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.3 <u>Configure Automatic Session Locking on Enterprise Assets</u> Configure automatic session locking on enterprise assets after a defined period of inactivity. For general purpose operating systems, the period must not exceed 15 minutes. For mobile end-user devices, the period must not exceed 2 minutes.			
v7	16.11 <u>Lock Workstation Sessions After Inactivity</u> Automatically lock workstation sessions after a standard period of inactivity.			

5.5.5 Ensure default user umask is configured (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The user file-creation mode mask (`umask`) is use to determine the file permission for newly created directories and files. In Linux, the default permissions for any newly created directory is 0777 (`rw-rw-rw-`), and for any newly created file it is 0666 (`rw-rw-rw-`). The `umask` modifies the default Linux permissions by restricting (masking) these permissions. The `umask` is not simply subtracted, but is processed bitwise. Bits set in the `umask` are cleared in the resulting file mode.

`umask` can be set with either `octal` or `Symbolic` values:

- **Octal (Numeric) Value** - Represented by either three or four digits. ie `umask 0027` or `umask 027`. If a four digit `umask` is used, the first digit is ignored. The remaining three digits effect the resulting permissions for user, group, and world/other respectively.
- **Symbolic Value** - Represented by a comma separated list for User `u`, group `g`, and world/other `o`. The permissions listed are not masked by `umask`. ie a `umask` set by `umask u=rwx,g=rx,o=` is the `Symbolic` equivalent of the `Octal` `umask 027`. This `umask` would set a newly created directory with file mode `drwxr-x---` and a newly created file with file mode `rw-r-----`.

The default `umask` can be set to use the `pam_umask` module or in a `System Wide Shell Configuration File`. The user creating the directories or files has the discretion of changing the permissions via the `chmod` command, or choosing a different default `umask` by adding the `umask` command into a `User Shell Configuration File`, (`.bash_profile` or `.bashrc`), in their home directory.

Setting the default umask:

- **pam_umask module:**
 - will set the umask according to the system default in `/etc/login.defs` and user settings, solving the problem of different `umask` settings with different shells, display managers, remote sessions etc.
 - `umask=<mask>` value in the `/etc/login.defs` file is interpreted as Octal
 - **Setting USERGROUPS_ENAB to yes in `/etc/login.defs` (default):**
 - will enable setting of the `umask` group bits to be the same as owner bits. (examples: 022 -> 002, 077 -> 007) for non-root users, if the `uid` is the same as `gid`, and `username` is the same as the `<primary group name>`
 - `userdel` will remove the user's group if it contains no more members, and `useradd` will create by default a group with the name of the user
- **System Wide Shell Configuration File:**
 - `/etc/profile` - used to set system wide environmental variables on users shells. The variables are sometimes the same ones that are in the `.bash_profile`, however this file is used to set an initial `PATH` or `PS1` for all shell users of the system. **is only executed for interactive login shells, or shells executed with the `--login` parameter.**
 - `/etc/profile.d` - `/etc/profile` will execute the scripts within `/etc/profile.d/*.sh`. It is recommended to place your configuration in a shell script within `/etc/profile.d` to set your own system wide environmental variables.
 - `/etc/bashrc` - System wide version of `.bashrc`. In Fedora derived distributions, `etc/bashrc` also invokes `/etc/profile.d/*.sh` if *non-login* shell, but redirects output to `/dev/null` if *non-interactive*. **Is only executed for interactive shells or if `BASH_ENV` is set to `/etc/bashrc`.**

User Shell Configuration Files:

- `~/.bash_profile` - Is executed to configure your shell before the initial command prompt. **Is only read by login shells.**
- `~/.bashrc` - Is executed for interactive shells. **only read by a shell that's both interactive and non-login**

Rationale:

Setting a secure default value for `umask` ensures that users make a conscious choice about their file permissions. A permissive `umask` value could result in directories or files with excessive permissions that can be read and/or written to by unauthorized users.

Audit:

Run the following to verify:

- A default user `umask` is set to enforce a newly created directories' permissions to be 750 (`drwxr-x---`), and a newly created file's permissions be 640 (`rw-r-----`), or more restrictive
- No less restrictive System Wide `umask` is set

Run the following script to verify that a default user `umask` is set enforcing a newly created directories' permissions to be 750 (`drwxr-x---`), and a newly created file's permissions be 640 (`rw-r-----`), or more restrictive:

```
#!/bin/bash

passing=""
grep -Eiq '^\\s*UMASK\\s+(0[0-7][2-7]7|[0-7][2-7]7)\\b' /etc/login.defs && grep
-Eqi '^\\s*USERGROUPS_ENAB\\s*"?no"?\\b' /etc/login.defs && grep -Eq
'^\\s*session\\s+(optional|required)\\s+pam_umask\\.so\\b'
/etc/pam.d/common-session && passing=true
grep -REiq '^\\s*UMASK\\s+\\s*(0[0-7][2-7]7|[0-7][2-
7]7|u=(r?|w?|x?) (r?|w?|x?) (r?|w?|x?),g=(r?x?|x?r?),o=)\\b' /etc/profile*
/etc/bashrc* && passing=true
[ "$passing" = true ] && echo "Default user umask is set"
```

Verify output is: "Default user umask is set"

Run the following to verify that no less restrictive system wide `umask` is set:

```
# grep -RPi '^(^|^[^#]*)\\s*umask\\s+([0-7][0-7][01][0-7]\\b|[0-7][0-7][0-7][0-
6]\\b|[0-7][01][0-7]\\b|[0-7][0-7][0-
6]\\b|(u=[rx]{0,3},)?(g=[rx]{0,3},)?o=[rx]+\\b|(u=[rx]{1,3},)?g=[^rx]{1,3}(
,o=[rx]{0,3})?\\b)' /etc/login.defs /etc/profile* /etc/bashrc*

No file should be returned
```

Remediation:

Review `/etc/bashrc`, `/etc/profile`, and all files ending in `*.sh` in the `/etc/profile.d/` directory and remove or edit all `umask` entries to follow local site policy. Any remaining entries should be: `umask 027`, `umask u=rwx,g=rx,o=` or more restrictive.

Configure `umask` in **one** of the following files:

- A file in the `/etc/profile.d/` directory ending in `.sh`
- `/etc/profile`
- `/etc/bashrc`

Example:

```
# vi /etc/profile.d/set_umask.sh

umask 027
```

Run the following command and remove or modify the `umask` of any returned files:

```
# grep -RPi ' (^|^ [^#]*) \s*umask\s+ ([0-7] [0-7] [01] [0-7] \b| [0-7] [0-7] [0-7] [0-7] \b| [0-7] [01] [0-7] \b| [0-7] [0-7] [0-6] \b| (u=[rwx]{0,3},)? (g=[rwx]{0,3},)? o=[rwx]+\b| (u=[rwx]{1,3},)? g=[^rx]{1,3} (,o=[rwx]{0,3})?\b)' /etc/login.defs /etc/profile* /etc/bashrc*
```

Follow one of the following methods to set the default user `umask`:

Edit `/etc/login.defs` and edit the `UMASK` and `USERGROUPS_ENAB` lines as follows:

```
UMASK 027

USERGROUPS_ENAB no
```

Edit the files `/etc/pam.d/password-auth` and `/etc/pam.d/system-auth` and add or edit the following:

```
session      optional      pam_umask.so
```

OR Configure `umask` in one of the following files:

- A file in the `/etc/profile.d/` directory ending in `.sh`
- `/etc/profile`
- `/etc/bashrc`

Example: `/etc/profile.d/set_umask.sh`

```
umask 027
```

Note: this method only applies to `bash` and `shell`. If other shells are supported on the system, it is recommended that their configuration files also are checked.

Default Value:

UMASK 022

Additional Information:

- Other methods of setting a default user umask exist
- If other methods are in use in your environment they should be audited
- The default user umask can be overridden with a user specific umask
- The user creating the directories or files has the discretion of changing the permissions:
 - Using the chmod command
 - Setting a different default umask by adding the umask command into a User Shell Configuration File, (.bashrc), in their home directory
 - Manually changing the umask for the duration of a login session by running the umask command

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 <u>Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v7	5.1 <u>Establish Secure Configurations</u> Maintain documented, standard security configuration standards for all authorized operating systems and software.			
v7	13 <u>Data Protection</u> Data Protection			

5.6 Ensure root login is restricted to system console (Manual)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The file `/etc/securetty` contains a list of valid terminals that may be logged in directly as root.

Rationale:

Since the system console has special properties to handle emergency situations, it is important to ensure that the console is in a physically secure location and that unauthorized consoles have not been defined.

Audit:

```
# cat /etc/securetty
```

Remediation:

Remove entries for any consoles that are not in a physically secure location.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.1 Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v8	<u>4.6 Securely Manage Enterprise Assets and Software</u> Securely manage enterprise assets and software. Example implementations include managing configuration through version-controlled-infrastructure-as-code and accessing administrative interfaces over secure network protocols, such as Secure Shell (SSH) and Hypertext Transfer Protocol Secure (HTTPS). Do not use insecure management protocols, such as Telnet (Teletype Network) and HTTP, unless operationally essential.			
v7	<u>5.1 Establish Secure Configurations</u> Maintain documented, standard security configuration standards for all authorized operating systems and software.			

5.7 Ensure access to the su command is restricted (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `su` command allows a user to run a command or shell as another user. The program has been superseded by `sudo`, which allows for more granular control over privileged access. Normally, the `su` command can be executed by any user. By uncommenting the `pam_wheel.so` statement in `/etc/pam.d/su`, the `su` command will only allow users in a specific groups to execute `su`. This group should be empty to reinforce the use of `sudo` for privileged access.

Rationale:

Restricting the use of `su`, and using `sudo` in its place, provides system administrators better control of the escalation of user privileges to execute privileged commands. The `sudo` utility also provides a better logging and audit mechanism, as it can log each command executed via `sudo`, whereas `su` can only record that a user executed the `su` program.

Audit:

Run the following command and verify the output matches the line:

```
# grep -Pi
'^\h*auth\h+(?:required|requisite)\h+pam_wheel\.so\h+(?:[^\#\n\r]+\h+)?((?!\\2)
(use_uid\b|group=\H+\b))\h+(?:[^\#\n\r]+\h+)?((?!\\1)(use_uid\b|group=\H+\b))(\
h+.*?)?$' /etc/pam.d/su

auth required pam_wheel.so use_uid group=<group_name>
```

Run the following command and verify that the group specified in `<group_name>` contains no users:

```
# grep <group_name> /etc/group

<group_name>:x:<GID>:
```

There should be no users listed after the Group ID field.

Remediation:

Create an empty group that will be specified for use of the `su` command. The group should be named according to site policy.

Example:

```
# groupadd sugroup
```

Add the following line to the `/etc/pam.d/su` file, specifying the empty group:

```
auth required pam_wheel.so use_uid group=sugroup
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 <u>Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v7	5.1 <u>Establish Secure Configurations</u> Maintain documented, standard security configuration standards for all authorized operating systems and software.			

6 System Maintenance

Recommendations in this section are intended as maintenance and are intended to be checked on a frequent basis to ensure system stability. Many recommendations do not have quick remediations and require investigation into the cause and best fix available and may indicate an attempted breach of system security.

6.1 System File Permissions

This section provides guidance on securing aspects of system files and directories.

6.1.1 Audit system file permissions (Manual)

Profile Applicability:

- Level 2 - Server
- Level 2 - Workstation

Description:

The RPM package manager has a number of useful options. One of these, the `-v` option, can be used to verify that system packages are correctly installed. The `v` option can be used to verify a particular package or to verify all system packages. If no output is returned, the package is installed correctly. The following table describes the meaning of output from the verify option:

Code	Meaning
S	File size differs.
M	File mode differs (includes permissions and file type).
5	The MD5 checksum differs.
D	The major and minor version numbers differ on a device file.
L	A mismatch occurs in a link.
U	The file ownership differs.
G	The file group owner differs.
T	The file time (mtime) differs.

The `rpm -qf` command can be used to determine which package a particular file belongs to. For example the following commands determines which package the `/etc/ssh/sshd_config` file belongs to:

```
# rpm -qf /etc/ssh/sshd_config
openssh-server-7.4p1-21.el7.x86_64
```

To verify the settings for the package that controls the `/bin/bash` file, run the following:

```
# rpm -V openssh-server-7.4p1-21.el7.x86_64
S.5....T. c /etc/ssh/sshd_config
```

Note: You can feed the output of the `rpm -qf` command to the `rpm -V` command:

```
# rpm -V $(rpm -qf /etc/ssh/sshd_config)
S.5....T. c /etc/ssh/sshd_config
```

Notes:

- *Since packages and important files may change with new updates and releases, it is recommended to verify everything, not just a finite list of files. This can be a time consuming task and results may depend on site policy therefore it is not a scorable benchmark item, but is provided for those interested in additional security measures.*
- *Some of the recommendations of this benchmark alter the state of files audited by this recommendation. The audit command will alert for all changes to a file permissions even if the new state is more secure than the default.*

Rationale:

It is important to confirm that packaged system files and directories are maintained with the permissions they were intended to have from the OS distributor or in accordance with local site policy.

Audit:

Run the following command to review all installed packages. Note that this may be very time consuming and may be best scheduled via the `cron` utility. It is recommended that the output of this command be redirected to a file that can be reviewed later. This command will ignore configuration files due to the extreme likelihood that they will change.

```
# rpm -Va --nomtime --nosize --nomd5 --nolinkto > <filename> | grep -vw c
```

Remediation:

Investigate the results to ensure any discrepancies found are understood and support proper secure operation of the system.

References:

1. <https://docs.fedoraproject.org/en-US/fedora/rawhide/system-administrators-guide/RPM/#s2-rpm-verifying>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 <u>Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v7	14.6 <u>Protect Information through Access Control Lists</u> Protect all information stored on systems with file system, network share, claims, application, or database specific access control lists. These controls will enforce the principle that only authorized individuals should have access to the information based on their need to access the information as a part of their responsibilities.			

6.1.2 Ensure permissions on /etc/passwd are configured (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `/etc/passwd` file contains user account information that is used by many system utilities and therefore must be readable for these utilities to operate.

Rationale:

It is critical to ensure that the `/etc/passwd` file is protected from unauthorized write access. Although it is protected by default, the file permissions could be changed either inadvertently or through malicious actions.

Audit:

Run the following command and verify `Uid` and `Gid` are both `0/root` and `Access` is `644` or more restrictive:

```
# stat /etc/passwd
Access: (0644/-rw-r--r--)  Uid: (    0/    root)   Gid: (    0/    root)
```

Remediation:

Run the following commands to set owner, group, and permissions on `/etc/passwd`:

```
# chown root:root /etc/passwd
# chmod u-x,g-wx,o-wx /etc/passwd
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 <u>Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v7	14.6 <u>Protect Information through Access Control Lists</u> Protect all information stored on systems with file system, network share, claims, application, or database specific access control lists. These controls will enforce the principle that only authorized individuals should have access to the information based on their need to access the information as a part of their responsibilities.			

6.1.3 Ensure permissions on /etc/passwd- are configured (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The /etc/passwd- file contains backup user account information.

Rationale:

It is critical to ensure that the /etc/passwd- file is protected from unauthorized access. Although it is protected by default, the file permissions could be changed either inadvertently or through malicious actions.

Audit:

Run the following command and verify Uid and Gid are both 0/root and Access is 644 or more restrictive:

```
# stat /etc/passwd-  
Access: (0644/-rw-----)  Uid: (    0/    root)   Gid: (    0/    root)
```

Remediation:

Run the following commands to set owner, group, and permissions on /etc/passwd- :

```
# chown root:root /etc/passwd-  
# chmod u-x,go-wx /etc/passwd-
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 <u>Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v7	14.6 <u>Protect Information through Access Control Lists</u> Protect all information stored on systems with file system, network share, claims, application, or database specific access control lists. These controls will enforce the principle that only authorized individuals should have access to the information based on their need to access the information as a part of their responsibilities.			

6.1.4 Ensure permissions on /etc/shadow are configured (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `/etc/shadow` file is used to store the information about user accounts that is critical to the security of those accounts, such as the hashed password and other security information.

Rationale:

If attackers can gain read access to the `/etc/shadow` file, they can easily run a password cracking program against the hashed password to break it. Other security information that is stored in the `/etc/shadow` file (such as expiration) could also be useful to subvert the user accounts.

Audit:

Run the following command and verify Uid and Gid are 0/root , and Access is 0000 :

```
# stat /etc/shadow
Access: (0000/-----)  Uid: (    0/    root)  Gid: (    0/    root)
```

Remediation:

Run the following commands to set owner, group, and permissions on `/etc/shadow` :

```
# chown root:root /etc/shadow
# chmod 0000 /etc/shadow
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 <u>Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v7	14.6 <u>Protect Information through Access Control Lists</u> Protect all information stored on systems with file system, network share, claims, application, or database specific access control lists. These controls will enforce the principle that only authorized individuals should have access to the information based on their need to access the information as a part of their responsibilities.			

6.1.5 Ensure permissions on /etc/shadow- are configured (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `/etc/shadow-` file is used to store backup information about user accounts that is critical to the security of those accounts, such as the hashed password and other security information.

Rationale:

It is critical to ensure that the `/etc/shadow-` file is protected from unauthorized access. Although it is protected by default, the file permissions could be changed either inadvertently or through malicious actions.

Audit:

Run the following command and verify Uid is 0/root, Gid is 0/root and Access is 0000 :

```
# stat /etc/shadow-  
Access: (0000/-----)  Uid: (    0/    root)  Gid: (    0/    root)
```

Remediation:

Run the following commands to set owner, group, and permissions on `/etc/shadow-` :

```
# chown root:root /etc/shadow-  
# chmod 0000 /etc/shadow-
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 <u>Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v7	14.6 <u>Protect Information through Access Control Lists</u> Protect all information stored on systems with file system, network share, claims, application, or database specific access control lists. These controls will enforce the principle that only authorized individuals should have access to the information based on their need to access the information as a part of their responsibilities.			

6.1.6 Ensure permissions on /etc/gshadow- are configured (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `/etc/gshadow-` file is used to store backup information about groups that is critical to the security of those accounts, such as the hashed password and other security information.

Rationale:

It is critical to ensure that the `/etc/gshadow-` file is protected from unauthorized access. Although it is protected by default, the file permissions could be changed either inadvertently or through malicious actions.

Audit:

Run the following command and verify `Uid` is `0/root`, `Gid` is `0/root` and `Access` is `0000` :

```
# stat /etc/gshadow-  
Access: (0000/-----)  Uid: (    0/    root)  Gid: (    0/    root)
```

Remediation:

Run the following commands to set owner, group, and permissions on `/etc/gshadow-` :

```
# chown root:root /etc/gshadow-  
# chmod 0000 /etc/gshadow-
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 <u>Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v7	16.4 <u>Encrypt or Hash all Authentication Credentials</u> Encrypt or hash with a salt all authentication credentials when stored.			

6.1.7 Ensure permissions on /etc/gshadow are configured (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `/etc/gshadow` file is used to store the information about groups that is critical to the security of those accounts, such as the hashed password and other security information.

Rationale:

If attackers can gain read access to the `/etc/gshadow` file, they can easily run a password cracking program against the hashed password to break it. Other security information that is stored in the `/etc/gshadow` file (such as group administrators) could also be useful to subvert the group.

Audit:

Run the following command and verify Uid is 0/root, Gid is 0/root and Access is 0000 :

```
# stat /etc/gshadow
Access: (0000/-----)  Uid: (    0/    root)   Gid: (    0/    root)
```

Remediation:

Run the following commands to set owner, group, and permissions on `/etc/gshadow` :

```
# chown root:root /etc/gshadow
# chmod 0000 /etc/gshadow
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 <u>Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v7	14.6 <u>Protect Information through Access Control Lists</u> Protect all information stored on systems with file system, network share, claims, application, or database specific access control lists. These controls will enforce the principle that only authorized individuals should have access to the information based on their need to access the information as a part of their responsibilities.			

6.1.8 Ensure permissions on /etc/group are configured (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `/etc/group` file contains a list of all the valid groups defined in the system. The command below allows read/write access for root and read access for everyone else.

Rationale:

The `/etc/group` file needs to be protected from unauthorized changes by non-privileged users, but needs to be readable as this information is used with many non-privileged programs.

Audit:

Run the following command and verify `Uid` and `Gid` are both `0/root` and `Access` is `644` or more restrictive:

```
# stat /etc/group
Access: (0644/-rw-r--r--)  Uid: (    0/    root)   Gid: (    0/    root)
```

Remediation:

Run the following commands to set owner, group, and permissions on `/etc/group`:

```
# chown root:root /etc/group
# chmod u-x,g-wx,o-wx /etc/group
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>3.3 Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v7	<u>14.6 Protect Information through Access Control Lists</u> Protect all information stored on systems with file system, network share, claims, application, or database specific access control lists. These controls will enforce the principle that only authorized individuals should have access to the information based on their need to access the information as a part of their responsibilities.			

6.1.9 Ensure permissions on /etc/group- are configured (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The /etc/group- file contains a backup list of all the valid groups defined in the system.

Rationale:

It is critical to ensure that the /etc/group- file is protected from unauthorized access. Although it is protected by default, the file permissions could be changed either inadvertently or through malicious actions.

Audit:

Run the following command and verify Uid and Gid are both 0/root and Access is 644 or more restrictive:

```
# stat /etc/group-  
Access: (0644/-rw-r--r--)  Uid: (   0/   root)   Gid: (   0/   root)
```

Remediation:

Run the following commands to set owner, group, and permissions on /etc/group-:

```
# chown root:root /etc/group-  
# chmod u-x,go-wx /etc/group-
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>3.3 Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v7	<u>14.6 Protect Information through Access Control Lists</u> Protect all information stored on systems with file system, network share, claims, application, or database specific access control lists. These controls will enforce the principle that only authorized individuals should have access to the information based on their need to access the information as a part of their responsibilities.			

6.1.10 Ensure no world writable files exist (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

Unix-based systems support variable settings to control access to files. World writable files are the least secure. See the `chmod(2)` man page for more information.

Rationale:

Data in world-writable files can be modified and compromised by any user on the system. World writable files may also indicate an incorrectly written script or program that could potentially be the cause of a larger compromise to the system's integrity.

Audit:

Run the following command and verify no files are returned:

```
# df --local -P | awk '{if (NR!=1) print $6}' | xargs -I '{}' find '{}' -xdev -type f -perm -0002
```

The command above only searches local filesystems, there may still be compromised items on network mounted partitions. Additionally the `--local` option to `df` is not universal to all versions, it can be omitted to search all filesystems on a system including network mounted filesystems or the following command can be run manually for each partition:

```
# find <partition> -xdev -type f -perm -0002
```

Remediation:

Removing write access for the "other" category (`chmod o-w <filename>`) is advisable, but always consult relevant vendor documentation to avoid breaking any application dependencies on a given file.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>3.3 Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v8	<u>4.1 Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	<u>5.1 Establish Secure Configurations</u> Maintain documented, standard security configuration standards for all authorized operating systems and software.			
v7	<u>13 Data Protection</u> Data Protection			

6.1.11 Ensure no unowned files or directories exist (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

Sometimes when administrators delete users from the password file they neglect to remove all files owned by those users from the system.

Rationale:

A new user who is assigned the deleted user's user ID or group ID may then end up "owning" these files, and thus have more access on the system than was intended.

Audit:

Run the following command and verify no files are returned:

```
# df --local -P | awk {'if (NR!=1) print $6'} | xargs -I '{}' find '{}' -xdev -nouser
```

The command above only searches local filesystems, there may still be compromised items on network mounted partitions. Additionally the `--local` option to `df` is not universal to all versions, it can be omitted to search all filesystems on a system including network mounted filesystems or the following command can be run manually for each partition:

```
# find <partition> -xdev -nouser
```

Remediation:

Locate files that are owned by users or groups not listed in the system configuration files, and reset the ownership of these files to some active user on the system as appropriate.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.1 Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	<u>13.2 Remove Sensitive Data or Systems Not Regularly Accessed by Organization</u> Remove sensitive data or systems not regularly accessed by the organization from the network. These systems shall only be used as stand alone systems (disconnected from the network) by the business unit needing to occasionally use the system or completely virtualized and powered off until needed.			

6.1.12 Ensure no ungrouped files or directories exist (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

Sometimes when administrators delete users or groups from the system they neglect to remove all files owned by those users or groups.

Rationale:

A new user who is assigned the deleted user's user ID or group ID may then end up "owning" these files, and thus have more access on the system than was intended.

Audit:

Run the following command and verify no files are returned:

```
# df --local -P | awk '{if (NR!=1) print $6}' | xargs -I '{}' find '{}' -xdev -nogroup
```

The command above only searches local filesystems, there may still be compromised items on network mounted partitions. Additionally the `--local` option to `df` is not universal to all versions, it can be omitted to search all filesystems on a system including network mounted filesystems or the following command can be run manually for each partition:

```
# find <partition> -xdev -nogroup
```

Remediation:

Locate files that are owned by users or groups not listed in the system configuration files, and reset the ownership of these files to some active user on the system as appropriate.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.1 Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	<u>13.2 Remove Sensitive Data or Systems Not Regularly Accessed by Organization</u> Remove sensitive data or systems not regularly accessed by the organization from the network. These systems shall only be used as stand alone systems (disconnected from the network) by the business unit needing to occasionally use the system or completely virtualized and powered off until needed.			

6.1.13 Audit SUID executables (Manual)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The owner of a file can set the file's permissions to run with the owner's or group's permissions, even if the user running the program is not the owner or a member of the group. The most common reason for a SUID program is to enable users to perform functions (such as changing their password) that require root privileges.

Rationale:

There are valid reasons for SUID programs, but it is important to identify and review such programs to ensure they are legitimate.

Audit:

Run the following command to list SUID files:

```
# df --local -P | awk '{if (NR!=1) print $6}' | xargs -I '{}' find '{}' -xdev -type f -perm -4000
```

The command above only searches local filesystems, there may still be compromised items on network mounted partitions. Additionally the `--local` option to `df` is not universal to all versions, it can be omitted to search all filesystems on a system including network mounted filesystems or the following command can be run manually for each partition:

```
# find <partition> -xdev -type f -perm -4000
```

Remediation:

Ensure that no rogue SUID programs have been introduced into the system. Review the files returned by the action in the Audit section and confirm the integrity of these binaries.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.1 Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	<u>5.1 Establish Secure Configurations</u> Maintain documented, standard security configuration standards for all authorized operating systems and software.			

6.1.14 Audit SGID executables (Manual)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The owner of a file can set the file's permissions to run with the owner's or group's permissions, even if the user running the program is not the owner or a member of the group. The most common reason for a SGID program is to enable users to perform functions (such as changing their password) that require root privileges.

Rationale:

There are valid reasons for SGID programs, but it is important to identify and review such programs to ensure they are legitimate. Review the files returned by the action in the audit section and check to see if system binaries have a different md5 checksum than what from the package. This is an indication that the binary may have been replaced.

Audit:

Run the following command to list SGID files:

```
# df --local -P | awk '{if (NR!=1) print $6}' | xargs -I '{}' find '{}' -xdev -type f -perm -2000
```

The command above only searches local filesystems, there may still be compromised items on network mounted partitions. Additionally the `--local` option to `df` is not universal to all versions, it can be omitted to search all filesystems on a system including network mounted filesystems or the following command can be run manually for each partition:

```
# find <partition> -xdev -type f -perm -2000
```

Remediation:

Ensure that no rogue SGID programs have been introduced into the system. Review the files returned by the action in the Audit section and confirm the integrity of these binaries.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.1 Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	<u>5.1 Establish Secure Configurations</u> Maintain documented, standard security configuration standards for all authorized operating systems and software.			

6.2 User and Group Settings

This section provides guidance on securing aspects of the users and groups.

Note: The recommendations in this section check local users and groups. Any users or groups from other sources such as LDAP will not be audited. In a domain environment similar checks should be performed against domain users and groups.

6.2.1 Ensure accounts in /etc/passwd use shadowed passwords (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

Local accounts can use shadowed passwords. With shadowed passwords, the passwords are saved in the shadow password file, `/etc/shadow`, encrypted by a salted one-way hash. Accounts with a shadowed password have an `x` in the second field in `/etc/passwd`.

Rationale:

The `/etc/passwd` file also contains information like user ID's and group ID's that are used by many system programs. Therefore, the `/etc/passwd` file must remain world readable. In spite of encoding the password with a randomly-generated one-way hash function, an attacker could still break the system if they got access to the `/etc/passwd` file. This can be mitigated by using shadowed passwords, thus moving the passwords in the `/etc/passwd` file to `/etc/shadow`. The `/etc/shadow` file is set so only root will be able to read and write. This helps mitigate the risk of an attacker gaining access to the encoded passwords with which to perform a dictionary attack.

Notes:

- *All accounts must have passwords or be locked to prevent the account from being used by an unauthorized user.*
- *A user account with an empty second field in `/etc/passwd` allows the account to be logged into by providing only the username.*

Audit:

Run the following command and verify that no output is returned:

```
# awk -F: '($2 != "x" ) { print $1 " is not set to shadowed passwords " }' /etc/passwd
```

Remediation:

If any accounts in the `/etc/passwd` file do not have a single `x` in the password field, run the following command to set these accounts to use shadowed passwords:

```
# sed -e 's/^\([a-zA-Z0-9_]*\) :[^:]*:/\1:x:/' -i /etc/passwd
```

Investigate to determine if the account is logged in and what it is being used for, to determine if it needs to be forced off.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.11 <u>Encrypt Sensitive Data at Rest</u> Encrypt sensitive data at rest on servers, applications, and databases containing sensitive data. Storage-layer encryption, also known as server-side encryption, meets the minimum requirement of this Safeguard. Additional encryption methods may include application-layer encryption, also known as client-side encryption, where access to the data storage device(s) does not permit access to the plain-text data.			
v7	4.4 <u>Use Unique Passwords</u> Where multi-factor authentication is not supported (such as local administrator, root, or service accounts), accounts will use passwords that are unique to that system.			

6.2.2 Ensure /etc/shadow password fields are not empty (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

An account with an empty password field means that anybody may log in as that user without providing a password.

Rationale:

All accounts must have passwords or be locked to prevent the account from being used by an unauthorized user.

Audit:

Run the following command and verify that no output is returned:

```
# awk -F: '($2 == "" ) { print $1 " does not have a password "}' /etc/shadow
```

Remediation:

If any accounts in the `/etc/shadow` file do not have a password, run the following command to lock the account until it can be determined why it does not have a password:

```
# passwd -l <username>
```

Also, check to see if the account is logged in and investigate what it is being used for to determine if it needs to be forced off.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	5.2 <u>Use Unique Passwords</u> Use unique passwords for all enterprise assets. Best practice implementation includes, at a minimum, an 8-character password for accounts using MFA and a 14-character password for accounts not using MFA.			
v7	4.4 <u>Use Unique Passwords</u> Where multi-factor authentication is not supported (such as local administrator, root, or service accounts), accounts will use passwords that are unique to that system.			

6.2.3 Ensure all groups in /etc/passwd exist in /etc/group (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

Over time, system administration errors and changes can lead to groups being defined in /etc/passwd but not in /etc/group.

Rationale:

Groups defined in the /etc/passwd file but not in the /etc/group file pose a threat to system security since group permissions are not properly managed.

Audit:

Run the following script and verify no results are returned:

```
#!/bin/bash

for i in $(cut -s -d: -f4 /etc/passwd | sort -u ); do
    grep -q -P "^.*?:[^:]*:$i:" /etc/group
    if [ $? -ne 0 ]; then
        echo "Group $i is referenced by /etc/passwd but does not exist in
/etc/group"
    fi
done
```

Remediation:

Analyze the output of the Audit step above and perform the appropriate action to correct any discrepancies found.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 <u>Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v7	16 <u>Account Monitoring and Control</u> Account Monitoring and Control			

6.2.4 Ensure shadow group is empty (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The shadow group allows system programs which require access the ability to read the `/etc/shadow` file. No users should be assigned to the shadow group.

Rationale:

Any users assigned to the shadow group would be granted read access to the `/etc/shadow` file. If attackers can gain read access to the `/etc/shadow` file, they can easily run a password cracking program against the hashed passwords to break them. Other security information that is stored in the `/etc/shadow` file (such as expiration) could also be useful to subvert additional user accounts.

Audit:

Run the following commands and verify no results are returned:

```
# awk -F: '($1=="shadow") {print $NF}' /etc/group
# awk -F: -v GID="$(awk -F: '($1=="shadow") {print $3}' /etc/group)"
'($4==GID) {print $1}' /etc/passwd
```

Remediation:

Run the following command to remove all users from the shadow group

```
# sed -ri 's/(^shadow:[^:]*:[^:]*:)([[:^:]]+$)/\1/' /etc/group
```

Change the primary group of any users with shadow as their primary group.

```
# usermod -g <primary group> <user>
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.1 Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	<u>5.1 Establish Secure Configurations</u> Maintain documented, standard security configuration standards for all authorized operating systems and software.			

6.2.5 Ensure no duplicate user names exist (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

Although the `useradd` program will not let you create a duplicate user name, it is possible for an administrator to manually edit the `/etc/passwd` file and change the user name.

Rationale:

If a user is assigned a duplicate user name, it will create and have access to files with the first UID for that username in `/etc/passwd`. For example, if "test4" has a UID of 1000 and a subsequent "test4" entry has a UID of 2000, logging in as "test4" will use UID 1000. Effectively, the UID is shared, which is a security problem.

Audit:

Run the following script and verify no results are returned:

```
#!/bin/bash

cut -d: -f1 /etc/passwd | sort | uniq -d | while read x; do
    echo "Duplicate login name ${x} in /etc/passwd"
done
```

Remediation:

Based on the results of the audit script, establish unique user names for the users. File ownerships will automatically reflect the change as long as the users have unique UIDs.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 <u>Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	16 <u>Account Monitoring and Control</u> Account Monitoring and Control			

6.2.6 Ensure no duplicate group names exist (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

Although the `groupadd` program will not let you create a duplicate group name, it is possible for an administrator to manually edit the `/etc/group` file and change the group name.

Rationale:

If a group is assigned a duplicate group name, it will create and have access to files with the first GID for that group in `/etc/group`. Effectively, the GID is shared, which is a security problem.

Audit:

Run the following script and verify no results are returned:

```
#!/bin/bash

cut -d: -f1 /etc/group | sort | uniq -d | while read -r x; do
    echo "Duplicate group name ${x} in /etc/group"
done
```

Remediation:

Based on the results of the audit script, establish unique names for the user groups. File group ownerships will automatically reflect the change as long as the groups have unique GIDs.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 <u>Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	16 <u>Account Monitoring and Control</u> Account Monitoring and Control			

6.2.7 Ensure no duplicate UIDs exist (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

Although the `useradd` program will not let you create a duplicate User ID (UID), it is possible for an administrator to manually edit the `/etc/passwd` file and change the UID field.

Rationale:

Users must be assigned unique UIDs for accountability and to ensure appropriate access protections.

Audit:

Run the following script and verify no results are returned:

```
#!/bin/bash

cut -f3 -d":" /etc/passwd | sort -n | uniq -c | while read -r x; do
    [ -z "$x" ] && break
    set - "$x"
    if [ "$1" -gt 1 ]; then
        users=$(awk -F: '($3 == n) { print $1 }' n="$2" /etc/passwd | xargs)
        echo "Duplicate UID ($2): $users"
    fi
done
```

Remediation:

Based on the results of the audit script, establish unique UIDs and review all files owned by the shared UIDs to determine which UID they are supposed to belong to.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.1 Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	<u>16 Account Monitoring and Control</u> Account Monitoring and Control			

6.2.8 Ensure no duplicate GIDs exist (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

Although the `groupadd` program will not let you create a duplicate Group ID (GID), it is possible for an administrator to manually edit the `/etc/group` file and change the GID field.

Note: You can also use the `grpck` command to check for other inconsistencies in the `/etc/group` file.

Rationale:

User groups must be assigned unique GIDs for accountability and to ensure appropriate access protections.

Audit:

Run the following script and verify no results are returned:

```
#!/bin/bash
cut -d: -f3 /etc/group | sort | uniq -d | while read -r x; do
    echo "Duplicate GID ($x) in /etc/group"
done
```

Remediation:

Based on the results of the audit script, establish unique GIDs and review all files owned by the shared GID to determine which group they are supposed to belong to.

Additional Information:

You can also use the `grpck` command to check for other inconsistencies in the `/etc/group` file.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.1 Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	<u>16 Account Monitoring and Control</u> Account Monitoring and Control			

6.2.9 Ensure root is the only UID 0 account (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

Any account with UID 0 has superuser privileges on the system.

Rationale:

This access must be limited to only the default `root` account and only from the system console. Administrative access must be through an unprivileged account using an approved mechanism as noted in Item 5.6 Ensure access to the `su` command is restricted.

Audit:

Run the following command and verify that only "root" is returned:

```
# awk -F: '($3 == 0) { print $1 }' /etc/passwd  
root
```

Remediation:

Remove any users other than `root` with UID 0 or assign them a new UID if appropriate.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 Establish and Maintain a Secure Configuration Process Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	5.1 Establish Secure Configurations Maintain documented, standard security configuration standards for all authorized operating systems and software.			

6.2.10 Ensure root PATH Integrity (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `root` user can execute any command on the system and could be fooled into executing programs unintentionally if the `PATH` is not set correctly.

Rationale:

Including the current working directory (`.`) or other writable directory in `root`'s executable path makes it likely that an attacker can gain superuser access by forcing an administrator operating as `root` to execute a Trojan horse program.

Audit:

Run the following script and verify no results are returned:

```
#!/bin/bash

RPCV="$(sudo -Hiu root env | grep '^PATH' | cut -d= -f2)"
echo "$RPCV" | grep -q "::-" && echo "root's path contains a empty directory (::)"
echo "$RPCV" | grep -q ":$" && echo "root's path contains a trailing (:)"
for x in $(echo "$RPCV" | tr ":" " "); do
    if [ -d "$x" ]; then
        ls -ldH "$x" | awk '$9 == "." {print "PATH contains current working directory (.)."}'
        $3 != "root" {print $9, "is not owned by root"}
        substr($1,6,1) != "-" {print $9, "is group writable"}
        substr($1,9,1) != "-" {print $9, "is world writable"}
    else
        echo "$x is not a directory"
    fi
done
```

Remediation:

Correct or justify any items discovered in the Audit step.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>4.1 Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	<u>5.1 Establish Secure Configurations</u> Maintain documented, standard security configuration standards for all authorized operating systems and software.			

6.2.11 Ensure all users' home directories exist (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

Users can be defined in `/etc/passwd` without a home directory or with a home directory that does not actually exist.

Rationale:

If the user's home directory does not exist or is unassigned, the user will be placed in `/` and will not be able to write any files or have local environment variables set.

Audit:

Run the following script and verify no results are returned:

```
#!/bin/bash

awk -F: '($1!~/ (halt|sync|shutdown|nfsnobody)/ &&
$7!~/^(\usr)?\sbin\|nologin(\|)?$/ && $7!~/(\usr)?\bin\|false(\|)?$/ ) {
print $1 " " $6 }' /etc/passwd | while read -r user dir; do
    if [ ! -d "$dir" ]; then
        echo "User: \"$user\" home directory: \"$dir\" does not exist."
    fi
done
```

Note: The audit script checks all users with interactive shells except `halt`, `sync`, `shutdown`, and `nfsnobody`.

Remediation:

If any users' home directories do not exist, create them and make sure the respective user owns the directory. Users without an assigned home directory should be removed or assigned a home directory as appropriate.

The following script will create a home directory for users with an interactive shell whose home directory doesn't exist:

```
#!/bin/bash

awk -F: '($1!~/ (halt|sync|shutdown|nfsnobody)/ &&
$7!~/^(\ /usr)?\ /sbin\ /nologin(\ /)?$/ && $7!~/(\ /usr)?\ /bin\ /false(\ /)?$/ ) {
print $1 " " $6 }' /etc/passwd | while read -r user dir; do
    if [ ! -d "$dir" ]; then
        mkdir "$dir"
        chmod g-w,o-wrx "$dir"
        chown "$user" "$dir"
    fi
done
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 <u>Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	5.1 <u>Establish Secure Configurations</u> Maintain documented, standard security configuration standards for all authorized operating systems and software.			

6.2.12 Ensure users own their home directories (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The user home directory is space defined for the particular user to set local environment variables and to store personal files.

Rationale:

Since the user is accountable for files stored in the user home directory, the user must be the owner of the directory.

Audit:

Run the following script and verify no results are returned:

```
#!/bin/bash

awk -F: '($1!~/ (halt|sync|shutdown|nfsnobody)/ &&
$7!~/^(\usr)?\sbin\|nologin(\|)?$/ && $7!~/(\usr)?\bin\|false(\|)?$/ ) {
print $1 " " $6 }' /etc/passwd | while read -r user dir; do
    if [ ! -d "$dir" ]; then
        echo "User: \"$user\" home directory: \"$dir\" does not exist."
    else
        owner=$(stat -L -c "%U" "$dir")
        if [ "$owner" != "$user" ]; then
            echo "User: \"$user\" home directory: \"$dir\" is owned by
\"$owner\""
        fi
    fi
done
```

Remediation:

Change the ownership of any home directories that are not owned by the defined user to the correct user.

The following script will create missing home directories, set the owner, and set the permissions for interactive users' home directories:

```
#!/bin/bash

awk -F: '($1!~/ (halt|sync|shutdown|nfsnobody)/ &&
$7!~/^(\usr)?\sbin\nologin(\/)?$/ && $7!~/(\usr)?\bin\false(\/)?$/ ) {
print $1 " " $6 }' /etc/passwd | while read -r user dir; do
    if [ ! -d "$dir" ]; then
        echo "User: \"$user\" home directory: \"$dir\" does not exist, creating
home directory"
        mkdir "$dir"
        chmod g-w,o-rwx "$dir"
        chown "$user" "$dir"
    else
        owner=$(stat -L -c "%U" "$dir")
        if [ "$owner" != "$user" ]; then
            chmod g-w,o-rwx "$dir"
            chown "$user" "$dir"
        fi
    fi
done
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 <u>Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v7	14.6 <u>Protect Information through Access Control Lists</u> Protect all information stored on systems with file system, network share, claims, application, or database specific access control lists. These controls will enforce the principle that only authorized individuals should have access to the information based on their need to access the information as a part of their responsibilities.			

6.2.13 Ensure users' home directories permissions are 750 or more restrictive (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

While the system administrator can establish secure permissions for users' home directories, the users can easily override these.

Rationale:

Group or world-writable user home directories may enable malicious users to steal or modify other users' data or to gain another user's system privileges.

Audit:

Run the following script and verify no results are returned:

```
#!/bin/bash

awk -F: '($1!~/ (halt|sync|shutdown|nfsnobody)/ &&
$7!~/^(\ /usr)?\ /sbin\ /nologin(\ /)?$/ && $7!~/^(\ /usr)?\ /bin\ /false(\ /)?$/ )
{print $1 " " $6}' /etc/passwd | while read -r user dir; do
    if [ ! -d "$dir" ]; then
        echo "User: \"$user\" home directory: \"$dir\" doesn't exist"
    else
        dirperm=$(stat -L -c "%A" "$dir")
        if [ "$(echo "$dirperm" | cut -c6)" != "-" ] || [ "$(echo "$dirperm" |
cut -c8)" != "-" ] || [ "$(echo "$dirperm" | cut -c9)" != "-" ] || [ "$(echo
"$dirperm" | cut -c10)" != "-" ]; then
            echo "User: \"$user\" home directory: \"$dir\" has permissions:
\"$(stat -L -c "%a" "$dir")\""
        fi
    fi
done
```

Remediation:

Making global modifications to user home directories without alerting the user community can result in unexpected outages and unhappy users. Therefore, it is recommended that a monitoring policy be established to report user file permissions and determine the action to be taken in accordance with site policy.

The following script can be used to remove permissions in excess of 750 from users' home directories:

```
#!/bin/bash

awk -F: '($1!~/ (halt|sync|shutdown|nfsnobody)/ &&
$7!~/^(\ /usr)?\ /sbin\ /nologin(\ /)?$/ && $7!~/^(\ /usr)?\ /bin\ /false(\ /)?$/ )
{print $6}' /etc/passwd | while read -r dir; do
    if [ -d "$dir" ]; then
        dirperm=$(stat -L -c "%A" "$dir")
        if [ "$(echo "$dirperm" | cut -c6)" != "-" ] || [ "$(echo "$dirperm" |
cut -c8)" != "-" ] || [ "$(echo "$dirperm" | cut -c9)" != "-" ] || [ "$(echo
"$dirperm" | cut -c10)" != "-" ]; then
            chmod g-w,o-rwx "$dir"
        fi
    fi
done
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 <u>Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v7	14.6 <u>Protect Information through Access Control Lists</u> Protect all information stored on systems with file system, network share, claims, application, or database specific access control lists. These controls will enforce the principle that only authorized individuals should have access to the information based on their need to access the information as a part of their responsibilities.			

6.2.14 Ensure users' dot files are not group or world writable (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

While the system administrator can establish secure permissions for users' "dot" files, the users can easily override these.

Rationale:

Group or world-writable user configuration files may enable malicious users to steal or modify other users' data or to gain another user's system privileges.

Audit:

Run the following script and verify no results are returned:

```
#!/bin/bash

awk -F: '($1!~/ (halt|sync|shutdown|nfsnobody)/ &&
$7!~/^(\usr)?\sbin\nologin(\)?$/ && $7!~/(\usr)?\bin\false(\)?$/ ) {
print $1 " " $6 }' /etc/passwd | while read -r user dir; do
    if [ -d "$dir" ]; then
        for file in "$dir"/*; do
            if [ ! -h "$file" ] && [ -f "$file" ]; then
                fileperm=$(stat -L -c "%A" "$file")
                if [ "$(echo "$fileperm" | cut -c6)" != "-" ] || [ "$(echo
"$fileperm" | cut -c9)" != "-" ]; then
                    echo "User: \"$user\" file: \"$file\" has permissions:
\"$fileperm\""
                fi
            fi
        done
    fi
done
```

Remediation:

Making global modifications to users' files without alerting the user community can result in unexpected outages and unhappy users. Therefore, it is recommended that a monitoring policy be established to report user dot file permissions and determine the action to be taken in accordance with site policy.

The following script will remove excessive permissions on dot files within interactive users' home directories.

```
#!/bin/bash

awk -F: '($1!~/ (halt|sync|shutdown|nfsnobody)/ &&
$7!~/^(\ /usr)?\ /sbin\ /nologin(\ /)?$/ && $7!~/(\ /usr)?\ /bin\ /false(\ /)?$/ ) {
print $6 }' /etc/passwd | while read -r dir; do
    if [ -d "$dir" ]; then
        for file in "$dir"/*.*; do
            if [ ! -h "$file" ] && [ -f "$file" ]; then
                fileperm=$(stat -L -c "%A" "$file")
                if [ "$(echo "$fileperm" | cut -c6)" != "-" ] || [ "$(echo
"$fileperm" | cut -c9)" != "-" ]; then
                    chmod go-w "$file"
                fi
            fi
        done
    fi
done
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 <u>Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v7	14.6 <u>Protect Information through Access Control Lists</u> Protect all information stored on systems with file system, network share, claims, application, or database specific access control lists. These controls will enforce the principle that only authorized individuals should have access to the information based on their need to access the information as a part of their responsibilities.			

6.2.15 Ensure no users have .forward files (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `.forward` file specifies an email address to forward the user's mail to.

Rationale:

Use of the `.forward` file poses a security risk in that sensitive data may be inadvertently transferred outside the organization. The `.forward` file also poses a risk as it can be used to execute commands that may perform unintended actions.

Audit:

Run the following script and verify no results are returned:

```
#!/bin/bash

awk -F: '($1~/^(root|halt|sync|shutdown|nfsnobody)/ &&
$7!~/^(\usr)?\sbin\nologin(\)?$/ && $7!~/(\usr)?\bin\false(\)?$/ ) {
print $1 " " $6 }' /etc/passwd | while read -r user dir; do
    if [ -d "$dir" ]; then
        file="$dir/.forward"
        if [ ! -h "$file" ] && [ -f "$file" ]; then
            echo "User: \"$user\" file: \"$file\" exists"
        fi
    fi
done
```

Remediation:

Making global modifications to users' files without alerting the user community can result in unexpected outages and unhappy users. Therefore, it is recommended that a monitoring policy be established to report user `.forward` files and determine the action to be taken in accordance with site policy.

The following script will remove `.forward` files from interactive users' home directories

```
#!/bin/bash

awk -F: '($1!~/ (root|halt|sync|shutdown|nfsnobody)/ &&
$7!~/^ (\/usr)?\/sbin\/nologin(\/)?$/ && $7!~/ (\/usr)?\/bin\/false(\/)?$/ ) {
print $6 }' /etc/passwd | while read -r dir; do
    if [ -d "$dir" ]; then
        file="$dir/.forward"
        [ ! -h "$file" ] && [ -f "$file" ] && rm -r "$file"
    fi
done
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 <u>Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	5.1 <u>Establish Secure Configurations</u> Maintain documented, standard security configuration standards for all authorized operating systems and software.			

6.2.16 Ensure no users have .netrc files (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

The `.netrc` file contains data for logging into a remote host for file transfers via FTP.

While the system administrator can establish secure permissions for users' `.netrc` files, the users can easily override these.

Rationale:

The `.netrc` file presents a significant security risk since it stores passwords in unencrypted form. Even if FTP is disabled, user accounts may have brought over `.netrc` files from other systems which could pose a risk to those systems.

If a `.netrc` file is required, and follows local site policy, it should have permissions of `600` or more restrictive.

Audit:

Run the following script. This script will return:

- **FAILED:** for any .netrc file with permissions less restrictive than 600
- **WARNING:** for any .netrc files that exist in interactive users' home directories.

```
#!/bin/bash

awk -F: '($1!~/ (halt|sync|shutdown|nfsnobody)/ &&
$7!~/^(\usr)?\sbin\nologin(\)?$/ && $7!~/(\usr)?\bin\false(\)?$/ ) {
print $1 " " $6 }' /etc/passwd | while read -r user dir; do
    if [ -d "$dir" ]; then
        file="$dir/.netrc"
        if [ ! -h "$file" ] && [ -f "$file" ]; then
            if stat -L -c "%A" "$file" | cut -c4-10 | grep -Eq '[-]+'; then
                echo "FAILED: User: \"$user\" file: \"$file\" exists with
permissions: \"$(stat -L -c "%a" "$file")\", remove file or excessive
permissions"
            else
                echo "WARNING: User: \"$user\" file: \"$file\" exists with
permissions: \"$(stat -L -c "%a" "$file")\", remove file unless required"
            fi
        fi
    fi
done
```

Verify:

- Any lines beginning with **FAILED:** - File should be removed unless deemed necessary, in accordance with local site policy, and permissions are updated to be 600 or more restrictive
- Any lines beginning with **WARNING:** - File should be removed unless deemed necessary, and in accordance with local site policy

Remediation:

Making global modifications to users' files without alerting the user community can result in unexpected outages and unhappy users. Therefore, it is recommended that a monitoring policy be established to report user `.netrc` files and determine the action to be taken in accordance with site policy.

The following script will remove `.netrc` files from interactive users' home directories

```
#!/bin/bash

awk -F: '($1!~/ (halt|sync|shutdown|nfsnobody)/ &&
$7!~/^(\usr)?\sbin\nologin(\)?$/ && $7!~/(\usr)?\bin\false(\)?$/ ) {
print $6 }' /etc/passwd | while read -r dir; do
    if [ -d "$dir" ]; then
        file="$dir/.netrc"
        [ ! -h "$file" ] && [ -f "$file" ] && rm -f "$file"
    fi
done
```

Additional Information:

While the complete removal of `.netrc` files is recommended, if any are required on the system secure permissions must be applied.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.6 Securely Manage Enterprise Assets and Software Securely manage enterprise assets and software. Example implementations include managing configuration through version-controlled-infrastructure-as-code and accessing administrative interfaces over secure network protocols, such as Secure Shell (SSH) and Hypertext Transfer Protocol Secure (HTTPS). Do not use insecure management protocols, such as Telnet (Teletype Network) and HTTP, unless operationally essential.			
v7	16.4 Encrypt or Hash all Authentication Credentials Encrypt or hash with a salt all authentication credentials when stored.			

6.2.17 Ensure no users have .rhosts files (Automated)

Profile Applicability:

- Level 1 - Server
- Level 1 - Workstation

Description:

While no `.rhosts` files are shipped by default, users can easily create them.

Rationale:

This action is only meaningful if `.rhosts` support is permitted in the file `/etc/pam.conf`. Even though the `.rhosts` files are ineffective if support is disabled in `/etc/pam.conf`, they may have been brought over from other systems and could contain information useful to an attacker for those other systems.

Audit:

Run the following script and verify no results are returned:

```
#!/bin/bash

awk -F: '($1!~/ (root|halt|sync|shutdown|nfsnobody)/ &&
$7!~/^ (\/usr)?\/sbin\/nologin(\/)?$/ && $7!~/ (\/usr)?\/bin\/false(\/)?$/ ) {
print $1 " " $6 }' /etc/passwd | while read -r user dir; do
    if [ -d "$dir" ]; then
        file="$dir/.rhosts"
        if [ ! -h "$file" ] && [ -f "$file" ]; then
            echo "User: \"$user\" file: \"$file\" exists"
        fi
    fi
done
```

Remediation:

Making global modifications to users' files without alerting the user community can result in unexpected outages and unhappy users. Therefore, it is recommended that a monitoring policy be established to report user `.rhosts` files and determine the action to be taken in accordance with site policy.

The following script will remove `.rhosts` files from interactive users' home directories

```
#!/bin/bash

awk -F: '($1!~/ (root|halt|sync|shutdown|nfsnobody)/ &&
$7!~/^ (\/usr)?\/sbin\/nologin(\/)?$/ && $7!~/ (\/usr)?\/bin\/false(\/)?$/ ) {
print $6 }' /etc/passwd | while read -r dir; do
    if [ -d "$dir" ]; then
        file="$dir/.rhosts"
        [ ! -h "$file" ] && [ -f "$file" ] && rm -r "$file"
    fi
done
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 <u>Establish and Maintain a Secure Configuration Process</u> Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	16.4 <u>Encrypt or Hash all Authentication Credentials</u> Encrypt or hash with a salt all authentication credentials when stored.			

Appendix: Recommendation Summary Table

Control		Set Correctly	
		Yes	No
1	Initial Setup		
1.1	Filesystem Configuration		
1.1.1	Disable unused filesystems		
1.1.1.1	Ensure mounting of cramfs filesystems is disabled (Automated)	☐	☐
1.1.1.2	Ensure mounting of squashfs filesystems is disabled (Automated)	☐	☐
1.1.1.3	Ensure mounting of udf filesystems is disabled (Automated)	☐	☐
1.1.2	Ensure /tmp is configured (Automated)	☐	☐
1.1.3	Ensure noexec option set on /tmp partition (Automated)	☐	☐
1.1.4	Ensure nodev option set on /tmp partition (Automated)	☐	☐
1.1.5	Ensure nosuid option set on /tmp partition (Automated)	☐	☐
1.1.6	Ensure /dev/shm is configured (Automated)	☐	☐
1.1.7	Ensure noexec option set on /dev/shm partition (Automated)	☐	☐
1.1.8	Ensure nodev option set on /dev/shm partition (Automated)	☐	☐
1.1.9	Ensure nosuid option set on /dev/shm partition (Automated)	☐	☐
1.1.10	Ensure separate partition exists for /var (Automated)	☐	☐
1.1.11	Ensure separate partition exists for /var/tmp (Automated)	☐	☐
1.1.12	Ensure /var/tmp partition includes the noexec option (Automated)	☐	☐
1.1.13	Ensure /var/tmp partition includes the nodev option (Automated)	☐	☐
1.1.14	Ensure /var/tmp partition includes the nosuid option (Automated)	☐	☐
1.1.15	Ensure separate partition exists for /var/log (Automated)	☐	☐
1.1.16	Ensure separate partition exists for /var/log/audit (Automated)	☐	☐
1.1.17	Ensure separate partition exists for /home (Automated)	☐	☐
1.1.18	Ensure /home partition includes the nodev option (Automated)	☐	☐
1.1.19	Ensure removable media partitions include noexec option (Automated)	☐	☐

1.1.20	Ensure nodev option set on removable media partitions (Automated)	☐	☐
1.1.21	Ensure nosuid option set on removable media partitions (Automated)	☐	☐
1.1.22	Ensure sticky bit is set on all world-writable directories (Automated)	☐	☐
1.1.23	Disable Automounting (Automated)	☐	☐
1.1.24	Disable USB Storage (Automated)	☐	☐
1.2	Configure Software Updates		
1.2.1	Ensure GPG keys are configured (Manual)	☐	☐
1.2.2	Ensure package manager repositories are configured (Manual)	☐	☐
1.2.3	Ensure gpgcheck is globally activated (Automated)	☐	☐
1.3	Filesystem Integrity Checking		
1.3.1	Ensure AIDE is installed (Automated)	☐	☐
1.3.2	Ensure filesystem integrity is regularly checked (Automated)	☐	☐
1.4	Secure Boot Settings		
1.4.1	Ensure bootloader password is set (Automated)	☐	☐
1.4.2	Ensure permissions on bootloader config are configured (Automated)	☐	☐
1.4.3	Ensure authentication required for single user mode (Automated)	☐	☐
1.5	Additional Process Hardening		
1.5.1	Ensure core dumps are restricted (Automated)	☐	☐
1.5.2	Ensure XD/NX support is enabled (Automated)	☐	☐
1.5.3	Ensure address space layout randomization (ASLR) is enabled (Automated)	☐	☐
1.5.4	Ensure prelink is not installed (Automated)	☐	☐
1.6	Mandatory Access Control		
1.6.1	Configure SELinux		
1.6.1.1	Ensure SELinux is installed (Automated)	☐	☐
1.6.1.2	Ensure SELinux is not disabled in bootloader configuration (Automated)	☐	☐
1.6.1.3	Ensure SELinux policy is configured (Automated)	☐	☐
1.6.1.4	Ensure the SELinux mode is enforcing or permissive (Automated)	☐	☐
1.6.1.5	Ensure the SELinux mode is enforcing (Automated)	☐	☐
1.6.1.6	Ensure no unconfined services exist (Automated)	☐	☐
1.6.1.7	Ensure SETroubleshoot is not installed (Automated)	☐	☐
1.6.1.8	Ensure the MCS Translation Service (mcstrans) is not installed (Automated)	☐	☐
1.7	Command Line Warning Banners		

1.7.1	Ensure message of the day is configured properly (Automated)	☐	☐
1.7.2	Ensure local login warning banner is configured properly (Automated)	☐	☐
1.7.3	Ensure remote login warning banner is configured properly (Automated)	☐	☐
1.7.4	Ensure permissions on /etc/motd are configured (Automated)	☐	☐
1.7.5	Ensure permissions on /etc/issue are configured (Automated)	☐	☐
1.7.6	Ensure permissions on /etc/issue.net are configured (Automated)	☐	☐
1.8	GNOME Display Manager		
1.8.1	Ensure GNOME Display Manager is removed (Manual)	☐	☐
1.8.2	Ensure GDM login banner is configured (Automated)	☐	☐
1.8.3	Ensure last logged in user display is disabled (Automated)	☐	☐
1.8.4	Ensure XDCMP is not enabled (Automated)	☐	☐
1.9	Ensure updates, patches, and additional security software are installed (Manual)	☐	☐
2	Services		
2.1	inetd Services		
2.1.1	Ensure xinetd is not installed (Automated)	☐	☐
2.2	Special Purpose Services		
2.2.1	Time Synchronization		
2.2.1.1	Ensure time synchronization is in use (Manual)	☐	☐
2.2.1.2	Ensure chrony is configured (Automated)	☐	☐
2.2.1.3	Ensure ntp is configured (Automated)	☐	☐
2.2.2	Ensure X11 Server components are not installed (Automated)	☐	☐
2.2.3	Ensure Avahi Server is not installed (Automated)	☐	☐
2.2.4	Ensure CUPS is not installed (Automated)	☐	☐
2.2.5	Ensure DHCP Server is not installed (Automated)	☐	☐
2.2.6	Ensure LDAP server is not installed (Automated)	☐	☐
2.2.7	Ensure DNS Server is not installed (Automated)	☐	☐
2.2.8	Ensure FTP Server is not installed (Automated)	☐	☐
2.2.9	Ensure HTTP server is not installed (Automated)	☐	☐
2.2.10	Ensure IMAP and POP3 server is not installed (Automated)	☐	☐
2.2.11	Ensure Samba is not installed (Automated)	☐	☐
2.2.12	Ensure HTTP Proxy Server is not installed (Automated)	☐	☐
2.2.13	Ensure net-snmp is not installed (Automated)	☐	☐
2.2.14	Ensure NIS server is not installed (Automated)	☐	☐
2.2.15	Ensure telnet-server is not installed (Automated)	☐	☐

2.2.16	Ensure mail transfer agent is configured for local-only mode (Automated)	☐	☐
2.2.17	Ensure nfs-utils is not installed or the nfs-server service is masked (Automated)	☐	☐
2.2.18	Ensure rpcbind is not installed or the rpcbind services are masked (Automated)	☐	☐
2.2.19	Ensure rsync is not installed or the rsyncd service is masked (Automated)	☐	☐
2.3	Service Clients		
2.3.1	Ensure NIS Client is not installed (Automated)	☐	☐
2.3.2	Ensure rsh client is not installed (Automated)	☐	☐
2.3.3	Ensure talk client is not installed (Automated)	☐	☐
2.3.4	Ensure telnet client is not installed (Automated)	☐	☐
2.3.5	Ensure LDAP client is not installed (Automated)	☐	☐
2.4	Ensure nonessential services are removed or masked (Manual)	☐	☐
3	Network Configuration		
3.1	Disable unused network protocols and devices		
3.1.1	Disable IPv6 (Manual)	☐	☐
3.1.2	Ensure wireless interfaces are disabled (Automated)	☐	☐
3.2	Network Parameters (Host Only)		
3.2.1	Ensure IP forwarding is disabled (Automated)	☐	☐
3.2.2	Ensure packet redirect sending is disabled (Automated)	☐	☐
3.3	Network Parameters (Host and Router)		
3.3.1	Ensure source routed packets are not accepted (Automated)	☐	☐
3.3.2	Ensure ICMP redirects are not accepted (Automated)	☐	☐
3.3.3	Ensure secure ICMP redirects are not accepted (Automated)	☐	☐
3.3.4	Ensure suspicious packets are logged (Automated)	☐	☐
3.3.5	Ensure broadcast ICMP requests are ignored (Automated)	☐	☐
3.3.6	Ensure bogus ICMP responses are ignored (Automated)	☐	☐
3.3.7	Ensure Reverse Path Filtering is enabled (Automated)	☐	☐
3.3.8	Ensure TCP SYN Cookies is enabled (Automated)	☐	☐
3.3.9	Ensure IPv6 router advertisements are not accepted (Automated)	☐	☐
3.4	Uncommon Network Protocols		
3.4.1	Ensure DCCP is disabled (Automated)	☐	☐
3.4.2	Ensure SCTP is disabled (Automated)	☐	☐
3.5	Firewall Configuration		
3.5.1	Configure firewalld		
3.5.1.1	Ensure firewalld is installed (Automated)	☐	☐
3.5.1.2	Ensure iptables-services not installed with firewalld (Automated)	☐	☐

3.5.1.3	Ensure nftables either not installed or masked with firewalld (Automated)	☐	☐
3.5.1.4	Ensure firewalld service enabled and running (Automated)	☐	☐
3.5.1.5	Ensure firewalld default zone is set (Automated)	☐	☐
3.5.1.6	Ensure network interfaces are assigned to appropriate zone (Manual)	☐	☐
3.5.1.7	Ensure firewalld drops unnecessary services and ports (Manual)	☐	☐
3.5.2	Configure nftables		
3.5.2.1	Ensure nftables is installed (Automated)	☐	☐
3.5.2.2	Ensure firewalld is either not installed or masked with nftables (Automated)	☐	☐
3.5.2.3	Ensure iptables-services not installed with nftables (Automated)	☐	☐
3.5.2.4	Ensure iptables are flushed with nftables (Manual)	☐	☐
3.5.2.5	Ensure an nftables table exists (Automated)	☐	☐
3.5.2.6	Ensure nftables base chains exist (Automated)	☐	☐
3.5.2.7	Ensure nftables loopback traffic is configured (Automated)	☐	☐
3.5.2.8	Ensure nftables outbound and established connections are configured (Manual)	☐	☐
3.5.2.9	Ensure nftables default deny firewall policy (Automated)	☐	☐
3.5.2.10	Ensure nftables service is enabled (Automated)	☐	☐
3.5.2.11	Ensure nftables rules are permanent (Automated)	☐	☐
3.5.3	Configure iptables		
3.5.3.1	Configure iptables software		
3.5.3.1.1	Ensure iptables packages are installed (Automated)	☐	☐
3.5.3.1.2	Ensure nftables is not installed with iptables (Automated)	☐	☐
3.5.3.1.3	Ensure firewalld is either not installed or masked with iptables (Automated)	☐	☐
3.5.3.2	Configure IPv4 iptables		
3.5.3.2.1	Ensure iptables loopback traffic is configured (Automated)	☐	☐
3.5.3.2.2	Ensure iptables outbound and established connections are configured (Manual)	☐	☐
3.5.3.2.3	Ensure iptables rules exist for all open ports (Automated)	☐	☐
3.5.3.2.4	Ensure iptables default deny firewall policy (Automated)	☐	☐
3.5.3.2.5	Ensure iptables rules are saved (Automated)	☐	☐
3.5.3.2.6	Ensure iptables is enabled and running (Automated)	☐	☐
3.5.3.3	Configure IPv6 ip6tables		
3.5.3.3.1	Ensure ip6tables loopback traffic is configured (Automated)	☐	☐
3.5.3.3.2	Ensure ip6tables outbound and established connections are configured (Manual)	☐	☐
3.5.3.3.3	Ensure ip6tables firewall rules exist for all open ports (Automated)	☐	☐

3.5.3.3.4	Ensure ip6tables default deny firewall policy (Automated)	☐	☐
3.5.3.3.5	Ensure ip6tables rules are saved (Automated)	☐	☐
3.5.3.3.6	Ensure ip6tables is enabled and running (Automated)	☐	☐
4	Logging and Auditing		
4.1	Configure System Accounting (auditd)		
4.1.1	Ensure auditing is enabled		
4.1.1.1	Ensure auditd is installed (Automated)	☐	☐
4.1.1.2	Ensure auditd service is enabled and running (Automated)	☐	☐
4.1.1.3	Ensure auditing for processes that start prior to auditd is enabled (Automated)	☐	☐
4.1.2	Configure Data Retention		
4.1.2.1	Ensure audit log storage size is configured (Automated)	☐	☐
4.1.2.2	Ensure audit logs are not automatically deleted (Automated)	☐	☐
4.1.2.3	Ensure system is disabled when audit logs are full (Automated)	☐	☐
4.1.2.4	Ensure audit_backlog_limit is sufficient (Automated)	☐	☐
4.1.3	Ensure events that modify date and time information are collected (Automated)	☐	☐
4.1.4	Ensure events that modify user/group information are collected (Automated)	☐	☐
4.1.5	Ensure events that modify the system's network environment are collected (Automated)	☐	☐
4.1.6	Ensure events that modify the system's Mandatory Access Controls are collected (Automated)	☐	☐
4.1.7	Ensure login and logout events are collected (Automated)	☐	☐
4.1.8	Ensure session initiation information is collected (Automated)	☐	☐
4.1.9	Ensure discretionary access control permission modification events are collected (Automated)	☐	☐
4.1.10	Ensure unsuccessful unauthorized file access attempts are collected (Automated)	☐	☐
4.1.11	Ensure use of privileged commands is collected (Automated)	☐	☐
4.1.12	Ensure successful file system mounts are collected (Automated)	☐	☐
4.1.13	Ensure file deletion events by users are collected (Automated)	☐	☐
4.1.14	Ensure changes to system administration scope (sudoers) is collected (Automated)	☐	☐
4.1.15	Ensure system administrator command executions (sudo) are collected (Automated)	☐	☐

4.1.16	Ensure kernel module loading and unloading is collected (Automated)	☐	☐
4.1.17	Ensure the audit configuration is immutable (Automated)	☐	☐
4.2	Configure Logging		
4.2.1	Configure rsyslog		
4.2.1.1	Ensure rsyslog is installed (Automated)	☐	☐
4.2.1.2	Ensure rsyslog Service is enabled and running (Automated)	☐	☐
4.2.1.3	Ensure rsyslog default file permissions configured (Automated)	☐	☐
4.2.1.4	Ensure logging is configured (Manual)	☐	☐
4.2.1.5	Ensure rsyslog is configured to send logs to a remote log host (Automated)	☐	☐
4.2.1.6	Ensure remote rsyslog messages are only accepted on designated log hosts. (Manual)	☐	☐
4.2.2	Configure journald		
4.2.2.1	Ensure journald is configured to send logs to rsyslog (Automated)	☐	☐
4.2.2.2	Ensure journald is configured to compress large log files (Automated)	☐	☐
4.2.2.3	Ensure journald is configured to write logfiles to persistent disk (Automated)	☐	☐
4.2.3	Ensure permissions on all logfiles are configured (Manual)	☐	☐
4.2.4	Ensure logrotate is configured (Manual)	☐	☐
5	Access, Authentication and Authorization		
5.1	Configure time-based job schedulers		
5.1.1	Ensure cron daemon is enabled and running (Automated)	☐	☐
5.1.2	Ensure permissions on /etc/crontab are configured (Automated)	☐	☐
5.1.3	Ensure permissions on /etc/cron.hourly are configured (Automated)	☐	☐
5.1.4	Ensure permissions on /etc/cron.daily are configured (Automated)	☐	☐
5.1.5	Ensure permissions on /etc/cron.weekly are configured (Automated)	☐	☐
5.1.6	Ensure permissions on /etc/cron.monthly are configured (Automated)	☐	☐
5.1.7	Ensure permissions on /etc/cron.d are configured (Automated)	☐	☐
5.1.8	Ensure cron is restricted to authorized users (Automated)	☐	☐
5.1.9	Ensure at is restricted to authorized users (Automated)	☐	☐
5.2	Configure sudo		
5.2.1	Ensure sudo is installed (Automated)	☐	☐
5.2.2	Ensure sudo commands use pty (Automated)	☐	☐

5.2.3	Ensure sudo log file exists (Automated)	☐	☐
5.3	Configure SSH Server		
5.3.1	Ensure permissions on /etc/ssh/sshd_config are configured (Automated)	☐	☐
5.3.2	Ensure permissions on SSH private host key files are configured (Automated)	☐	☐
5.3.3	Ensure permissions on SSH public host key files are configured (Automated)	☐	☐
5.3.4	Ensure SSH access is limited (Automated)	☐	☐
5.3.5	Ensure SSH LogLevel is appropriate (Automated)	☐	☐
5.3.6	Ensure SSH X11 forwarding is disabled (Automated)	☐	☐
5.3.7	Ensure SSH MaxAuthTries is set to 4 or less (Automated)	☐	☐
5.3.8	Ensure SSH IgnoreRhosts is enabled (Automated)	☐	☐
5.3.9	Ensure SSH HostbasedAuthentication is disabled (Automated)	☐	☐
5.3.10	Ensure SSH root login is disabled (Automated)	☐	☐
5.3.11	Ensure SSH PermitEmptyPasswords is disabled (Automated)	☐	☐
5.3.12	Ensure SSH PermitUserEnvironment is disabled (Automated)	☐	☐
5.3.13	Ensure only strong Ciphers are used (Automated)	☐	☐
5.3.14	Ensure only strong MAC algorithms are used (Automated)	☐	☐
5.3.15	Ensure only strong Key Exchange algorithms are used (Automated)	☐	☐
5.3.16	Ensure SSH Idle Timeout Interval is configured (Automated)	☐	☐
5.3.17	Ensure SSH LoginGraceTime is set to one minute or less (Automated)	☐	☐
5.3.18	Ensure SSH warning banner is configured (Automated)	☐	☐
5.3.19	Ensure SSH PAM is enabled (Automated)	☐	☐
5.3.20	Ensure SSH AllowTcpForwarding is disabled (Automated)	☐	☐
5.3.21	Ensure SSH MaxStartups is configured (Automated)	☐	☐
5.3.22	Ensure SSH MaxSessions is limited (Automated)	☐	☐
5.4	Configure PAM		
5.4.1	Ensure password creation requirements are configured (Automated)	☐	☐
5.4.2	Ensure lockout for failed password attempts is configured (Automated)	☐	☐
5.4.3	Ensure password hashing algorithm is SHA-512 (Automated)	☐	☐
5.4.4	Ensure password reuse is limited (Automated)	☐	☐
5.5	User Accounts and Environment		
5.5.1	Set Shadow Password Suite Parameters		
5.5.1.1	Ensure password expiration is 365 days or less (Automated)	☐	☐

5.5.1.2	Ensure minimum days between password changes is configured (Automated)	☐	☐
5.5.1.3	Ensure password expiration warning days is 7 or more (Automated)	☐	☐
5.5.1.4	Ensure inactive password lock is 30 days or less (Automated)	☐	☐
5.5.1.5	Ensure all users last password change date is in the past (Automated)	☐	☐
5.5.2	Ensure system accounts are secured (Automated)	☐	☐
5.5.3	Ensure default group for the root account is GID 0 (Automated)	☐	☐
5.5.4	Ensure default user shell timeout is configured (Automated)	☐	☐
5.5.5	Ensure default user umask is configured (Automated)	☐	☐
5.6	Ensure root login is restricted to system console (Manual)	☐	☐
5.7	Ensure access to the su command is restricted (Automated)	☐	☐
6	System Maintenance		
6.1	System File Permissions		
6.1.1	Audit system file permissions (Manual)	☐	☐
6.1.2	Ensure permissions on /etc/passwd are configured (Automated)	☐	☐
6.1.3	Ensure permissions on /etc/passwd- are configured (Automated)	☐	☐
6.1.4	Ensure permissions on /etc/shadow are configured (Automated)	☐	☐
6.1.5	Ensure permissions on /etc/shadow- are configured (Automated)	☐	☐
6.1.6	Ensure permissions on /etc/gshadow- are configured (Automated)	☐	☐
6.1.7	Ensure permissions on /etc/gshadow are configured (Automated)	☐	☐
6.1.8	Ensure permissions on /etc/group are configured (Automated)	☐	☐
6.1.9	Ensure permissions on /etc/group- are configured (Automated)	☐	☐
6.1.10	Ensure no world writable files exist (Automated)	☐	☐
6.1.11	Ensure no unowned files or directories exist (Automated)	☐	☐
6.1.12	Ensure no ungrouped files or directories exist (Automated)	☐	☐
6.1.13	Audit SUID executables (Manual)	☐	☐
6.1.14	Audit SGID executables (Manual)	☐	☐
6.2	User and Group Settings		
6.2.1	Ensure accounts in /etc/passwd use shadowed passwords (Automated)	☐	☐

6.2.2	Ensure /etc/shadow password fields are not empty (Automated)	☐	☐
6.2.3	Ensure all groups in /etc/passwd exist in /etc/group (Automated)	☐	☐
6.2.4	Ensure shadow group is empty (Automated)	☐	☐
6.2.5	Ensure no duplicate user names exist (Automated)	☐	☐
6.2.6	Ensure no duplicate group names exist (Automated)	☐	☐
6.2.7	Ensure no duplicate UIDs exist (Automated)	☐	☐
6.2.8	Ensure no duplicate GIDs exist (Automated)	☐	☐
6.2.9	Ensure root is the only UID 0 account (Automated)	☐	☐
6.2.10	Ensure root PATH Integrity (Automated)	☐	☐
6.2.11	Ensure all users' home directories exist (Automated)	☐	☐
6.2.12	Ensure users own their home directories (Automated)	☐	☐
6.2.13	Ensure users' home directories permissions are 750 or more restrictive (Automated)	☐	☐
6.2.14	Ensure users' dot files are not group or world writable (Automated)	☐	☐
6.2.15	Ensure no users have .forward files (Automated)	☐	☐
6.2.16	Ensure no users have .netrc files (Automated)	☐	☐
6.2.17	Ensure no users have .rhosts files (Automated)	☐	☐

Appendix: Change History

Date	Version	Changes for this version
Jun 30, 2020	3.0.0	Published
Sep 17, 2020	3.1.0	UPDATE - Ensure sudo commands use pty - update AAC (Ticket 11385)
Sep 17, 2020	3.1.0	UPDATE - Ensure core dumps are restricted - updated audit (Ticket 11386)
Sep 17, 2020	3.1.0	UPDATE - Ensure IP forwarding is disabled - correct bug in audit script (Ticket 11388)
Sep 17, 2020	3.1.0	UPDATE - Ensure ICMP redirects are not accepted - correct bug in audit script (Ticket 11390)
Sep 17, 2020	3.1.0	UPDATE - Ensure IPv6 router advertisements are not accepted - correct bug in audit script (Ticket 11391)
Sep 17, 2020	3.1.0	UPDATE - Ensure chrony is configured - correct bug in AAC (Ticket 11387)
Sep 17, 2020	3.1.0	UPDATE - Ensure loopback traffic is configured - correct bug in audit script (Ticket 11392)
Sep 17, 2020	3.1.0	UPDATE - Ensure IPv6 default deny firewall policy - correct bug in audit script (Ticket 11393)
Sep 17, 2020	3.1.0	UPDATE - Ensure IPv6 loopback traffic is configured - correct bug in audit script (Ticket 11394)
Sep 17, 2020	3.1.0	UPDATE - Ensure IPv6 outbound and established connections are configured - correct bug in audit script (Ticket 11395)
Sep 17, 2020	3.1.0	UPDATE - Ensure IPv6 firewall rules exist for all open ports- correct bug in audit script (Ticket 11396)
Sep 17, 2020	3.1.0	UPDATE - Ensure ip6tables rules are saved - correct bug in audit script (Ticket 11397)

Sep 17, 2020	3.1.0	UPDATE - Ensure ip6tables is enabled and running - correct bug in audit script (Ticket 11398)
Sep 17, 2020	3.1.0	UPDATE - Ensure source routed packets are not accepted - correct bug in audit script (Ticket 11389)
Sep 17, 2020	3.1.0	UPDATE - Ensure events that modify date and time information are collected - Corrected audit (Ticket 11399)
Sep 17, 2020	3.1.0	UPDATE - Ensure events that modify the system's network environment are collected - Corrected audit (Ticket 11400)
Sep 17, 2020	3.1.0	UPDATE - Ensure unsuccessful unauthorized file access attempts are collected - Corrected audit (Ticket 11401)
Sep 18, 2020	3.1.0	UPDATE - Ensure SSH LogLevel is appropriate - Updated audit (Ticket 11402)
Sep 18, 2020	3.1.0	UPDATE - Ensure SSH X11 forwarding is disabled - Updated audit (Ticket 11403)
Sep 18, 2020	3.1.0	UPDATE - Ensure SSH MaxAuthTries is set to 4 or less - Updated audit (Ticket 11404)
Sep 18, 2020	3.1.0	UPDATE - Ensure SSH IgnoreRhosts is enabled - Updated audit (Ticket 11405)
Sep 18, 2020	3.1.0	UPDATE - Ensure SSH HostbasedAuthentication is disabled - Updated audit (Ticket 11406)
Sep 18, 2020	3.1.0	UPDATE - Ensure SSH root login is disabled - Updated audit (Ticket 11407)
Sep 18, 2020	3.1.0	UPDATE - Ensure SSH PermitEmptyPasswords is disabled - Updated audit (Ticket 11408)
Sep 18, 2020	3.1.0	UPDATE - Ensure SSH PermitUserEnvironment is disabled - Updated audit (Ticket 11409)
Sep 18, 2020	3.1.0	UPDATE - Ensure only strong Ciphers are used - Updated audit (Ticket 11410)

Sep 18, 2020	3.1.0	UPDATE - Ensure only strong MAC algorithms are used - Updated audit (Ticket 11411)
Sep 18, 2020	3.1.0	UPDATE - Ensure only strong Key Exchange algorithms are used - Updated audit (Ticket 11412)
Sep 18, 2020	3.1.0	UPDATE - Ensure SSH Idle Timeout Interval is configured - Updated audit (Ticket 11413)
Sep 18, 2020	3.1.0	UPDATE - Ensure SSH LoginGraceTime is set to one minute or less - Updated audit (Ticket 11414)
Sep 18, 2020	3.1.0	UPDATE - Ensure SSH warning banner is configured - Updated audit (Ticket 11415)
Sep 18, 2020	3.1.0	UPDATE - Ensure SSH PAM is enabled - Updated audit (Ticket 11416)
Sep 18, 2020	3.1.0	UPDATE - Ensure SSH AllowTcpForwarding is disabled - Updated audit (Ticket 11417)
Sep 18, 2020	3.1.0	UPDATE - Ensure SSH MaxStartups is configured - Updated audit (Ticket 11418)
Sep 18, 2020	3.1.0	UPDATE - Ensure SSH MaxSessions is limited - Updated audit (Ticket 11419)
Sep 18, 2020	3.1.0	UPDATE - Ensure all users' home directories exist - Update AAC (Ticket 11420)
Sep 18, 2020	3.1.0	UPDATE - Ensure users' home directories permissions are 750 or more restrictive - Update Audit, Remediation, and AAC (Ticket 11421)
Sep 18, 2020	3.1.0	UPDATE - Ensure users own their home directories - Update audit, remediation, and AAC (Ticket 11422)
Sep 18, 2020	3.1.0	UPDATE - Ensure users' dot files are not group or world writable - Update audit, remediation, and AAC (Ticket 11423)
Sep 18, 2020	3.1.0	UPDATE - Ensure no users have .forward files - Update audit, remediation, and AAC (Ticket 11424)
Sep 18, 2020	3.1.0	UPDATE - Ensure users' .netrc Files are not group or world accessible - Update AAC (Ticket 11426)

Sep 18, 2020	3.1.0	UPDATE - Ensure no users have .netrc files - Update audit, remediation, and AAC (Ticket 11425)
Sep 18, 2020	3.1.0	UPDATE - Ensure no users have .rhosts files - Update audit, remediation, and AAC (Ticket 11427)
Sep 18, 2020	3.1.0	UPDATE - Ensure all groups in /etc/passwd exist in /etc/group - Update AAC (Ticket 11428)
Sep 21, 2020	3.1.0	Published
Mar 17, 2021	3.1.0	UPDATE - Ensure password reuse is limited (Ticket 12098)
Mar 19, 2021	3.1.0	Redundancy. (Ticket 8514)
Mar 22, 2021	3.1.0	UPDATE - Ensure separate partition exists for /var/log/audit (Ticket 11303)
Mar 22, 2021	3.1.0	UPDATE - Ensure separate partition exists for /home (Ticket 12456)
Mar 22, 2021	3.1.0	UPDATE - Configure sudo - Move Sub-section to section 5 (Ticket 12495)
Mar 22, 2021	3.1.0	CIS-CAT Pro v3.0.69 fails this auditing for different options (Ticket 11444)
Mar 22, 2021	3.1.0	UPDATE - Ensure filesystem integrity is regularly checked - regex in AAC too restrictive (Ticket 11467)
Mar 26, 2021	3.1.0	UPDATE - Ensure filesystem integrity is regularly checked (Ticket 12554)
Mar 26, 2021	3.1.0	Ensure prelink is disabled - Change title (Ticket 12555)
Mar 26, 2021	3.1.0	UPDATE - Ensure GDM is removed or login is configured (Ticket 12556)
Mar 26, 2021	3.1.0	DELETE - Ensure GDM is removed or login is configured (Ticket 12557)
Mar 26, 2021	3.1.0	2.1 inetd Services (Ticket 6523)
Mar 26, 2021	3.1.0	Audit for "2.2.1.4 Ensure chrony is configured" needs to be updated (Ticket 8554)

Mar 26, 2021	3.1.0	UPDATE - Special Purpose Services (Ticket 12558)
Mar 26, 2021	3.1.0	UPDATE - Ensure rsync is not installed or the rsyncd service is masked (Ticket 12560)
Mar 26, 2021	3.1.0	UPDATE - Ensure nfs-utils is not installed or the nfs-server service is masked (Ticket 12561)
Mar 26, 2021	3.1.0	UPDATE - Ensure rpcbind is not installed or the rpcbind services are masked (Ticket 12562)
Mar 29, 2021	3.1.0	UPDATE - Ensure shadow group is empty (Ticket 12578)
Apr 1, 2021	3.1.0	UPDATE - Configure SSH Server - Effects all recommendations in this section (Ticket 12610)
Apr 1, 2021	3.1.0	Audit for "5.2.11 Ensure SSH PermitUserEnvironment is disabled" needs to be updated (Ticket 8578)
Apr 1, 2021	3.1.0	5.2.11 Ensure only approved MAC algorithms are used (Ticket 6474)
Apr 1, 2021	3.1.0	UPDATE Ensure SSH access is limited - provide connection data to ``ssh -T (Ticket 11441)
Apr 1, 2021	3.1.0	Alternate way of enforcing wireless (Ticket 5932)
Apr 1, 2021	3.1.0	Alternate way of enforcing wireless (Ticket 5930)
Apr 1, 2021	3.1.0	Disabling vfat Kernel Module breaks boot on UEFI Systems (Ticket 5719)
Apr 1, 2021	3.1.0	list only removable media (Ticket 10627)
Apr 1, 2021	3.1.0	Propose automated test for removable media (Ticket 10446)
Apr 1, 2021	3.1.0	UPDATE - Disable Automounting (Ticket 12494)
Apr 1, 2021	3.1.0	1.4.2 Ensure bootloader password is set (Ticket 6341)
Apr 1, 2021	3.1.0	Check returning incorrect result (Ticket 5730)
Apr 1, 2021	3.1.0	UPDATE - Ensure bootloader password is set (Ticket 12547)
Apr 2, 2021	3.1.0	UPDATE - Ensure IP forwarding is disabled (Ticket 12613)

Apr 2, 2021	3.1.0	UPDATE - Ensure source routed packets are not accepted (Ticket 12614)
Apr 2, 2021	3.1.0	Duplicate artifact (Ticket 6815)
Apr 2, 2021	3.1.0	UPDATE - Ensure ICMP redirects are not accepted (Ticket 12615)
Apr 2, 2021	3.1.0	UPDATE - Ensure IPv6 router advertisements are not accepted (Ticket 12616)
Apr 2, 2021	3.1.0	UPDATE - Ensure password reuse is limited - update remediation (Ticket 12617)
Apr 2, 2021	3.1.0	UPDATE - Ensure default user shell timeout is configured - correct audit and AAC (Ticket 12619)
Apr 2, 2021	3.1.0	UPDATE - Ensure access to the su command is restricted (Ticket 12621)
Apr 2, 2021	3.1.0	UPDATE - System File Permissions - Update ordering (Ticket 12622)
Apr 2, 2021	3.1.0	UPDATE - User and Group Settings - Update recommendation order (Ticket 12623)
Apr 2, 2021	3.1.0	UPDATE - Ensure root PATH Integrity - update audit (Ticket 12601)
Apr 2, 2021	3.1.0	UPDATE - Ensure all users' home directories exist (Ticket 12602)
Apr 2, 2021	3.1.0	DELETE - Ensure users' .netrc Files are not group or world accessible (Ticket 12600)
Apr 2, 2021	3.1.0	UPDATE - Ensure no users have .rhosts files (Ticket 12609)
Apr 2, 2021	3.1.0	UPDATE - Ensure no users have .netrc files (Ticket 12599)
Apr 2, 2021	3.1.0	UPDATE - Ensure no users have .forward files (Ticket 12607)
Apr 2, 2021	3.1.0	UPDATE - Ensure users' dot files are not group or world writable (Ticket 12605)
Apr 2, 2021	3.1.0	UPDATE - Ensure users' home directories permissions are 750 or more restrictive (Ticket 12603)
Apr 2, 2021	3.1.0	UPDATE - Ensure users own their home directories (Ticket 12596)

Apr 2, 2021	3.1.0	UPDATE - User and Group Settings (Ticket 12549)
Apr 2, 2021	3.1.0	UPDATE - Partition option checks (Ticket 12481)
Apr 2, 2021	3.1.0	UPDATE - Filesystem Configuration (Ticket 12457)
Apr 2, 2021	3.1.0	UPDATE - Ensure ntp is configured (Ticket 12611)
Apr 2, 2021	3.1.0	UPDATE - Ensure ntp is configured (Ticket 12624)
Apr 2, 2021	3.1.0	UPDATE - Network Parameters (Host and Router) (Ticket 12626)
Apr 2, 2021	3.1.0	UPDATE - Network Parameters (Host Only) (Ticket 12625)
Apr 2, 2021	3.1.0	UPDATE - Network parameters (Ticket 11710)
Apr 2, 2021	3.1.0	#620 4.2 Configure Logging - add section about journalctl (Ticket 1897)
Apr 2, 2021	3.1.0	Audit of 6.2.11 failing when it should be passing (no .forward files exist) (Ticket 8191)
Apr 2, 2021	3.1.0	Failing on '1m' for rule Ensure 'LoginGraceTime' sshd config parameter less than or equal 60 (Ticket 9529)
Apr 2, 2021	3.1.0	New scored item "5.2.19 Ensure SSH MaxSessions is set to 4 or less" should set MaxSessions value to 10, not 4 (Ticket 9430)
Apr 2, 2021	3.1.0	CIS test 6.1.6 indicates that perms for /etc/passwd should be 600, but RH Linux indicates it should be readable by all users. (Ticket 6247)
Apr 5, 2021	3.1.0	Is the chmod for directories correct? (Ticket 12009)
Apr 5, 2021	3.1.0	UPDATE - Disable Automounting (Ticket 12546)
Apr 5, 2021	3.1.0	UPDATE - Ensure iptables-services package is not installed (Ticket 12629)
Apr 5, 2021	3.1.0	UPDATE - Ensure nftables is not installed or stopped and masked - Update title (Ticket 12630)
Apr 5, 2021	3.1.0	UPDATE - Ensure firewalld service enabled and running (Ticket 12631)
Apr 6, 2021	3.1.0	UPDATE - Ensure default zone is set (Ticket 12633)

Apr 6, 2021	3.1.0	UPDATE - Ensure unnecessary services and ports are not accepted (Ticket 12635)
Apr 6, 2021	3.1.0	systemctl status iptables will return Active: active (exited) and not active (active) (Ticket 11607)
Apr 6, 2021	3.1.0	4.1.3 may refer to wrong grub config file (Ticket 6407)
Apr 6, 2021	3.1.0	UPDATE - Ensure firewalld is installed (Ticket 12646)
Apr 6, 2021	3.1.0	UPDATE - Ensure network interfaces are assigned to appropriate zone (Ticket 12634)
Apr 6, 2021	3.1.0	UPDATE - Ensure nftables is installed (Ticket 12647)
Apr 6, 2021	3.1.0	UPDATE - Ensure firewalld is not installed or stopped and masked (Ticket 12636)
Apr 6, 2021	3.1.0	UPDATE - Ensure iptables-services package is not installed (Ticket 12639)
Apr 6, 2021	3.1.0	UPDATE - Ensure iptables are flushed - Change title (Ticket 12640)
Apr 6, 2021	3.1.0	UPDATE - Ensure loopback traffic is configured (Ticket 12439)
Apr 6, 2021	3.1.0	3.5.6 assessor v4 false positive : incorrect parsing (Ticket 9669)
Apr 6, 2021	3.1.0	UPDATE - Ensure iptables rules are saved (Ticket 12641)
Apr 6, 2021	3.1.0	UPDATE - Ensure iptables is enabled and running (Ticket 11905)
Apr 6, 2021	3.1.0	The Sub State Of Iptables Service (Ticket 11811)
Apr 6, 2021	3.1.0	UPDATE - Ensure IPv6 loopback traffic is configured (Ticket 12642)
Apr 6, 2021	3.1.0	UPDATE - Ensure IPv6 outbound and established connections are configured (Ticket 12643)
Apr 6, 2021	3.1.0	UPDATE - Ensure IPv6 firewall rules exist for all open ports (Ticket 12644)
Apr 6, 2021	3.1.0	UPDATE - Ensure IPv6 default deny firewall policy (Ticket 12645)
Apr 6, 2021	3.1.0	systemctl status ip6tables will return Active: active (exited) (Ticket 11608)

Apr 6, 2021	3.1.0	Inconsistencies Between Suggested Remediation and Automated Assessment Content (Ticket 11812)
Apr 6, 2021	3.1.0	4.1.14 Failing Audit (Ticket 6226)
Apr 6, 2021	3.1.0	Different methods for sending logs to remote host (Ticket 10965)
Apr 6, 2021	3.1.0	UPDATE - Ensure password hashing algorithm is SHA-512 (Ticket 12548)
Apr 7, 2021	3.1.0	Default umask logic error? - CentOS7 2.2.0.1 (Ticket 6431)
Apr 7, 2021	3.1.0	UPDATE - Ensure default user umask is configured (Ticket 12618)
Apr 7, 2021	3.1.0	6.1.9 fails audit with Assessor v3.0.46 (Ticket 6239)
Apr 7, 2021	3.1.0	6.2.7 AAC seems incorrect in RHEL benchmark v3.0.1 (Ticket 11909)
Apr 8, 2021	3.1.0	DELETE - Command Line Warning Banners (Ticket 12663)
Apr 8, 2021	3.1.0	UPDATE - Warning Banners (Ticket 12664)
Apr 9, 2021	3.1.0	1.4.1 Ensure permissions on bootloader config are configured (Ticket 6454)
Apr 9, 2021	3.1.0	Incorrect way of checking Bootloader path in EFI enabled hardware - CIS Red Hat Enterprise Linux 7 Benchmark v2.1.1 (Ticket 5292)
Apr 9, 2021	3.1.0	UPDATE - Ensure SELinux is not disabled in bootloader configuration (Ticket 12670)
Apr 9, 2021	3.1.0	UPDATE - Ensure auditing for processes that start prior to auditd is enabled (Ticket 12671)
Apr 9, 2021	3.1.0	UPDATE - Ensure audit_backlog_limit is sufficient (Ticket 12672)
Apr 9, 2021	3.1.0	Logic to check if gpgcheck is enabled (Ticket 11895)
Apr 9, 2021	3.1.0	1.2.3.1 pattern match fails if spaces are present (Ticket 11706)
Apr 9, 2021	3.1.0	Regex for gpgcheck isn't quite right (Ticket 11577)
Apr 9, 2021	3.1.0	v3.0.0 Audit failures due to use of UEFI, can eval scripts actually eval UEFI correctly (Ticket 11012)

Apr 30, 2021	3.1.0	PUBLISHED
May 21, 2021	3.1.1	Mapped CIS Controls v8
May 21, 2021	3.1.1	PUBLISHED
Aug 25, 2021	3.1.2	UPDATE - nix_bootloader_password_chk.sh (Ticket 13588)
Aug 25, 2021	3.1.2	UPDATE - nix_grub_option_exist_chk.sh (Ticket 13589)
Aug 25, 2021	3.1.2	UPDATE - nix_grub_option_not_exist_chk.sh (Ticket 13590)
Aug 25, 2021	3.1.2	CentOS7 3.1.1 Duplicate Grub Directories - (nix_grub_does_not_exist.sh) (Ticket 13245)
Aug 25, 2021	3.1.2	UPDATE - sce script ipv6_disabled_vs.sh (Ticket 13587)
Aug 31, 2021	3.1.2	PUBLISHED